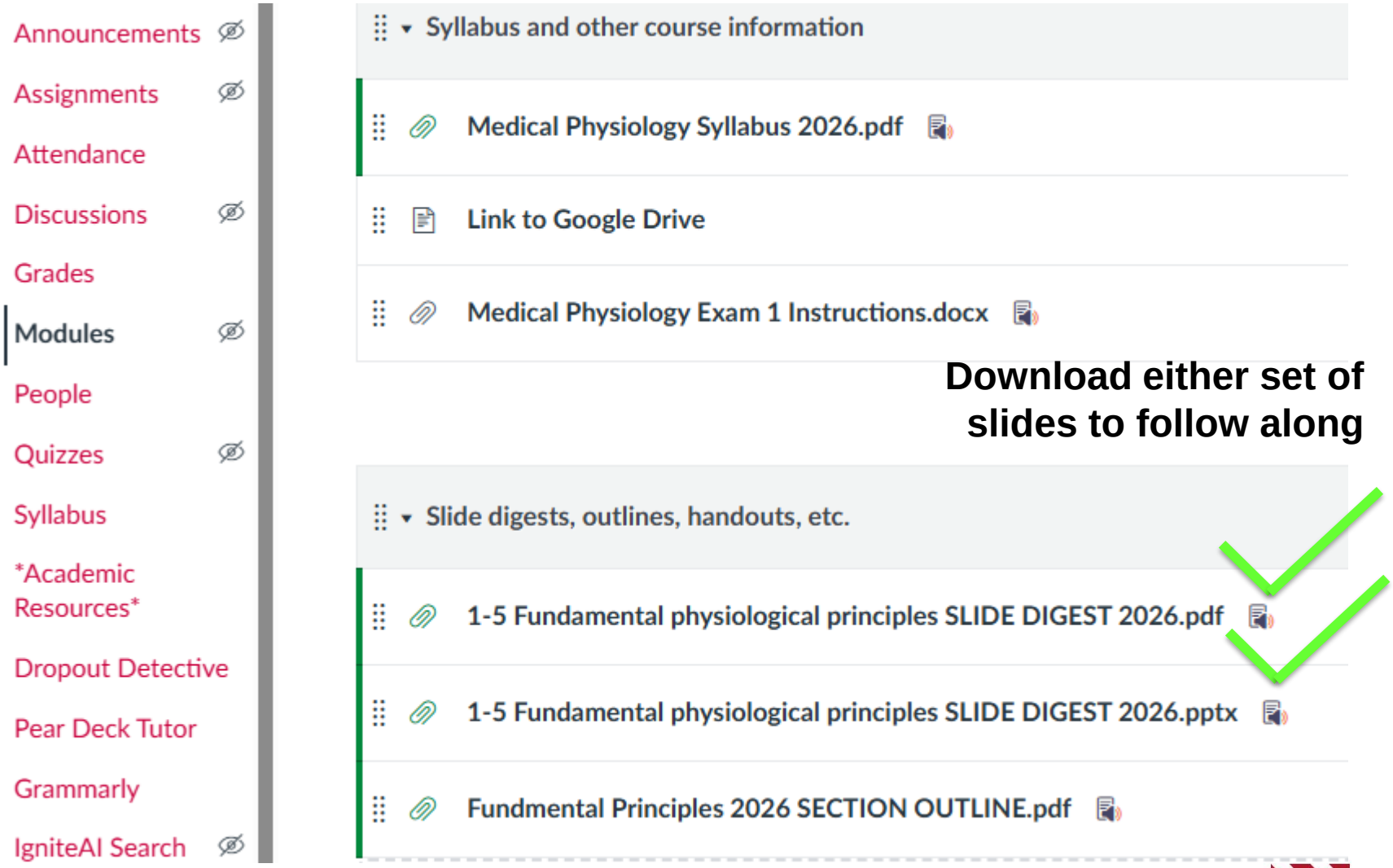
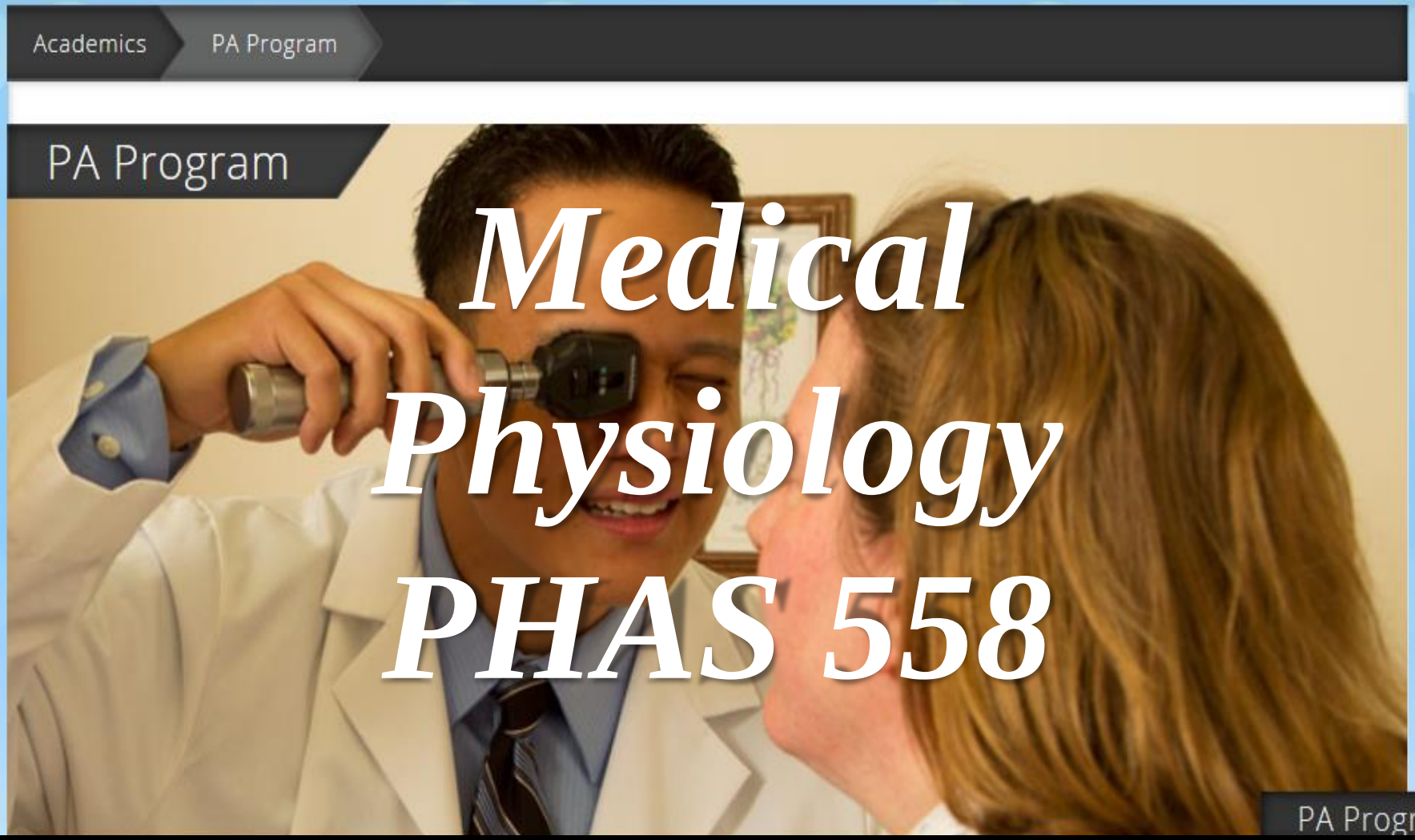
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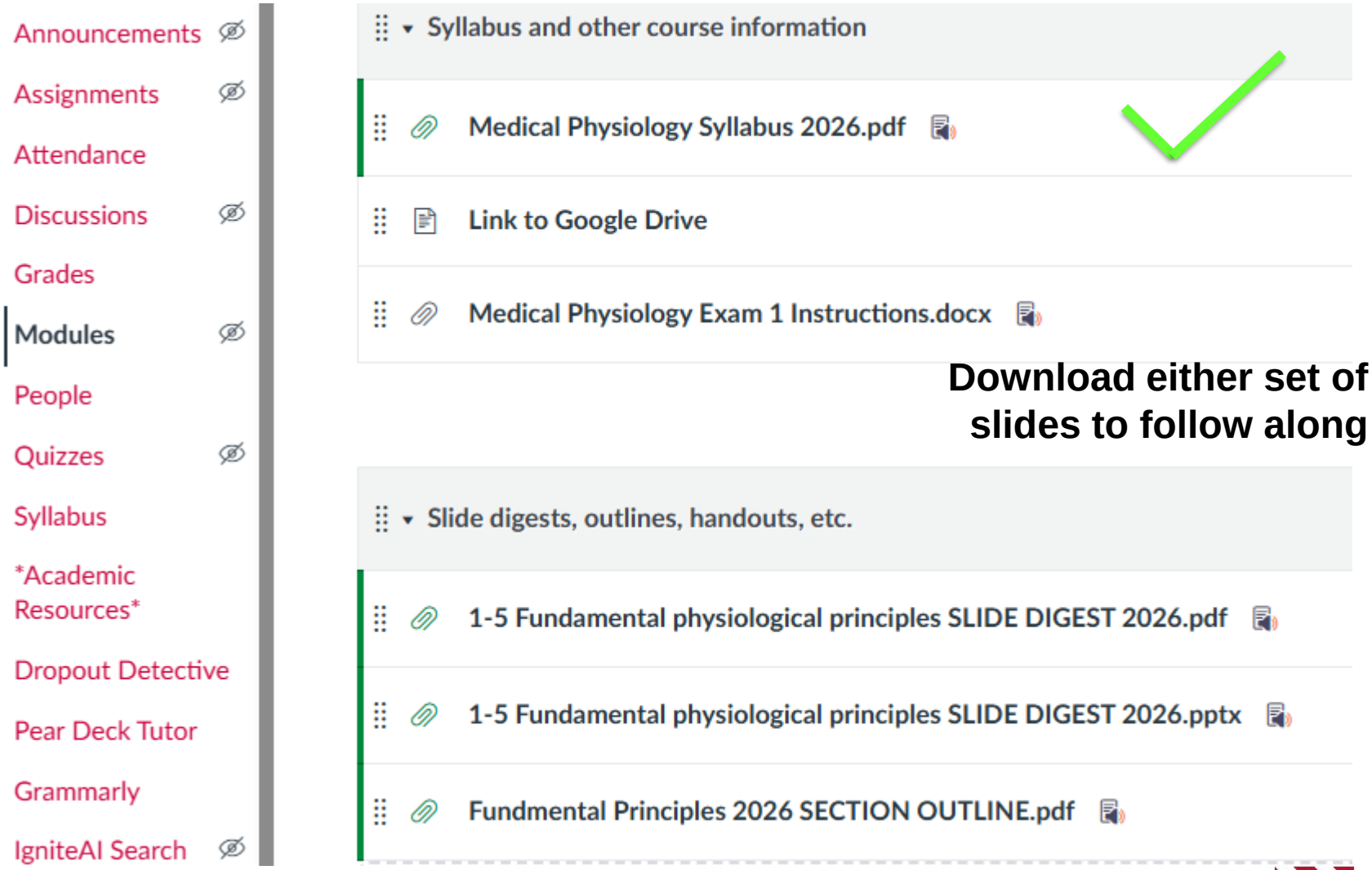
Larry D. Crouch, PhD
lcrouch@unmc.edu (please use this email and put Union in subject line)
402-560-5472 (text and cell)



Larry D. Crouch, PhD
Department of Oral Biology
UNMC College of Dentistry
Lcrouch@unmc.edu (Subject: Union)

University of Nebraska Medical Center

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UNION ADVENTIST UNIVERSITY | PHYSICIAN ASSISTANT PROGRAM

UNION ADVENTIST UNIVERSITY
PHYSICIAN ASSISTANT PROGRAM

PHAS 558 Medical Physiology
Fall 2026

Instructor Larry Crouch, Ph.D.

Office location: PA building (LS), Room 204
Office hours: Monday, Wednesday, Friday, 7:00-7:30 am, or by appointment
Office phone: (402) 486-2527
Cell phone: (402) 560-5472 (voice and text)
Email: lcrouch@unmc.edu
Fax: (402) 486-2559

Class Location
LS 214

Class Time
Mondays, Wednesdays, Fridays, 7:30-8:50 am

Credit
5 semester hours

Course Goals
The goal of this course is to develop an advanced, integrative understanding of human physiological systems and their application to clinical practice.

Course Description
This course is designed to provide students with an advanced understanding of physiology at the graduate level. Instruction will provide an in-depth study of the physiology of the cardiovascular, respiratory, renal, gastrointestinal, endocrine, musculoskeletal, nervous, integumentary, and reproductive systems, with emphasis on concepts applicable to the practice of medicine. In-class presentations, group exercises, online audiovisual units, and independent study modules provide instructional learning opportunities.

Program Learning Outcomes
1. Demonstrate interpersonal communication skills that result in effective information exchange with patients, physicians, professional associates, and other individuals within the healthcare system to promote optimal patient-centered care.
2. Utilize information technology to critically analyze medical literature to make evidence-based medical decisions that maximize diagnostic and therapeutic effectiveness.
3. Apply medical knowledge to evaluate and treat patients across the lifespan by

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5. Provide counseling, patient and family education, interventions, and appropriate referral for promotion, restoration, and maintenance of optimal levels of health for patients.
6. Exhibit ethical behavior and professional conduct expected of the PA in a medical practice setting while acknowledging professional and personal limitations.
7. Demonstrate the delivery of healthcare to disadvantaged and underserved populations.
8. Commit to professional and public service by carrying out compassionate and wholistic medical practice in light of a Christian faith and biblical perspective.
9. Demonstrate proficient clinical reasoning and problem-solving abilities to form an accurate diagnosis and appropriate patient management.
10. Integrate essential aspects of value-based health care and apply this understanding to the delivery of safe and quality patient care.

Course Learning Outcomes*

On completion of this course, students will be able to:

1. Demonstrate foundational comprehension of major organ systems and apply normal physiologic processes to establish a strong basis for clinical reasoning and patient care. (B2.02b) (PLO 3)
2. Differentiate and correlate normal and altered physiologic states to support clinical reasoning and disease analysis, building a foundational understanding for future application in patient evaluation and care. . (B2.02b-c) (PLO 3)

*When applicable, course learning outcomes are followed by the relevant ARC-PA *Standard* (e.g., "B2.09"), and the relevant program learning outcome (e.g., "PLO 1").

Instructional Objectives

See Appendix 2 for specific instructional objectives.

Teaching Strategies

- Lecture
- Independent study modules
- Group exercises
- Online audiovisual units

Course Texts

Required:

McCance, K. L., Huether, S. E., & Brashers, V. L. (2022). *Pathophysiology: The biologic basis for disease in adults and children* (9th ed.). Elsevier. ISBN-13: 978-0323789875

Required: (choose one):

Widmaier, E. P., Raff, H., & Strang, K. T. (2022). *Vander's human physiology: The - mechanisms of body function* (16th ed.). McGraw Hill. ISBN-13: 978-1265131814 - paperback version

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Widmaier, E. P., Raff, H., & Strang, K. T. (2022). *Vander's human physiology: The mechanisms of body function* (16th ed.). McGraw Hill. ISBN-13: 978-1264451128 - looseleaf version 1.4.2022

NOTE: Both the physiology and the pathophysiology textbooks will also be required for the first-year spring semester Clinical Pathophysiology course.

Course Requirements

1. Class attendance and participation are mandatory. Students are expected to arrive on time and actively engage in all class sessions.
2. Cooperative participation by each student is mandatory during group learning activities.
3. Complete all independent study modules, exams, assignments, and quizzes.

Attendance

TARDINESS Class attendance is required. Students are expected to arrive on time, if not early, for all scheduled didactic sessions and clinical activities. Tardiness is considered unprofessional behavior and may be reflected in evaluations during both didactic and clinical training. In the event of an anticipated or unanticipated tardiness, didactic students must notify the course instructor and academic director, and clinical students must notify the clinical director and preceptor as soon as possible. Repeated tardiness may result in grade lowering, course failure, and/or other disciplinary action at the discretion of the course director, program director, and/or AAPC.

EXCUSED ABSENTEEISM Serious illness of the student or critical illness or death of an immediate family member constitutes an excused absence. Students must notify the course instructor by email as soon as possible, including the date and reason for the absence. Documentation (e.g., a note from a healthcare provider) may be required. While the response to any absence is at the discretion of the instructor, reasonable efforts will be made to provide equivalent instructional opportunities when feasible. Students are responsible for obtaining missed material, including notes or recordings, when available, at the discretion of the instructor.

UNEXCUSED ABSENTEEISM Absences for reasons other than those outlined above—including, but not limited to, vacations, weddings, family events, elective appointments, or conferences—will be considered unexcused. More than two (2) unexcused absences per course, per semester will result in a reduction of the final course grade by one-third of a letter grade (e.g., A- to B+). More than three (3) unexcused absences may result in a full letter grade reduction or withdrawal from the course. Additional disciplinary action may be taken at the discretion of course instructor, program director, and/or AAPC.

Evaluation and Grading

Grades will be determined from the assessment components listed below.

Exams

There will be three in-class unit examinations administered throughout the

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semester and one comprehensive final examination administered at the end of the semester. Each unit examination is worth 100 points, with 60 minutes allotted for completion. The comprehensive final examination is worth 200 points, with 120 minutes allotted for completion. Unit examinations are cumulative; therefore, material from previous units may be included on subsequent unit and final examinations. For example, Unit I content may appear on the Unit II, Unit III, and final examinations.

Quizzes

Throughout the semester, there will be eleven quizzes covering material from preceding lectures and assigned readings. Each quiz is worth 5 points, with the ten highest quiz scores counted toward the final quiz grade (50 points maximum). The remaining quiz score will be added as bonus points to the student's total course points earned. One quiz will be completed in class as a group activity with a 15-minute time limit. The remaining ten quizzes will be completed online outside of class and will be available for a minimum of four (4) days. Quiz dates listed in the course schedule are tentative and subject to change based on course needs.

Written Assignments

There will be four writing assignments throughout the course. Three assignments will consist of item analysis papers completed following Unit Exams I, II, and III and are worth 10 points each for a total of 30 points (See Appendix 3). The fourth assignment will consist of a review of an online report and the utilization of online resources related to medical conditions and diseases and is worth 20 points (See Appendix 4). The total possible points for all written assignments combined is 50 points.

Independent Study Modules

Independent study modules must be completed and will be assessed on unit and final examinations. Modules may include a combination of videos, PowerPoint presentations, outlines, and PDF materials. Refer to the course schedule for assigned modules and due dates.

Points Available

Description	Points Possible
Three unit exams @100 points each	300
Quizzes @5 points each	50
Assignments (3 @10 points each, 1 @20 pts)	50
Final exam	200

600 points

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Grade Scale

A	93-100	B+	87-89	C+	77-79	D+	67-69	F	<60
A-	90-92	B	83-86	C	73-76	D	63-66	FA*	<60
		B-	80-82	C-	70-72	D-	60-62		

*A grade of "FA" is a federal requirement. If the student receives an "F" due to not attending class, as well as not completing or failing work, the grade "FA" is assigned. This has a dire impact on a student's financial aid and may impact their ability to continue their education.

Guest Lecturer Assessment

Due to the importance of student feedback, the program requests that students complete brief evaluations of all guest speakers. These will be made available following the relevant class session and will be completed anonymously online.

Course Progression

An overall grade of 73% or higher (a letter grade of C or above) must be achieved in order to pass this course.

Make-up Work/Exams

All late work will have 5% per school day deducted (up to 50%).

Rounding

For the final letter grade, the final course grade is rounded up to the next percentage or whole number if above 0.5.

Course Policies

Below is a list of specific policies that pertain to the PA Program and this course. Many have been excerpted from the Union Adventist University Graduate Academic Bulletin and the Union Adventist University PA Student Handbook; links are provided to both documents. Inclusion in the syllabus is for emphasis only.
Union Adventist University Graduate Academic Bulletin 2025-2027, vol. 1 (Bulletin)
Union Adventist University PA Student Handbook, 2026-2027, vol. 33 (PASH)

- **Integrity of Scholarship/Academic Dishonesty** (Bulletin, page 37; PASH, section 9.13.1)
  - Note: The PA program academic integrity policy is in alignment and compliance with the college's policy. However, as a graduate program, the PA program exercises an internal process for the management of issues involving academic integrity.
- **Professional Behavior Expectations** (PASH, section 9.1)
- **Electronic Devices and Social Media** (PASH, section 9.3)

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- **Policy Review for Testing, Exemplify, and Examination Review** (PASH, section 9.6)
- **Policy Review for Academic Standing** (PASH, section 9.11)
- **Policy Review for Attendance (includes Unexcused Absenteeism** (PASH, section 9.12)
- **Accessibility Accommodations for Students with Disabilities** (Bulletin, page 27)
- **Final Examination Schedule and Exceptions** (Bulletin, page 35)
- **Prep Days for Finals** (Bulletin, pages 35-36) *Note: Observation of "prep days" or "dead days" is an undergraduate policy, which is not observed in the graduate program.*
- **Generative AI** (Bulletin, page 36 and see below)
- **Intellectual Property** (PASH, section 9.13)

Disability Information and Access (DIA)

Disability Information and Access (DIA) – Your access is our focus. Teaching, Learning, Collaborating (TLC) is an initiative through Disability Information and Access (DIA) to create partnerships for an accessible university environment. An important step in this partnership includes providing access for students with disabilities through accommodations or strategies. Students who would like to inquire more about establishing accommodations for the classroom, campus, or experiential learning (e.g., internships), please contact DIA Director Debbie Forshee-Sweeney at 402.486.2600, ext. 2080 or email debbie.forsheesweeney@uau.edu. Find out more about accessibility services on Union's website.

Generative AI

Union Adventist University recognizes the promise and challenge of generative artificial intelligence (AI), machines and programs that learn from input and are capable of producing text, image, audio, code, and more. Examples include ChatGPT, Copilot, Gemini, Open AI Codex, DALL-E, Runway, etc.*

Generative AI is an emerging and swiftly evolving technology. At Union, we aim to support students' generative AI literacy while maintaining the highest standards of ethical conduct. Syllabi provide course-specific policies for use or non-use, varying from discipline to discipline and even assignment to assignment. However, students may encounter guidelines such as the following:

· Use prohibited—no generative AI may be used in the course. Students are expected to complete all assignments without significant support from others, including AI, except those technologies supplied through formal accommodations from the director of Disability Information and Access.

Unauthorized use of generative AI violates the University's Academic Integrity Policy. See that policy for an explanation of violation levels and consequences.

* Note that the names of software and platforms may change with time. Those listed above are merely examples. The policy applies to all forms of generative AI.

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APPENDIX 1
PHAS 558 Medical Physiology
Fall 2026 Course Schedule

Wk.	Mo.	Day	Topic	Quiz
1	Aug Aug Aug	24 26 28	Syllabus/Fundamental Principles Fundamental Principles Fundamental Principles **Independent Study:** -Fundamentals: Decide NOW to SUCCEED -Fundamentals: Diffusion and transporter clinical correlations	
2	Aug Sept Sept	31 2 4	Fundamental Principles Fundamental Principles Fundamental Principles **Independent Study:** -Fundamentals: G-protein mediated cAMP signal transduction -Fundamentals: Eicosanoid metabolism	Quiz 1*
3	Sept Sept Sept	7 9 11	**NO CLASS (Labor Day)** Fundamental Principles Fundamental Principles	Quiz 2*
4	Sept Sept Sept	14 16 18	Fundamental Principles Fundamental Principles **EXAM #1**	
5	Sept Sept Sept	21 23 25	Neurophysiology Neurophysiology Neurophysiology **Independent Study:** -Neurophysiology: Vision, Taste and Olfaction -Neurophysiology: Audition & Equilibrium	Quiz 3*
6	Sept Sept Oct	28 30 2	Neurophysiology Autonomic Nervous System Skeletal Muscle Contraction	Quiz 4*
7	Oct Oct Oct	5 7 9	Cardiac & Smooth Muscle Cont. Cardiovascular Physiology Cardiovascular Physiology	Quiz 5*

The syllabus is REQUIRED READING

			Independent Study: -Cardiovascular: Blood and its components -Cardiovascular: PREVENT Online Calculator	
8	Oct Oct Oct	12 14 16	Cardiovascular Physiology **EXAM #2** **Fall break (no classes)**	
9	Oct Oct Oct	19 21 23	Cardiovascular Physiology Cardiovascular Physiology Cardiovascular Physiology **Independent Study:** -Cardiovascular: Lipoprotein metabolism -Cardiovascular: Integumentary System Physiology	Quiz 6*
10	Oct Oct Oct	26 28 30	Respiratory Physiology Respiratory Physiology Respiratory Physiology	Quiz 7*
11	Nov Nov Nov	2 4 6	Respiratory Physiology Respiratory Physiology Respiratory Physiology	Quiz 8*
12	Nov Nov Nov	9 11 13	Renal Physiology Renal Physiology Renal Physiology **Independent Study:** -Renal: Renal clearance and glucose handling -Renal: Acid-base homeostasis	Quiz 9*
13	Nov Nov Nov	16 18 20	Glycemic Control and Diabetes Glycemic Control and Diabetes **EXAM #3**	
14	Nov Nov Nov	23 25 27	**NO CLASS (Thanksgiving break)** **NO CLASS (Thanksgiving break)** **NO CLASS (Thanksgiving break)**	
15	Nov Dec Dec	30 2 4	Gastrointestinal (GI) Physiology GI and Endocrine Physiology Endocrine Physiology	Quiz 10*
16	Dec Dec Dec	7 9 11	Endocrine Physiology Reproductive Physiology Reproductive Physiology	Quiz 11*
17	Dec	14	**FINAL EXAM 7:30-9:30**	

The syllabus is REQUIRED READING

*Quiz dates are tentative to allow better assessment of specific concept areas; actual quiz dates will be announced at class meetings prior to quizzes.

APPENDIX 2
PHAS 558 Medical Physiology

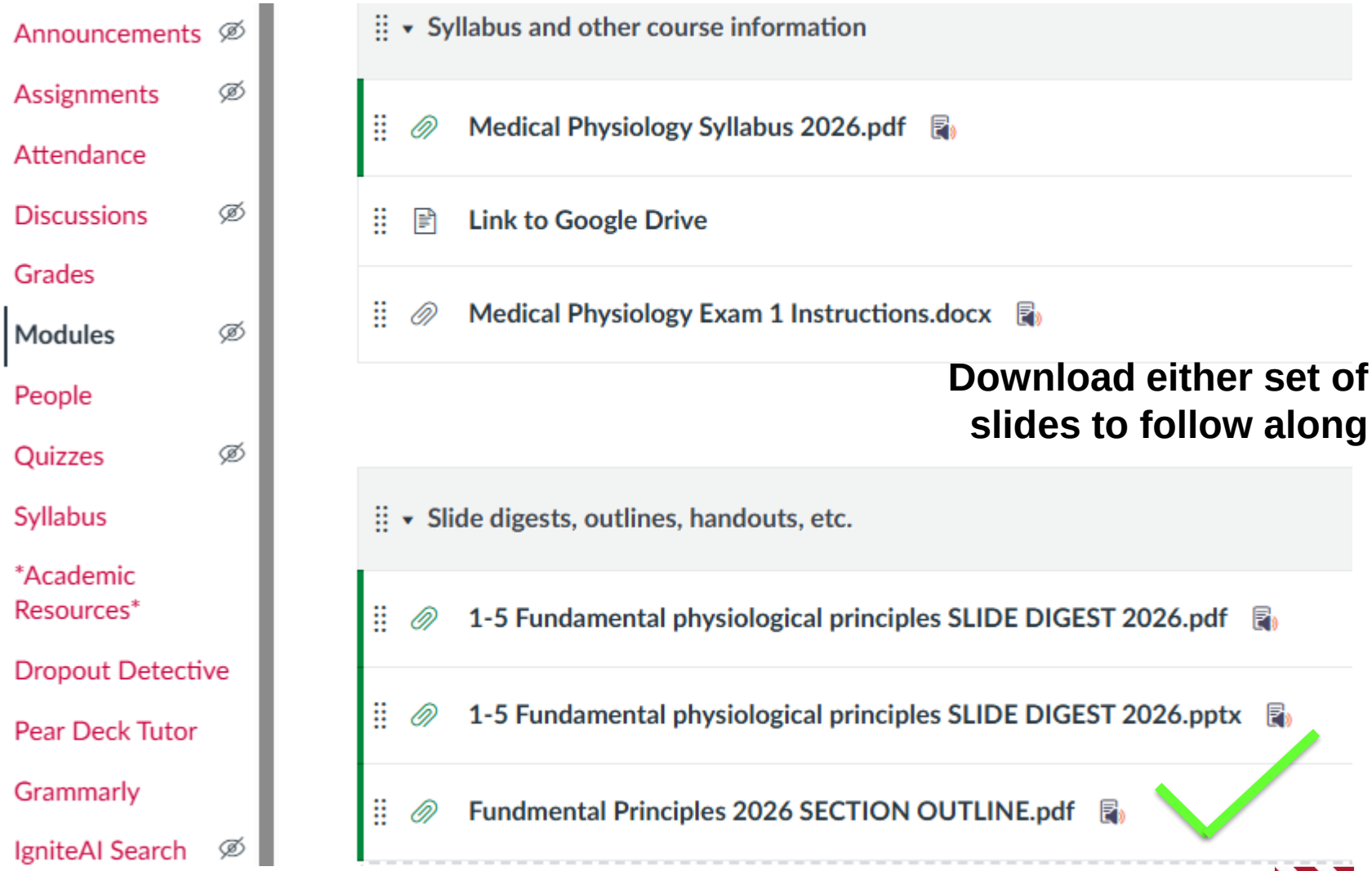
Topic 01: Introduction to Physiology and Homeostasis

Following this section, the successful learner will be able to:

1. Define and differentiate physiology and pathophysiology
2. Describe how graphs and equations function as summaries of physiological relationships
3. Describe how to identify and utilize the individual elements of specific graphs and equations to understand the contribution of the elements to the physiological relationship
4. Differentiate between direct and inverse relationships
5. Contrast and compare, by noting similarities and differences, between the functions, the limitations and the specific attributes associated with nervous, humeral and cell-to-cell communication with regard to physiological regulation
6. Differentiate between telocrine, paracrine, autocrine, juxtacrine and gap junction communication and identify the functions, advantages and limitations of each
7. Define homeostasis and the internal environment, and describe the relationship of homeostasis to the internal environment
8. Describe the fluid compartments of the body and the contribution of each to the body's total body water content
9. Calculate estimates of each fluid compartment given bodyweight and/or clinical fluid compartment information
10. Calculate specific medication dosing given bodyweight and/or clinical fluid compartment information
11. Provide multiple examples of homeostatically regulated factors, and explain how the regulation of the factors impact protein and bodily function
12. Discuss the function of each element of a sensor-effector-negative feedback loop, while also identifying the afferent and efferent limbs of the loop
13. Discuss the meaning and necessity of the bidirectional control of a homeostatically regulated factor in maintaining homeostasis
14. Compare positive and negative feedback, and discuss the extent for which each is utilized in the body
15. Graph the normal range for fasting plasma glucose, along with plasma concentration ranges for hypoglycemia and hyperglycemia
16. Explain differences between typical meal-associated hyperglycemia and chronic hyperglycemia
17. Discuss pathological consequences of for hypoglycemia and chronic hyperglycemia
18. Diagram and identify the specific tissue elements and hormonal mediators utilized for the bidirectional, sensor-effect-negative feedback control of plasma glucose
19. Provide an explanation and examples of physiological reserve
20. Describe how loss of physiological reserve or homeostatic control leads to signs and symptoms of disease processes; provide examples

The syllabus is REQUIRED READING

Go to CANVAS page for Med Phys course



I. INTRODUCTION TO PHYSIOLOGY AND HOMEOSTASIS

A. Physiology

1. The study of an organism's function (molecules to cells, tissues, organs, systems & organism).
2. The study of intercellular and intracellular communication and control.

We have more than 40,000,000,000,000 cells in our body, all communicating, interacting and controlling the functions of one another

(1) Nervous system communication (via nerves)
(2) Humoral communication (via the bloodstream and extracellular fluid)
(3) Cell-to-cell communication (via adhesion molecules (e.g., integrins) and gap junctions)

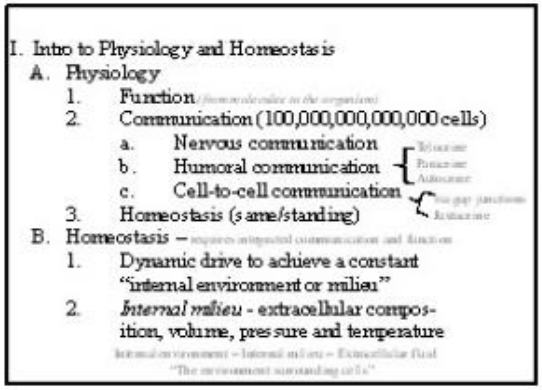
3. The study of the body's drive to achieve **homeostasis**.
4. All the above

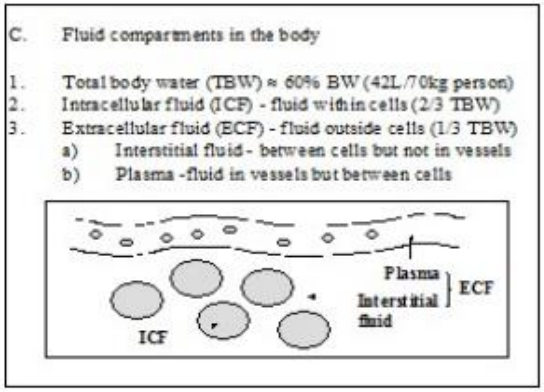
B. Homeostasis

1. The dynamic drive to achieve and maintain a constant internal environment or "internal milieu"
2. "**Internal milieu**" = extracellular fluid composition, volume, pressure and temperature
3. Requires finely integrated communication and function

C. **Fluid compartments** in the body

1. Total body water (**TBW**) ≈ 60% of bodyweight (42 L in 70 kg person) (Impact of aging, obesity)
2. Intracellular fluid (**ICF**) - fluid within cells (2/3 of TBW or 28 L)
3. Extracellular fluid (**ECF**) – fluid outside of cells (1/3 of TBW or 14 L)
   a) **Interstitial** fluid – between cells but not in vasculature (11 L)
   b) Plasma - fluid between cells within the vasculature (3 L)





D. Homeostatically regulated factors in ECF (in no particular order)

1. pH, temperature, Na^+, Ca^{++}, K^+, PO_4^-, CO_2, O_2, osmolarity, ECF pressure and volume, etc.
2. There are literally hundreds of factors requiring homeostatic control for optimal function

E. Why/how are the homeostatically controlled factors regulated?

1. Many physiological processes require several controlled factors to be regulated within very narrow ranges
2. Recall from biochemistry: *Enzymes****, as a general rule, function optimally only within narrow ranges of pH, temperature, salt concentration, etc.: these **set-points** are often specific for each enzyme

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- When Hospital Medicine Becomes Air Traffic Control
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The Gut Remembers: One Course, Years of Echo

Medscape Medical News | August 21, 2026 |

A single antibiotic course may alter gut diversity for years. Find out which drugs left the biggest — and most surprising — traces.

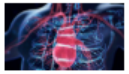
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Medscape Medical News | August 21, 2026 |

Inflammatory Hypothesis Proponents Undaunted by ZEUS Flop



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F. Perry Wilson, MD, MSCE, Associate Professor, Department of Medicine; Interim Director, Program of Applied Translational Research, Yale School of Medicine, New Haven, Connecticut

Pregnant Women Are Ignoring the 'Don't Drink' Message

New research shows that many women, despite knowing better, aren't eschewing alcohol during pregnancy.

Medscape, July 27, 2026

AHA Finally Comes Out in Favor of Coffee

For decades, doctors have advised patients with heart conditions to avoid caffeine. The tide, much to the relief of coffee lovers everywhere, has shifted.

Medscape, July 20, 2026 20

STEMI Patients Are Awash in Microplastics

Notwithstanding definitive proof of cause and effect, micro- and nanoplastics could have a major role in STEMI.

Medscape, July 14, 2026

Should GLP-1 Drugs Be Over the Counter?

Dr F. Perry Wilson comments on research showing that it's shockingly easy to get GLP-1 drugs online without obtaining a provider's prescription first.

Medscape, July 06, 2026 4

Can AI Save Doctors From Missed Diagnoses?

For AI to be useful, we must also tackle the problem of alert fatigue.

Medscape, June 30, 2026 6

The COVID Vaccine Study the CDC Doesn't Want You to See

Higher-ups at the CDC blocked publication of a COVID vaccine effectiveness study, citing methodologic weaknesses. Now the study has been published for all to read.

Medscape, June 23, 2026 50



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Year of Profess School

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Oral Biology

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Decide Now to Succeed

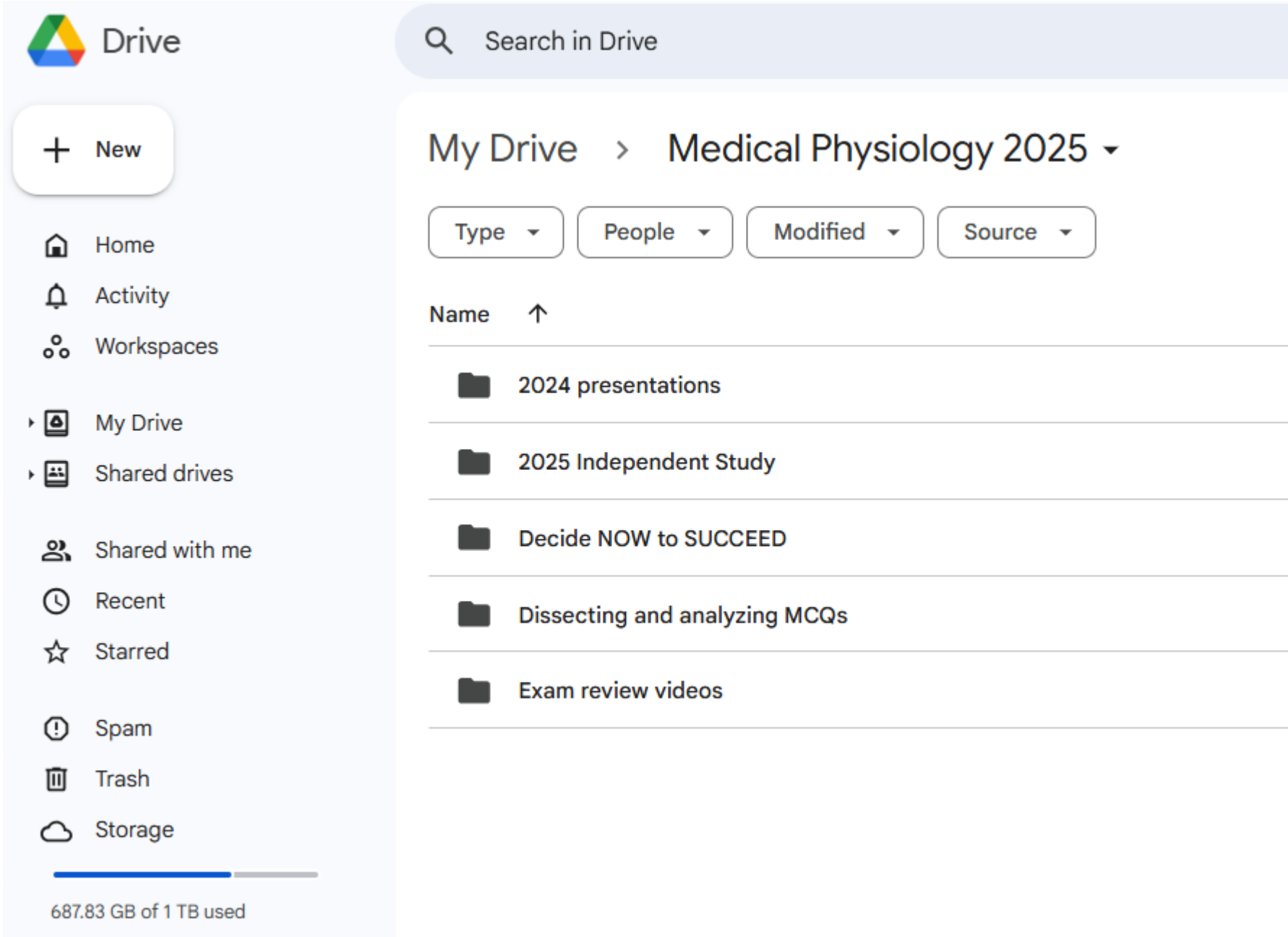
(Observations of successful and less successful first year professional students)



Decide NOW to Succeed

https://www.youtube.com/watch?v=HXr9n642nA0&feature=youtu.be%E2%80%AC

Google Drive page for Med Phys course



Decide Now to Succeed

(Observations of successful and less successful first year professional students)

This presentation will focus primarily on those things students have taught me in how you can have a SUCCESSFUL and PRODUCTIVE:

First week

First month

First semester

Developing a solid foundation

Maintaining momentum

Decide Now to Succeed

(Observations of successful and less successful first year professional students)

1. What is the one thing that surprised you most about your first semester of health care professions school? Please explain.
2. Describe one thing you did that helped you greatly during your first semester. Please explain.
3. Knowing what you know now, describe one thing you would do differently if you were to repeat the first semester. Please explain.
4. Offer your two best pieces of advice to future students on how to be more successful and less stressed during the first semester of health care professions school. Please explain.
5. What will you do differently during your first day, first week and first month of second semester than you did during your first semester here? Please explain.

Decide Now to Succeed

(Observations of successful and less successful first year professional students)

The need for group study. In undergrad I hardly utilized group studying just because more often than not they weren't as productive or efficient as they could have been. I have discovered this past semester, however, that it's group study or die. Independent studying is essential and should not be replaced by group studying, however, other people will often offer bits of information that you didn't catch, correct a concept you learned incorrectly (happened to me a lot), or even just hold one another accountable.

One thing that helped greatly was finding good study groups for classes. This helped a lot because you are able to bounce questions off each other to figure out questions you don't know. It also helps because they may see something that you would have missed.

I really enjoy having a small class because I am able to lean on them when times get hard. We are all in it together and are happy to help one another. We have become such a close-knit group and I feel comfortable asking any of them for help when I am struggling with a class. Because the entire school is composed of small classes, I have found myself asking older students for help when it comes to dental techniques.

Decide Now to Succeed

(Observations of successful and less successful first year professional students)

I was also very shocked at how much the professors really cared about our success and our wellbeing in general. They are more than happy to sit down with you to talk about school or any other challenges that you are facing with your first semester. This was something that I was not used to following my undergraduate program.

Ask questions!!! Don't be shy like I was in the beginning. If you are struggling, just ASK questions!

I am going to really try to go to professors more when I am needing clarification on a topic. They are all here to help us, and I know I need to use them as resources more.

I was surprised by just how approachable the professors were and willing to go above and beyond for their student's success. I would tell everyone to utilize their professors if they are struggling in the class or are having trouble understanding a topic in lecture or can't get something to work properly in lab. Every time myself or a classmate had issues in either lecture or lab, a professor was greatly willing to help us workout whatever problem we were having. And this can save you valuable time instead of trying to figure things out when you have no clue what is going on.

Decide Now to Succeed

(Observations of successful and less successful first year professional students)

- "Doc, do you have a minute? I need to talk with you about something …. I've discovered that I've never really learned how to …. study."
- Hasn't yet learned how to be an extremely EFFICIENT and EFFECTIVE learner
  - *Realization #1:* "This material isn't conceptually any more for difficult than my undergraduate courses."
  - *Realization #2:* "I didn't notice at first but there is *VOLUME, VOLUME, VOLUME!*"
  - Realization #3: "I'm expected to comprehend, organize, apply and retain *ALL* of this information."

"Professional school starts on Day One, and that is when you need to start studying."
Numerous health care professional students the summer following 1st year

- You have a decision. Will you:
  - Binge and purge your first semester information?
  - Actually learn and retain your course information by learning now how to learn efficiently and effectively?

Decide Now to Succeed

(Observations of successful and less successful first year professional students)

- **Learn now how to learn; learn now how to study**
  - **Be a humble, open learner (learn from everyone)**
  - **Develop several study groups & study partners**

    There are things that you know
    There are things that you know that you don't know
    There are things that you don't know that you don't know*

    ****Can only be resolved by external feedback****
  - **Ask your professors "good" questions, early and often**

    <u>Always</u> bring a friend or two with you
  - **Be an "active" learner**

    Don't just think it out

    Write it out, draw it out, talk it out, walk it out; "lecture to yourself"
  - *****Teach to learn*****

Learn to teach in two languages: Scientific/Medical and Living Room language

Decide Now to Succeed

(Observations of successful and less successful first year professional students)

- **Conquer the VOLUME, VOLUME, VOLUME and retain the information**
- **Study/review DAILY from DAY ONE**

"Professional school starts on Day One, and that is when you need to start studying."
Numerous health care professional students the summer following 1st year

- **Study for maximum comprehension and retention**

  Brief, rapid, DAILY reviews ("Give me fifteen!") NEUROPLASTICITY
  Constantly write notes and questions to your future self
  Color-code your notes for yourself in your books, handouts, computer
  Work hard to understand underlying concepts and principles rather than just memorizing facts (diagrams, equations, graphs, etc.)
  Carefully manage your information inflow channels while learning (including friends and family)

- **Manage your sleep and exercise to your cognitive advantage**

Decide Now to Succeed

(Observations of successful and less successful first year professional students)

- Some serious considerations regarding your mental health
- Do NOT hesitate to ask for help; ask for help sooner rather than later
- In the health professions, asking for help is generally regarded as a sign of maturity, self-confidence and strength of personality rather than a weakness

Decide Now to Succeed
(Observations of successful and less successful first year professional students)

- **Decide now to:**
  - **Become a highly efficient and effective learner**
  - **Review early, review often; conquer the VOLUME**
  - **NEVER, NEVER fall behind**
  - **Learn actively**
    - **Write it out, draw it out, talk it out, walk it out and TEACH**
  - **Identify "What don't I know that I don't know" with colleagues**
  - **Ask for help sooner rather than later**
  - **Sleep well**
  - **Exercise well**
  - **Eat well**
  - **Find joy in working hard, learning and succeeding with your new colleagues**

Decide Now to Succeed

(Observations of successful and less successful first year professional students)

1. What is the one thing that surprised you most about your first semester of health care professions school? Please explain. **VOLUME; speed in which information is presented; professors are extremely supportive**
2. Describe one thing you did that helped you greatly during your first semester. Please explain. **Partner/group study; regular reviews; asking questions; set aside time for myself**
3. Knowing what you know now, describe one thing you would do differently if you were to repeat the first semester. Please explain.
4. Offer your two best pieces of advice to future students on how to be more successful and less stressed during the first semester of health care professions school. Please explain.
5. What will you do differently during your first day, first week and first month of second semester than you did during your first semester here? Please explain.

Start studying day one/NEVER fall behind; engage with professors more often; getting to know my classmates sooner; prioritize sleep; set aside time to exercise/be with friends/relax/binge on mindless TV shows, podcasts, etc.

"The very fact that we remain alive is almost beyond our control, for hunger makes us seek food and fear makes us seek refuge. Sensations of cold make us provide warmth, and other forces cause us to seek fellowship and reproduce. Thus, the human being is actually on automation, and the fact that we are sensing, feeling, and knowledgeable beings is part of the automatic sequence of life; these special attributes allow us to exist under widely varying conditions that otherwise would make life impossible."

Guyton and Hall, *Human Physiology and Mechanisms of Disease*, 6th edition, pg. 3

“(T)he human being is actually on automation”

The statement above provides the framework for this course

We will study:

Physiology

- what are these automatic processes?

- how is this automation achieved?

Pathophysiology

- what happens when automation processes fail or become dysregulated?

Throughout the semester:

1. **Practice quickly sketching out physiological processes (link overview to detailed mechanism)**

2. **Strive to understand each elements of and the information provided by:**

    a. **equations**
    b. **graphs**
    c. **diagrams**

Diffusion equation (Fick's Law of Diffusion)

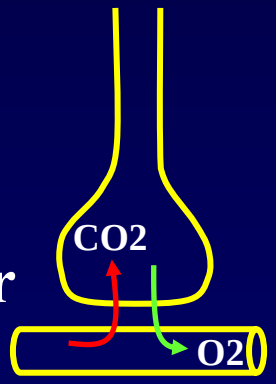
SA – surface area for diffusion
Diff – diffusivity
P1-P2 – partial pressure gradient
T – thickness of diffusion barrier

$$V_{gas} = \frac{SA \times Diff \times (P1 - P2)}{T}$$

Directly
Inversely

- **FACTORS THAT AFFECT DIFFUSION RATE**

  a. **Exercise** ↑ SA
  b. **Emphysema** ↓ SA diff impair
  c. **Pulmonary edema** ↑ T diff impair
  d. **Alveolar fibrosis** ↑ T diff impair
  e. **Supplemental oxygen** ↑ P1-P2



Pharmacological Management of Hypertension

MABP = (HR x SV) x TPR

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

Antihypertensive medications function to control mean arterial blood pressure (MABP) by regulating heart rate (HR), stroke volume (SV) and total peripheral resistance (TPR); MABP = CO x TPR; Cardiac output (CO) = HR x SV;

therefore, MABP = (HR x SV) x TPR

Pharmacological Management of Hypertension

MABP = (HR x SV) x TPR

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

DIURETICS

Pharmacological Management of Hypertension

MABP = (HR x SV) x TPR

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility

SymNS
(β-receptors)

Preload

Venous compliance
SymNS (α-receptors)

Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction

SymNS
(α -receptors)
Angiotensin II
(AII receptors)

β-BLOCKERS

Pharmacological Management of Hypertension

MABP = (HR x SV) x TPR

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

ACE INHIBITORS

Pharmacological Management of Hypertension

MABP = (HR x SV) x TPR

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

ANGIOTENSION II RECEPTOR BLOCKERS

Pharmacological Management of Hypertension

$$MABP = (HR \times SV) \times TPR$$

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

ALDOSTERONE RECEPTOR BLOCKERS

Pharmacological Management of Hypertension

MABP = (HR x SV) x TPR

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α-receptors)
Angiotensin II
(AII receptors)

CALCIUM CHANNEL BLOCKERS

Pharmacological Management of Hypertension

MABP = (HR x SV) x TPR

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

α BLOCKERS

Management of Acute Angina or Hypertensive Crisis

MABP = (HR x SV) x TPR

SymNS
(β-receptors)

Parasymp NS
(muscarinic R)

Contractility
SymNS
(β-receptors)

Preload
Venous compliance
SymNS (α-receptors)
Plasma volume
Water volume
Sodium retention
Renin-Angio-Aldo

Arterial vaso-constriction
SymNS
(α -receptors)
Angiotensin II
(AII receptors)

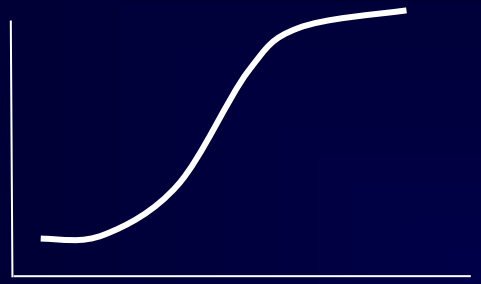
NITRIC OXIDE DONORS
(direct vasodilation)

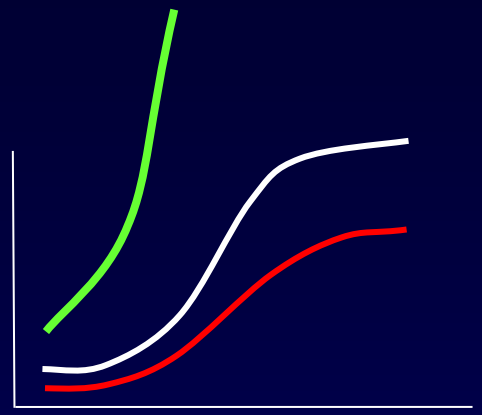
Throughout the semester:

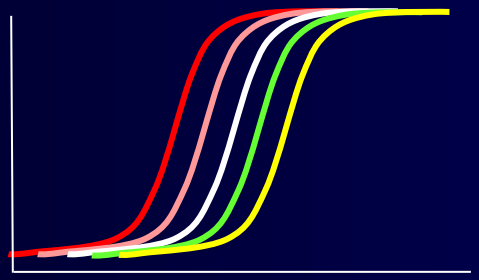
1. **Practice quickly sketching out physiological processes (link overview to detailed mechanism)**

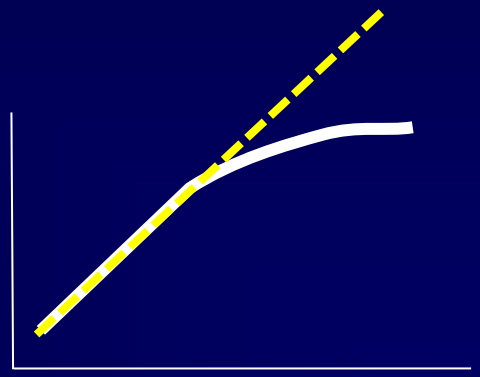
2. **Strive to understand each elements of and the information provided by:**

    a. **equations**
    b. **graphs**
    c. **diagrams**







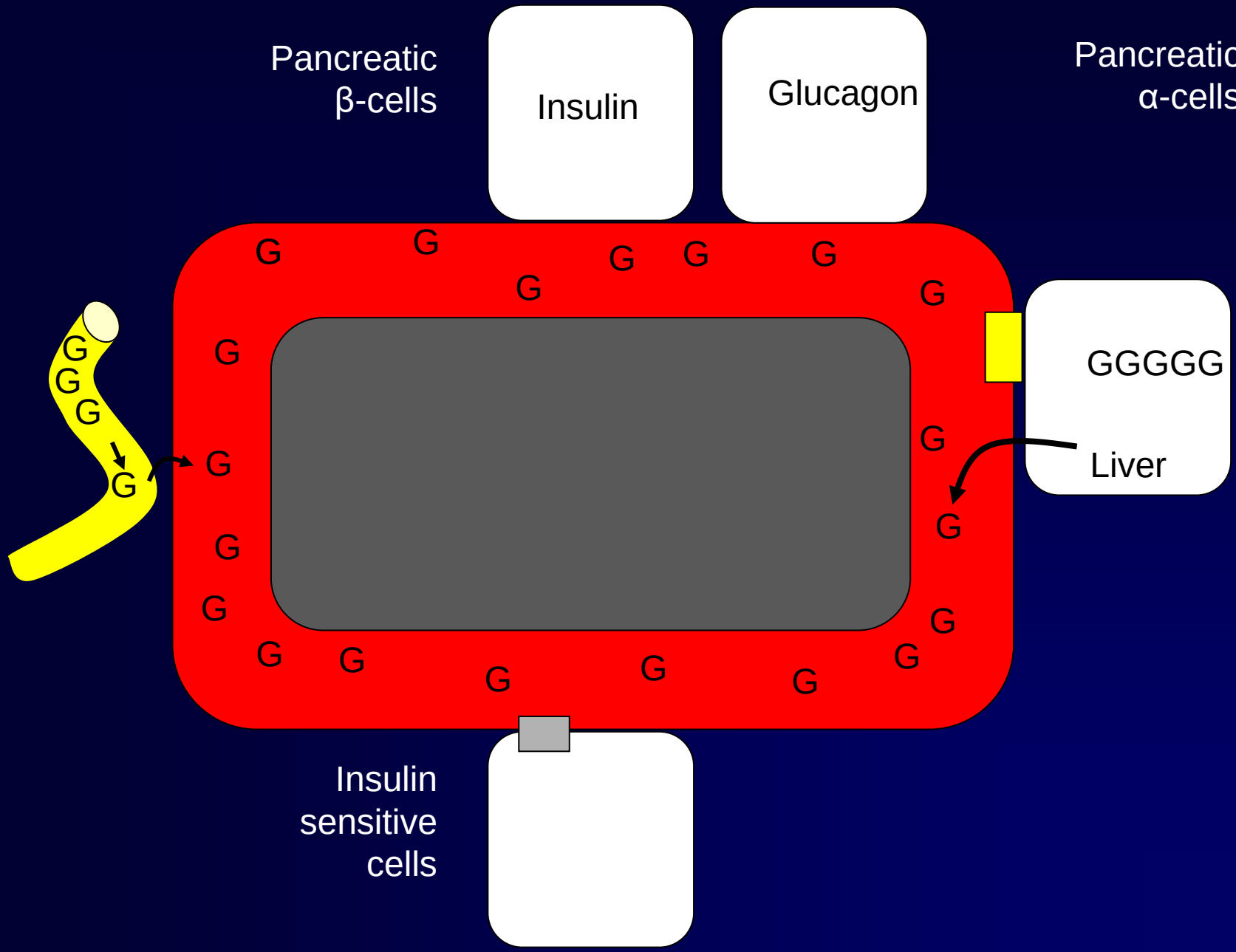


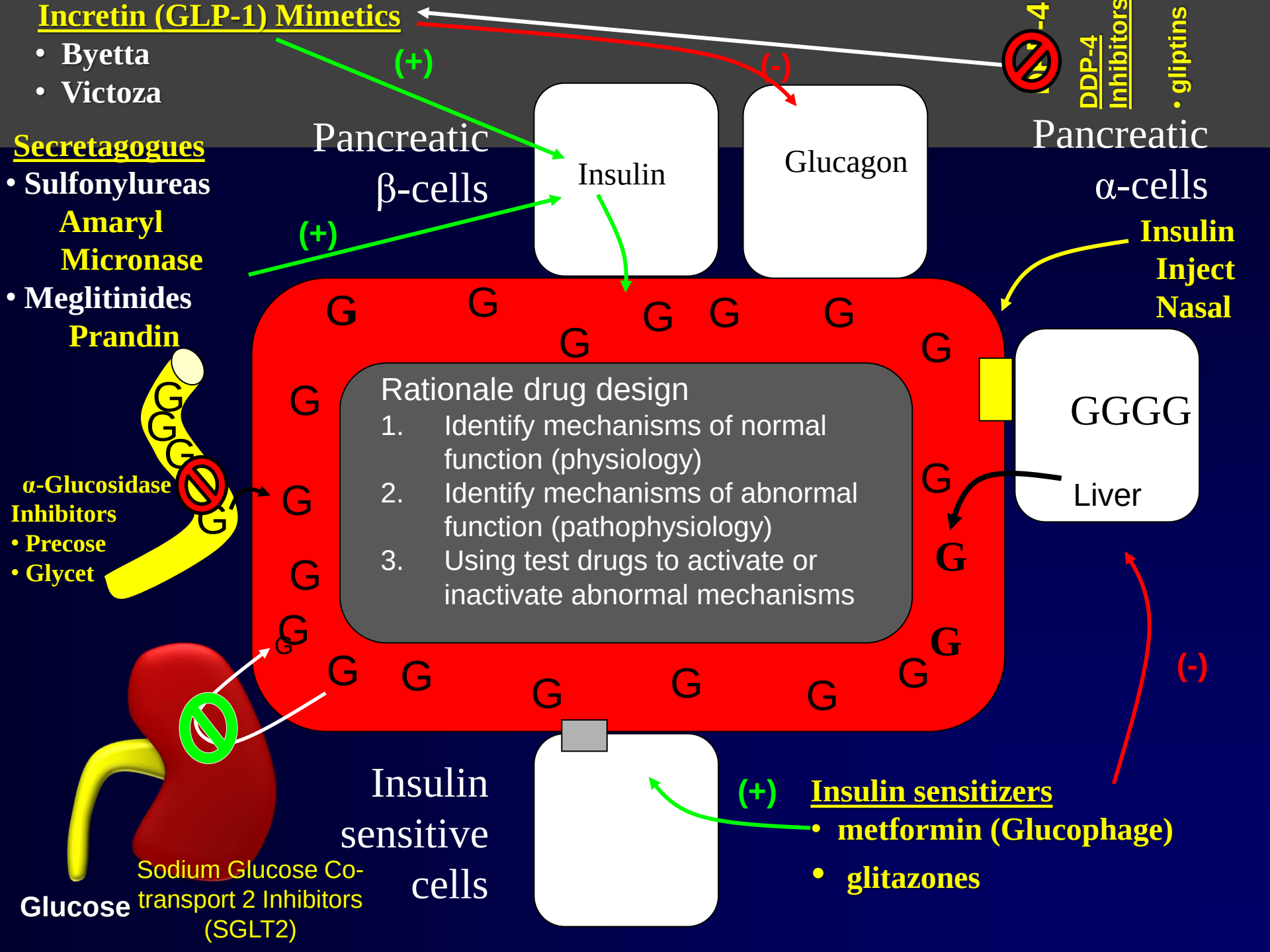
Throughout the semester:

1. **Practice quickly sketching out physiological processes (link overview to detailed mechanism)**

2. **Strive to understand each elements of and the information provided by:**

    a. **equations**
    b. **graphs**
    c. **diagrams**


Pancreatic
β-cells
Insulin
Glucagon
Pancreatic
α-cells
GGGGG
Liver
Insulin
sensitive
cells


Incretin (GLP-1) Mimetics
• Byetta
• Victoza
(+)
(-)
DDP-4 Inhibitors
• gliptins
Secretagogues
• Sulfonylureas
Amaryl
Micronase
• Meglitinides
Prandin
Pancreatic β-cells
Insulin
Glucagon
Pancreatic α-cells
(+)
Insulin
Inject
Nasal
Rationale drug design
1. Identify mechanisms of normal function (physiology)
2. Identify mechanisms of abnormal function (pathophysiology)
3. Using test drugs to activate or inactivate abnormal mechanisms
GGGG
Liver
α-Glucosidase Inhibitors
• Precose
• Glycet
(-)
Insulin sensitive cells
(+)
Insulin sensitizers
• metformin (Glucophage)
• glitazones
Glucose
Sodium Glucose Co-transport 2 Inhibitors (SGLT2)

In biochemistry you learned that all of the metabolic pathways are linked and interdependent.

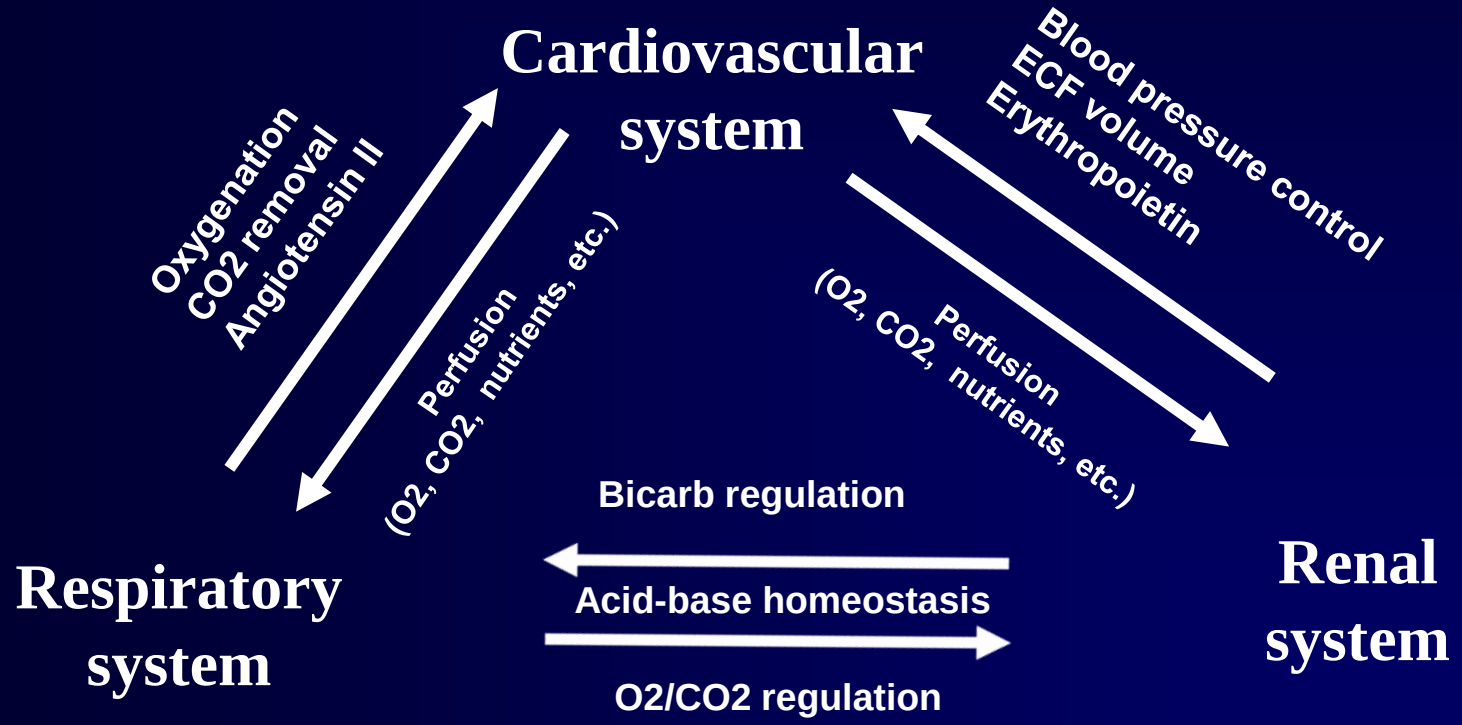
Dysfunction in one pathway leads to compensation and/or dysfunction in other pathways.

In physiology you will learn that all of the systems are linked and interdependent.

Dysfunction in one system leads to compensation and/or dysfunction in other systems.

In physiology you will learn that all of the systems are linked and interdependent.

Dysfunction in one system leads to compensation and/or dysfunction in other systems.



FUNDAMENTAL PHYSIOLOGICAL PRINCIPLES (Part I)

I.	**Introduction to physiology and homeostasis**	READ pp 1-12	Vander's
	A. Physiology – Definitions		
	B. Homeostasis	STUDY Figs 1.8, 1.9	
	C. Fluid compartments		
	D. Homeostatically regulated factors		
	E. How/why are factors regulated		
	F. Homeostatic hmmmmms		
II.	**Membrane dynamics (structure, function, signaling)**		
	A. Gradients drive physiological processes		
	B. Membrane organization	Study pp 46-52	
	C. Membrane transport and translocation	Study pp 95-110	
	D. Signal transduction	READ Chapter 5	
III.	**Membrane potentials**	Study 148-170	
	A. Membrane potentials and gradients		
	B. Types of membrane potentials		
	C. Resting membrane potentials		
	D. Communication using membrane potentials		
	E. Determinates of membrane and equilibrium potentials		
	F. Graded and action potentials		
IV.	**Nerve impulse propagation, synapses and neural modulation**		
	A. Think communication!		
	B. Nerve impulse propagation and conduction		
	C. Synapses		
	D. Neural modulation		

I. Intro to Physiology and Homeostasis

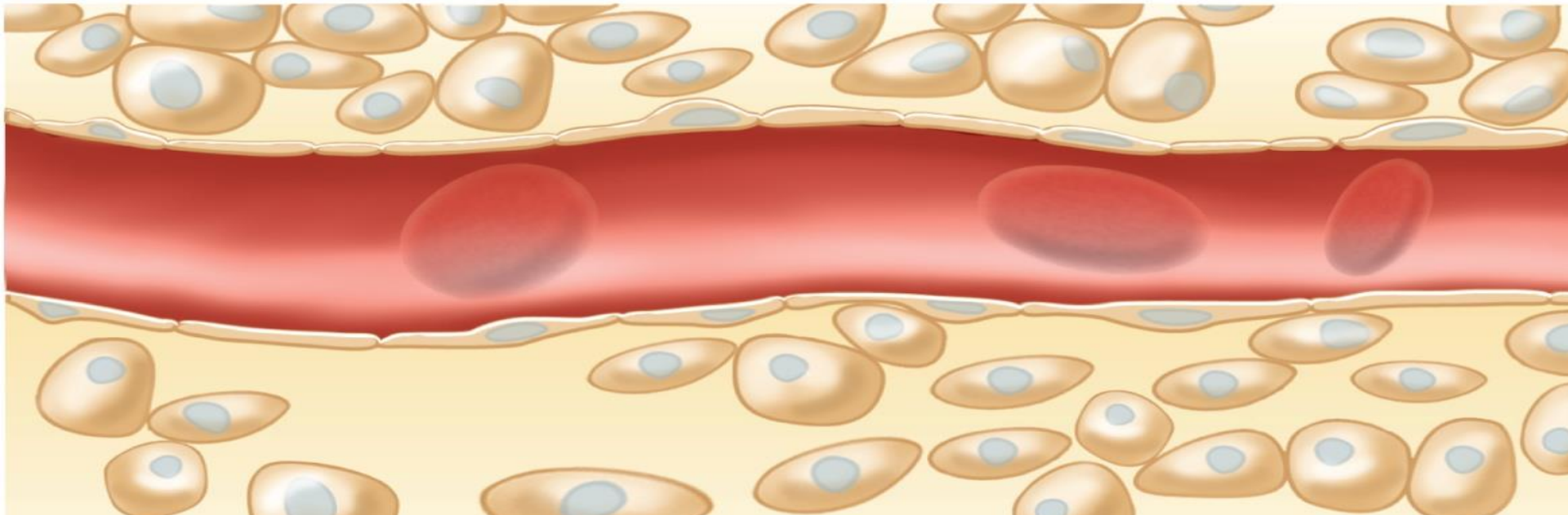
A. Physiology – What is it?

1. Function *(from molecules to the organism)*

2. Communication (40,000,000,000,000 cells)

a. Nervous communication

b. Humoral communication
- Telocrine
- Paracrine
- Autocrine



I. Intro to Physiology and Homeostasis

A. Physiology – What is it?

1. Function *(from molecules to the organism)*

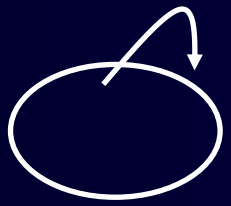
2. Communication (40,000,000,000,000 cells)

a. Nervous communication

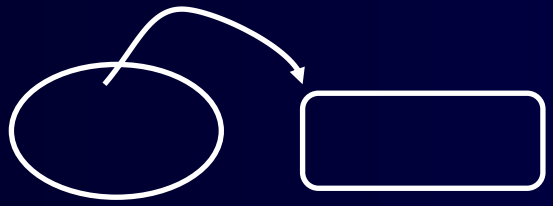
b. Humoral communication
- Telocrine
- Paracrine
- Autocrine

c. Cell-to-cell communication
- Juxtacrine
- Gap junctions

Exocrine Secretions –travel through ducts to an epithelial surface
Endocrine Secretions – released into extracellular space for movement in interstitial fluid and bloodstream (ductless)



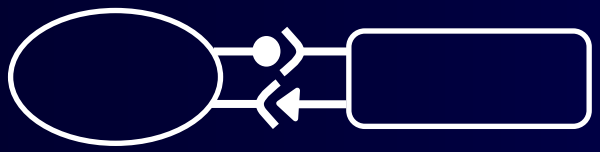
Autocrine communication – a cell releases a substance that acts on the same cell



Paracrine communication – a cell releases a substance that acts on a nearby cell

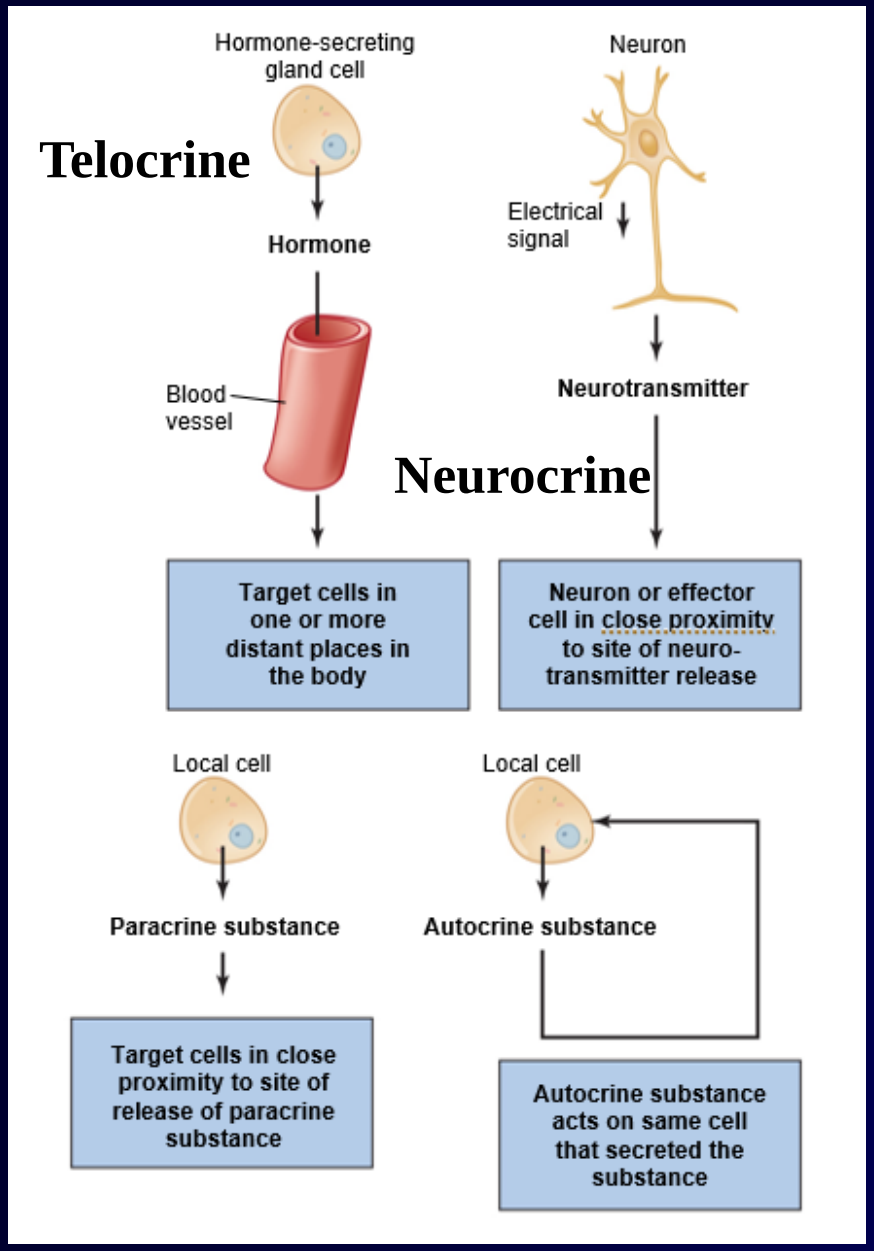


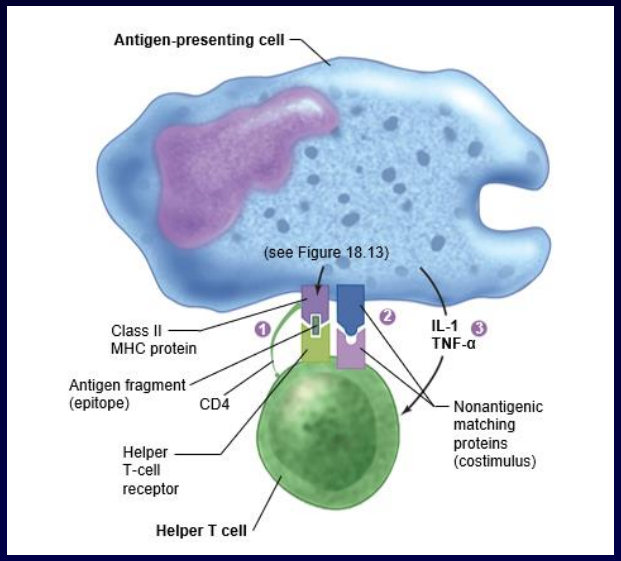
Telocrine communication – a cell releases a substance that travels in the bloodstream to act on a distant cell



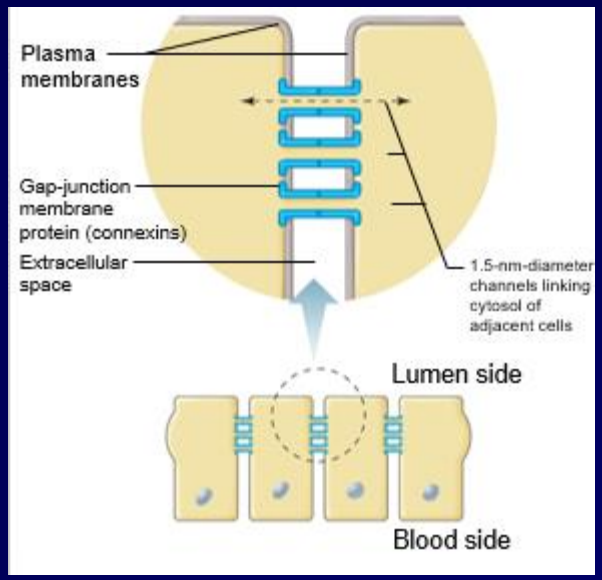
Juxtacrine communication – cell-to-cell communication by interaction of surface molecules

Fig. 1-10 Vander





Juxtacrine Fig. 18-15



Gap junctions Fig. 3-9

I. Intro to Physiology and Homeostasis

A. Physiology – What is it?

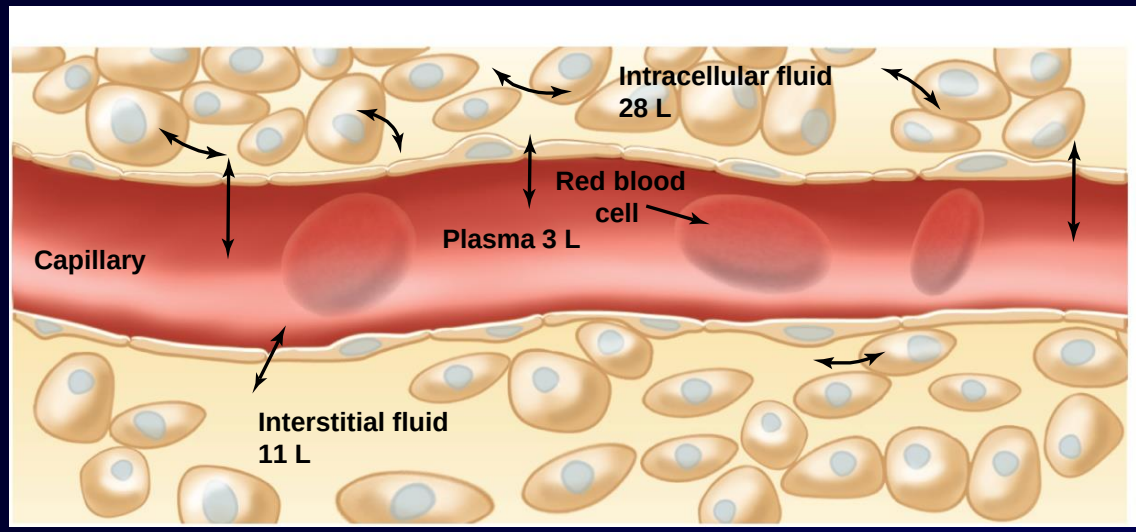
1. **Function** *(from molecules to the organism)*
2. **Communication (40,000,000,000,000 cells)**
   - a. **Nervous communication**
   - b. **Humoral communication** — Telocrine, Paracrine, Autocrine
   - c. **Cell-to-cell communication** — Juxtacrine, Gap junctions
3. **Homeostasis (same/standing)**
   - a. Requires integrated communication and function
   - b. Dynamic drive to achieve a constant "internal environment or milieu"
   - c. *Internal milieu* - extracellular composition, volume, pressure, temperature, pH, etc.

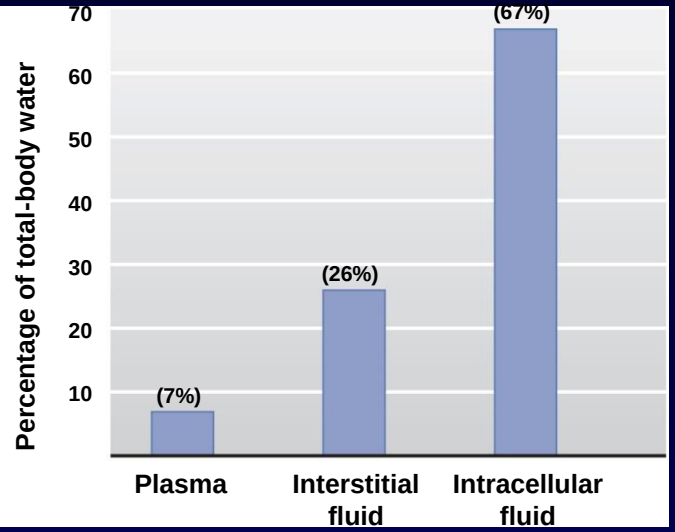
Internal environment = Internal milieu = Extracellular fluid

"The environment surrounding cells"

C. Fluid compartments in the body

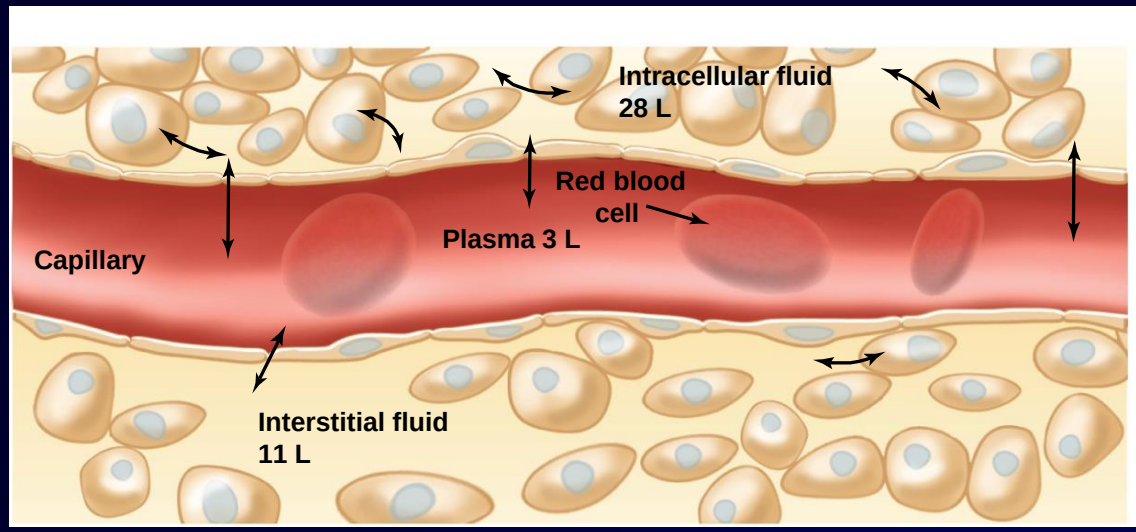
Fig. 1-3

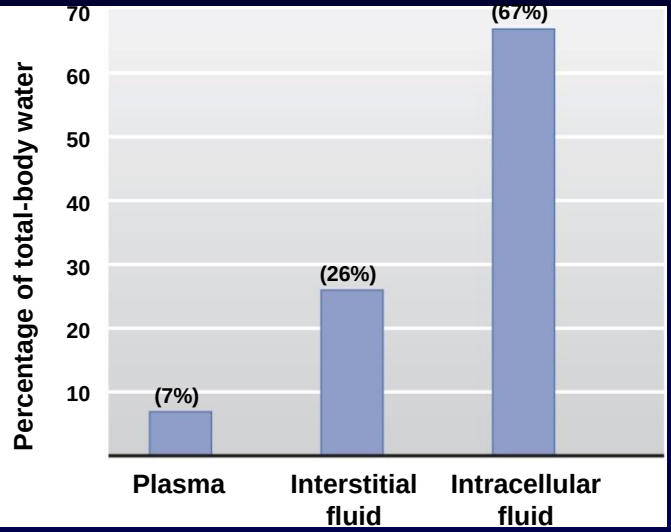




1. Total body water (TBW) ≈ 60% BW (42L/70kg person)
2. Intracellular fluid (ICF) - fluid within cells (2/3 TBW)
3. Extracellular fluid (ECF) - fluid outside cells (1/3 TBW)
   a) Interstitial fluid - between cells but not in vessels
   b) Plasma -fluid in vessels but between cells

C. Fluid compartments in the body (for a 70 kg person)





Estimates

TBW volume is 60% of bodyweight ≈ 42 liters (TBW)

ICF volume is 2/3 of TBW ≈ 28 liters

ECF volume is 1/3 of TBW ≈ 14 liters

Plasma volume ≈ 3 liters

Interstitial fluid volume ≈ 11 liters

60:40:20 rule

Impact of obesity and aging

C. Fluid compartments in the body

1. Total body water (TBW) ≈ 60% BW (42L/70kg person)
2. Intracellular fluid (ICF) - fluid within cells (2/3 TBW)
3. Extracellular fluid (ECF) - fluid outside cells (1/3 TBW)
   a) Interstitial fluid - between cells but not in vessels
   b) Plasma -fluid in vessels but between cells

A 100 kg male is prescribed a drug that distributes itself to the ECF. Assuming no immediate metabolism or clearance, how much drug must be administered if the effective dose is 1 mg of drug for every 100 ml of ECF?

TBW = 0.60 x 100 kg = 60 L
ECF volume = TBW x 0.333 = 60 L x 0.333 = 20 L
Total dose = 1 mg/100 ml x 20,000 ml = 200 mg

D. Hundreds of factors are homeostatically regulated in the ECF to maintain a constant internal environment and optimal body function

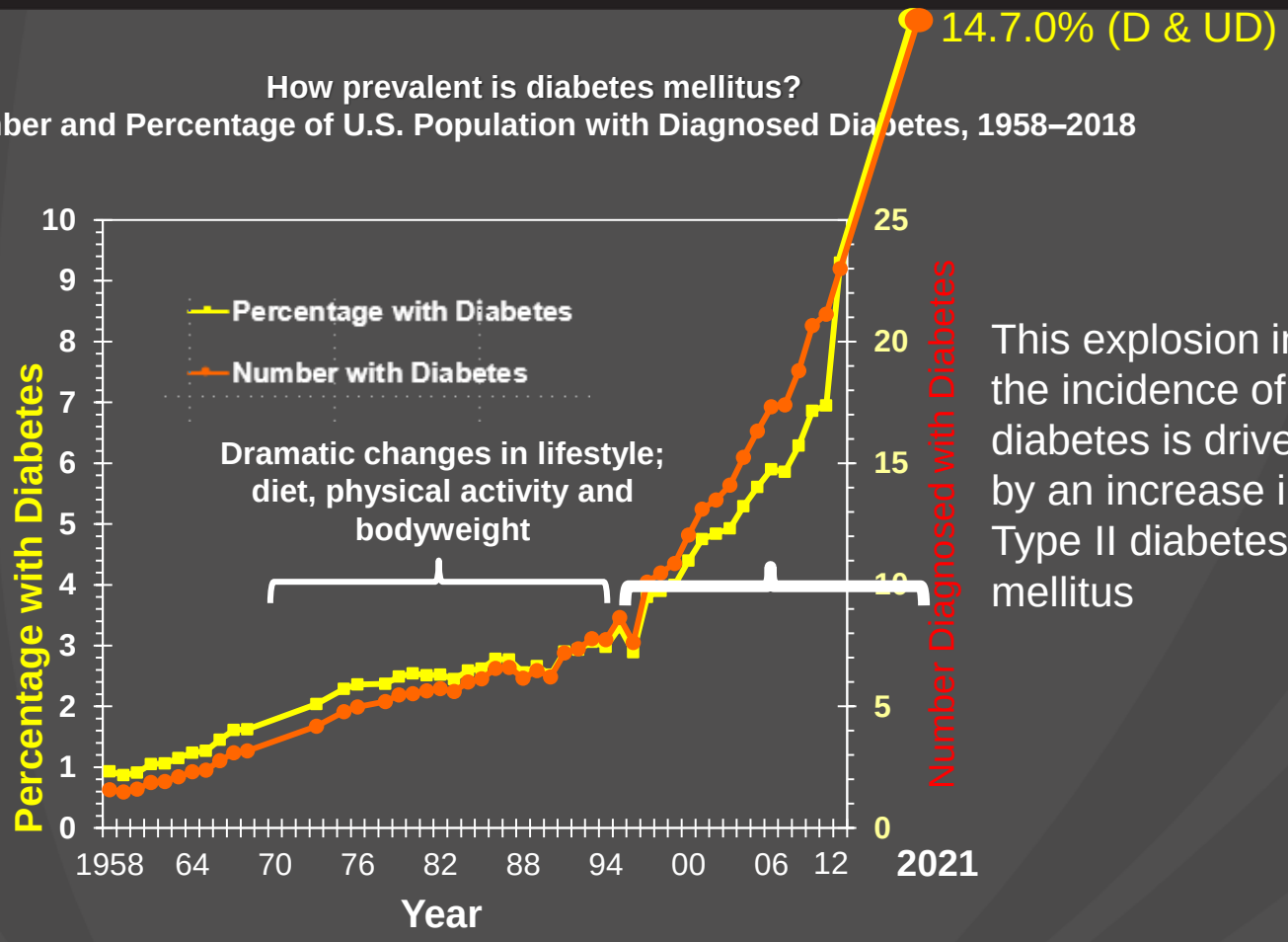
Examples: pH, pressure, volume, osmolarity, glucose, amino acids, fatty acids, Na^+, K^+, Ca^{++}, PO_4^-, O_2, CO_2, etc., etc.

E. Why/how are homeostatically controlled factors regulated?

1. **Proteins function by binding things**
2. **Proteins function optimally within narrow ranges of pH, temperature, salt concentration, etc.; set-points**
3. **Therefore, anything that affects the binding characteristics of a protein will affect its function or activity**
4. **Enzymes, membrane and nuclear receptors, channel pores, antibodies, etc., etc., etc., are proteins**
5. **Sensor-effector-negative feedback loops**

70%-80% of Type II diabetes in this country is associated with:

- Bodyweight control
- Diet (quantity and quality)
- Physical activity

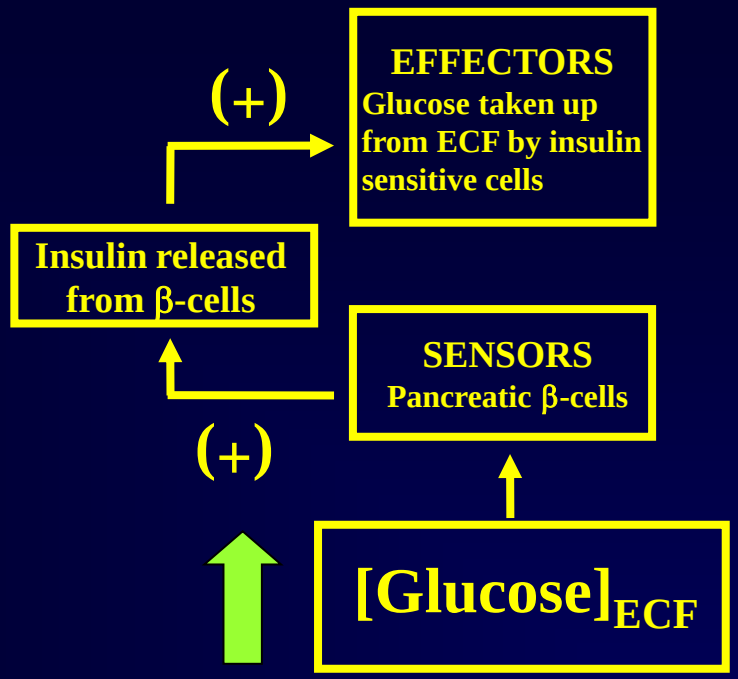


This explosion in the incidence of diabetes is driven by an increase in Type II diabetes mellitus

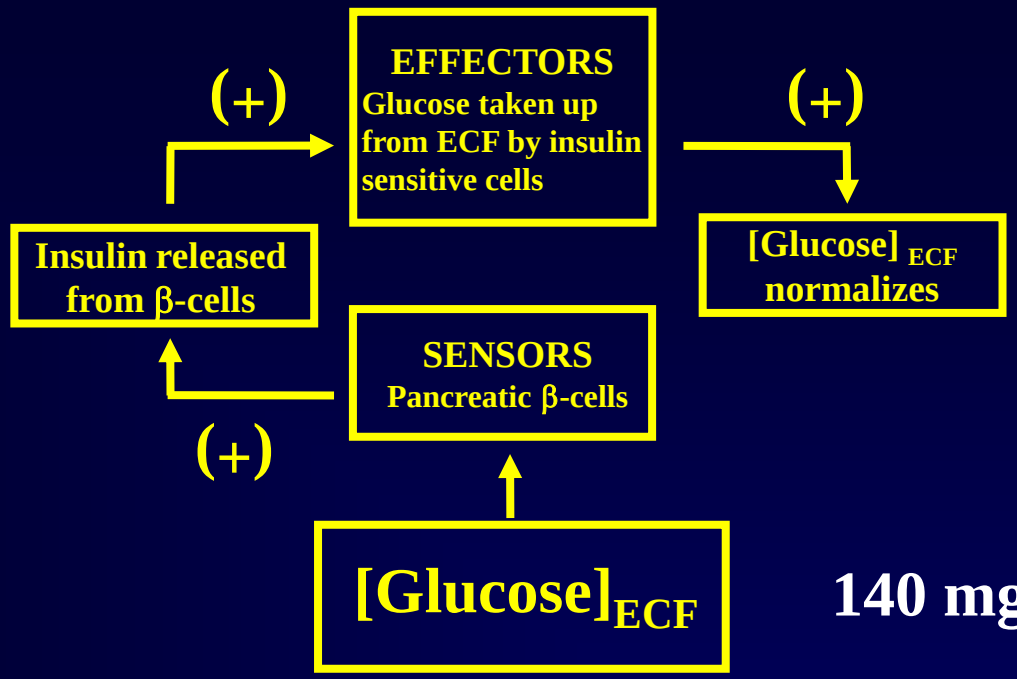
CDC's Division of Diabetes Translation. National Diabetes Surveillance System available at http://www.cdc.gov/diabetes/statistics

Centers for Disease Control National Diabetes Statistics Report 2020

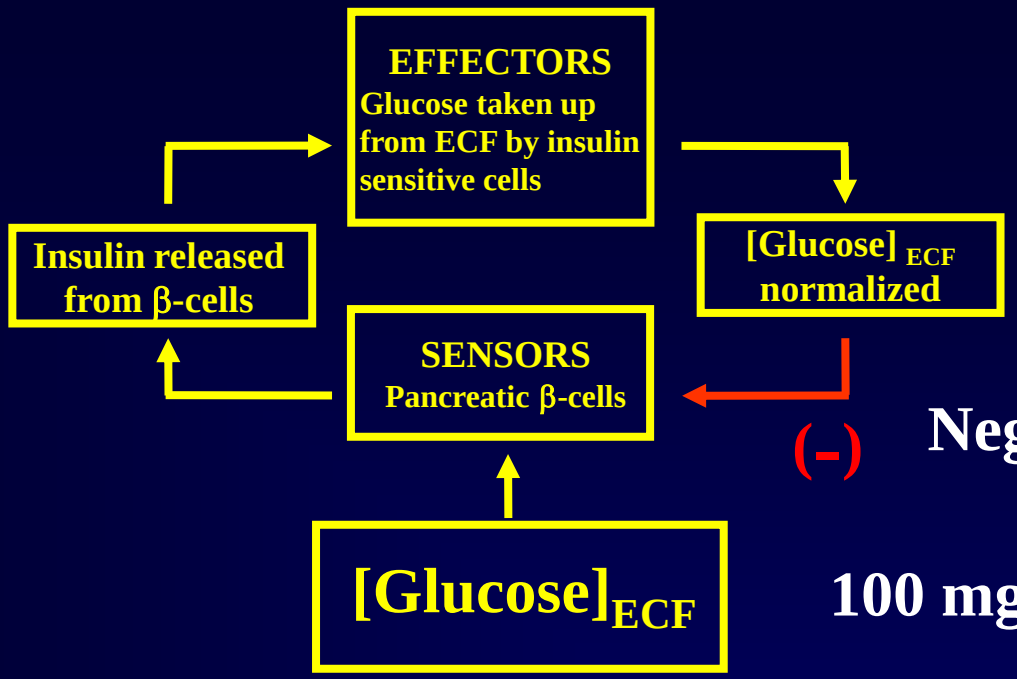
$[Glucose]_{ECF}$ **100 mg/100 ml**


EFFECTORS
Glucose taken up from ECF by insulin sensitive cells
(+)
Insulin released from β-cells
SENSORS
Pancreatic β-cells
(+)
[Glucose]ECF

180 mg/100 ml


EFFECTORS
Glucose taken up from ECF by insulin sensitive cells
(+)
(+)
Insulin released from β-cells
[Glucose] ECF normalizes
SENSORS
Pancreatic β-cells
(+)
[Glucose]ECF

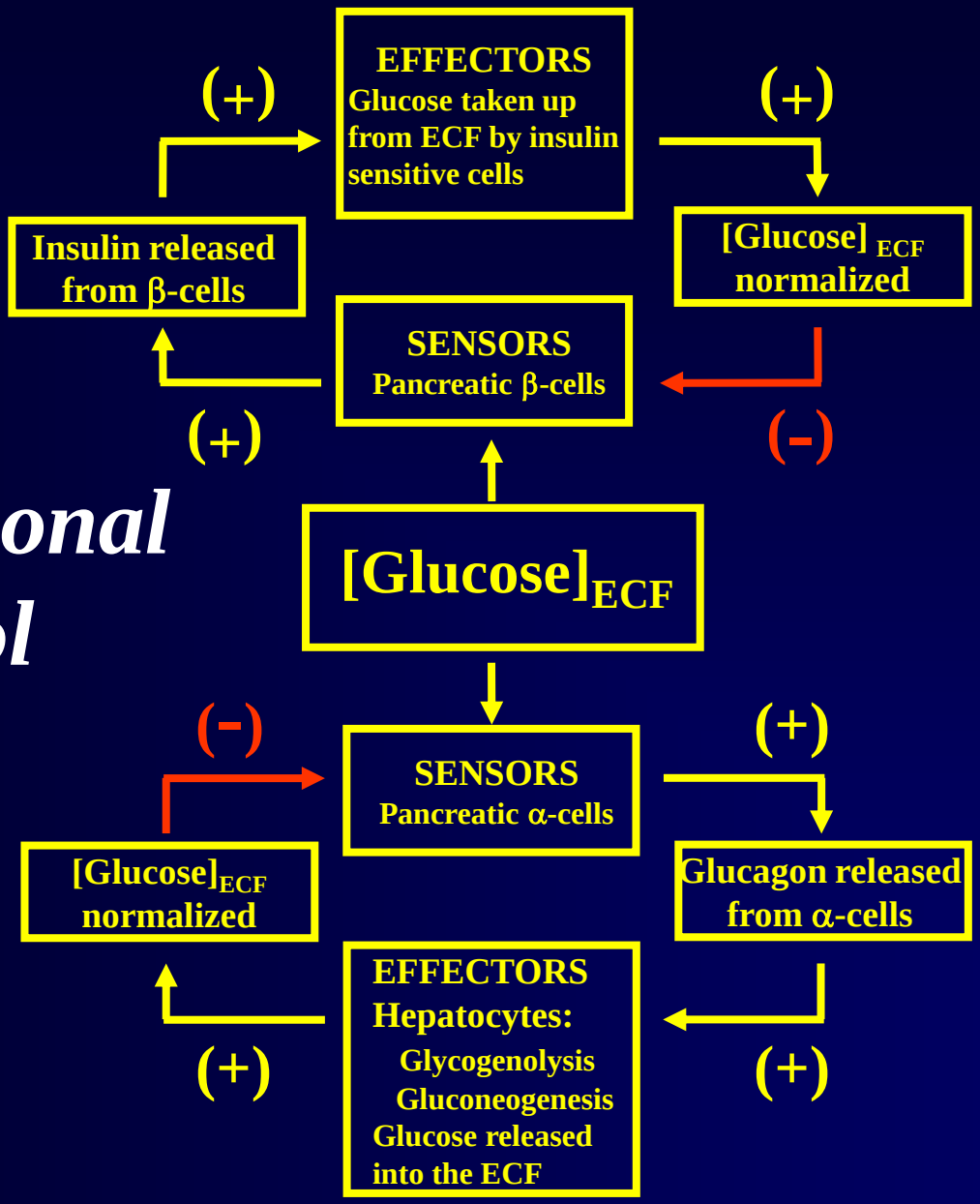
140 mg/100 ml

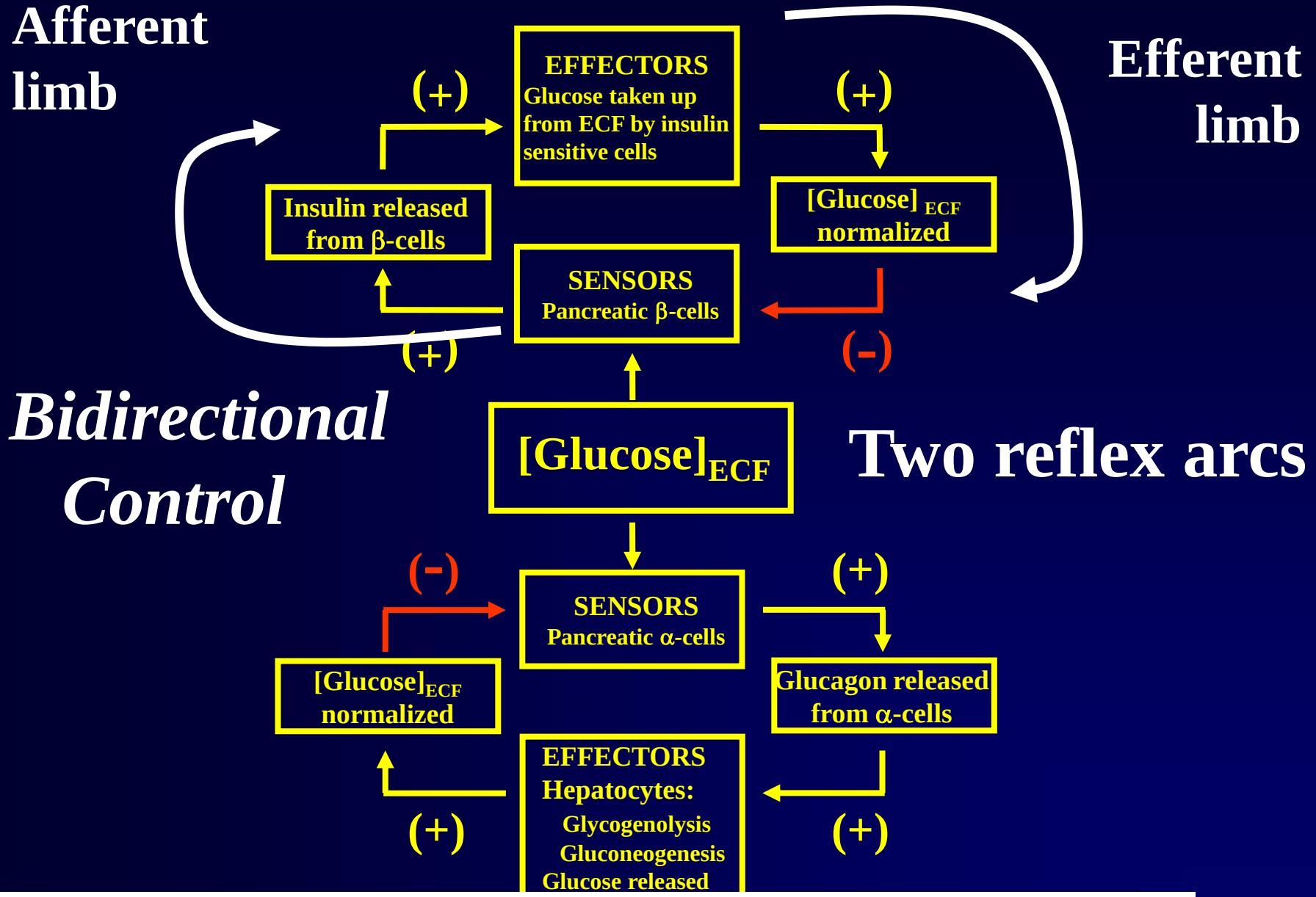


Negative feedback

100 mg/100 ml

Bidirectional Control



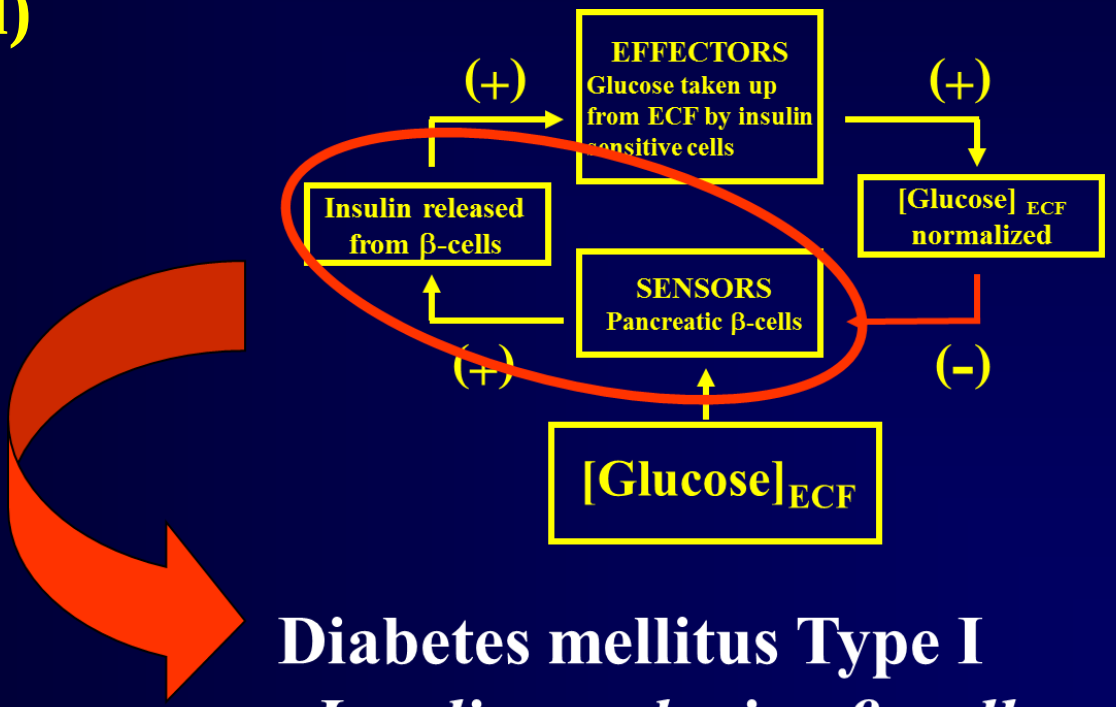


Positive feedback – when an initial alteration or disturbance actives events that increases the alteration or disturbance even further

Feedforward Regulation – some systems can anticipate or prepare for alterations or disturbances BEFORE they occur

F. Homeostatic Hmmmmmms

1. Some set-points can be dynamic rather than static
2. Many diseases processes, signs and symptoms due to homeostatic dysregulation (loss of homeostatic control)



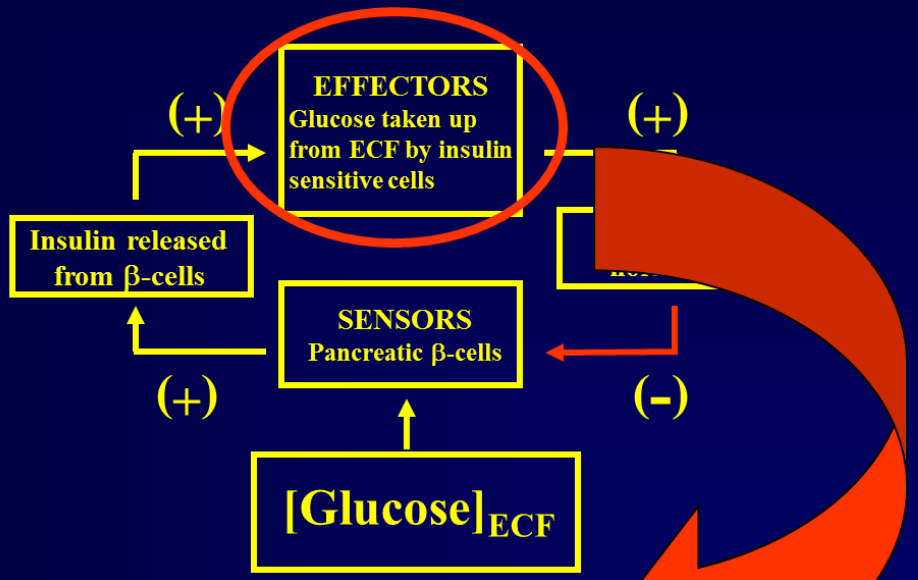
Diabetes mellitus Type I

Insulin producing β–cells are lost

Therefore, lack of insulin

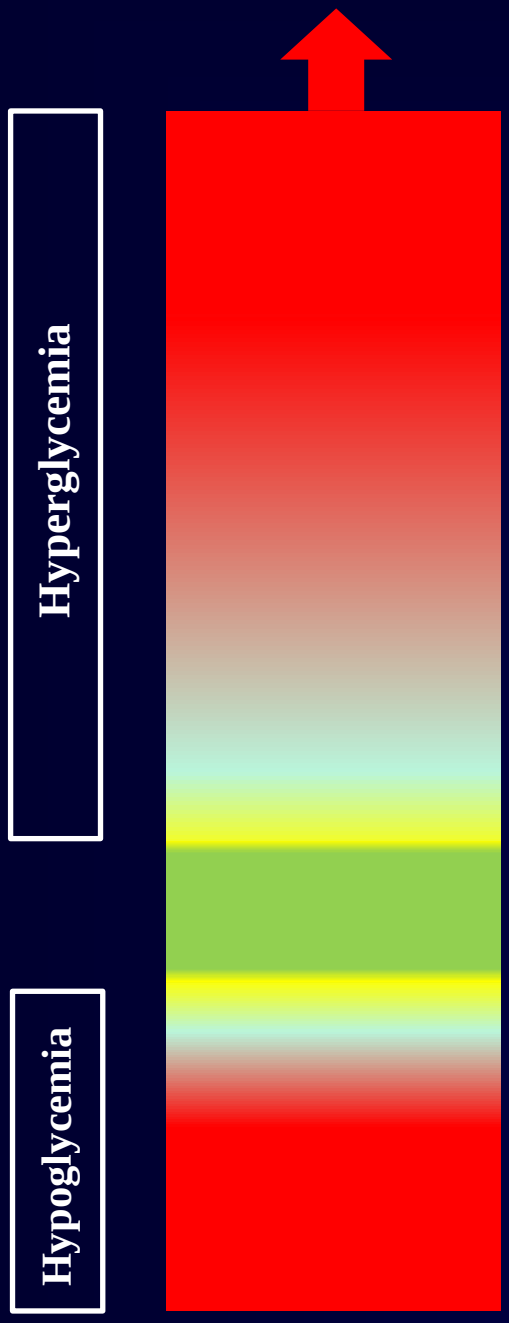
F. Homeostatic Hmmmmmms

1. Some set-points can be dynamic rather than static
2. Many diseases processes, signs and symptoms due to homeostatic dysregulation (loss of homeostatic control)



Diabetes mellitus Type II

Effector cells are insensitive or resistant to insulin



Gradual but progressive damage to blood vessels and tissues; severity dependent on duration and degree of hyperglycemia

SLOW – occurs over years

Shakiness, dizziness, nauseated, sweaty, irritable, weakness, nervousness, headache

Neural dysfunction, loss of consciousness, seizure, coma, death

RAPID – can occur within minutes

F. Homeostatic Hmmmmmms

1. **Some set-points can be dynamic rather than static**
2. **Many diseases processes, signs and symptoms due to homeostatic dysregulation (loss of homeostatic control)**
3. **Maxed out or loss of <u>physiological reserve</u>**

Physiological reserve

Cardiac output (CO)= HR x SV

Rest: *70 beats/min x 70 ml/beat = 4.9 L/min*

Max Ex: *200 beats/min x 140 ml/beat = 28 L/min*

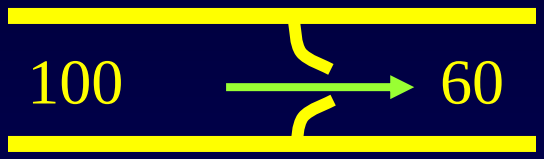
FUNDAMENTAL PHYSIOLOGICAL PRINCIPLES (Part I)

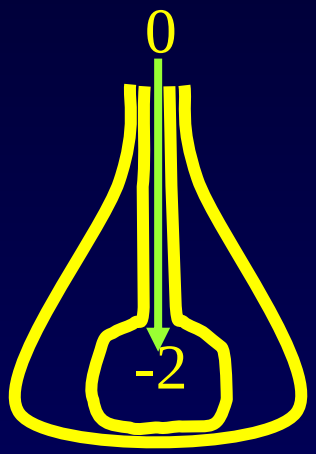
I.	**Introduction to physiology and homeostasis**	**READ pp 1-12**	**Vander's**
	A. Physiology – Definitions		
	B. Homeostasis	**STUDY Figs 1.8, 1.9**	
	C. Fluid compartments		
	D. Homeostatically regulated factors		
	E. How/why are factors regulated		
	F. Homeostatic hmmmmms		
II.	**Membrane dynamics (structure, function, signaling)**		
	A. Gradients drive physiological processes		
	B. Membrane organization	**Study pp 46-52**	
	C. Membrane transport and translocation	**Study pp 95-110**	
	D. Signal transduction	**READ Chapter 5**	
III.	**Membrane potentials**	**Study 148-170**	
	A. Membrane potentials and gradients		
	B. Types of membrane potentials		
	C. Resting membrane potentials		
	D. Communication using membrane potentials		
	E. Determinates of membrane and equilibrium potentials		
	F. Graded and action potentials		
IV.	**Nerve impulse propagation, synapses and neural modulation**		
	A. Think communication!		
	B. Nerve impulse propagation and conduction		
	C. Synapses		
	D. Neural modulation		

GRADIENTS drive physiological processes!

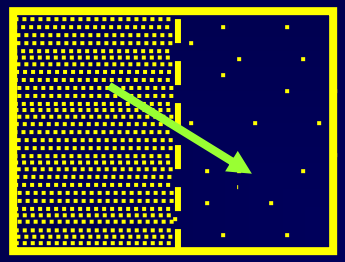
Gradient - when a difference exists between two compartments with respect to the pressure, concentration or electrical potential of a given variable

Pressure gradients





Concentration gradients



Electrical gradients

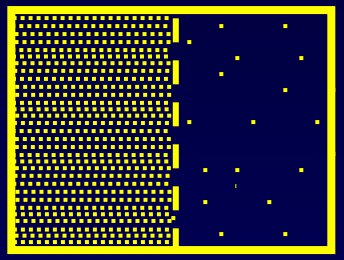
0 mV



GRADIENTS drive physiological processes!

Gradient - when a difference exists between two compartments with respect to the pressure, concentration or electrical potential of a given variable

Concentration gradients



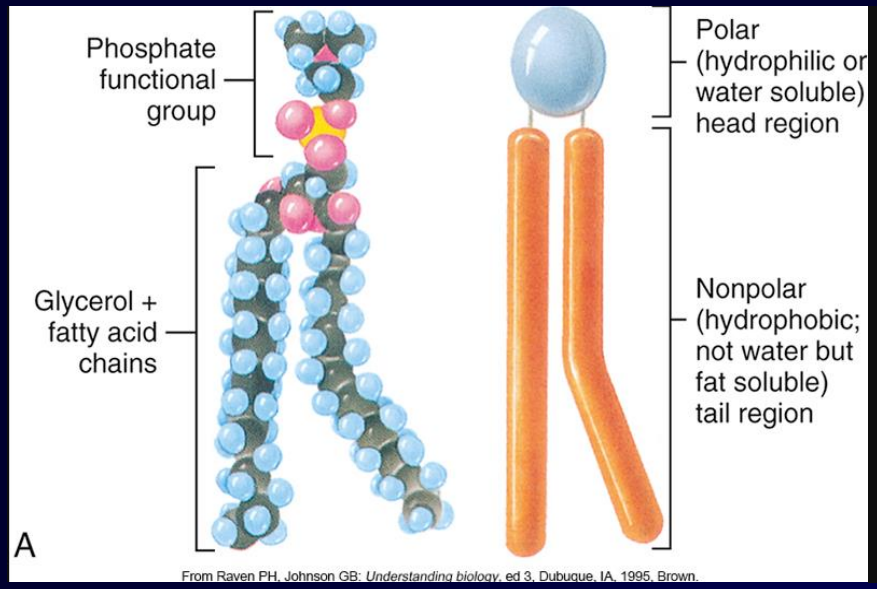
	ECF	ICF	
Na+	150	15	mM
K+	4	140	



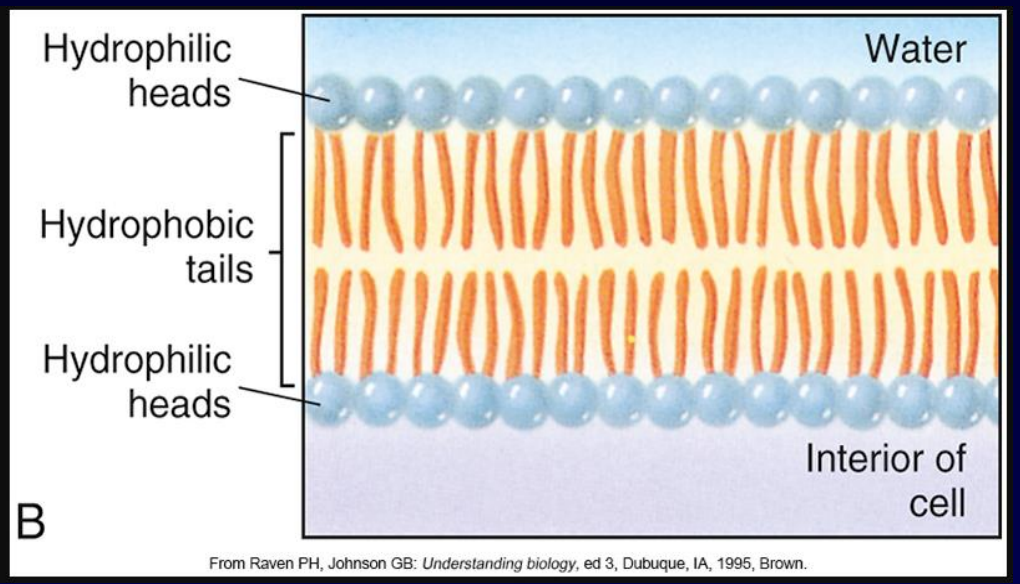
Olenich, Katelyn

The Ocean Banana

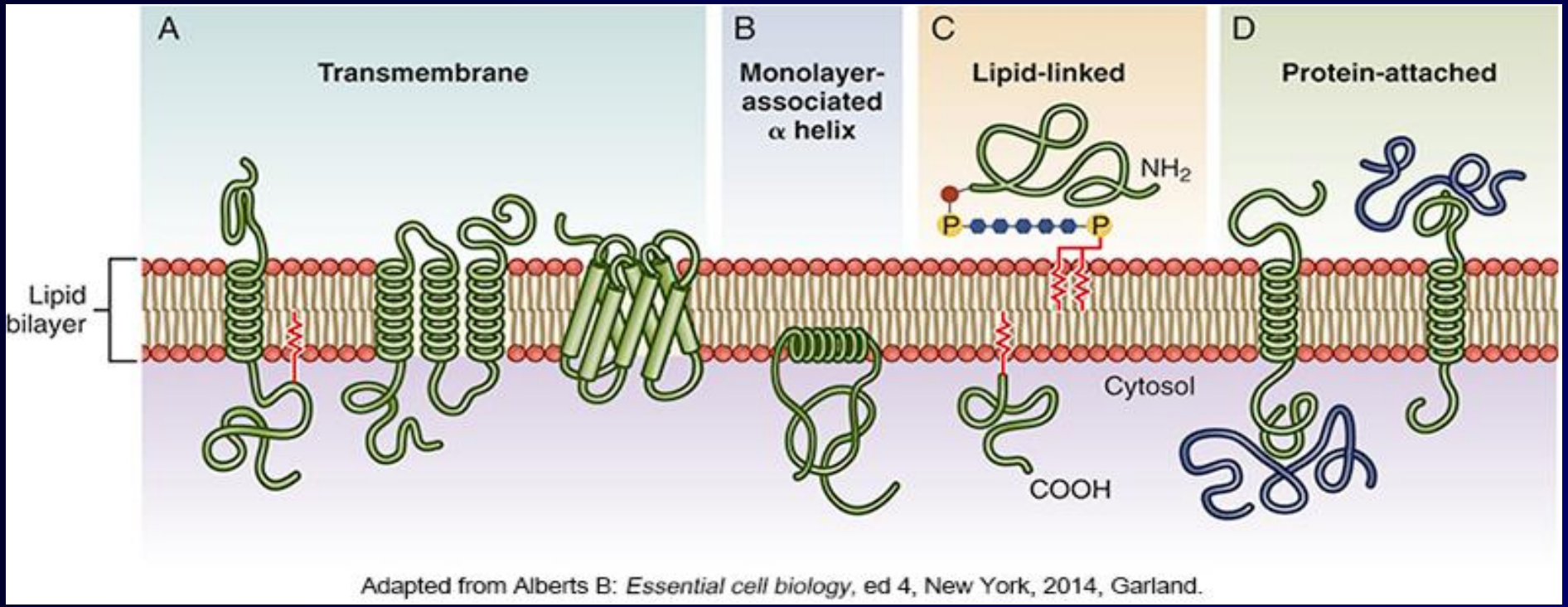
Fig 1-12, 1-13 Membrane organization Clin Path textbook



From Raven PH, Johnson GB: *Understanding biology*, ed 3, Dubuque, IA, 1995, Brown.



From Raven PH, Johnson GB: *Understanding biology*, ed 3, Dubuque, IA, 1995, Brown.



Adapted from Alberts B: *Essential cell biology*, ed 4, New York, 2014, Garland.

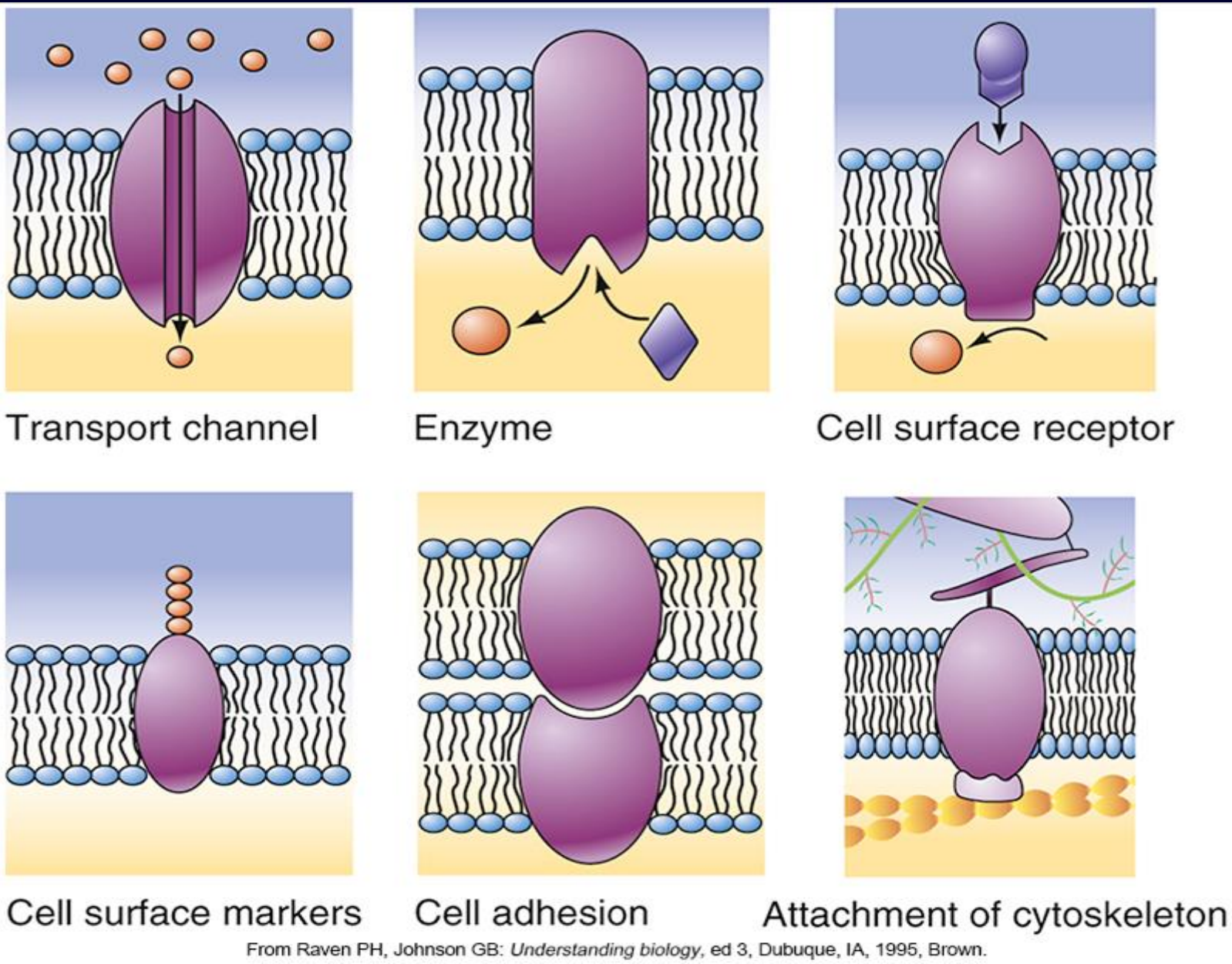
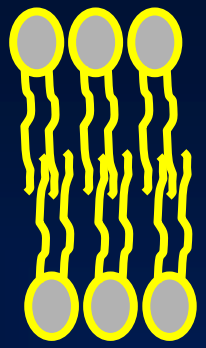


Fig 1-09 Functions of plasma membrane proteins

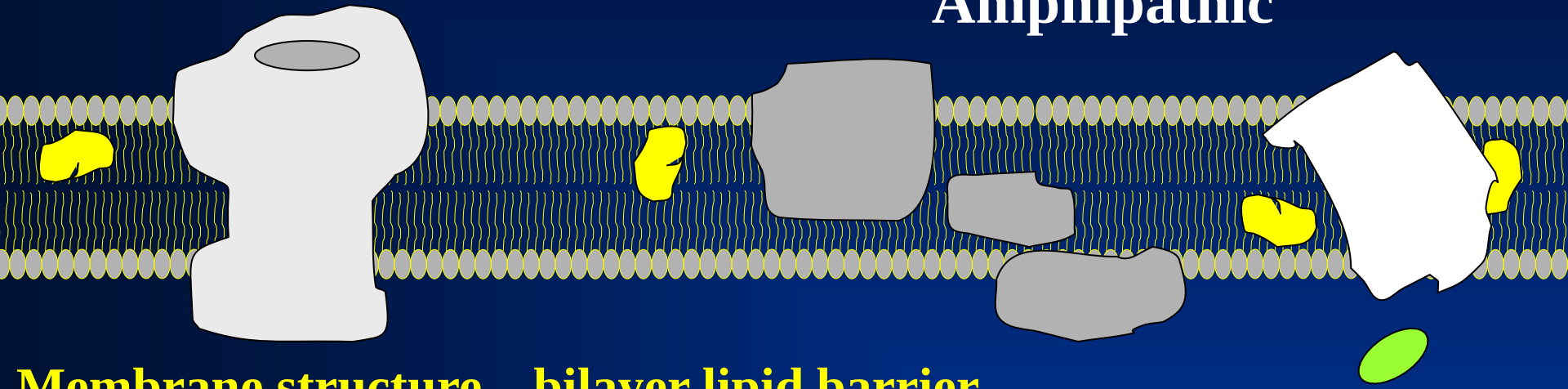
Hydrophobic permeability barrier



Phospholipid subunits are polar (charged)/hydrophilic/lipophobic

Fatty acid tails are neutral (uncharged)/hydrophobic/lipophilic, Creating a barrier against hydrophilic/lipophobic charged molecules

Amphipathic



Membrane structure – bilayer lipid barrier

- **Repels hydrophilic/lipophobic/polar/charged molecules**
- **Admits hydrophobic/lipophilic/nonpolar/uncharged molecules**

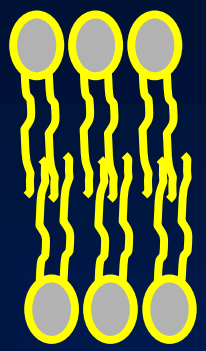
Communication across the membrane

- **Membrane transport proteins**
- **Membrane translocation channels**

} **Molecules go across**

- **Signal transduction processes – information goes across**

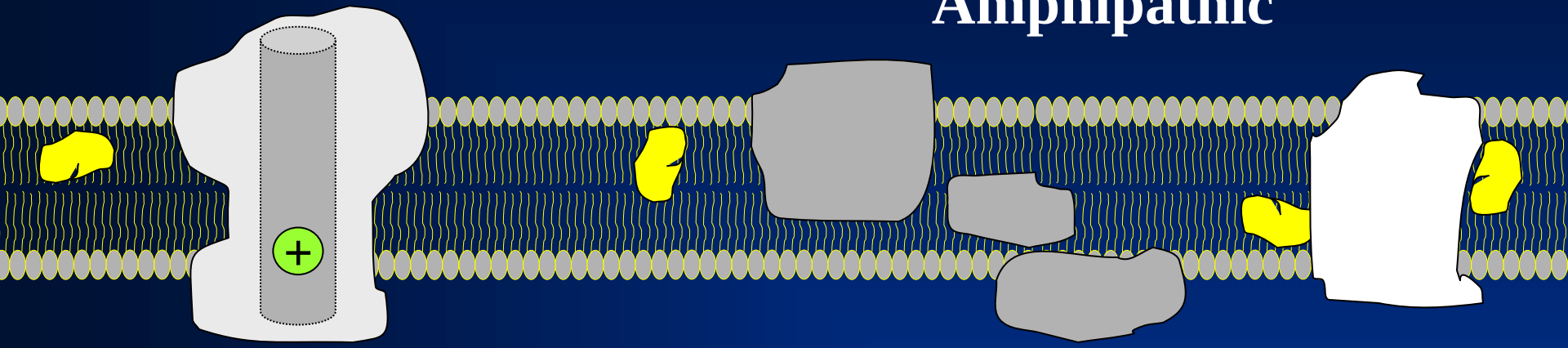
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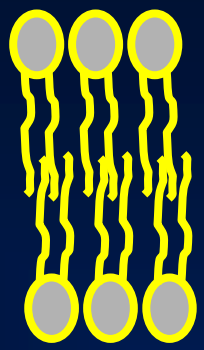
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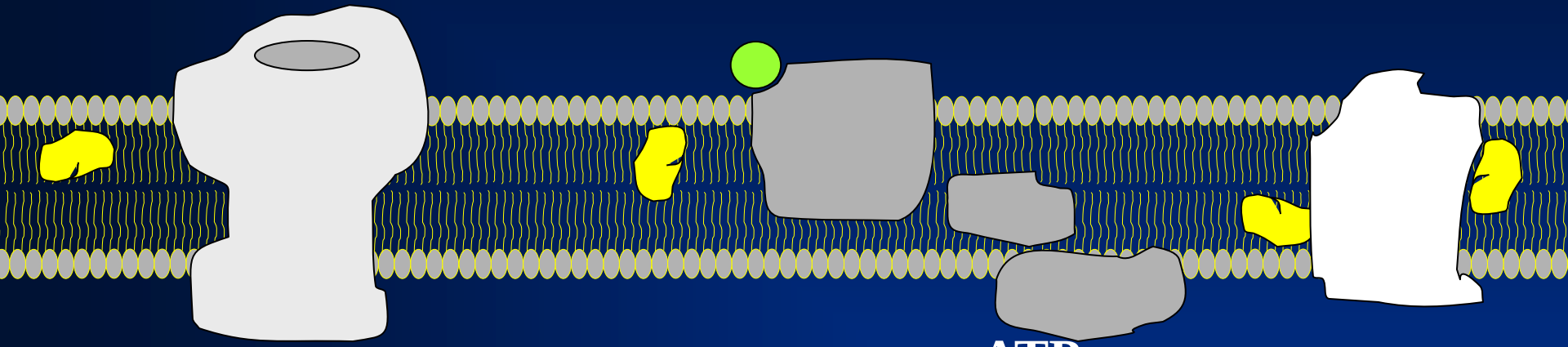
- **Signal transduction processes – information goes across**

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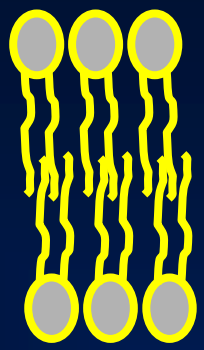
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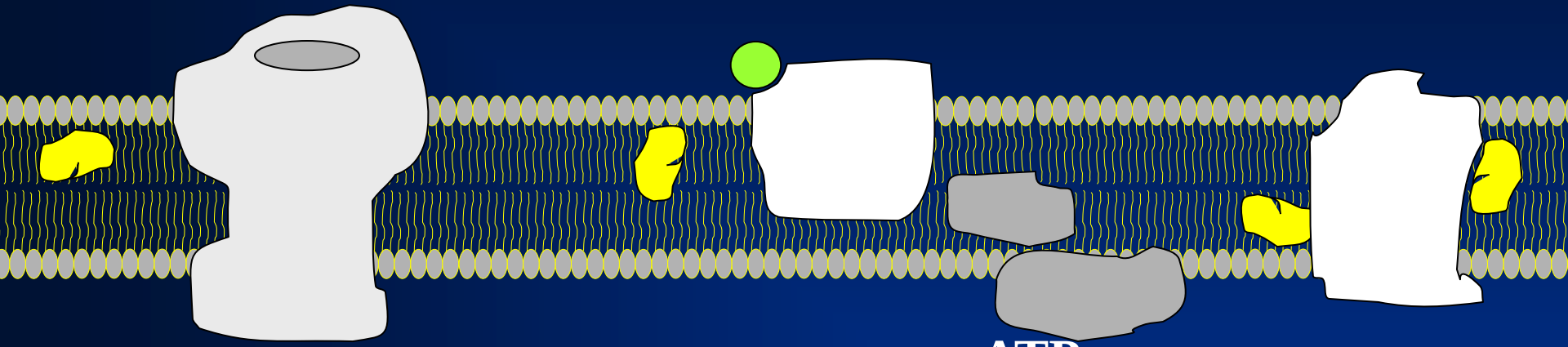
- **Signal transduction processes – information goes across**

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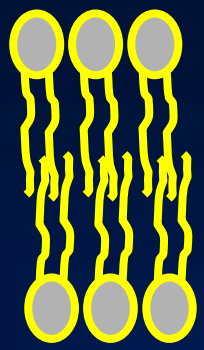
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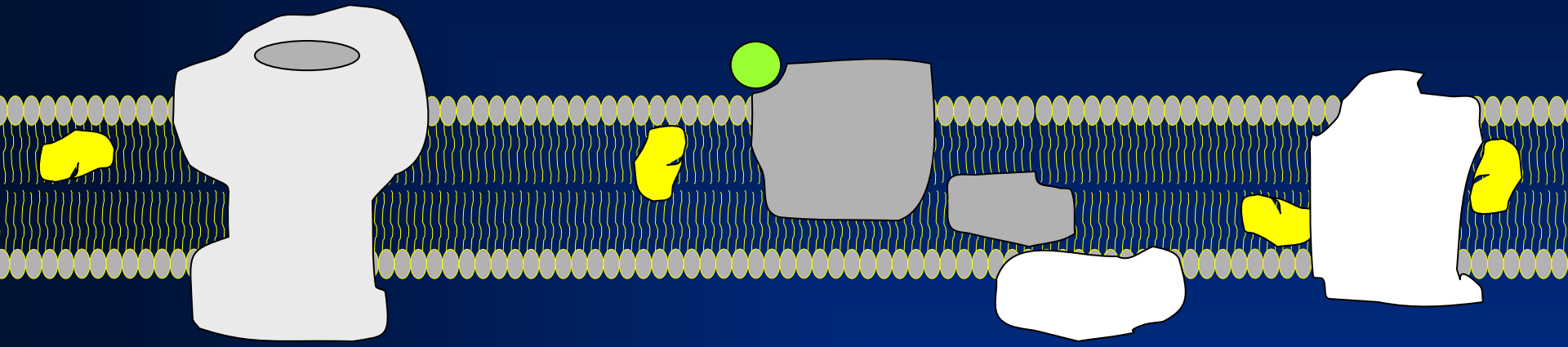
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cAMP

Membrane structure – bilayer lipid barrier

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Communication across the membrane

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- **Signal transduction processes – information goes across**

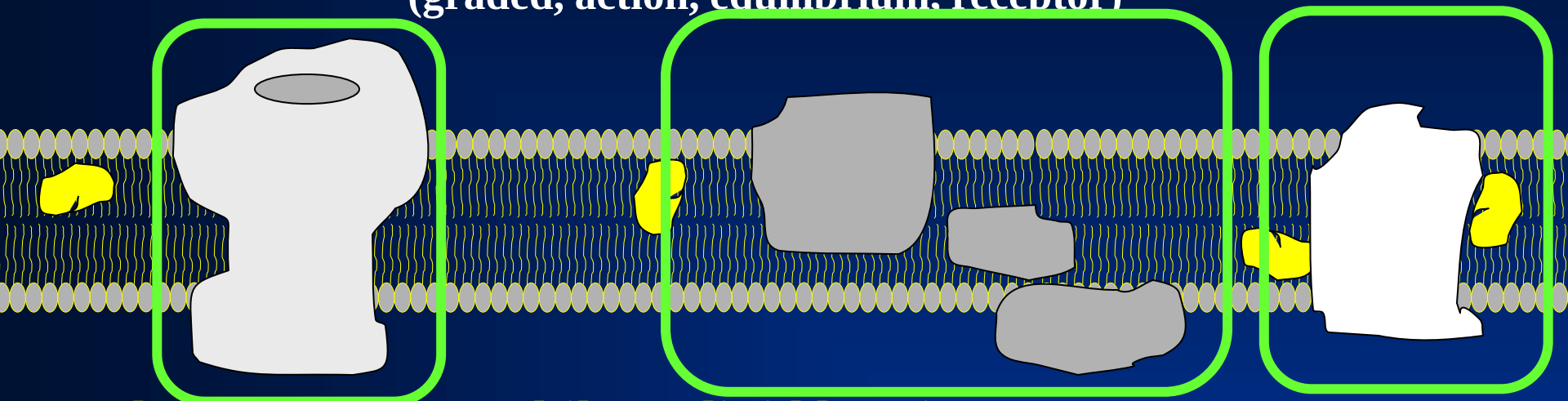
Membrane function

(1.) Membrane dynamics

(transport, translocation, signal transduction)

(2.) Membrane potentials

(graded, action, equilibrium, receptor)



Membrane structure – bilayer lipid barrier

- **Repels hydrophilic/lipophobic/polar/charged molecules**
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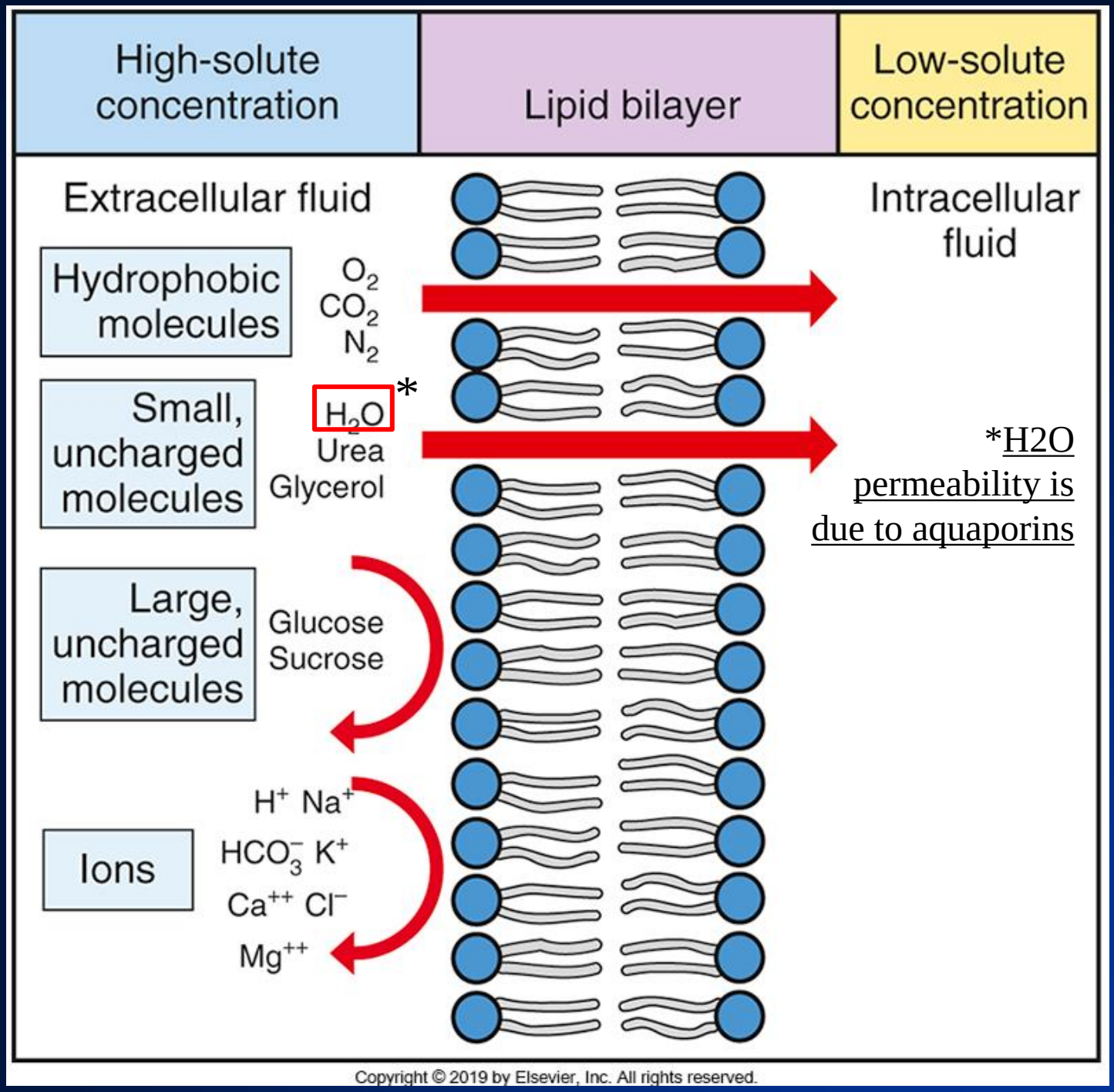
Communication across the membrane

- **Membrane transport proteins**
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} **Molecules go across**

- **Signal transduction processes – information goes across**

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ECF (aqueous)

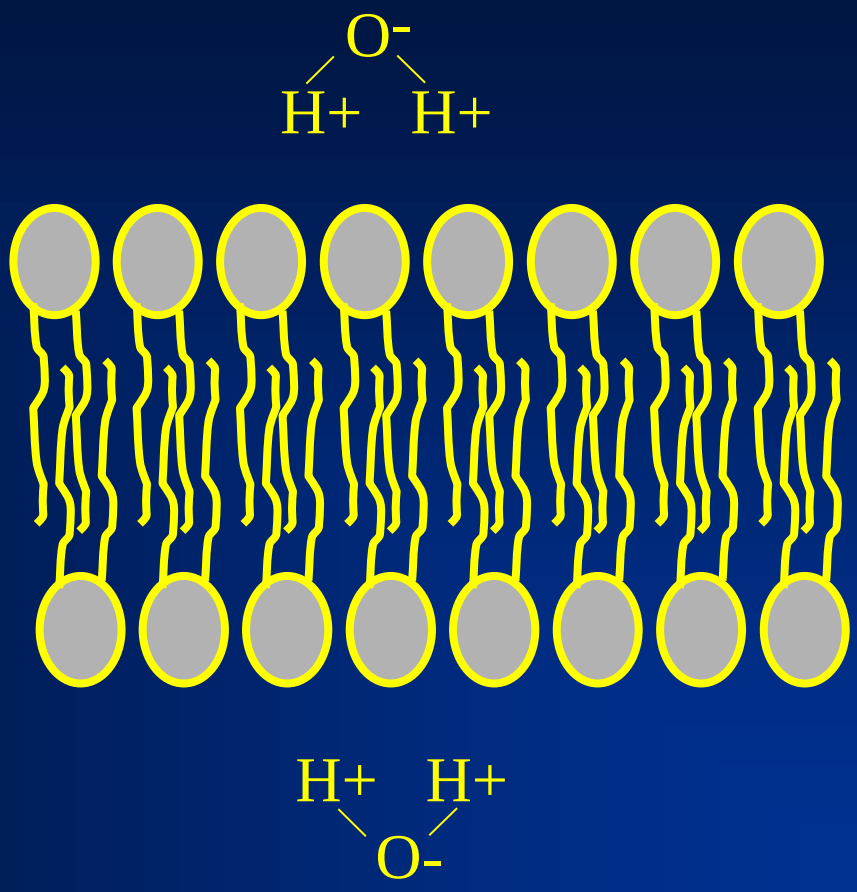
Charged

Charged

Neutral

Charged

Charged



ICF (aqueous)

WHAT TYPES OF MOLECULES ARE ABLE TO FREELY CROSS THE MEMBRANE? WHAT TYPES CANNOT?

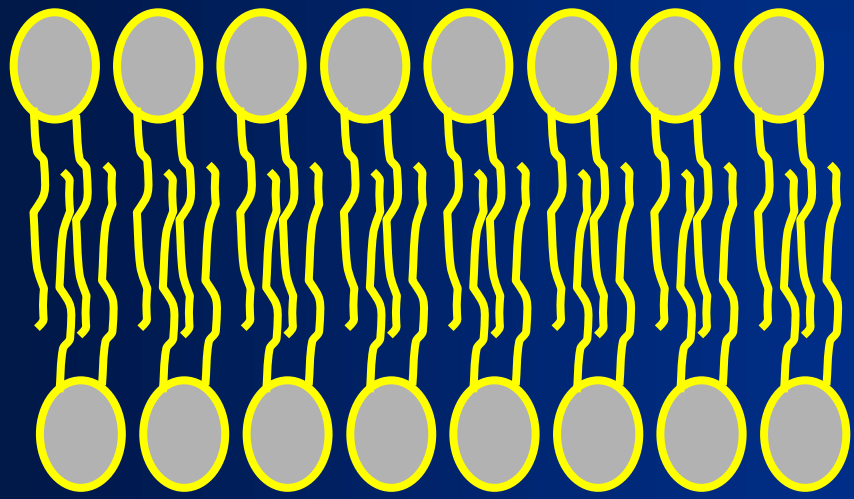
REMEMBER

Charged molecules like to hang out with charged molecules

Charged molecules are repelled by neutral molecules

Neutral molecules like to hang out with neutral molecules

Neutral molecules are repelled by charged molecules



Charged

Neutral (BARRIER)

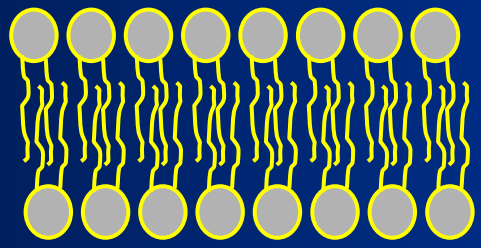
Charged

TERMS AND PHYSICAL INTERRELATIONSHIPS

Charged
- Polar
  - Hydrophilic
  - Lipophobic
- Cannot cross freely

Neutral (uncharged)
- Nonpolar
  - Hydrophobic
  - Lipophilic
- Can cross freely

Charged heads are not to scale; they are relatively small and do not present much of a barrier to neutral molecules



TERMS AND PHYSICAL INTERRELATIONSHIPS

Charged
- Polar
  - Hydrophilic
  - Lipophobic

Cannot cross freely

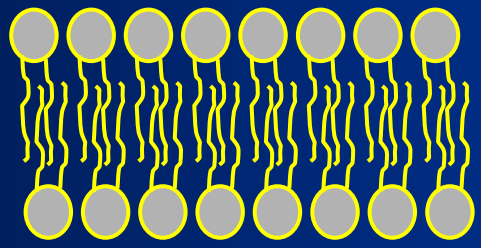
Ions, glucose, amino acids, neurotrans-mitters, etc.

Neutral (uncharged)
- Nonpolar
  - Hydrophobic
  - Lipophilic

Can cross freely

Fatty acids, cholesterol, steroid hormones, O2, CO2, etc.

<u>Determinates</u>
- Polarity
- Lipid solubility
- Size



TERMS AND PHYSICAL INTERRELATIONSHIPS

Charged

Polar

Hydrophilic

Lipophobic

Cannot cross freely

Ions, glucose, amino acids, neurotrans-mitters, etc.

Neutral (uncharged)

Nonpolar

Hydrophobic

Lipophilic

Can cross freely

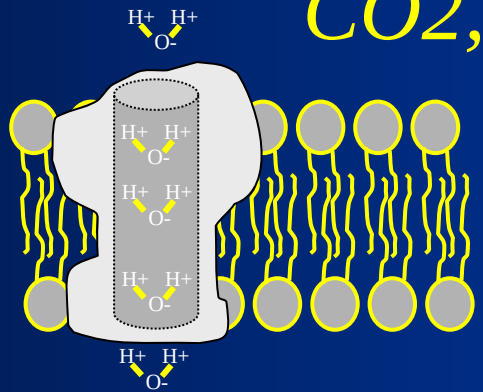
Fatty acids, cholesterol, steroid hormones, O2, CO2, etc.

<u>Determinates</u>

Polarity

Lipid solubility

Size



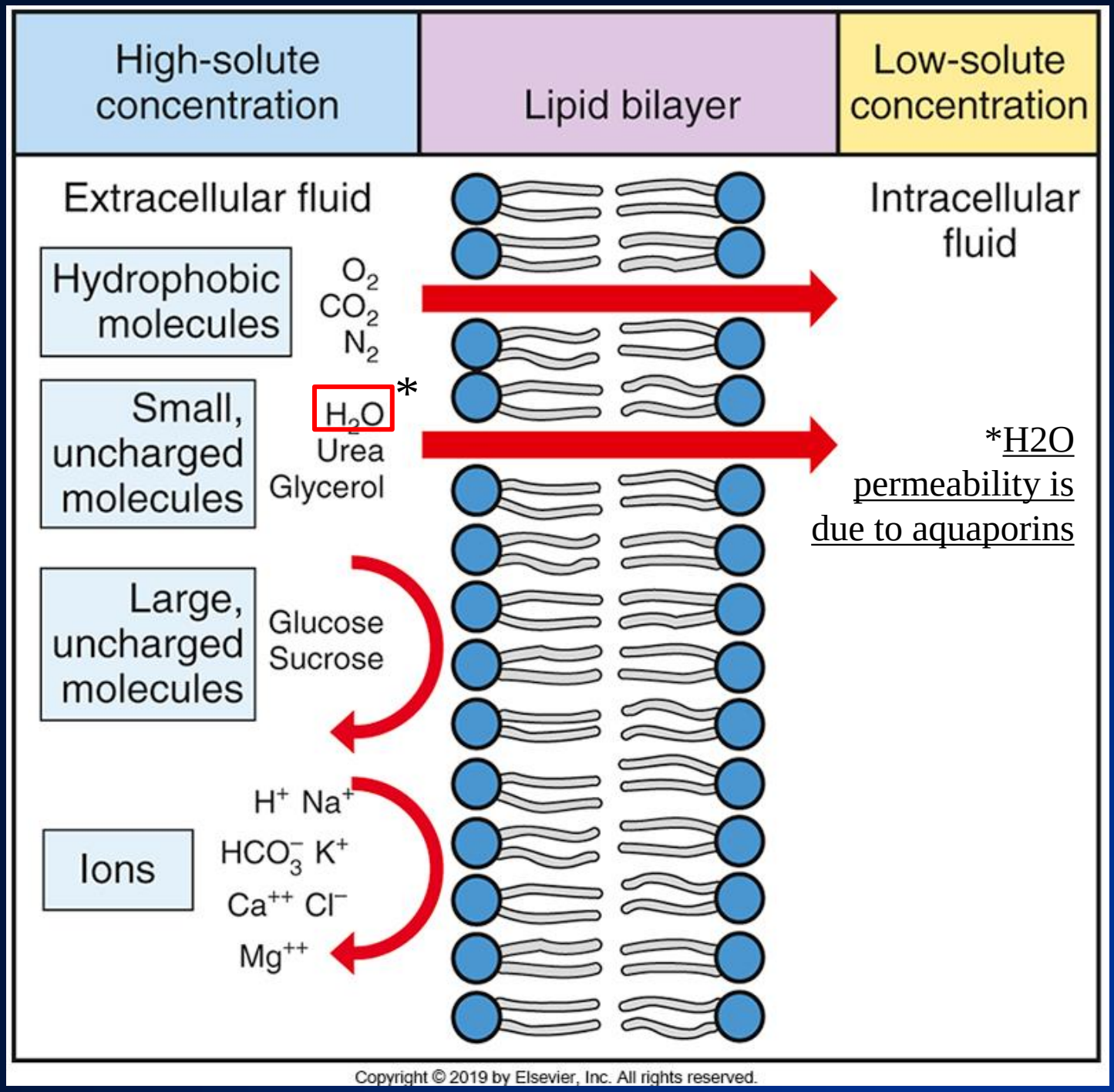
H+ H+

O-

WATER

(aquaporins)

Clin Path textbook



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FUNDAMENTAL PRINCIPLES (Part I)

I. Introduction to physiology and homeostasis
   - A. Physiology – Definitions
   - B. Homeostasis
   - C. Fluid compartments
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   - A. Gradients drive physiological processes
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III. Membrane potentials
   - A. Membrane potentials and gradients
   - B. Types of membrane potentials
   - C. Resting membrane potentials
   - D. Communication using membrane potentials
   - E. Determinates of membrane and equilibrium potentials
   - F. Graded and action potentials

IV. Nerve impulse propagation, synapses and neural modulation
   - A. Think communication!
   - B. Nerve impulse propagation and conduction
   - C. Synapses
   - D. Neural modulation

MOVEMENT ACROSS THE MEMBRANE

Molecules

Transport

- **Passive processes (no ATP energy required)**
  - **Diffusion**
  - **Facilitated diffusion**
- **Active processes (ATP energy required)**
  - **Primary active transport**
  - **Secondary active transport**

(Membrane Carrier Protein) Mediated transport

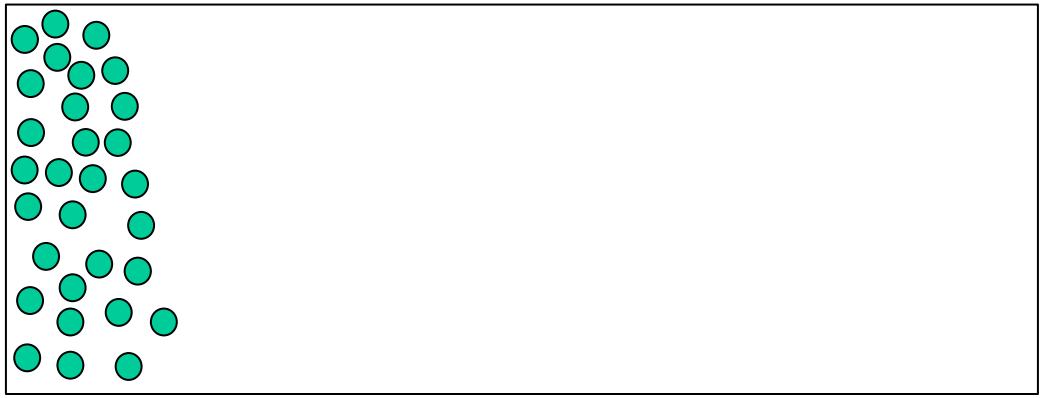
Translocation

- **Ligand-gated channels**
- **Voltage-gated channels**
- **Many other channel proteins**

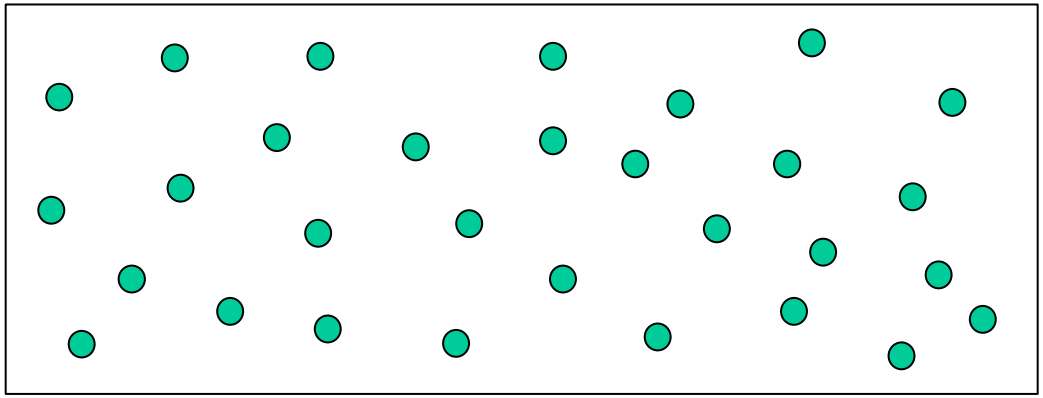
Information

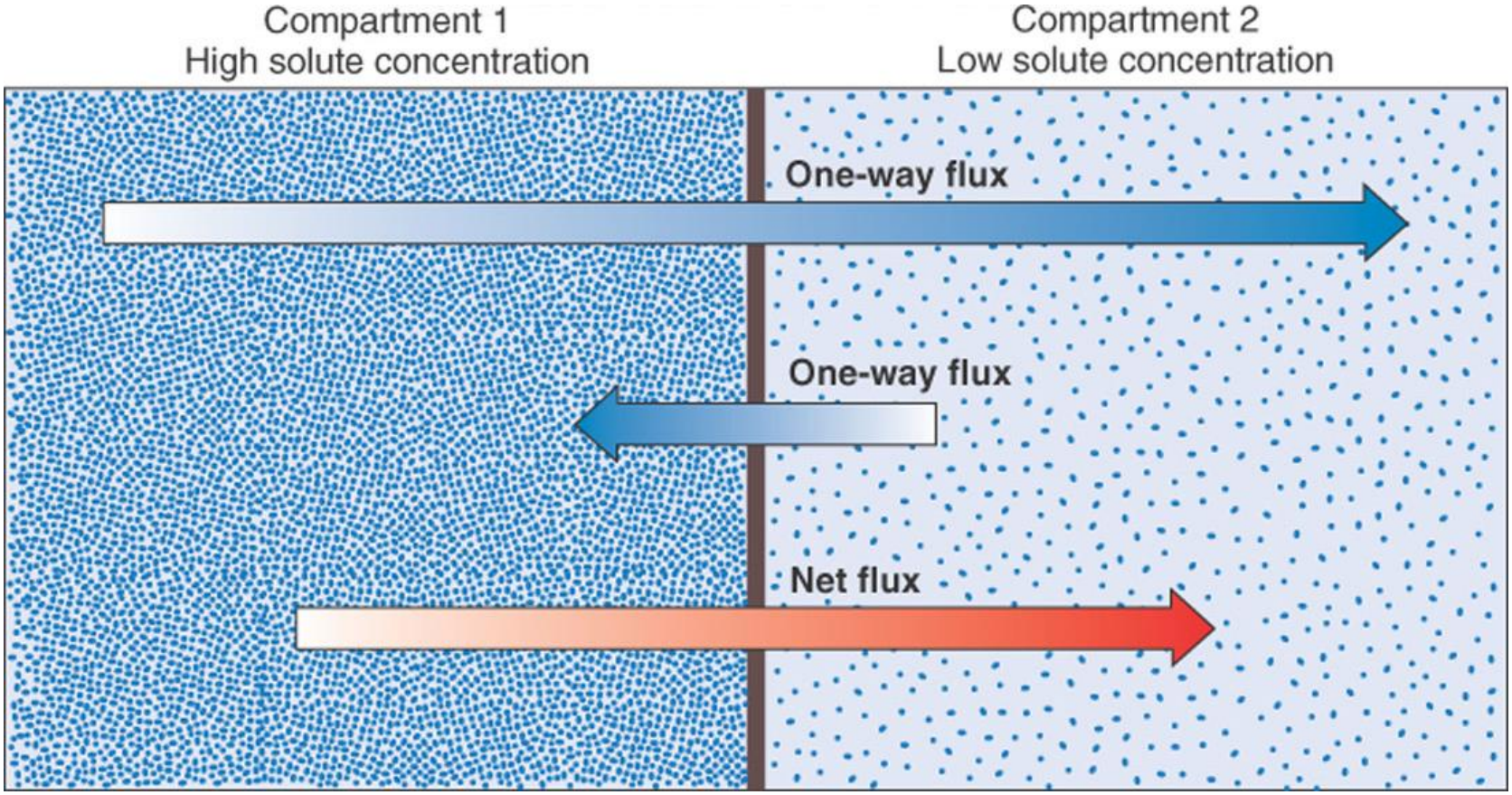
- **Signal transduction**

Solute molecules placed in a solvent will evenly distribute themselves over time, generally moving from an area of higher concentration to lower.



The particles will eventually be evenly distributed, achieving a distribution equilibrium.



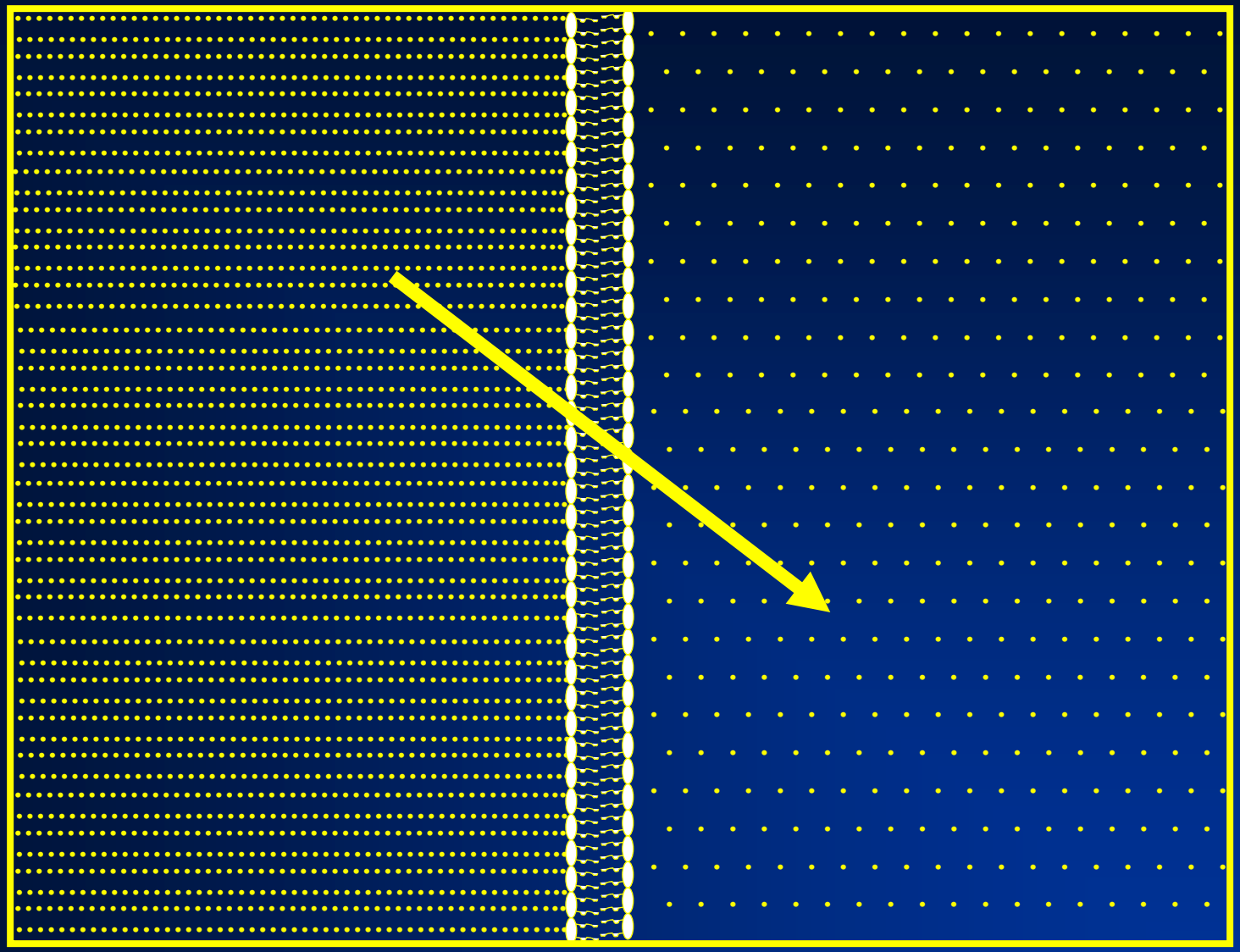


Net flux is the difference between the two movements

Semipermeable membrane that allows the solute particle to pass through

Adapted from Vander's Physiology

The driving force for diffusion is the chemical concentration gradient



Simple Diffusion

Diffusion rate = [SA * D * (C(in) – C(out))]/T

Fick's Law of Diffusion

$$\text{Diffusion rate} = \frac{(D)^{\text{Lipid sol}} [SA * \sqrt{MW} * (C(in) - C(out))]}{T}$$

SA = surface area available for diffusion

D = diffusivity of solute = Lipid solubility/$\sqrt{MW}$

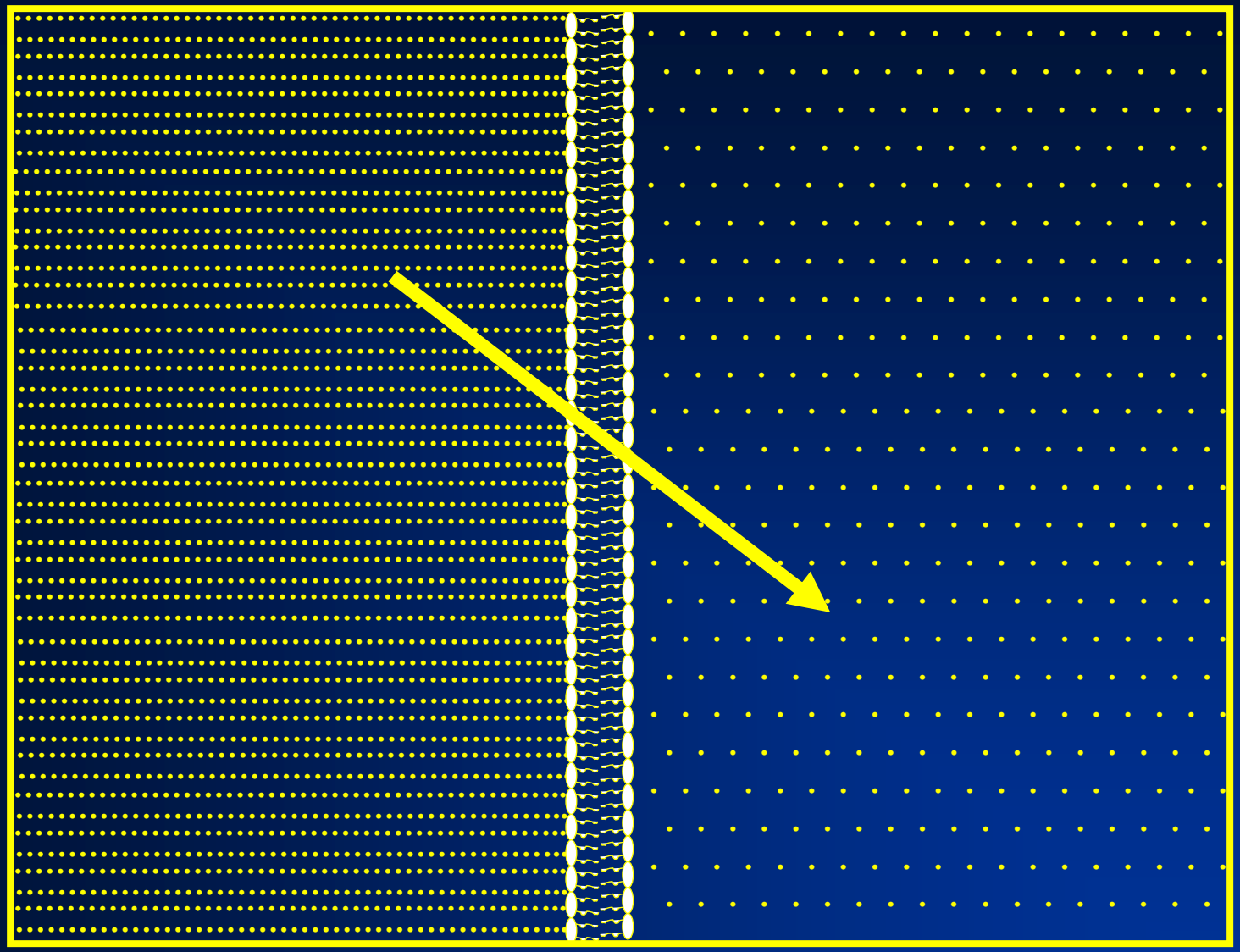
C = concentration of solute

T = thickness of the diffusion barrier

Which factors are directly related to diffusion?

Which factors are inversely related to diffusion?

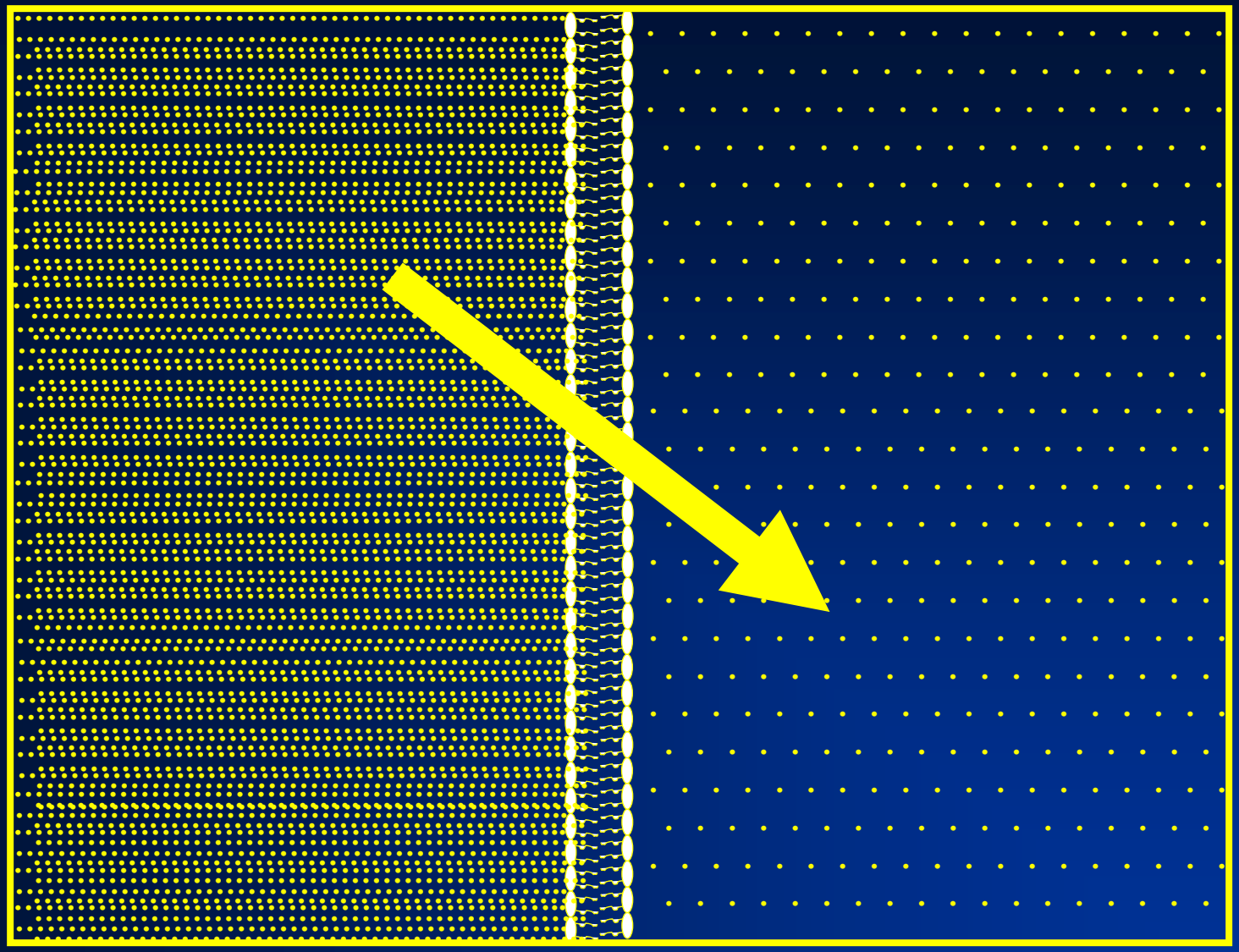
The driving force for diffusion is the chemical concentration gradient



Simple Diffusion

Diffusion rate = [SA * D * (C(in) – C(out))]/T

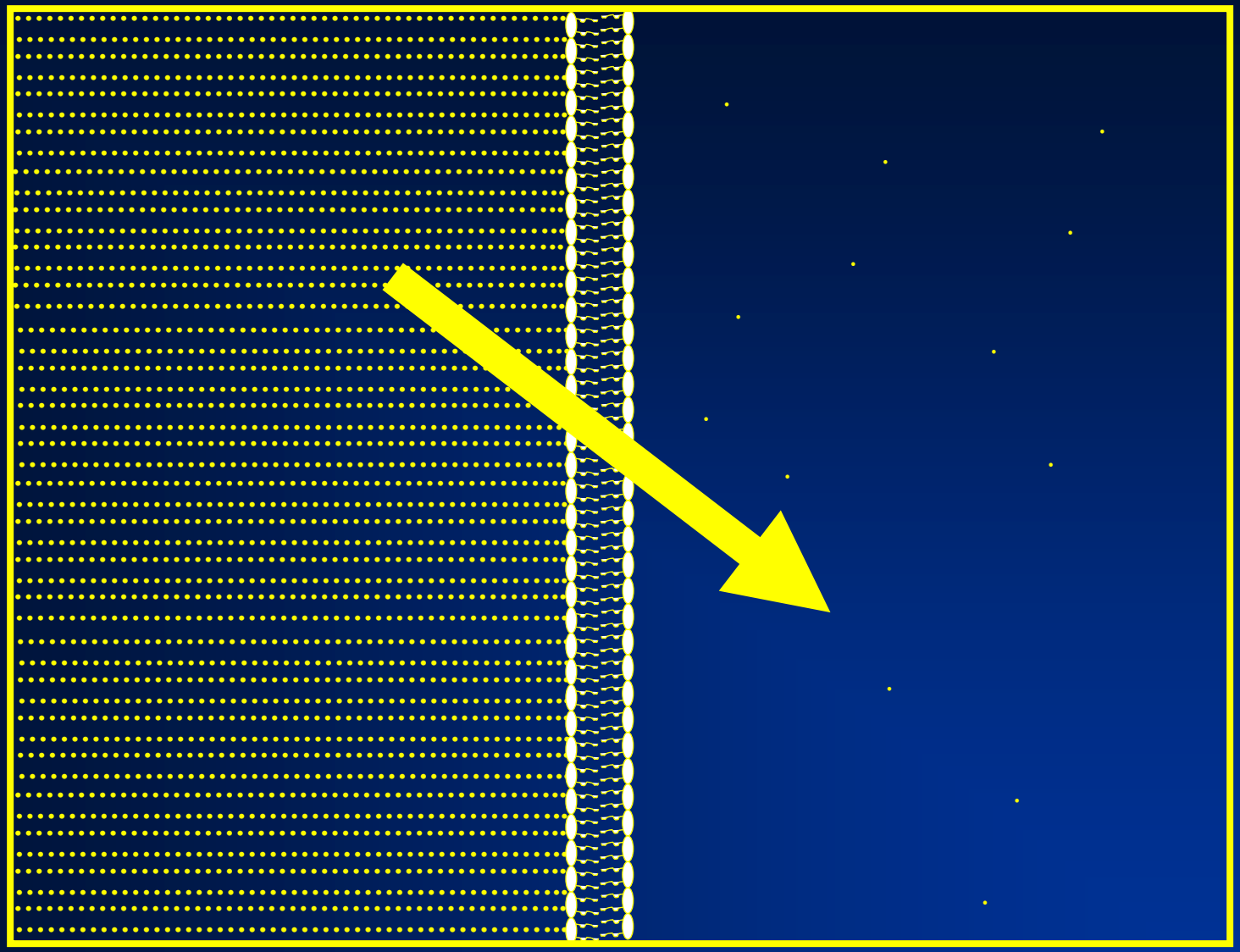
The driving force for diffusion is the chemical concentration gradient



Simple Diffusion

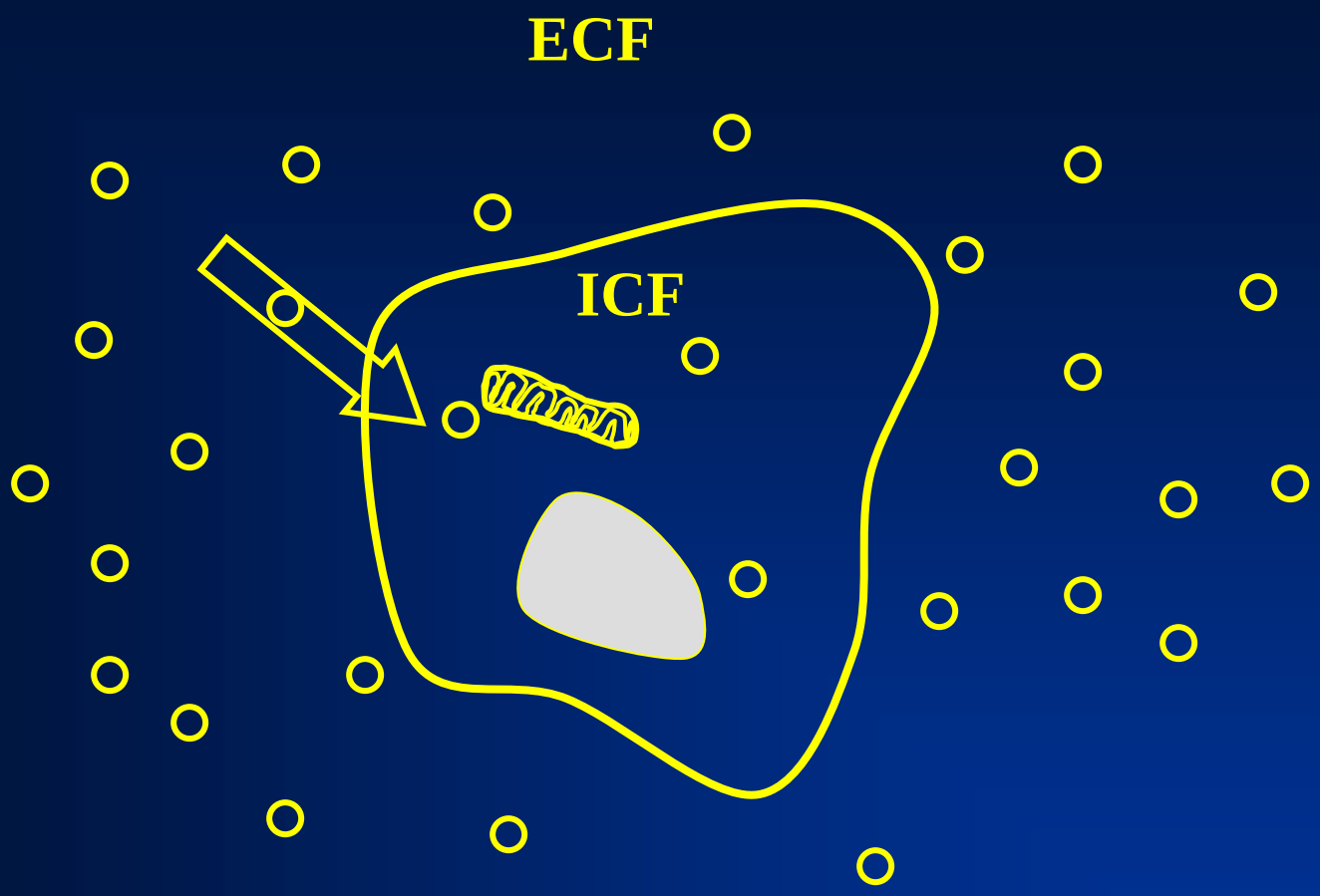
Diffusion rate = [SA * D * (C(in) – C(out))]/T

The driving force for diffusion is the chemical concentration gradient



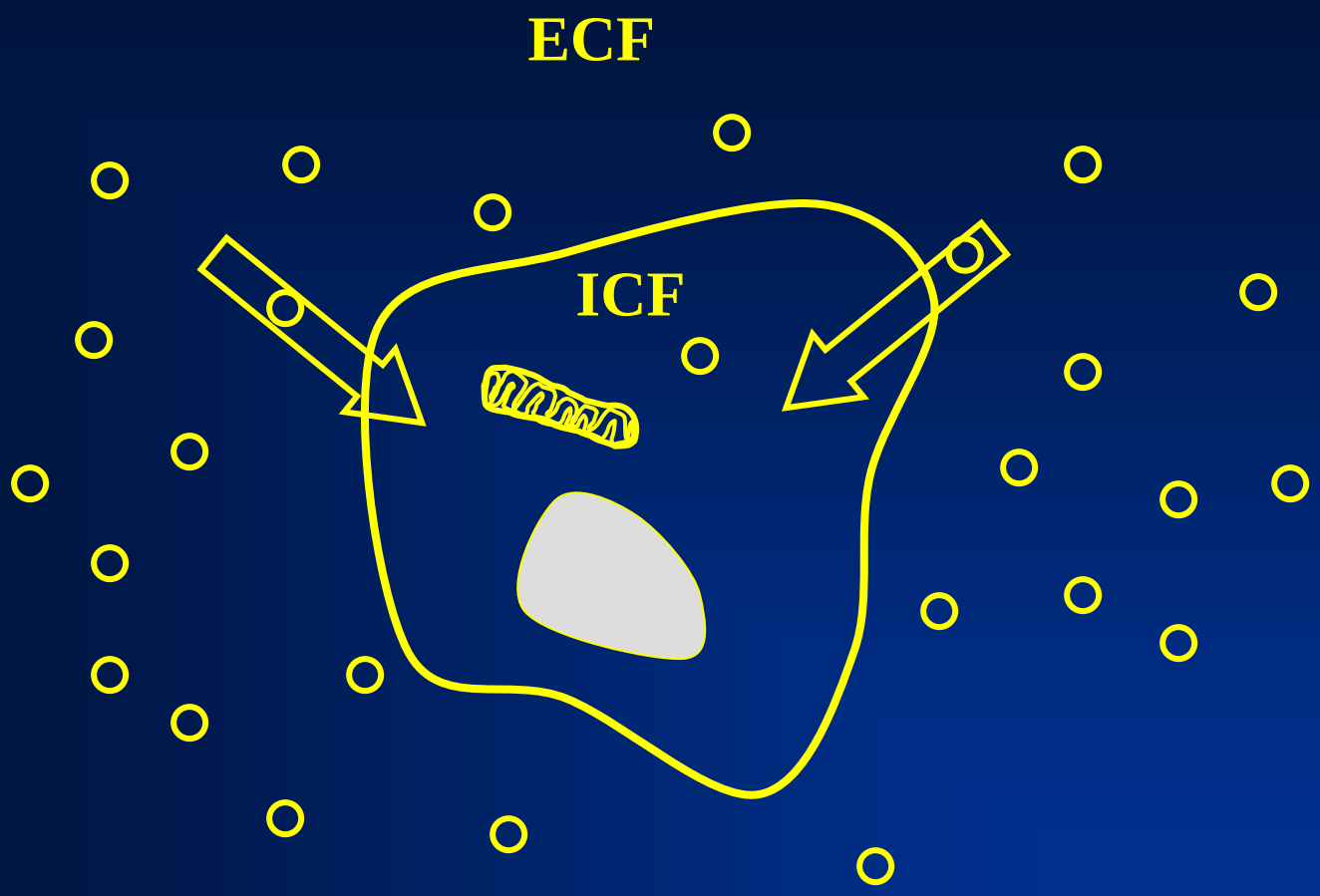
Simple Diffusion

Diffusion rate = [SA * D * (C(in) – C(out))]/T



- Oxygen diffuses into the cell

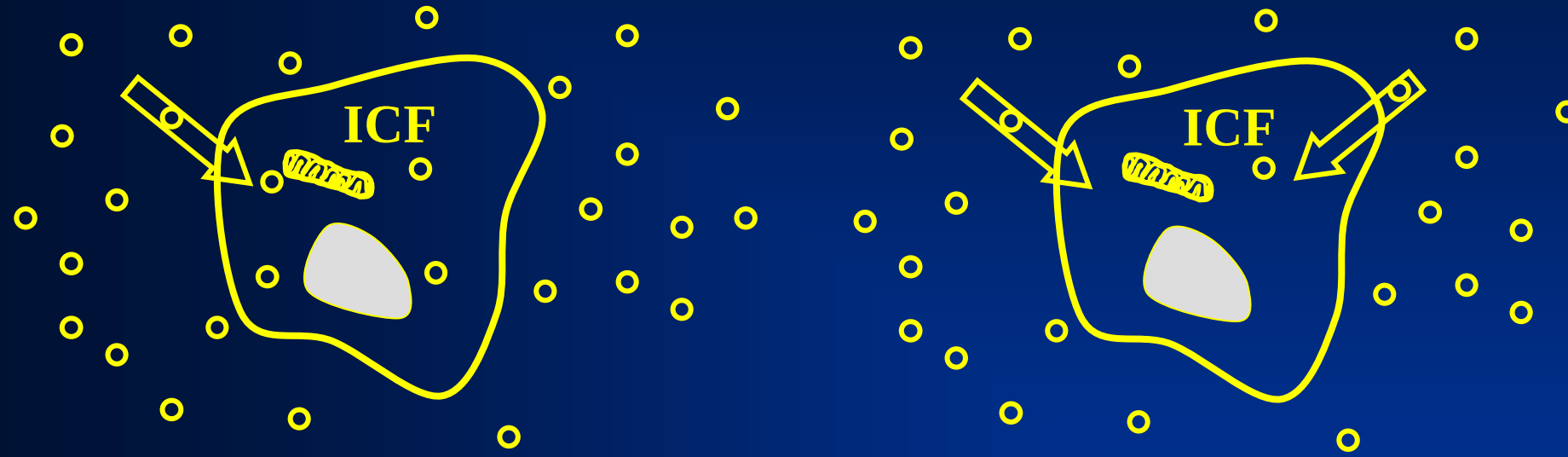
What happens to intracellular oxygen content when a cell becomes metabolically active?



- Oxygen diffuses into the cell

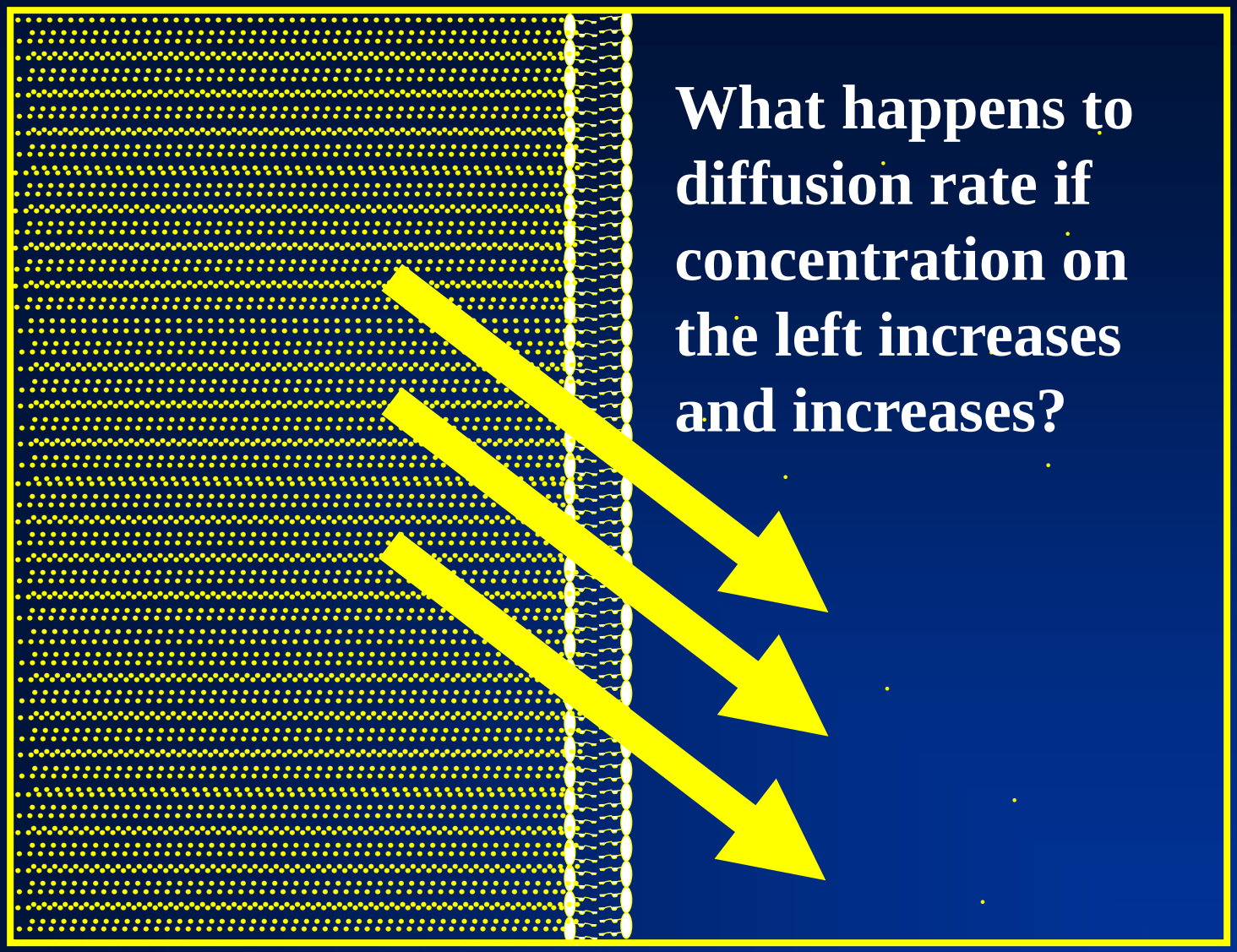
What happens to diffusion rate when a cell becomes metabolically active?

ECF



- Oxygen diffuses into the cell

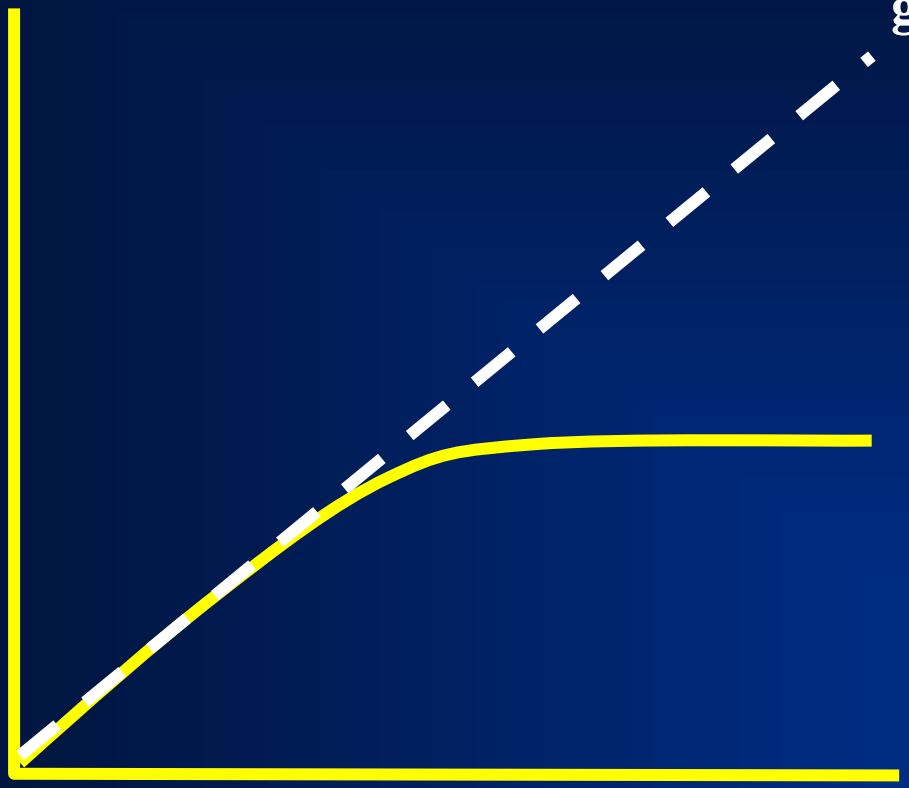
What happens to diffusion rate when a cell becomes metabolically active?



Simple Diffusion

Diffusion rate = [SA * D * (C(in) – C(out))]/T

With simple diffusion, flux rate and capacity is dependent on concentration gradient



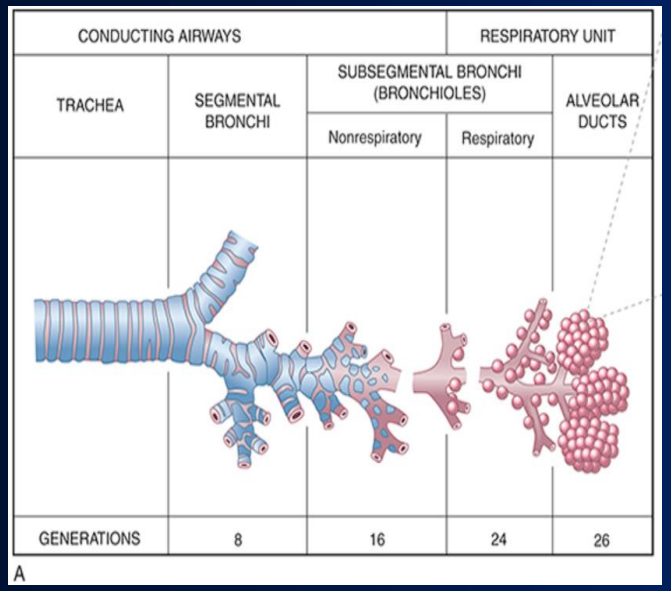
With mediated transport, maximum flux rate is limited by the number of carriers

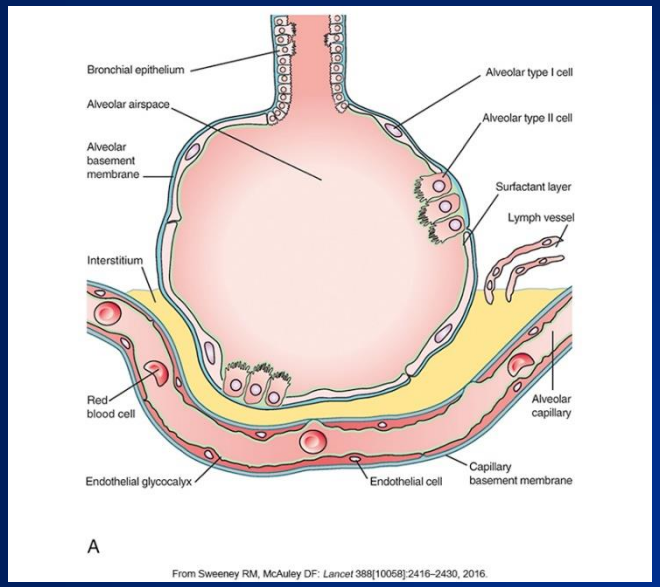
Diffusion rate = [SA * D * (C(in) – C(out))]/T

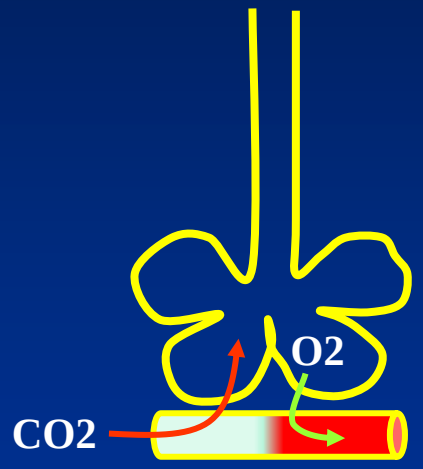
Clinical Correlation:
Gas diffusion in the lung

$$\dot{V}_{gas} = \frac{SA \times Diff \times (C1 - C2)}{T}$$

(Diff = Solubility/ √ MW)







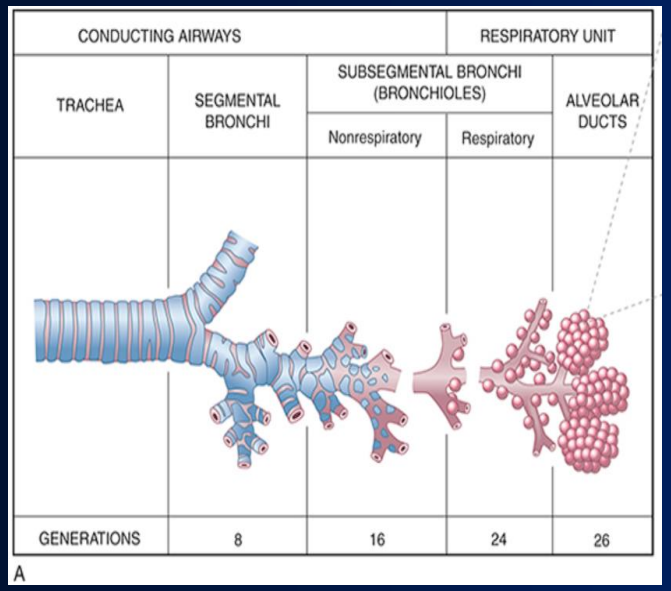
Normally, there are about 1000 capillaries embedded in the wall of each alveolus

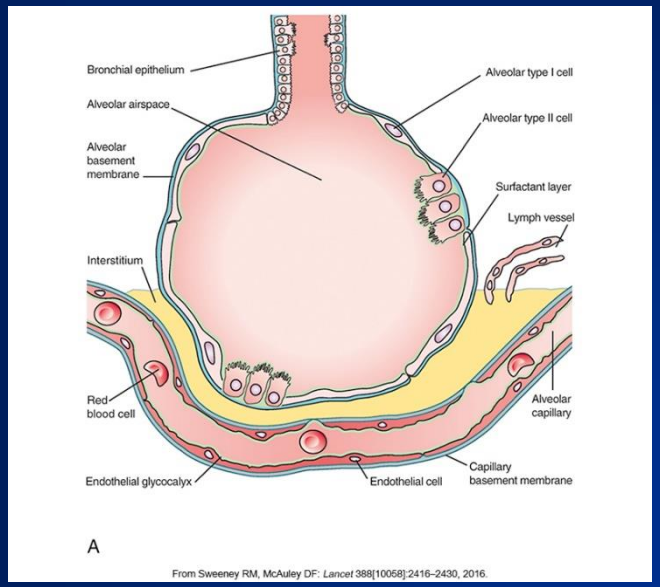
Fig. 35.3, 36.09A Clin Path textbook

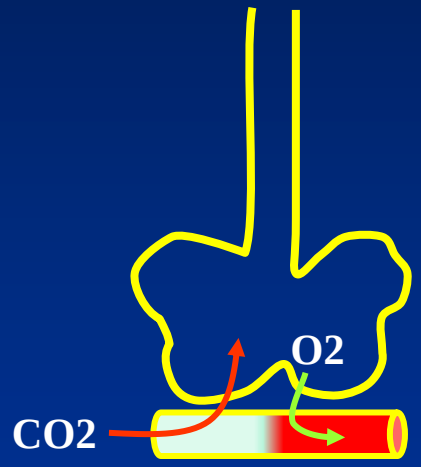
Clinical Correlation:
Gas diffusion in the lung

$$\dot{V}_{gas} = \frac{SA \times Diff \times (C1 - C2)}{T}$$

(Diff = Solubility/ √ MW)





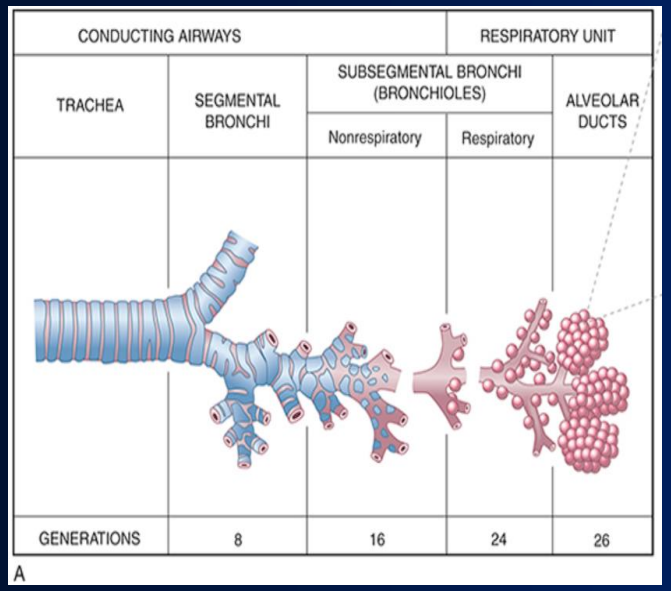


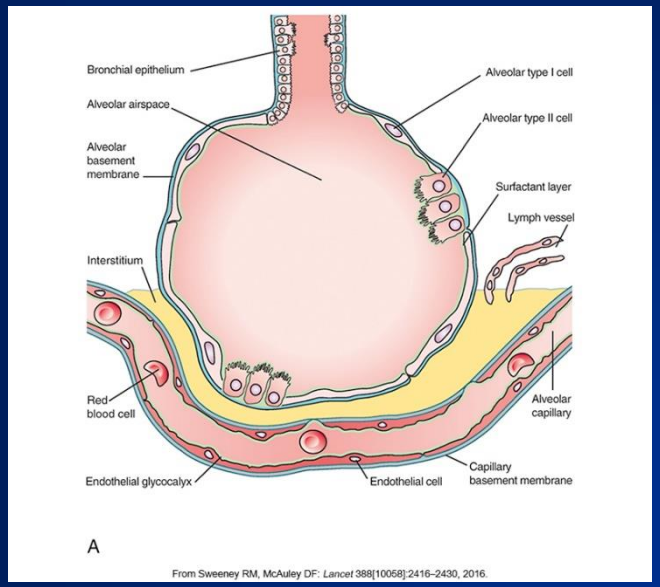
Emphysema results in septal wall breakdown and a decrease in effective surface are for diffusion

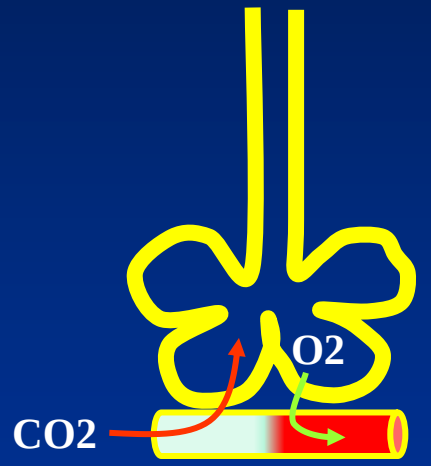
Fig. 35.3, 36.09A

Clinical Correlation:
Gas diffusion in the lung

$$\dot{V}_{gas} = \frac{SA \times Diff \times (C1 - C2)}{T} \qquad (Diff = Solubility/\sqrt{MW})$$







Interstitial pulmonary fibrosis is the result of the proliferation of scar tissue in the small airways and alveolar walls

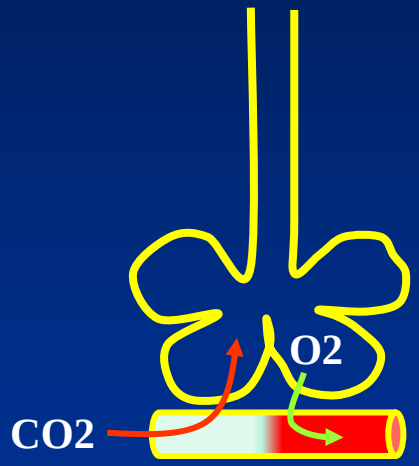
Fig. 35.3, 36.09A Clin Path textbook

Clinical Correlation:
Gas diffusion in the lung

$$\dot{V}_{gas} = \frac{SA \times Diff \times (C1 - C2)}{T}$$

(Diff = Solubility/ $\sqrt{}$ MW)

a. Exercise
b. Emphysema
c. Pulmonary edema
d. Alveolar fibrosis
e. Supplemental oxygen



What part of the equation is affected when supplemental oxygen is used?

Video is Posted on Google Drive in the Independent Study folder and Handout is Posted on CANVAS

Is simple diffusion really simple?
Or is it simply elegant, effective and efficient?

Giving proper respect to simple diffusion

MOVEMENT ACROSS THE MEMBRANE

Molecules

Transport

- **Passive processes (no ATP energy required)**
  - **Diffusion**
  - **Facilitated diffusion**
- **Active processes (ATP energy required)**
  - **Primary active transport**
  - **Secondary active transport**

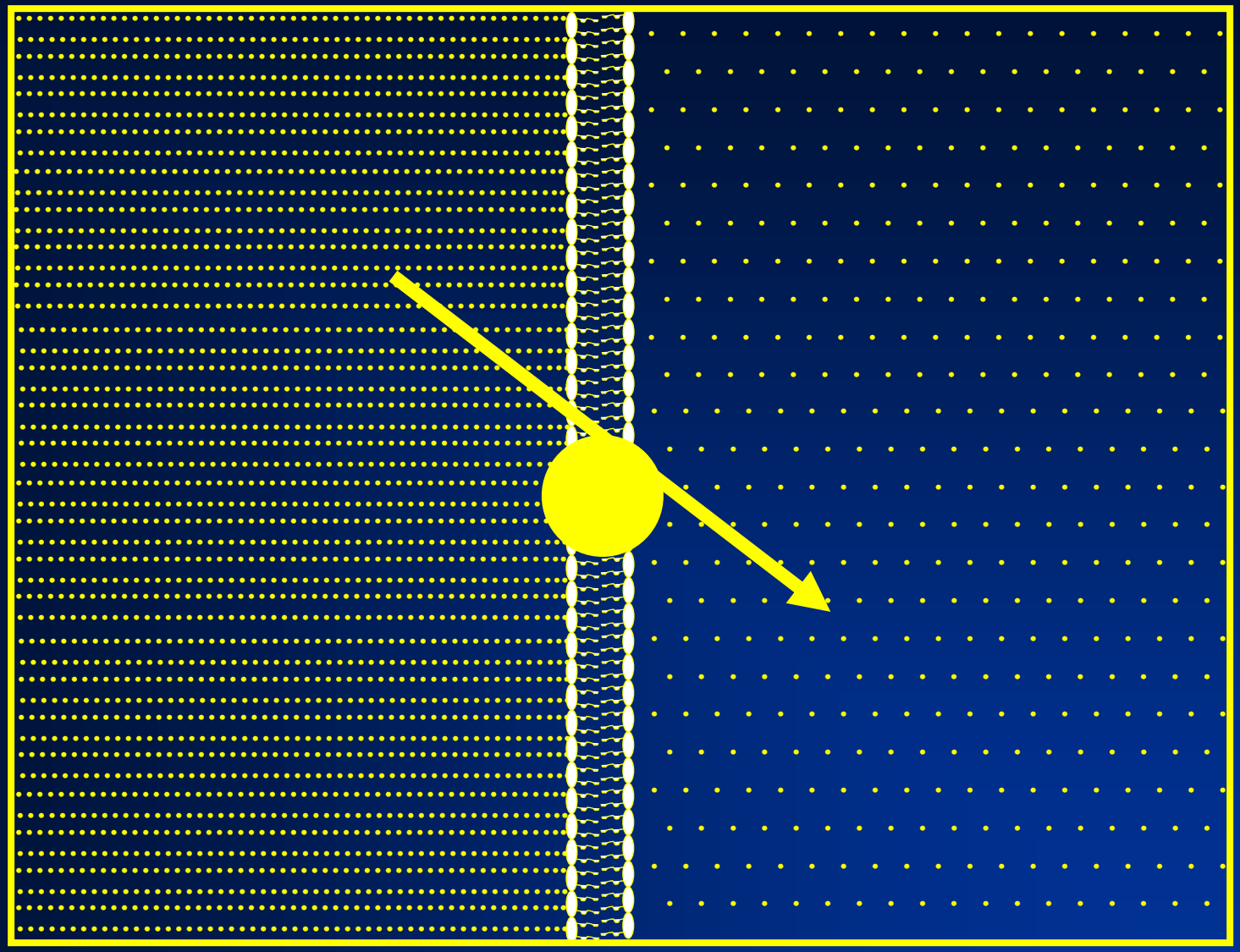
(Membrane Carrier Protein) Mediated transport

Translocation

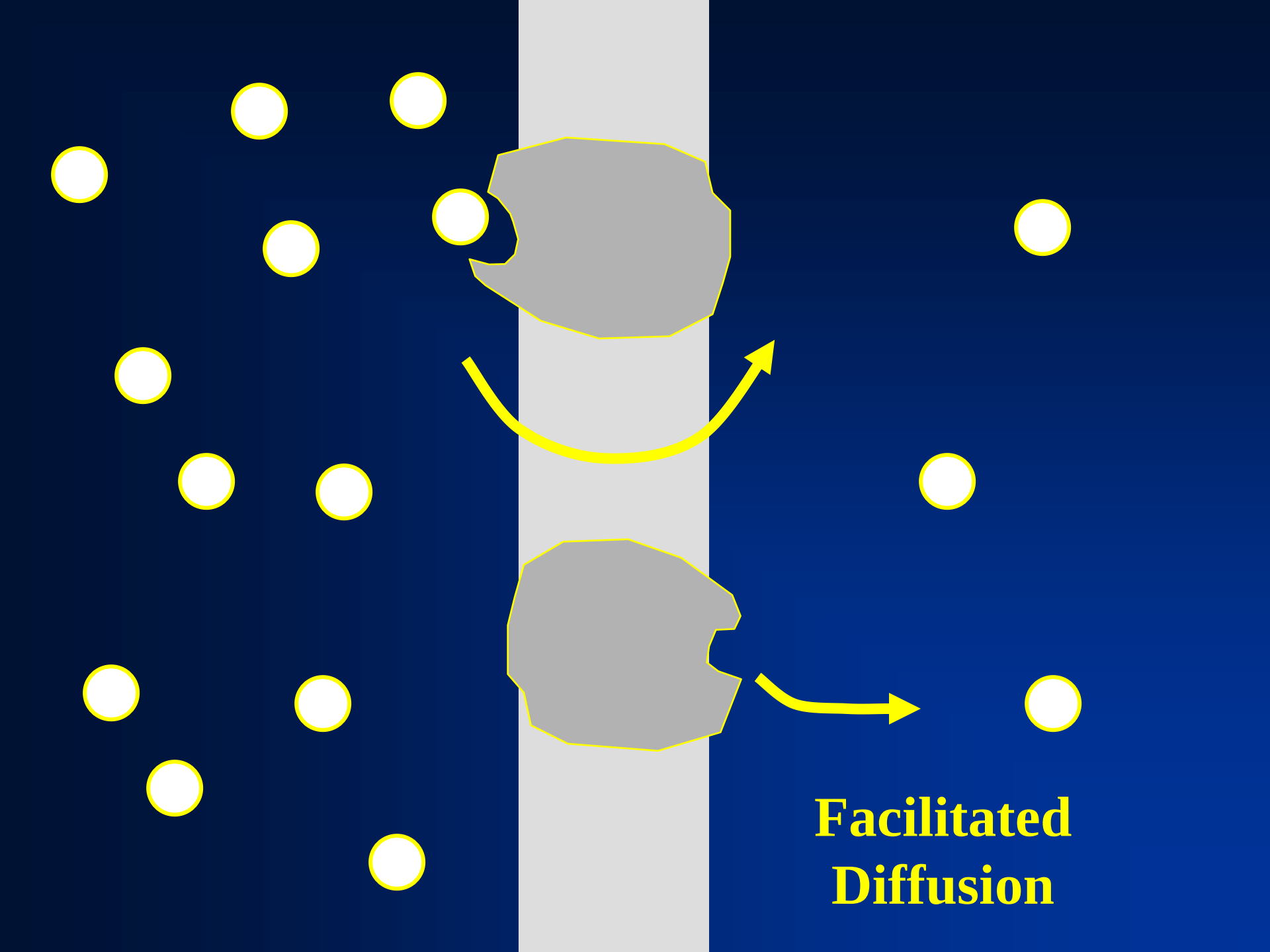
- **Ligand-gated channels**
- **Voltage-gated channels**
- **Many other channel proteins**

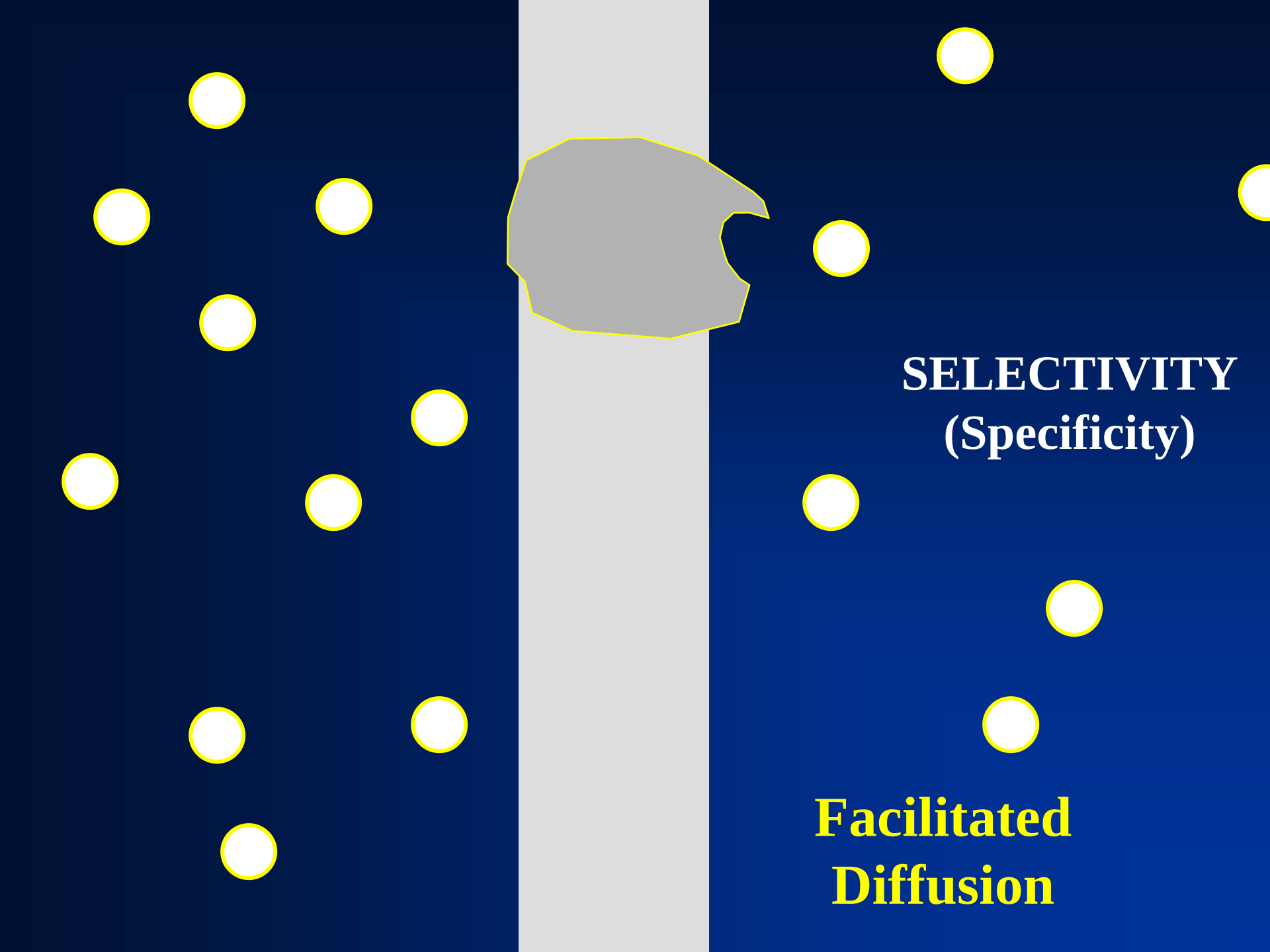
Information

- **Signal transduction**

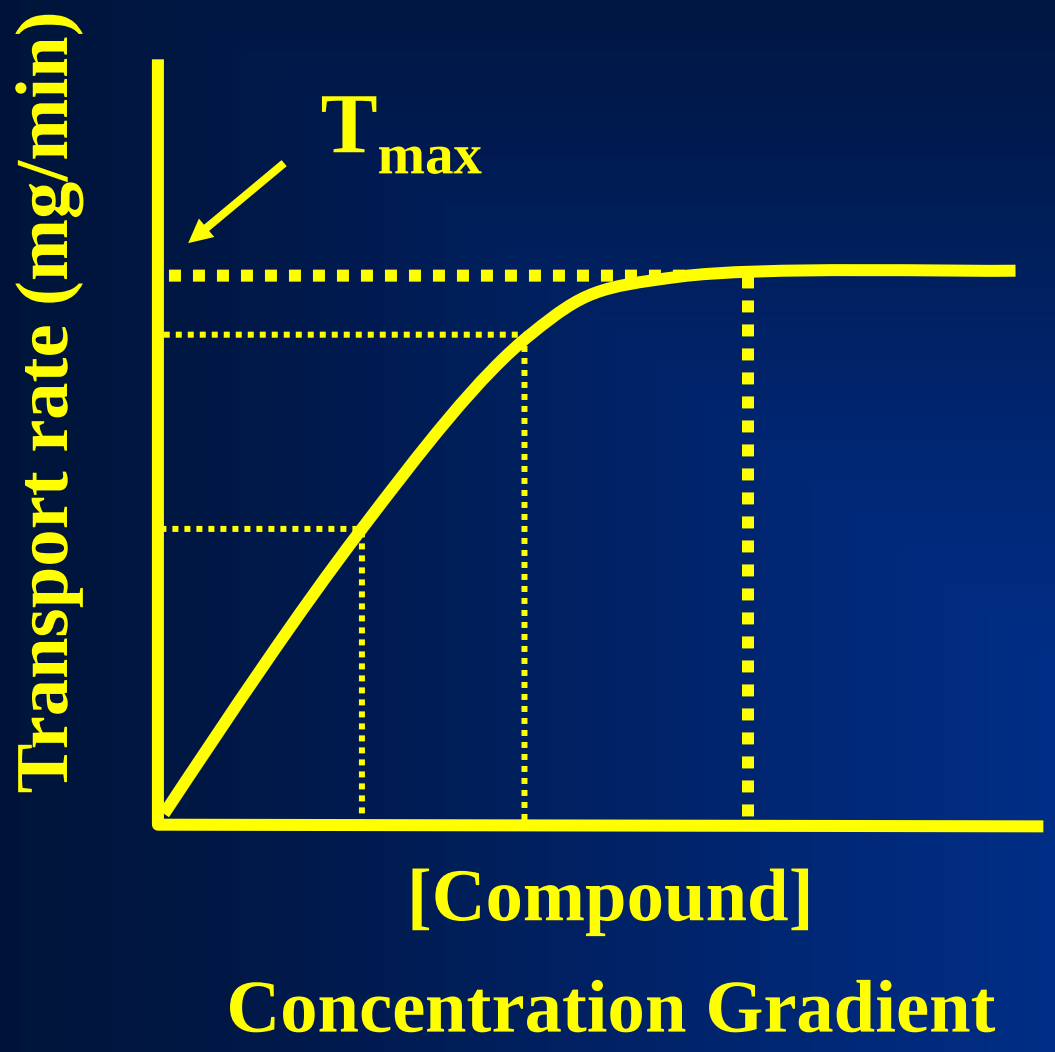


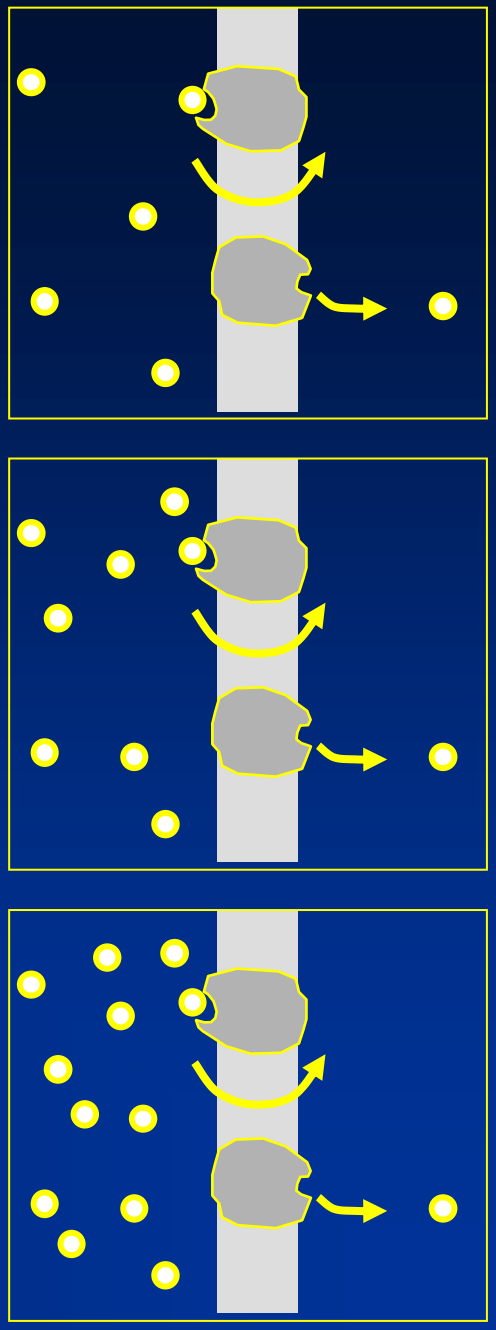
Carrier Facilitated Diffusion


Facilitated
Diffusion

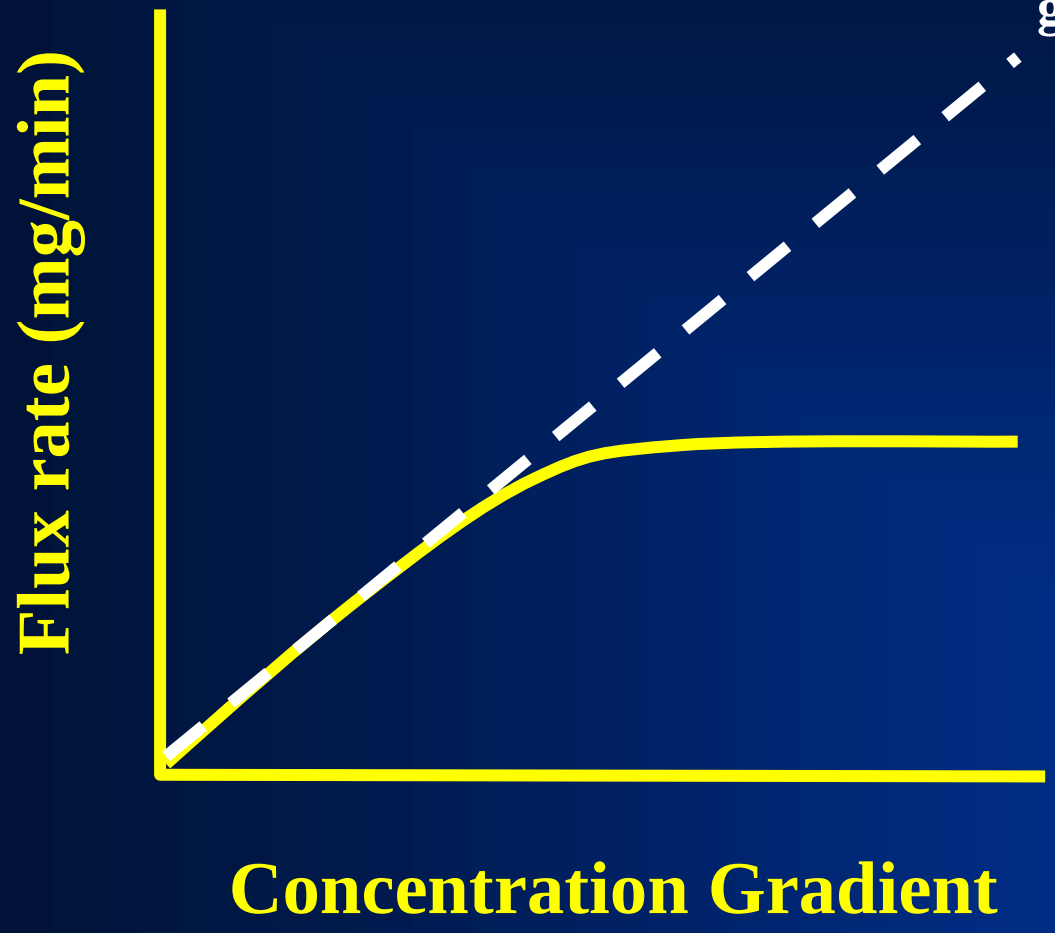

SELECTIVITY
(Specificity)
Facilitated
Diffusion

Saturation kinetics and T_{max}



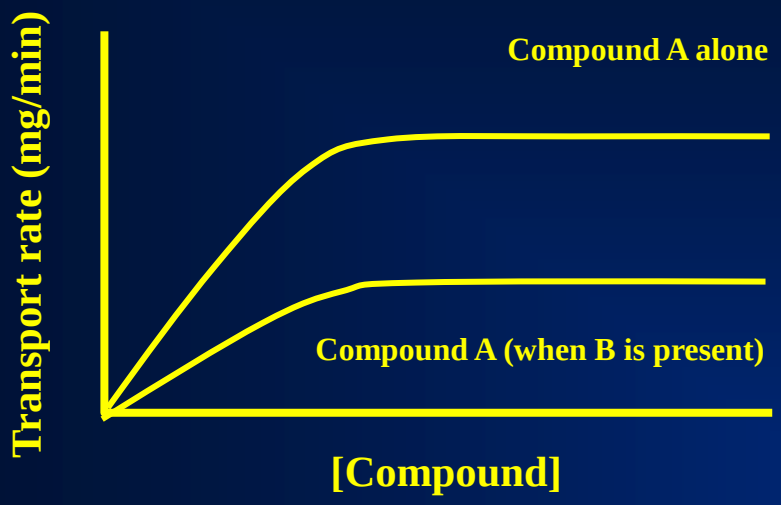


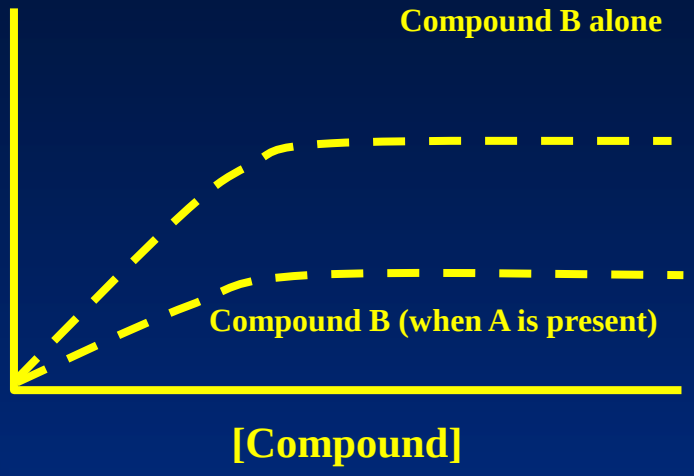
With simple diffusion, flux rate and capacity is dependent on concentration gradient

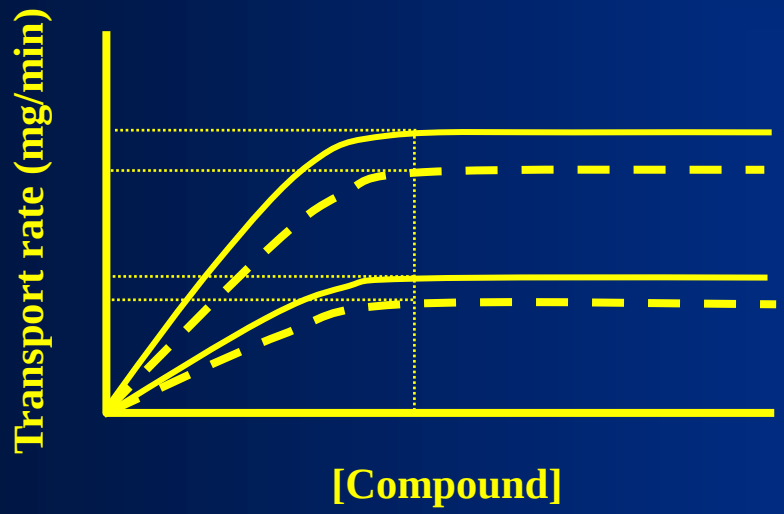


With mediated transport, maximum flux rate is limited by the number of carriers

Competition



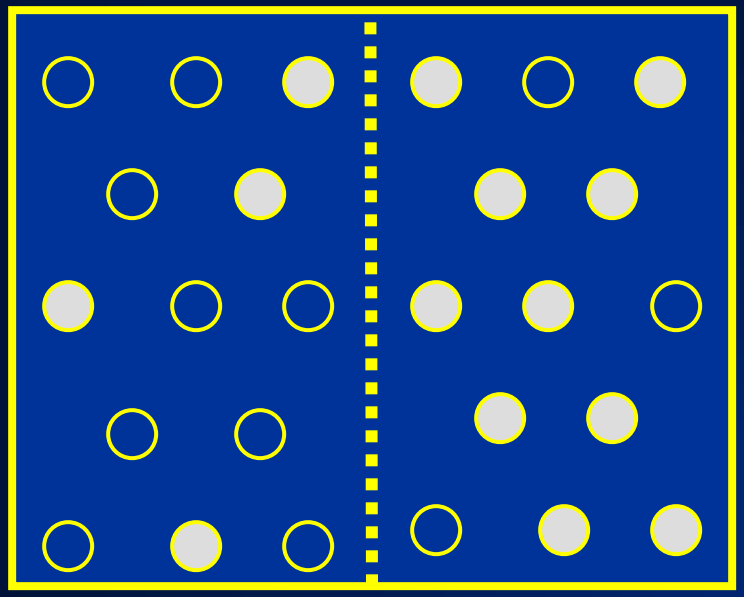




Note changes in Tmax due to competition

Water molecule

Solute molecule

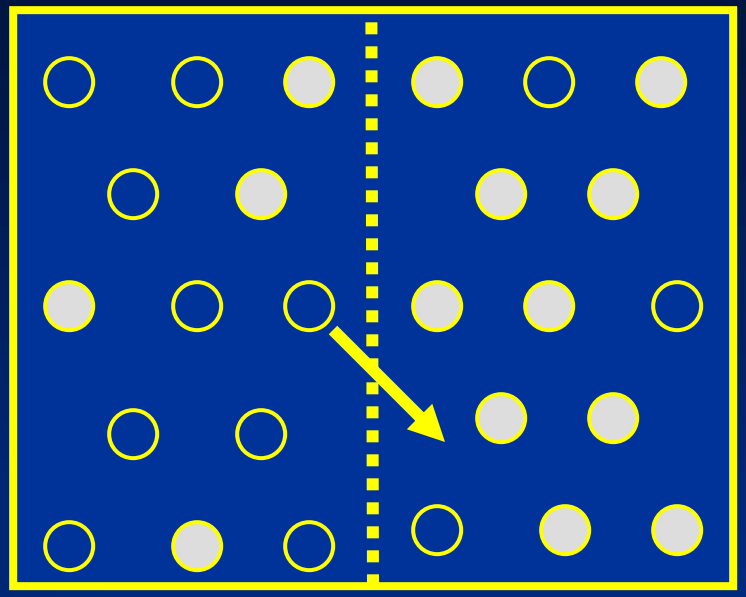


A

High [water]
Low [solute]

B

Low [water]
High [solute]



Water will diffuse down its concentration gradient from compartment A → B

Osmosis

MOVEMENT ACROSS THE MEMBRANE

Molecules

Transport

- **Passive processes (no ATP energy required)**
  - **Diffusion**
  - **Facilitated diffusion**
- **Active processes (ATP energy required)**
  - **Primary active transport**
  - **Secondary active transport**

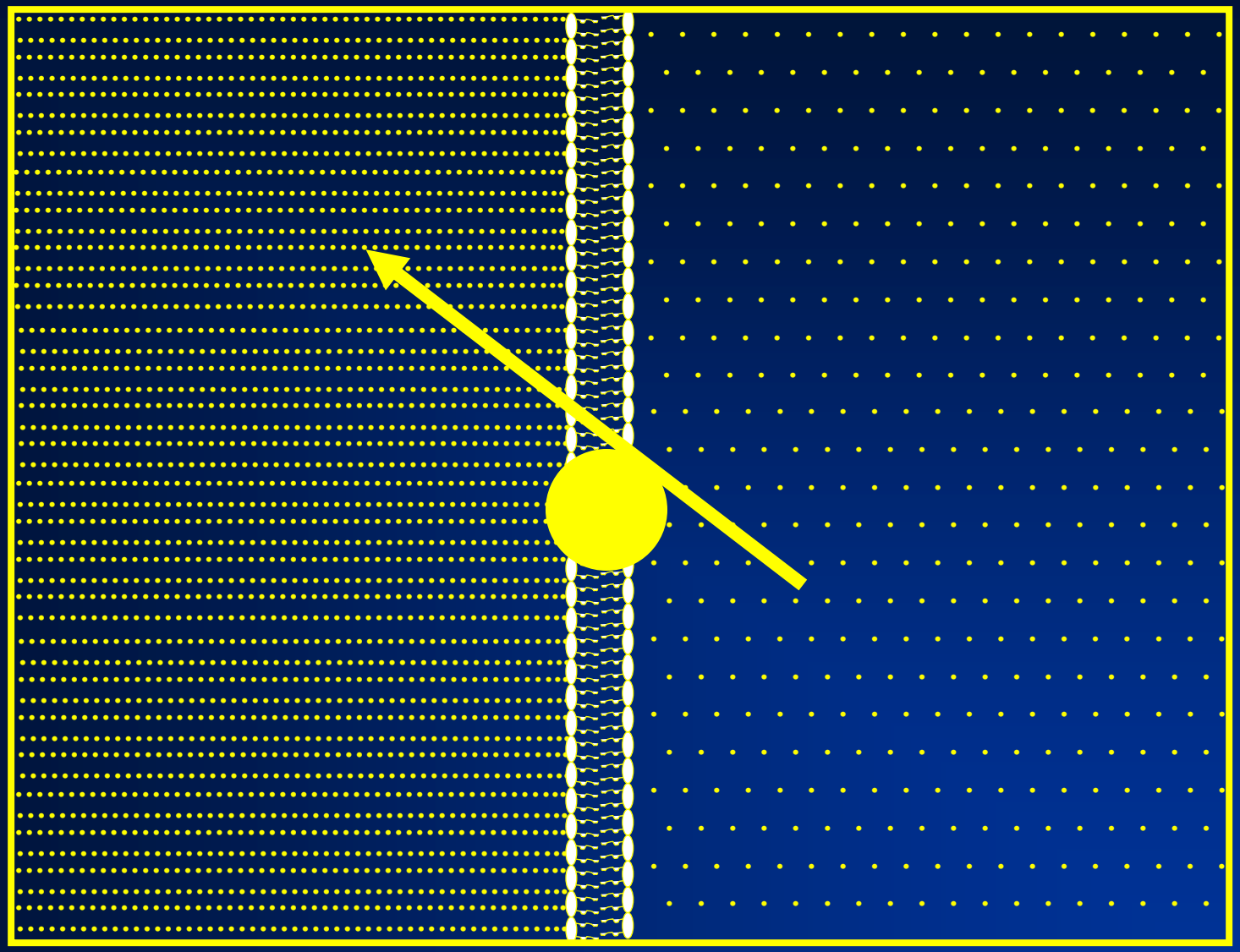
(Carrier) Mediated transport

Translocation

- **Ligand-gated channels**
- **Voltage-gated channels**

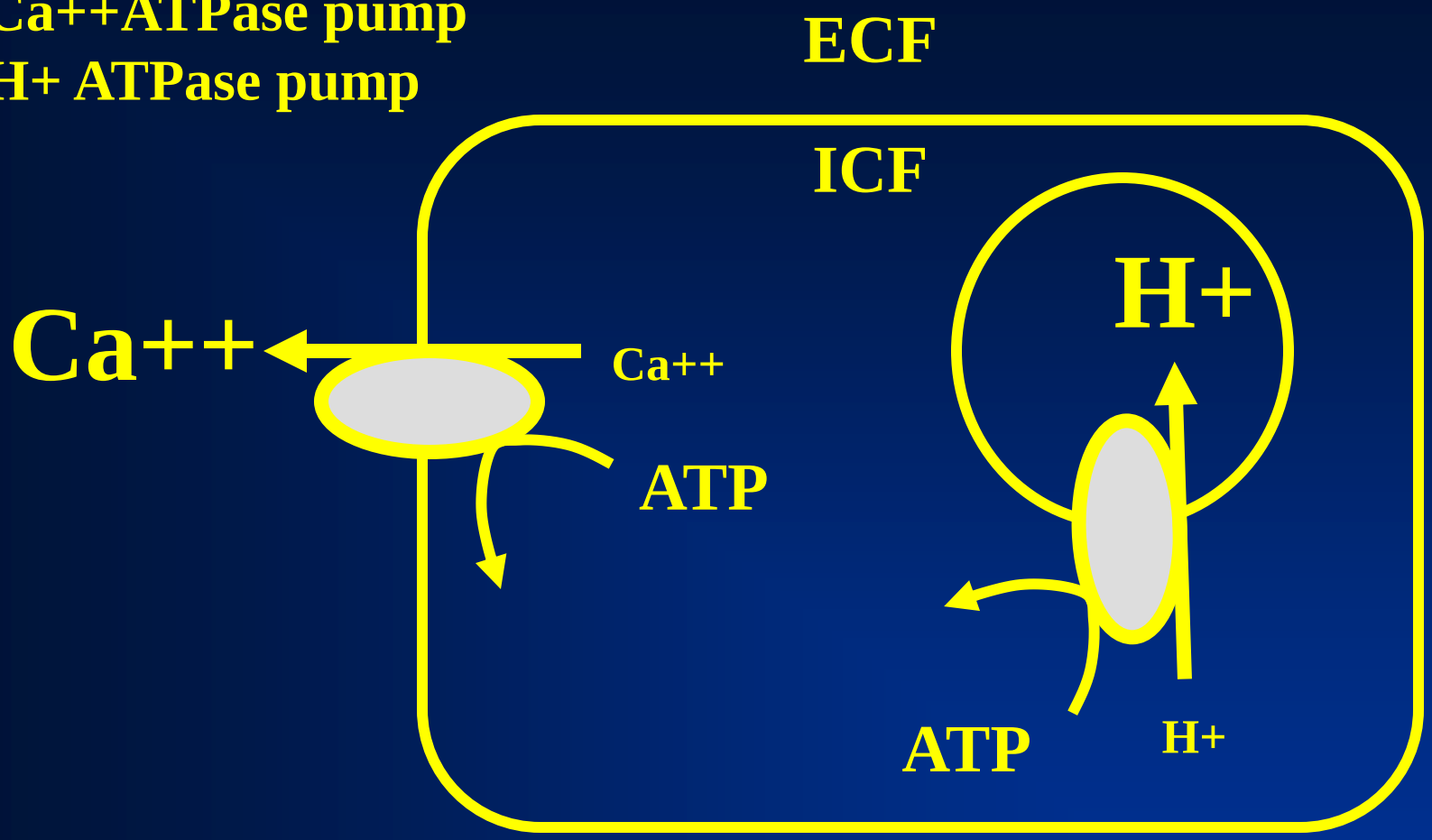
Information

- **Signal transduction**



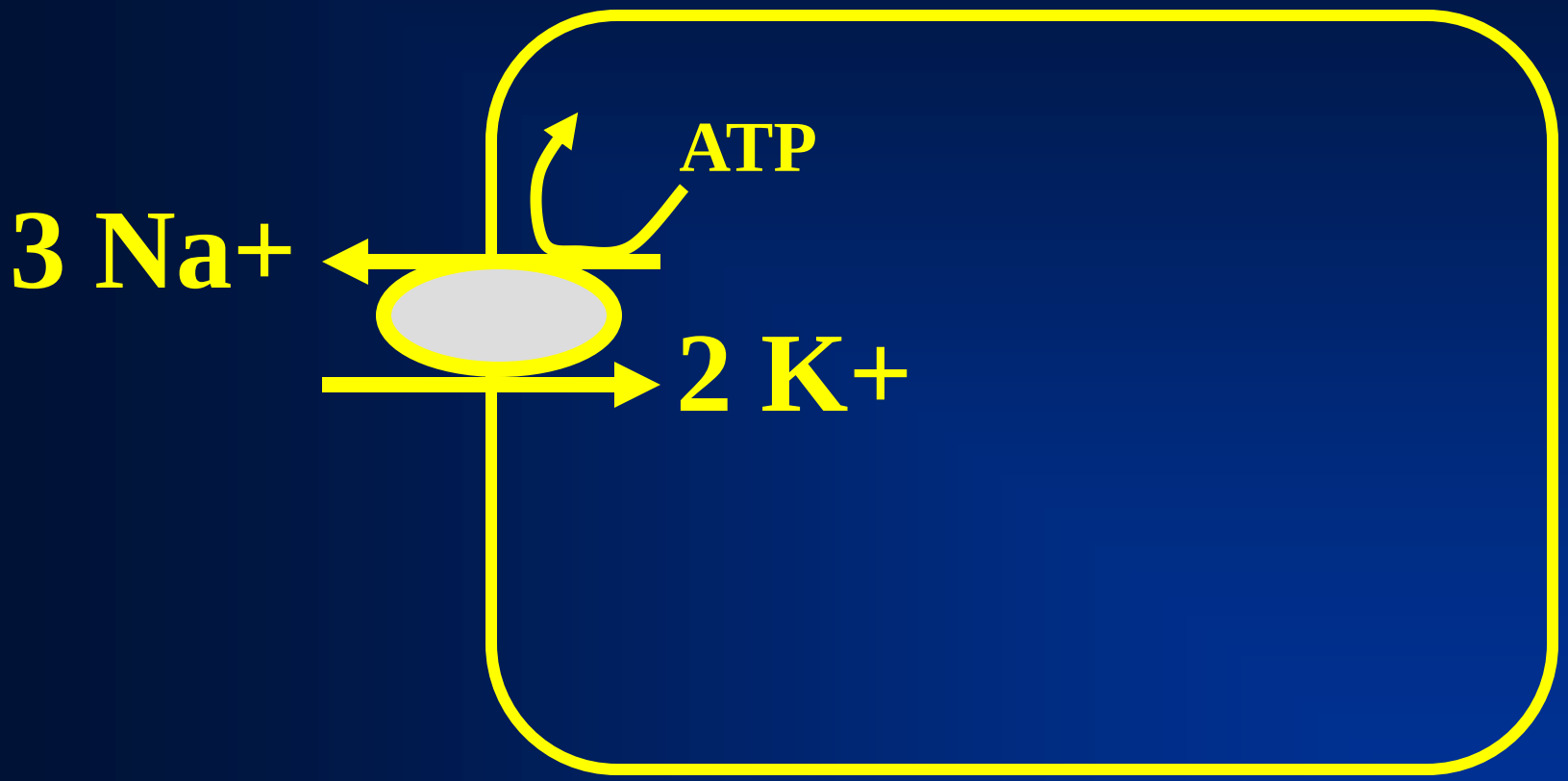
Active Transport

Ca++ATPase pump
H+ ATPase pump



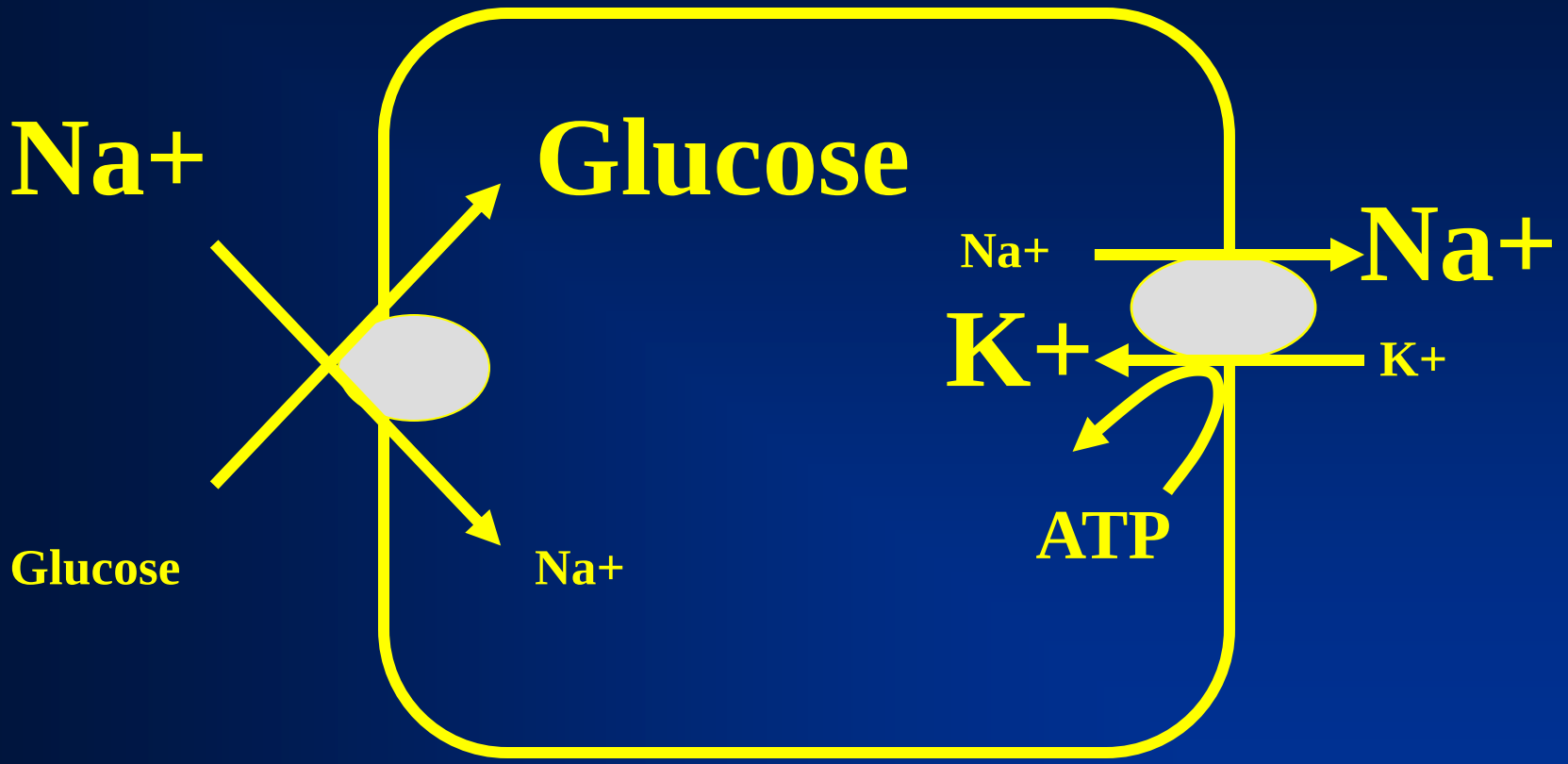
Primary (1º) Active Transport

Na+-K+ ATPase exchange pump



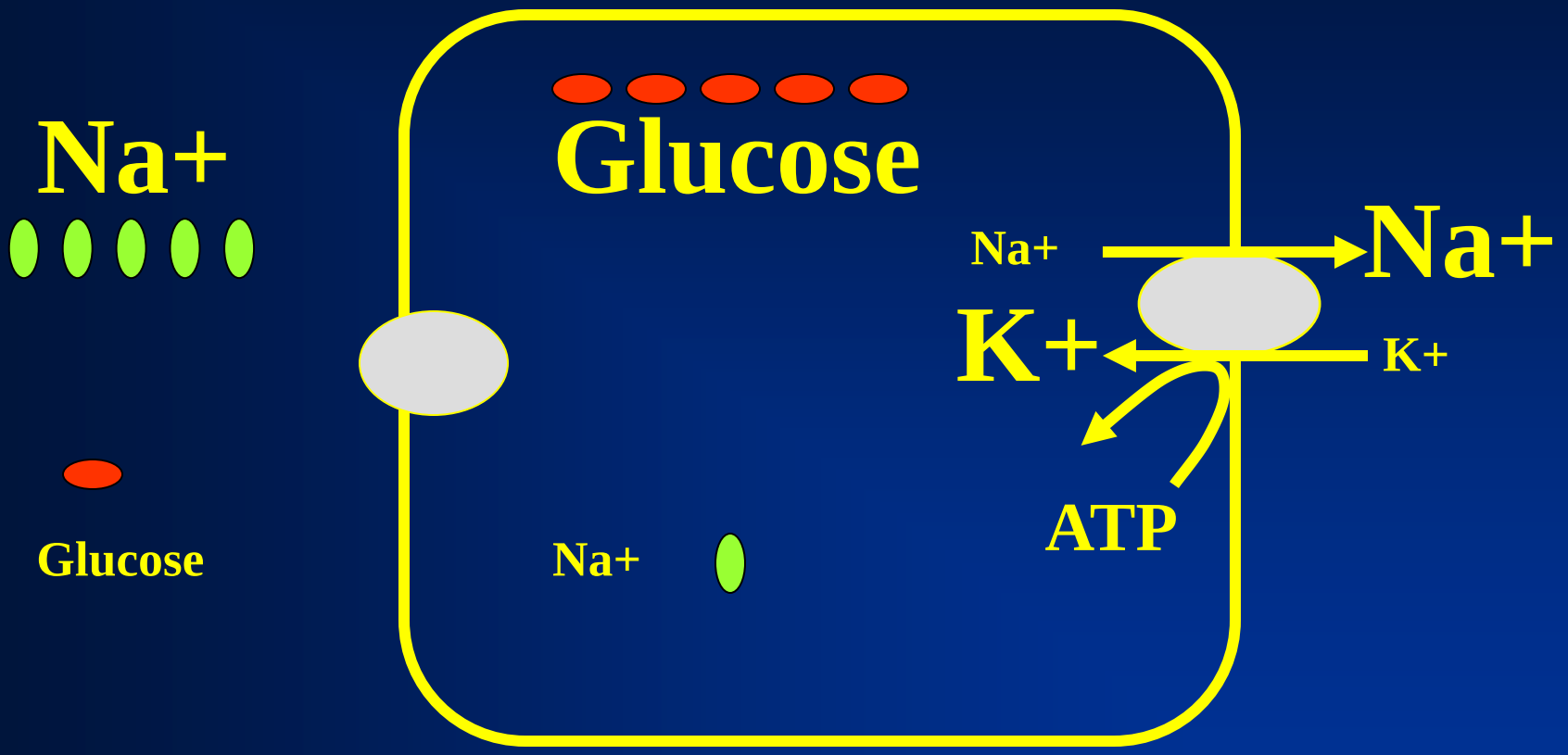
Primary (1º) Active Transport

Co-transport (symport)



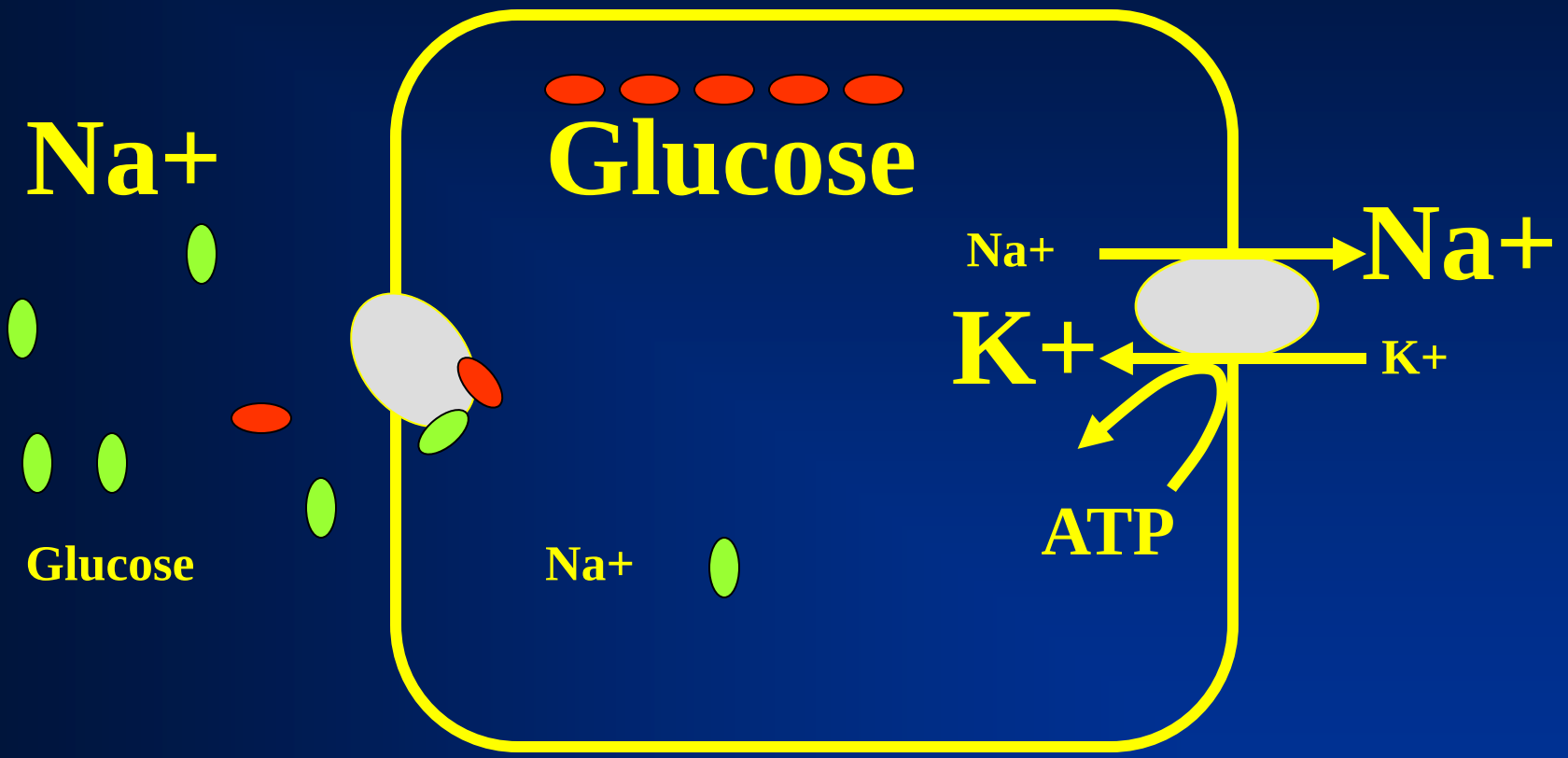
Secondary (2º) Active Transport

Co-transport (symport)



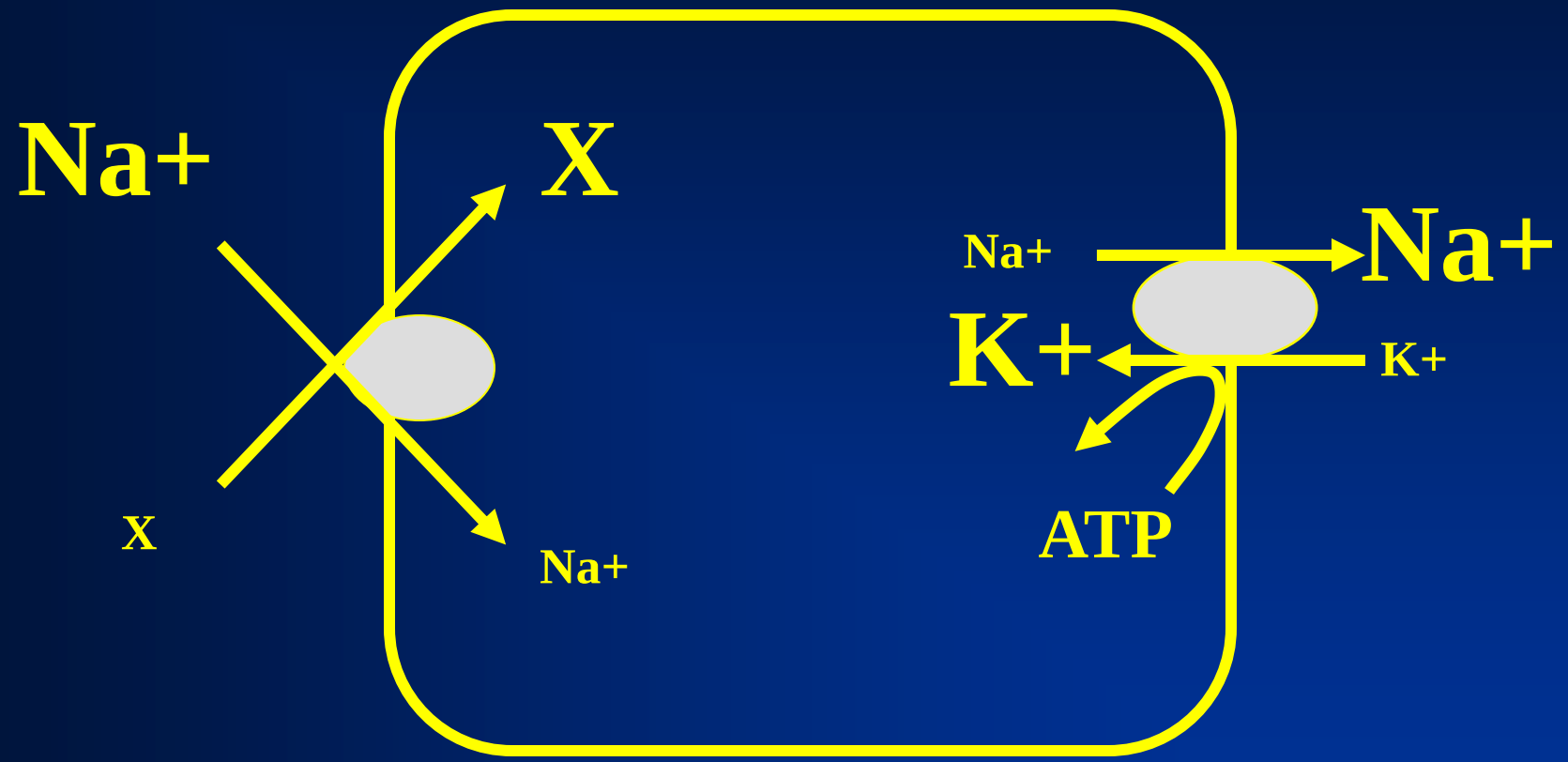
Secondary (2º) Active Transport

Co-transport (symport)



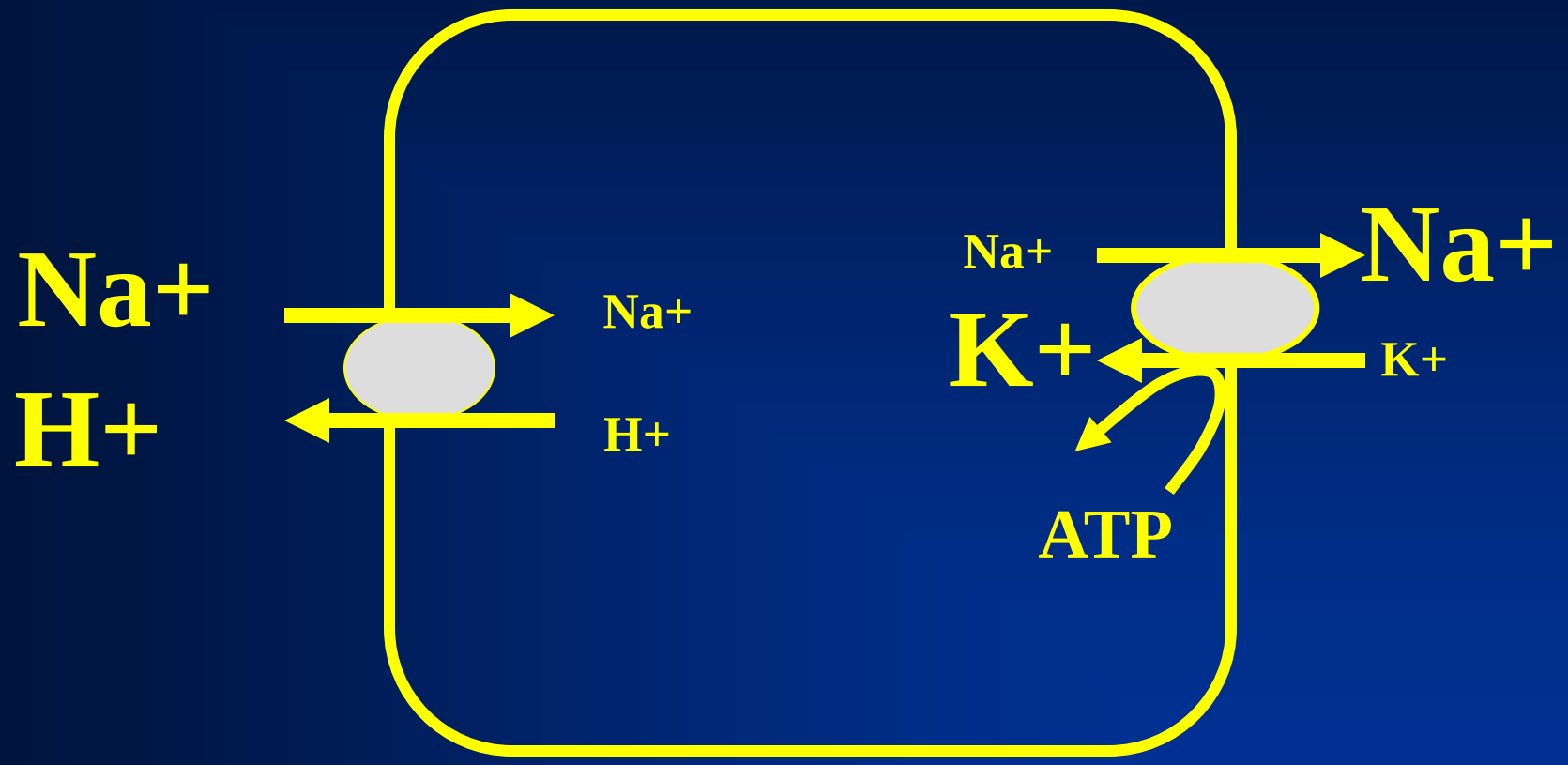
Secondary (2º) Active Transport

Many different co-transport proteins utilize the sodium concentration gradient



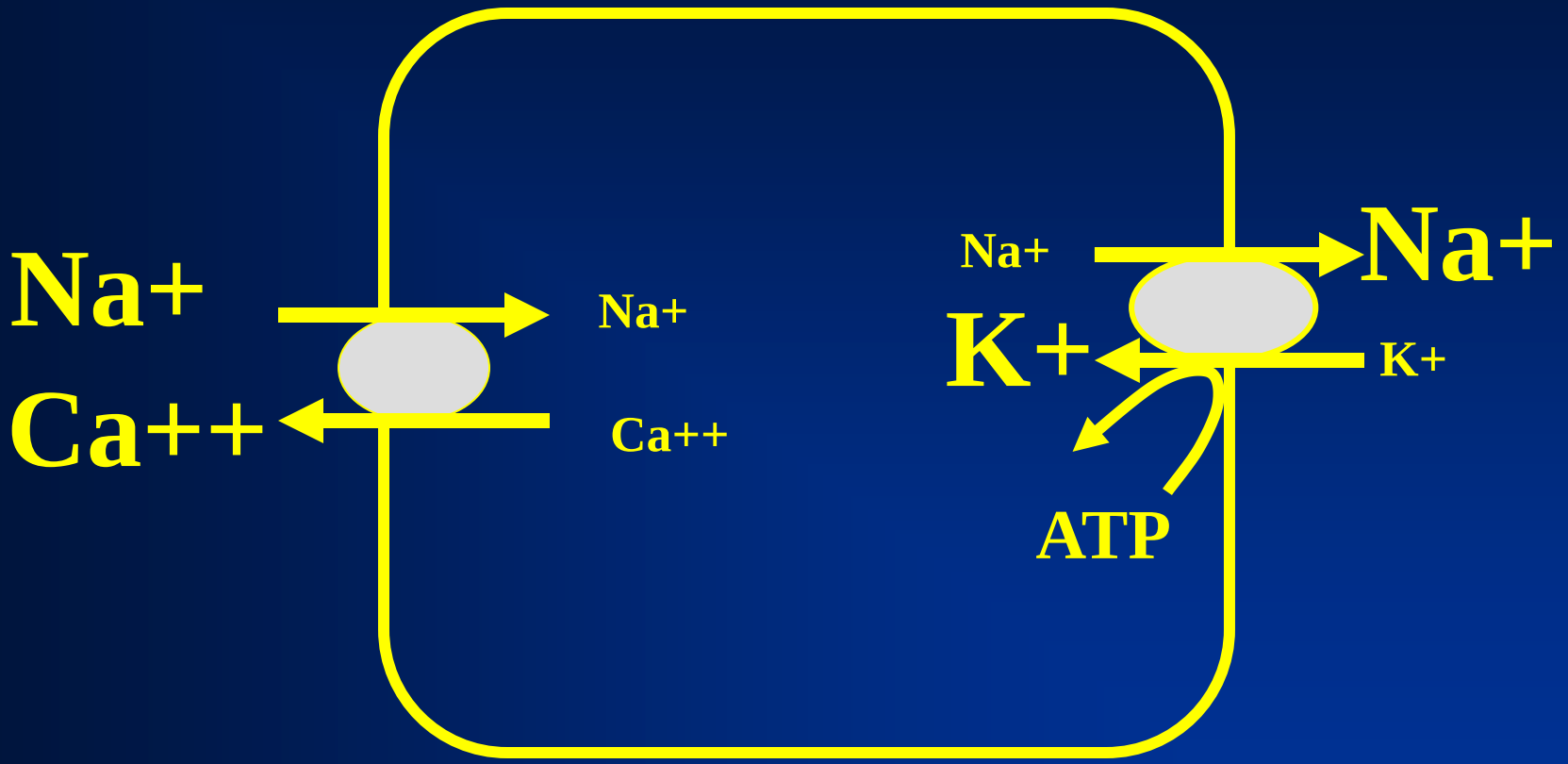
Secondary (2º) Active Transport

Counter-transport (antiport, exchanger)



Secondary (2º) Active Transport

Counter-transport (antiport, exchanger)



Secondary (2º) Active Transport

An important point to remember:

Transport and transporters

- **Passive processes (no ATP energy required)**
  - **Diffusion**
  - **Facilitated diffusion**
- **Active processes (ATP energy required)**
  - **Primary active transport**
  - **Secondary active transport**

There are MANY transporters that do not precisely fit into one of these categories; they share characteristics of two or more categories and ofter have other specific attributes, as well

Video Posted on Google Drive and Handout is Posted on Moodle

"Why should I care about this transport stuff?"

Clinical Correlation: Transporters

- **Transporters and medications**
- **Transporters and diseases**

Clinical Correlation:Co-transport (symport)

"Why should I care about this transport stuff?"

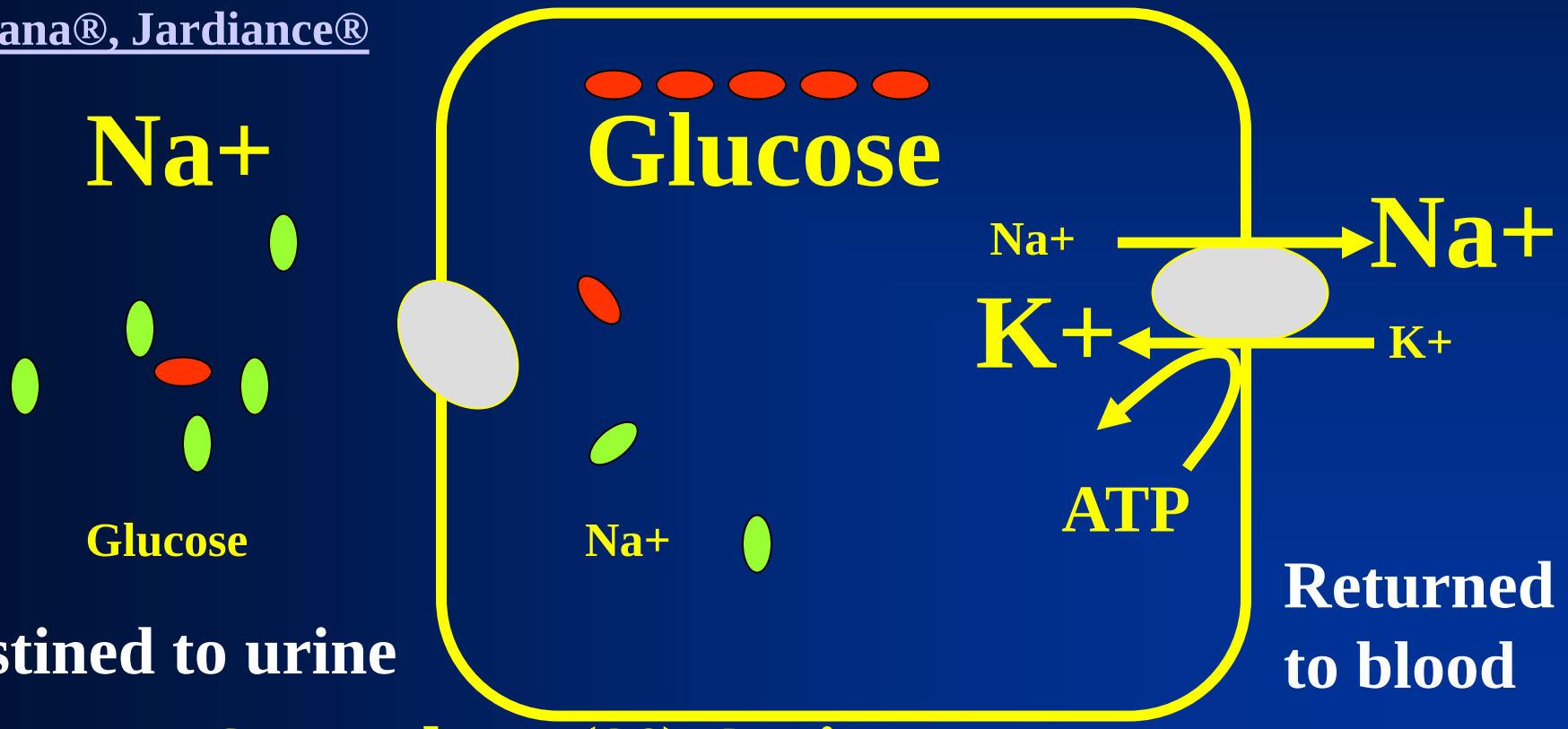
Clinical correlation (Sodium-dependent glucose transporters)

Small intestine enterocytes- SGLT1

Renal proximal tubule – SGLT1, SGLT2

SGLT2 inhibitors are used in the treatment of type II diabetes

Invokana®, Jardiance®



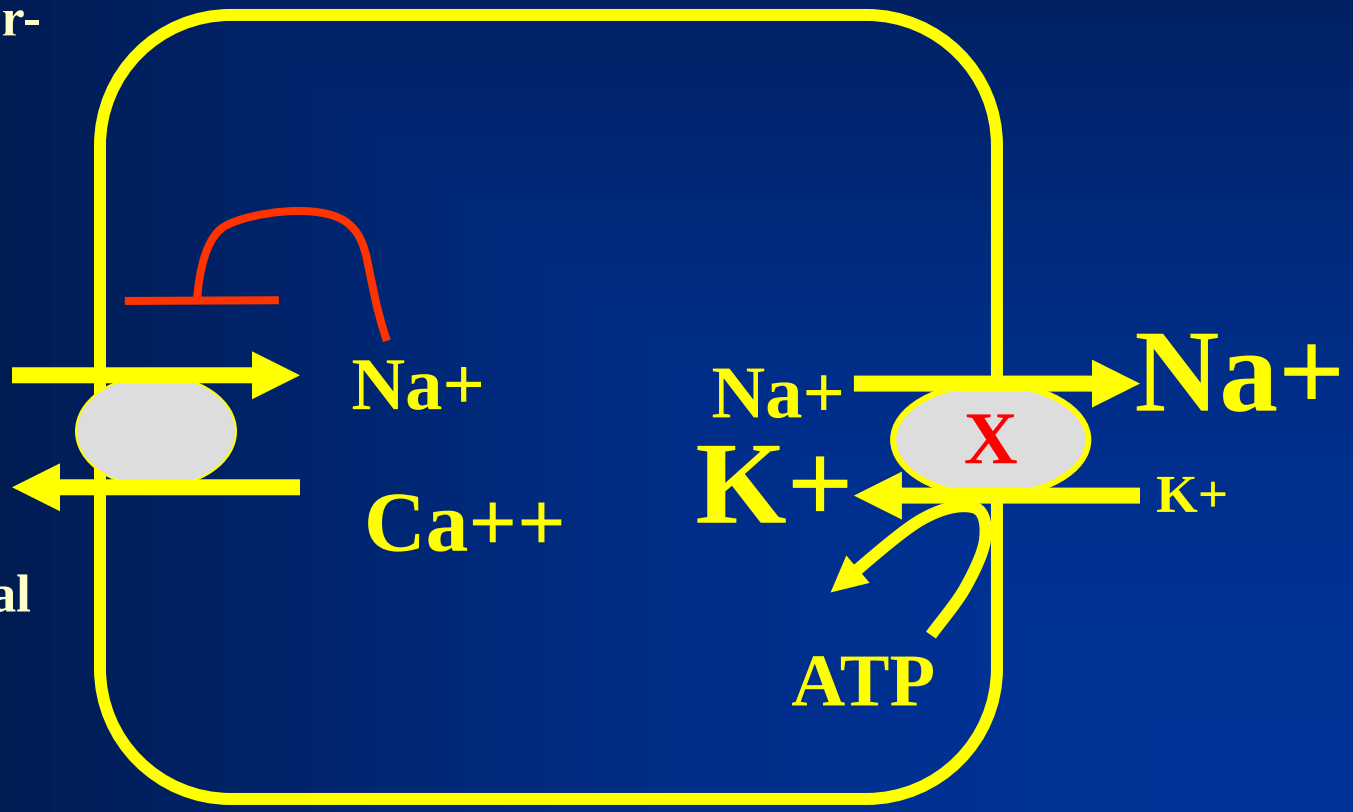
Secondary (2º) Active Transport

"Why should I care about this transport stuff?"

Clinical Correlation

Cardiac Glycosides

- Foxglove plant (w/ digitalis) used ~230 years ago
- Digoxin used for specific heart failures (also atrial fibrillation, not shown here)
- Digoxin decreases Na+/K+ exchange pump activity
- ↑ cellular Na+
- ↓ Na+/Ca++ counter-transport activity
- ↑ cellular Ca++
- ↑ myocardial ino-tropy



- Stronger myocardial contractions
- ↑ stroke volume

Co-transport (symport)

"Why should I care about this transport stuff?"

"Are any diseases associated with transporter dysfunction?"

MOVEMENT ACROSS THE MEMBRANE

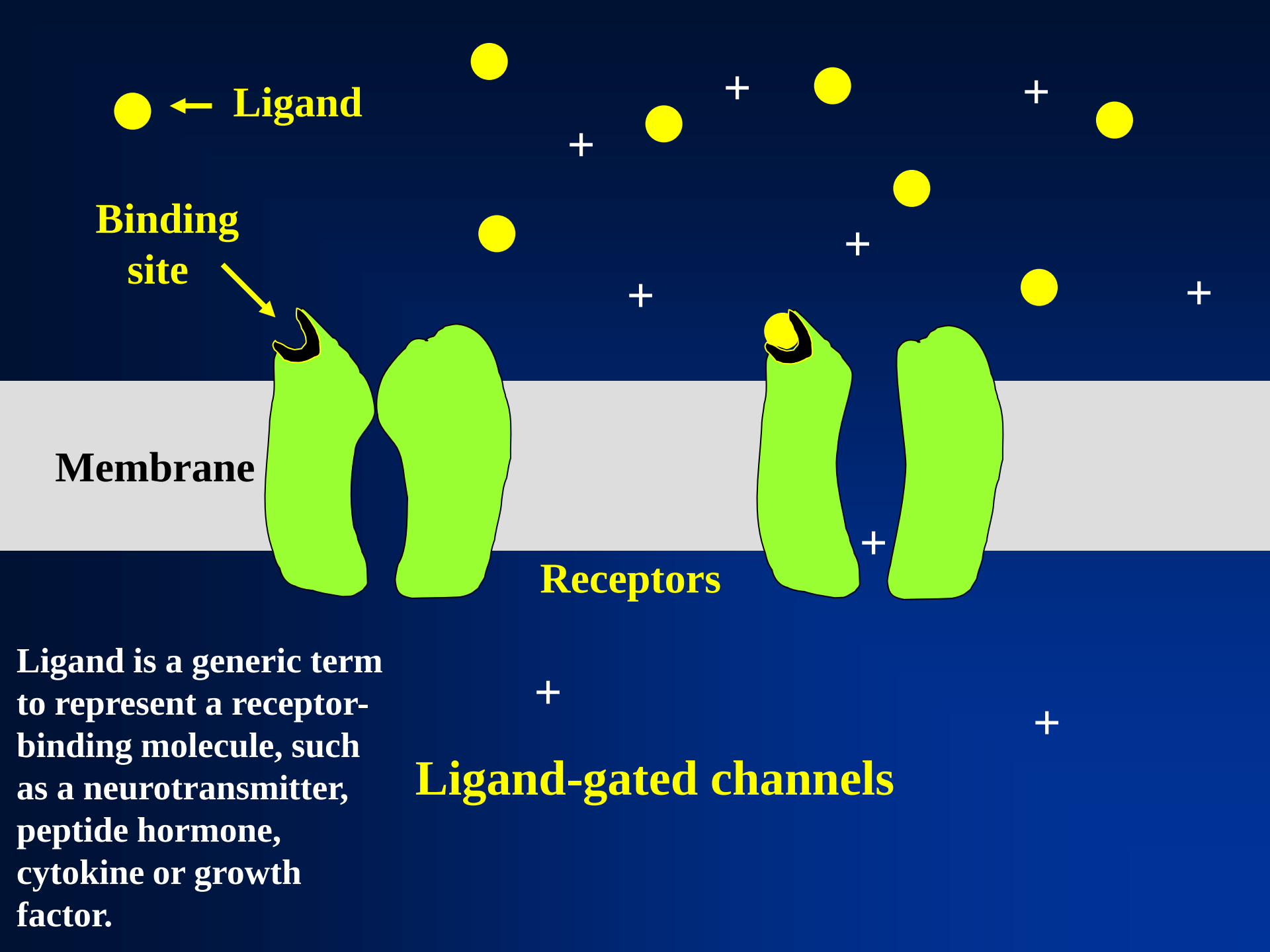
Molecules

Transport

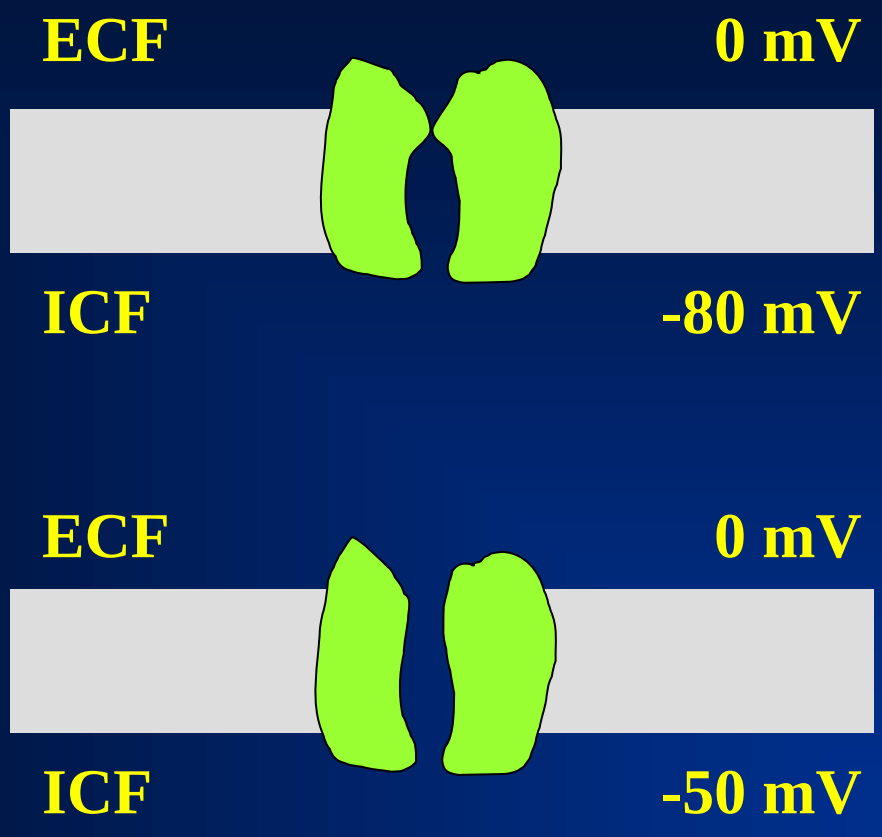
- **Passive processes (no energy required)**
  - **Diffusion**
  - **Facilitated diffusion**
- **Active processes (energy required)**
  - **Primary active transport**
  - **Secondary active transport**

Translocation

- **Ligand-gated (-sensitive) channels**
- **Voltage-gated (-sensitive) channels**



Ligand is a generic term to represent a receptor-binding molecule, such as a neurotransmitter, peptide hormone, cytokine or growth factor.



Voltage-gated channels

MOVEMENT ACROSS THE MEMBRANE

Molecules

Transport

- **Passive processes (no ATP energy required)**
  - **Diffusion**
  - **Facilitated diffusion**
- **Active processes (ATP energy required)**
  - **Primary active transport**
  - **Secondary active transport**

(Carrier) Mediated transport

Translocation

- **Ligand-gated channels**
- **Voltage-gated channels**
- **Many other gated and non-gated channel proteins**
  - **Mechanically-gated (sensitive) channel**
  - **Temperature-gated (sensitive) channel**
  - **pH-gated (sensitive) channel**
  - **Aquaporin (non-gated) channel proteins**
  - **Etc, etc, etc.....**

Summary of Transmembrane Movements of Targeted Solutes

	Simple diffusion-like kinetics		Mediated transport kinetics		
	Thru membrane lipid bilayer	**Thru membrane translocation channel protein**	**Facilitated diffusion**	**Primary active transport**	**Secondary active transport**
Nature of movement?	Passive	Passive	Passive	Active	Active
Direction of movement for the targeted solute	From high to low; down chemical concentration gradient	From high to low; down chemical concentration gradient	From high to low; down chemical concentration gradient	From low to high; against chemical concentration gradient	From low to high; against chemical concentration gradient
Utilization of membrane protein?	No	Yes	Yes	Yes	Yes
Solute selectivity?	Yes	Yes	Yes	Yes	Yes
ATP utilized?	No	No	No	Yes, directly	Yes, in second step
Can be saturated with physiological conditions? Tmax?	No	No	Yes	Yes	Yes
Characteristics of solutes and examples	Nonpolar, lipophilic, small; fatty acids, steroid hormones, O2, CO2	Ions: Na+, K+, Cl-, Ca++	Polar, lipophobic: glucose, amino acids	Ions: Na+, K+, H+, Ca++	Polar, lipophobic: glucose, amino acids

Adapted from Vander's Physiology

FUNDAMENTAL PRINCIPLES (Part I)

I. Introduction to physiology and homeostasis
   - A. Physiology – Definitions
   - B. Homeostasis
   - C. Fluid compartments
   - D. Homeostatically regulated factors
   - E. How/why are factors regulated
   - F. Homeostatic hmmmmms

II. Membrane dynamics (structure, function, signaling)
   - A. Gradients drive physiological processes
   - B. Membrane organization
   - C. Membrane transport and translocation
   - D. Signal transduction

III. Membrane potentials
   - A. Membrane potentials and gradients
   - B. Types of membrane potentials
   - C. Resting membrane potentials
   - D. Communication using membrane potentials
   - E. Determinates of membrane and equilibrium potentials
   - F. Graded and action potentials

IV. Nerve impulse propagation, synapses and neural modulation
   - A. Think communication!
   - B. Nerve impulse propagation and conduction
   - C. Synapses
   - D. Neural modulation

MOVEMENT ACROSS THE MEMBRANE

Molecules

Transport

- **Passive processes (no energy required)**
  - **Diffusion**
  - **Facilitated diffusion**
- **Active processes (energy required)**
  - **Primary active transport**
  - **Secondary active transport**

Translocation

- **Ligand-gated channels**
- **Voltage-gated channels**

Information

- **Signal transduction**

Signal Transduction Pathways

1. **Pathways initiated by lipid-soluble messengers** – can cross plasma membrane (e.g., steroid and thyroid hormones, vitamin D, etc.)
2. **Pathways initiated by water-soluble messengers** – cannot cross plasma membrane (Membrane signal transduction; e.g., peptide hormones, cytokines, neurotransmitters, growth factors, etc.)

---

1. **Pathways initiated by lipid-soluble messengers**

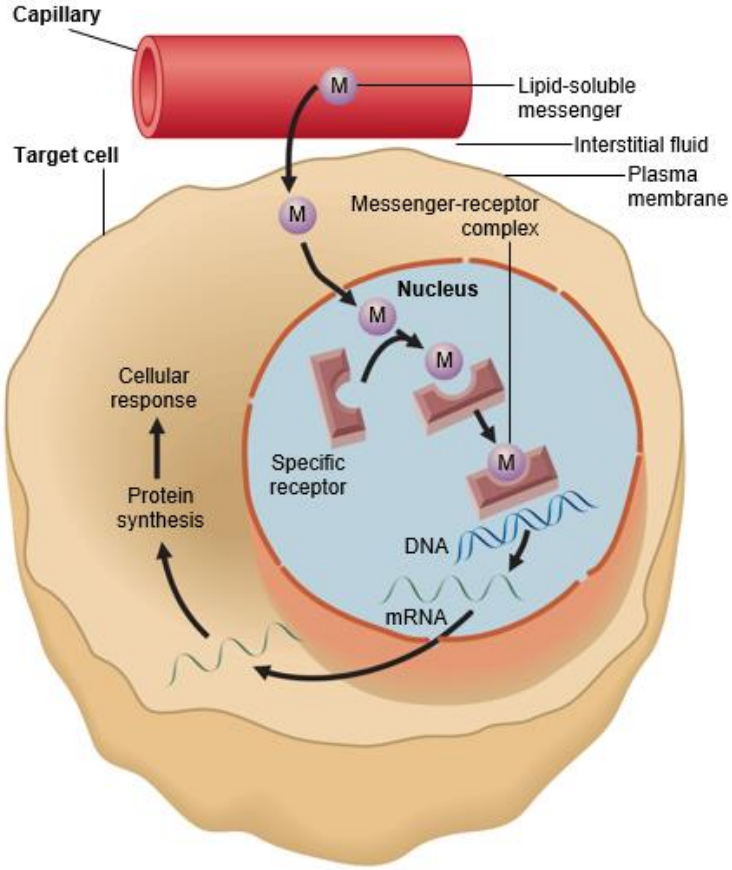
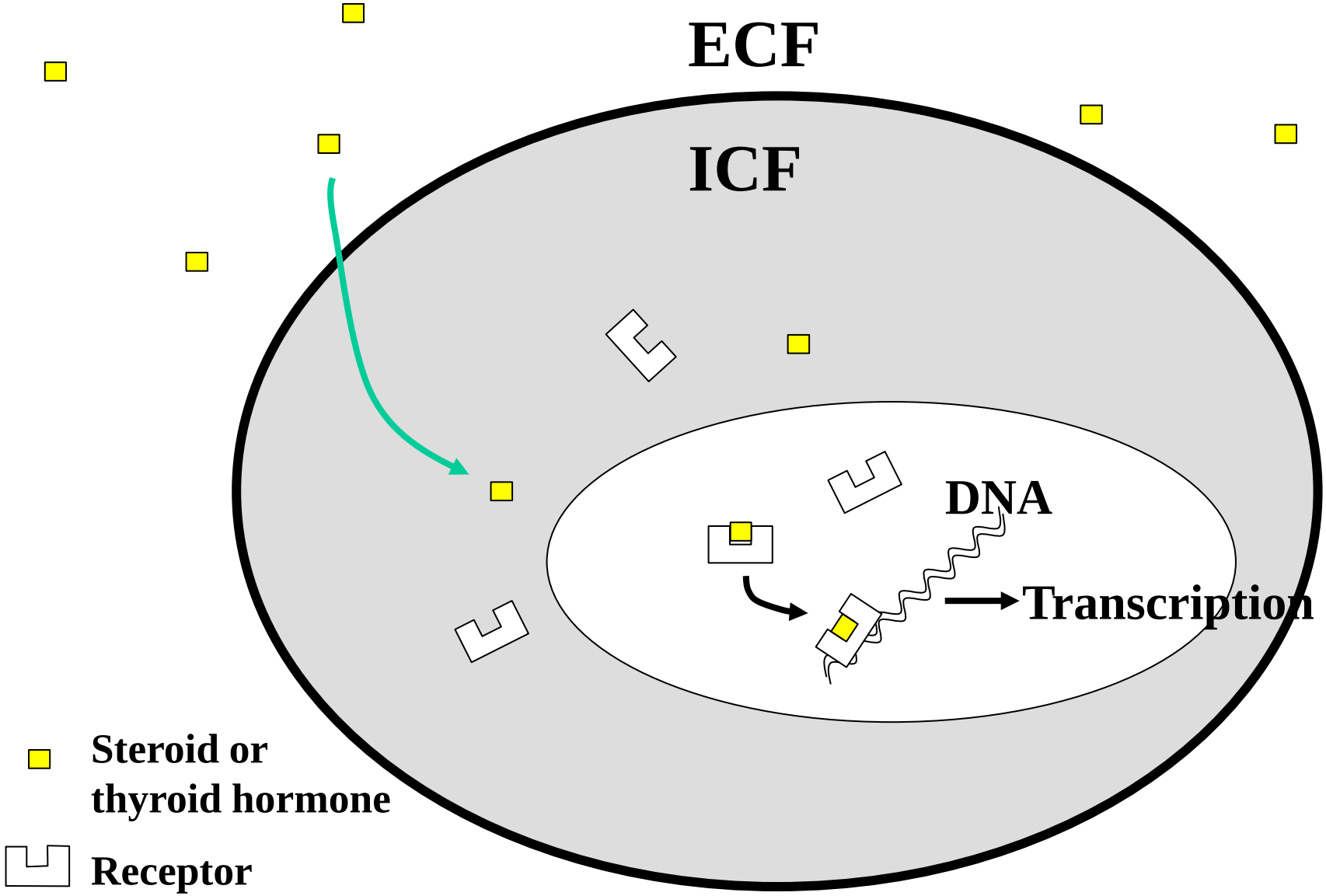


Fig. 5-4



Steroid and thyroid hormones bind cytosolic and/or nuclear receptors and modulate gene transcription

Signal Transduction Pathways

1. **Pathways initiated by lipid-soluble messengers** – can cross plasma membrane (e.g., steroid and thyroid hormones, vitamin D, etc.)
2. **Pathways initiated by water-soluble messengers** – cannot cross plasma membrane (Membrane signal transduction; e.g., peptide hormones, cytokines, neurotransmitters, growth factors, etc.)

2. **Pathways initiated by water-soluble messengers (ligands)**

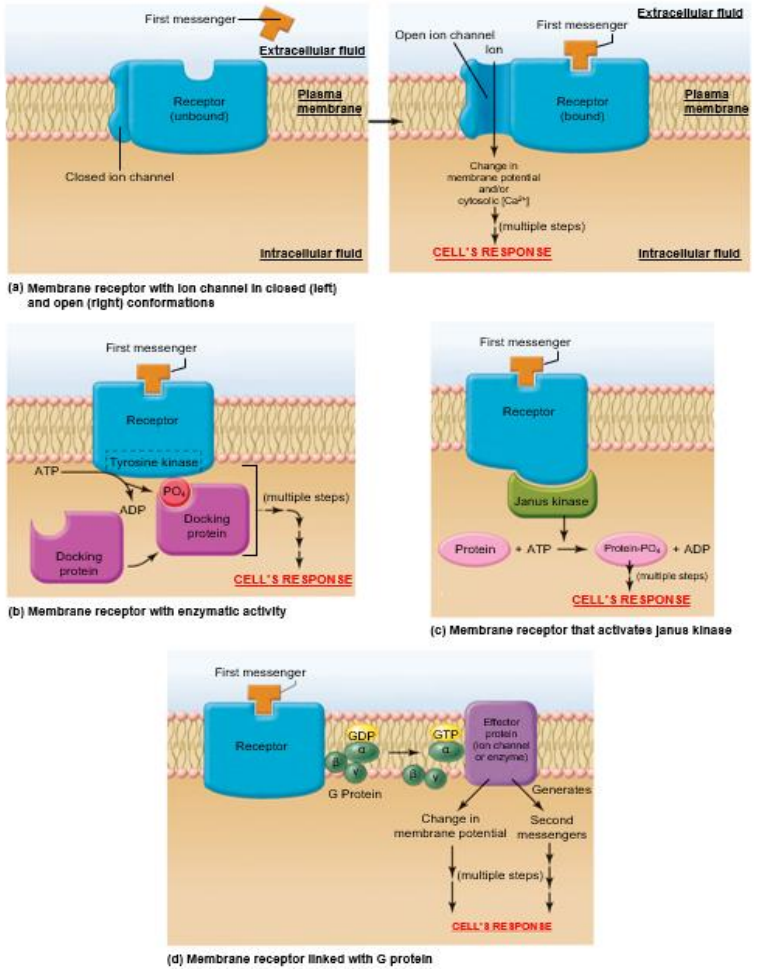
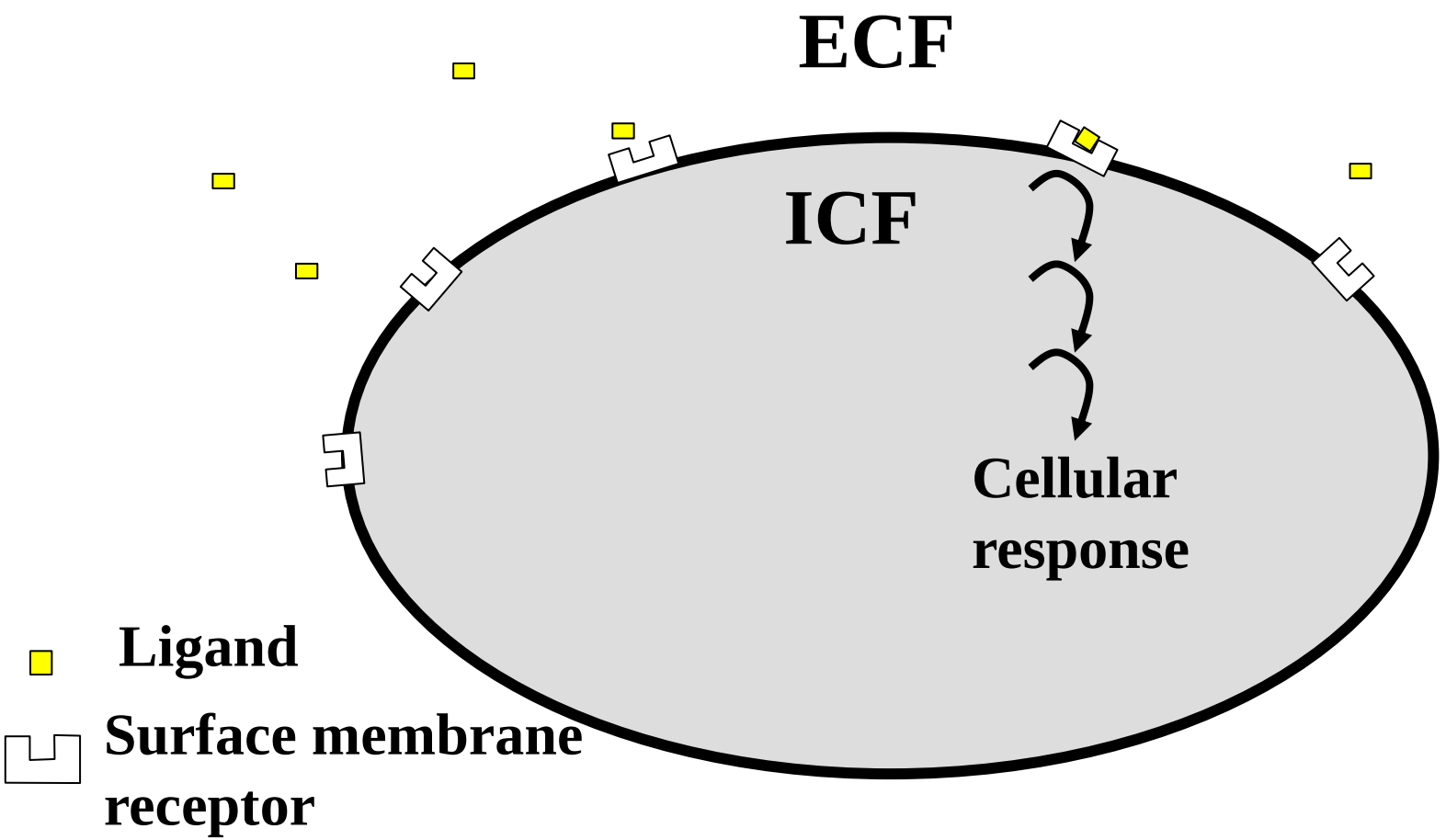
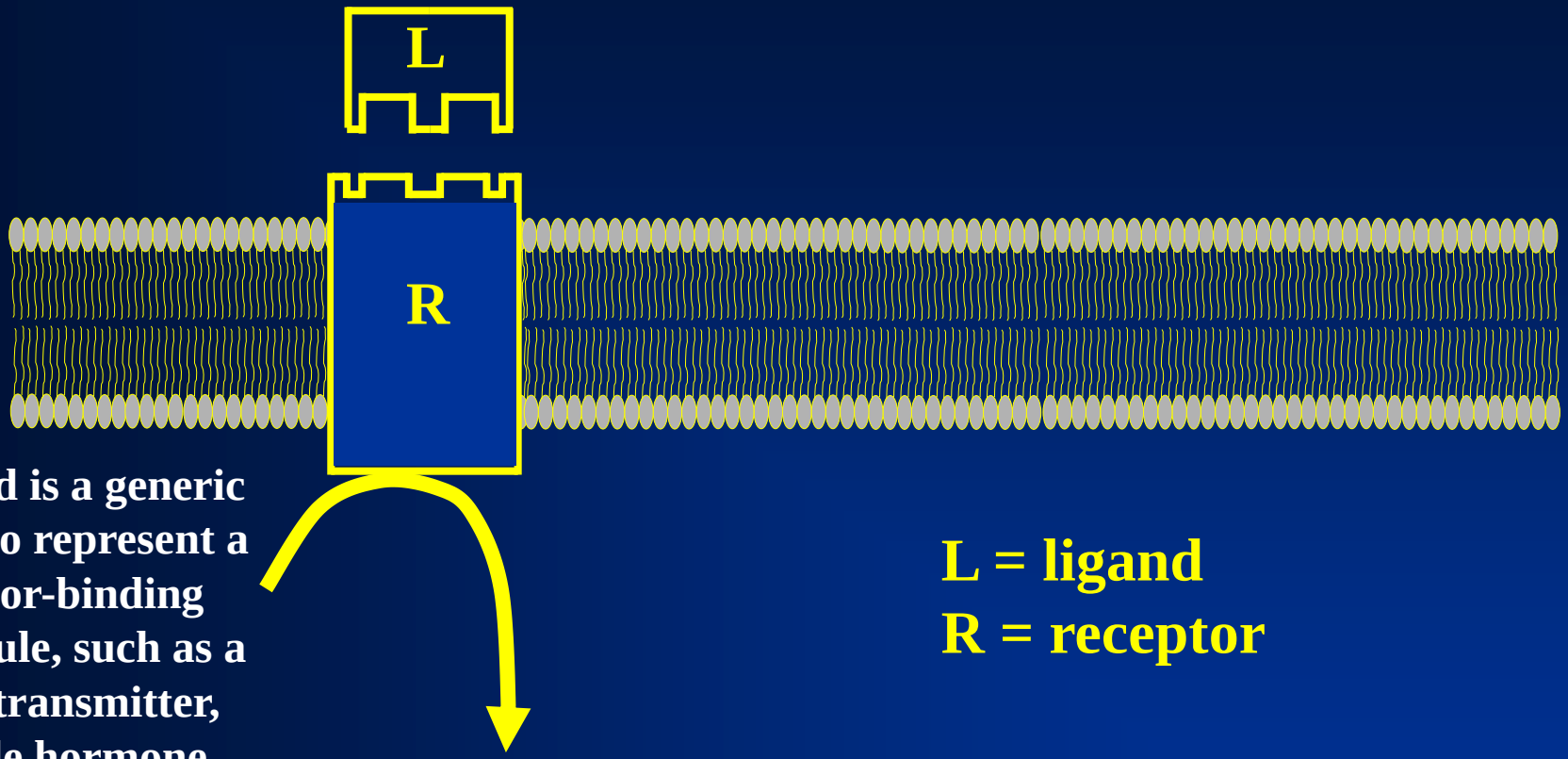


Fig. 5-5



Many ligands such as peptide hormones and neurotransmitters communicate with their target cells via surface membrane receptors; ligand-receptor binding activates a transmembrane signal transduction process that elicits a cellular response

MEMBRANE RECEPTOR-MEDIATED SIGNAL TRANSDUCTION



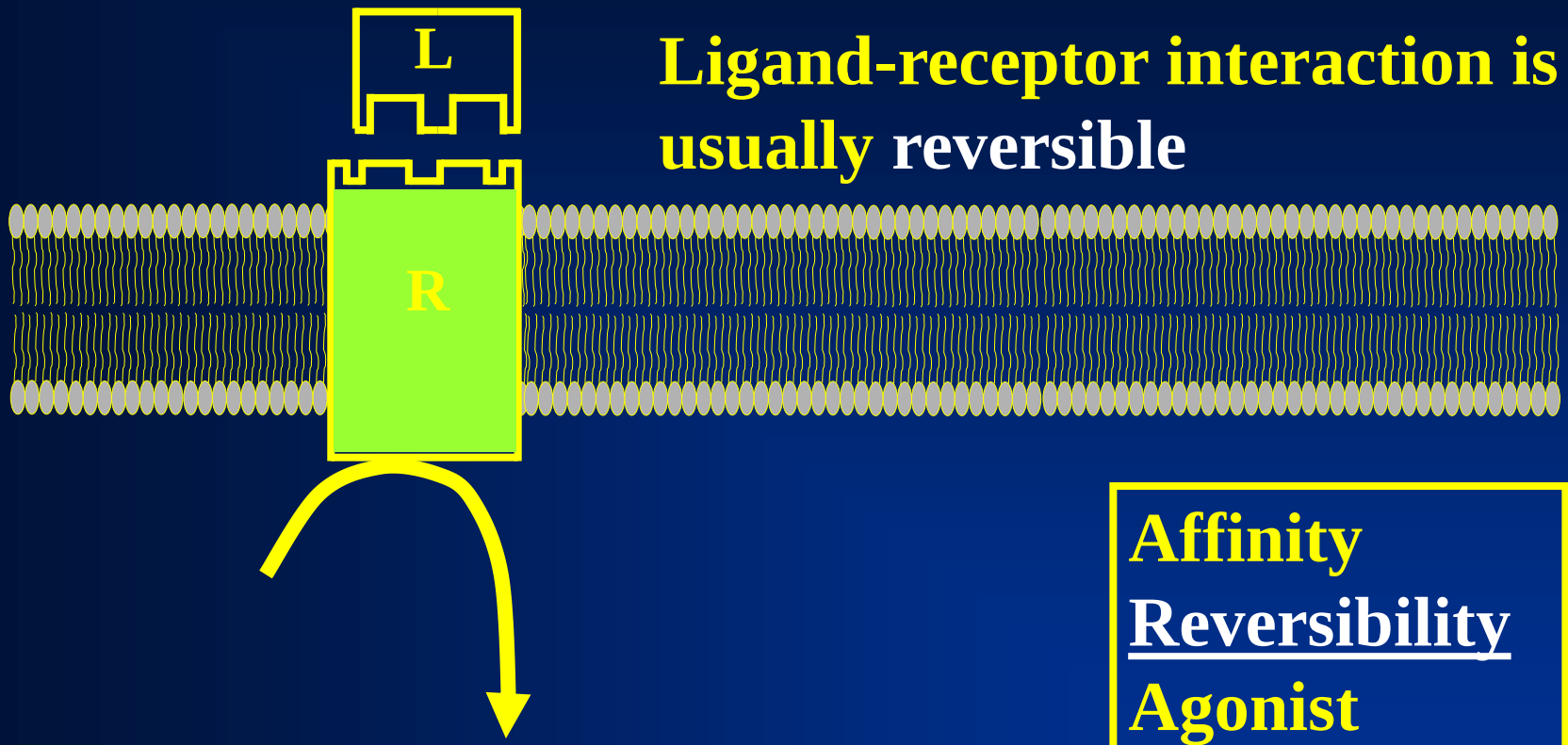
Ligand is a generic term to represent a receptor-binding molecule, such as a neurotransmitter, peptide hormone, cytokine or growth factor. For membrane signal transduction, the ligand is the first messenger.

L = ligand
R = receptor

Information

Sending a signal across the membrane

Fundamental ligand-receptor pharmacology

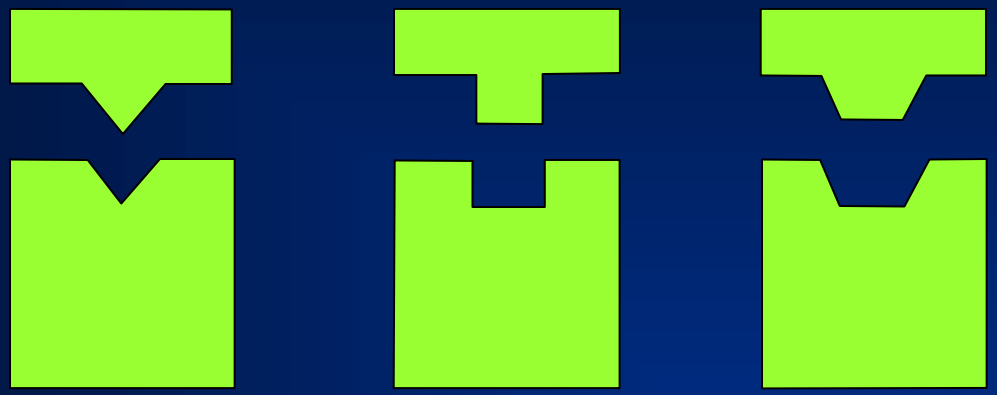


Ligand-receptor interaction is usually reversible

Information (leads to cellular action)

Affinity
Reversibility
Agonist
Antagonist
Tissue specificity

Fundamental ligand-receptor pharmacology

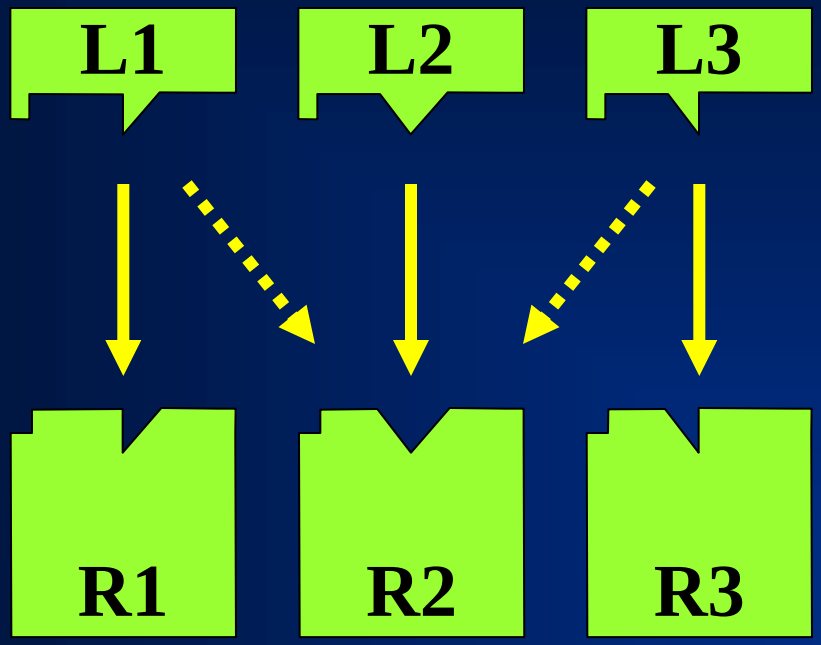


Ligands

Receptors

Ligand-receptor binding is usually very specific...

Agonists



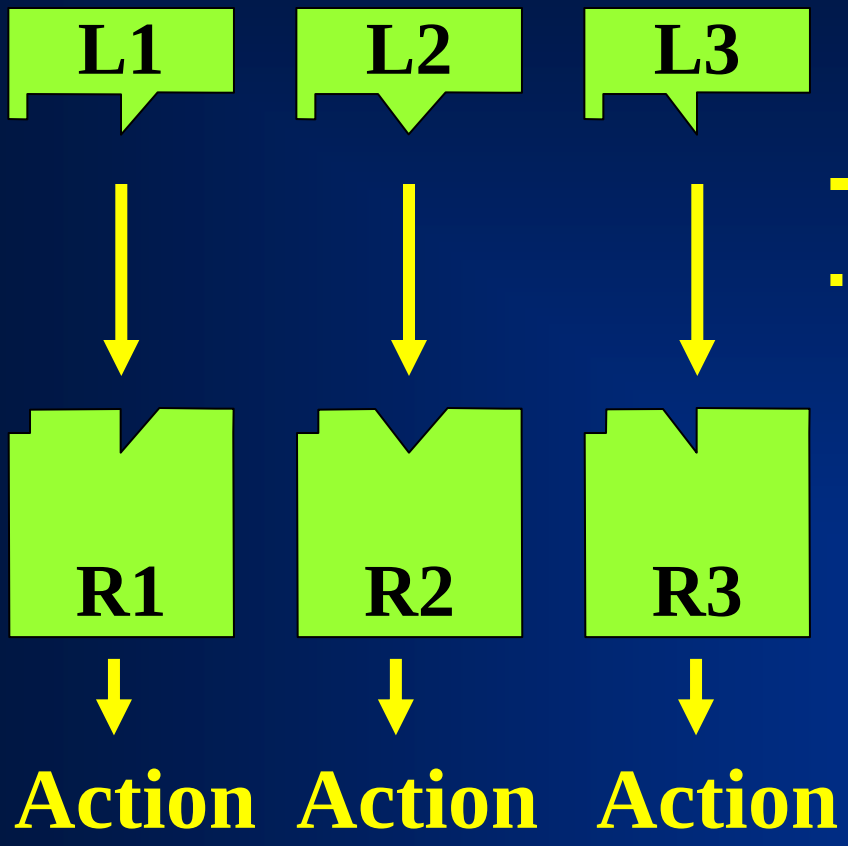
High affinity binding

Lower affinity binding

…but important cross-reactivities do occur.

Affinity
Reversibility
Agonist
Antagonist
Tissue specificity

Agonists

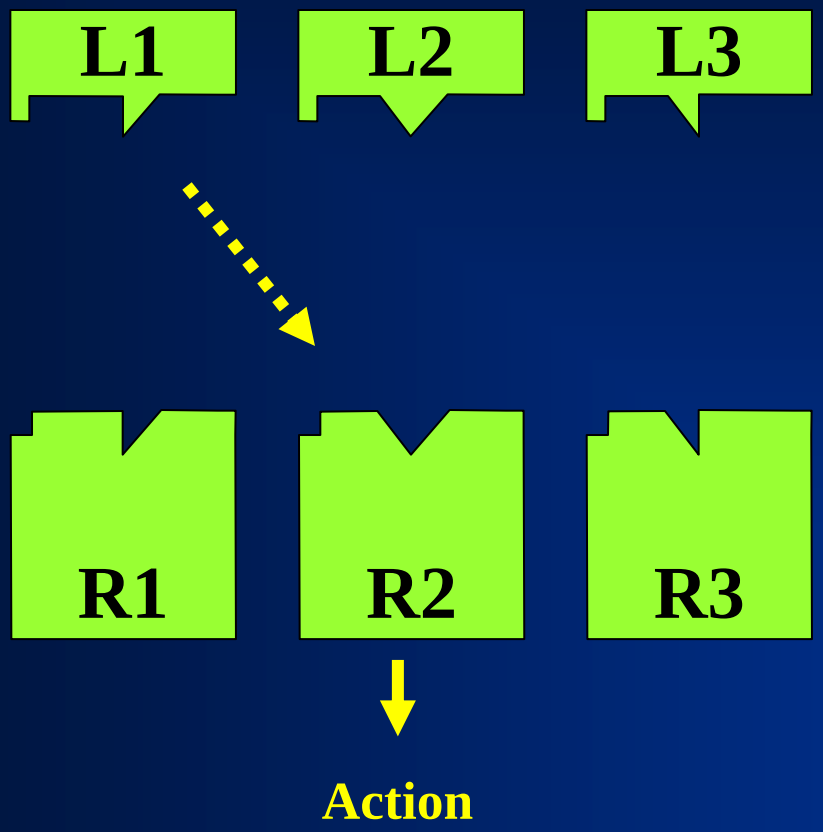


High affinity binding

Lower affinity binding

Affinity
Reversibility
Agonist
Antagonist
Tissue specificity

Agonists



High affinity binding

Lower affinity binding

Affinity
Reversibility
Agonist
Antagonist
Tissue specificity

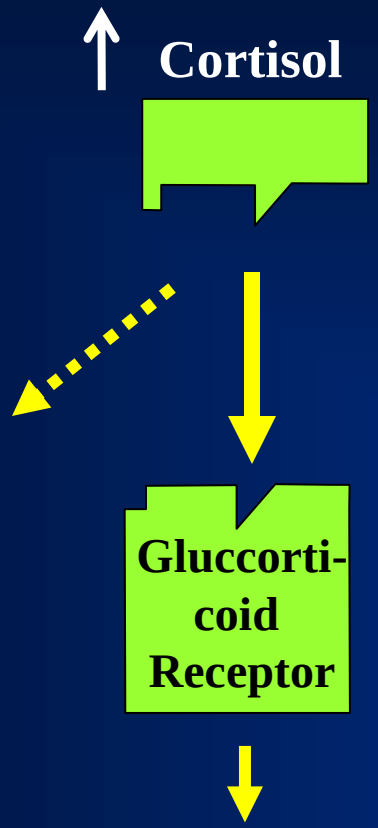
An agonist (L1) binds a receptor and stimulates the same cellular actions as the native ligand (L2). The degree of simulation is related to affinity to active site.

Cushing's disease
and
Cushing's syndrome

Clinical Correlation

Agonists

Fundamental ligand-receptor pharmacology



High affinity binding

Lower affinity binding

Affinity
Reversibility
Agonist
Antagonist
Tissue speci-ficity

Iatrogenic effects

Clinical Correlation

Agonists

Fundamental ligand-receptor pharmacology



High affinity binding

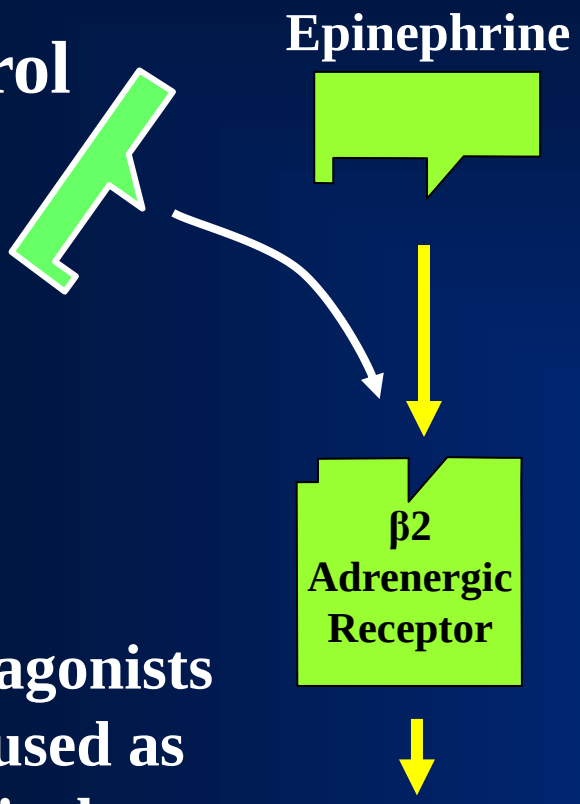
Lower affinity binding

Affinity
Reversibility
Agonist
Antagonist
Tissue speci-ficity

Clinical Correlation

Agonists

Fundamental ligand-receptor pharmacology



High affinity binding

Lower affinity binding

Receptor agonists are often used as therapeutic drugs

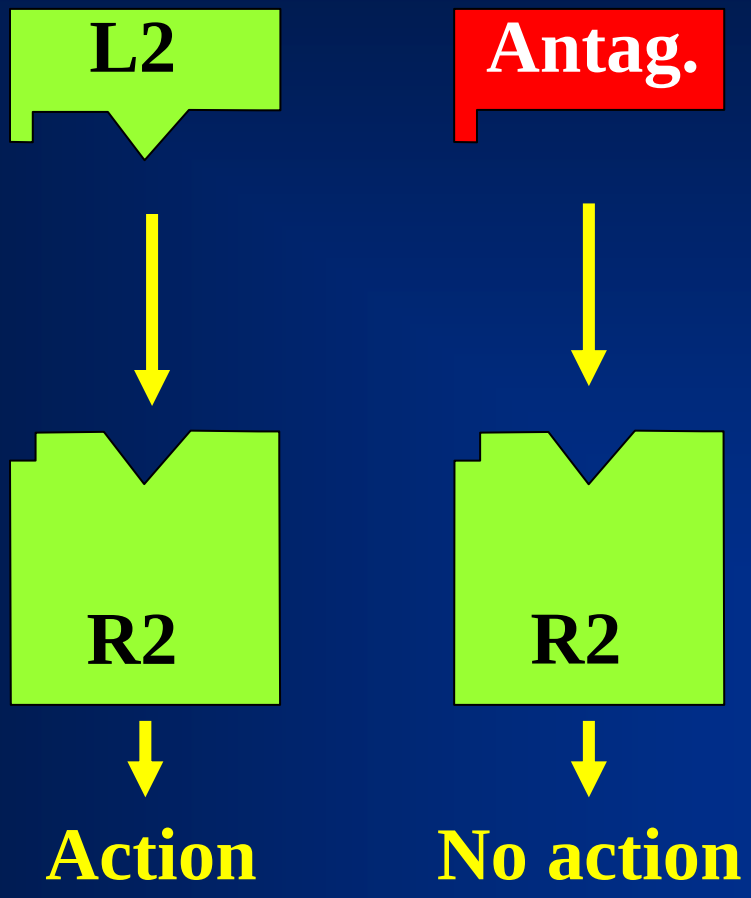
- Natural or Endogenous (native) agonists
- Full agonists
- Partial agonists
- Superagonists
- Selective agonists



Mylan is now in the crosshairs over severe price hikes for its EpiPen. (AP Photo/Mark Zaleski, File)

Washington Post 8/30/2016

Receptor antagonists (blockers)

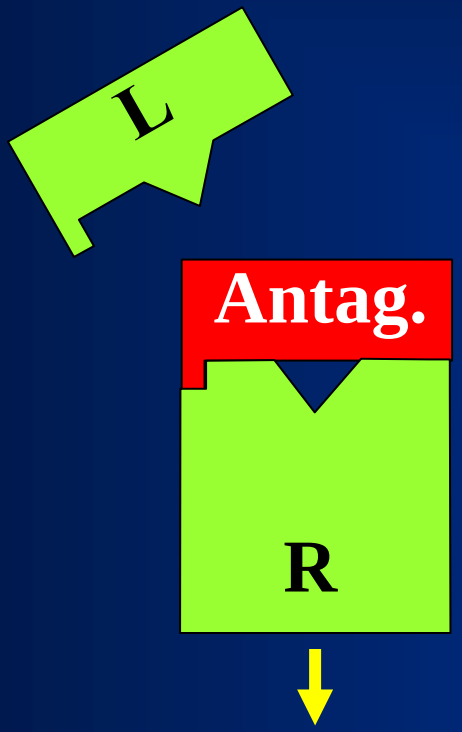


Receptor antagonists bind the receptor without engaging the receptor active site and therefore cause no cellular action.

- Affinity
- Reversibility
- Agonist
- Antagonist
- Tissue specificity

Receptor antagonists (blockers)

Receptor antagonists block agonistic ligands from binding to the receptor active site thereby blocking activation of the receptor



- Affinity
- Reversibility
- Agonist
- Antagonist
- Tissue specificity

Receptor antagonists (blockers)

Receptor antagonists block agonistic ligands from binding to and activating the receptor

β-blocker

Epinephrine



Receptor antagonists are often used as therapeutic drugs

Excite[illegible]rdia[illegible]muscle
Increase he[illegible] ra[illegible]an[illegible]contractility
Increase[illegible]myoc[illegible]ial work

Affinity
Reversibility
Agonist
Antagonist
Tissue speci-ficity

Receptor antagonists (blockers)

Food and Drug Administration
USP Therapeutic Categories Model Guidelines
Formulary Key Drug Types

Opioid Receptor Antagonists
5-Hydroxytryptamine 3 (5-HT3) Antagonists
Neurokinin 1 (NK1) Receptor Antagonists
Alpha-beta–adrenergic Blocking Agents
Cardio-selective Beta-adrenergic Blocking Agents
Aldosterone Receptor Antagonists
Angiotensin II Receptor Antagonists (ARBs)
Endothelin Receptor Antagonists
Alpha 1-adrenergic Blocking Agents
Growth Hormone Antagonists
H1 Blocking Agents, Sedating
GI Histamine2 (H2) Blocking Agents
Leukotriene Receptor Antagonists

Affinity
Reversibility
Agonist
Antagonist
Tissue specificity

The lesson of this story?

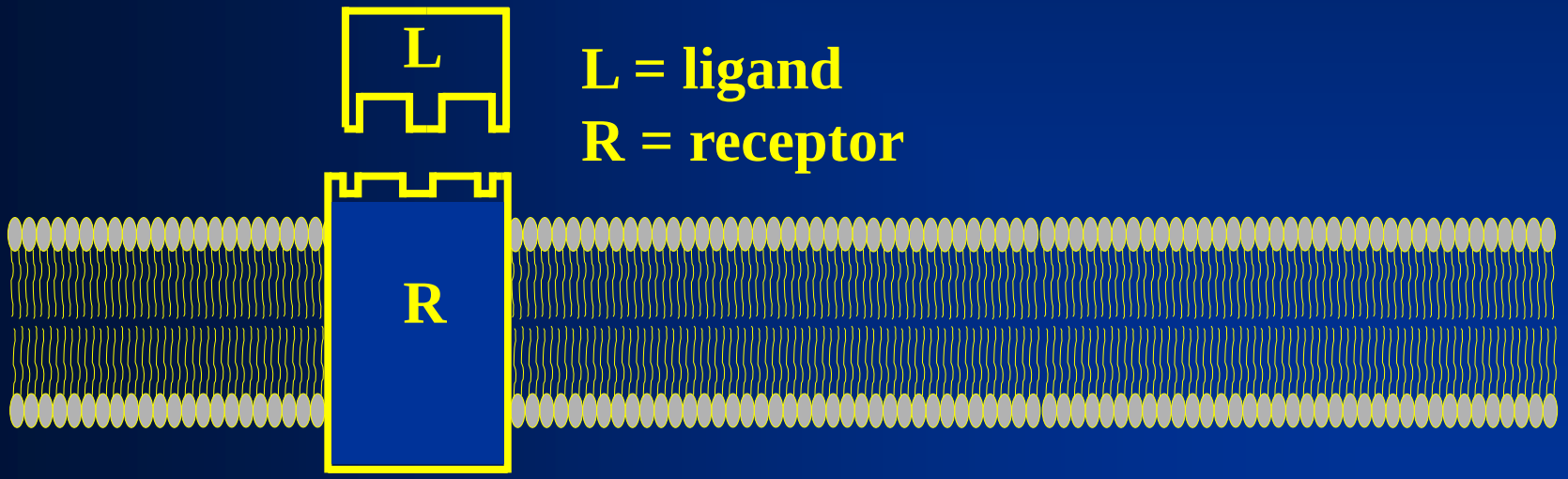
All tissues do not have all receptors

Affinity
Reversibility
Agonist
Antagonist
Tissue speci-ficity

SOME SURFACE MEMBRANE RECEPTORS:

- **have enzymatic activity (often tyrosine kinase activity)**
- **are linked to enzymes**
- **are linked to G proteins**
- **are ligand-gated ion channels**

ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR

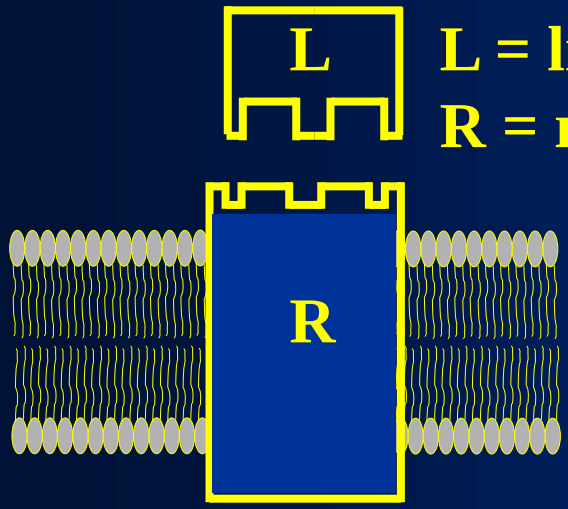


L = ligand
R = receptor

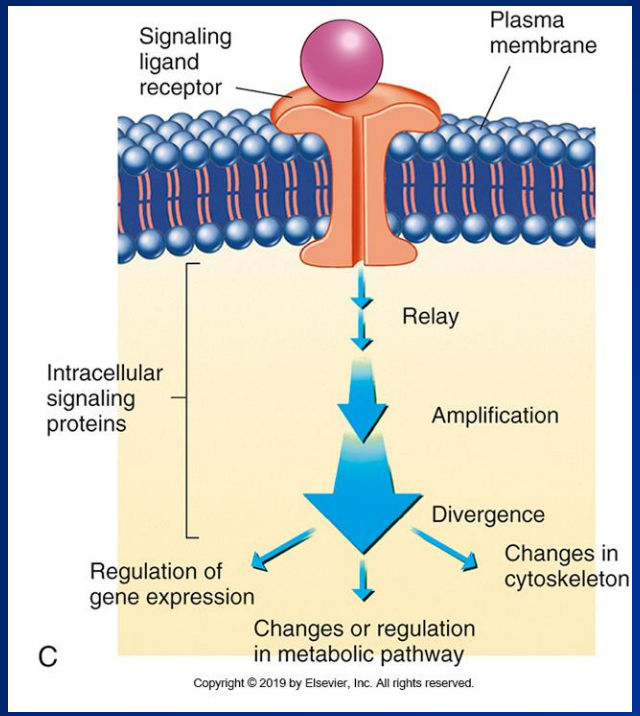
SOME SURFACE MEMBRANE RECEPTORS:

- have enzymatic activity
- are linked to enzymes
- are linked to G proteins
- are ligand-gated ion channels

ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR



L = ligand
R = receptor



Signaling molecule

Relay

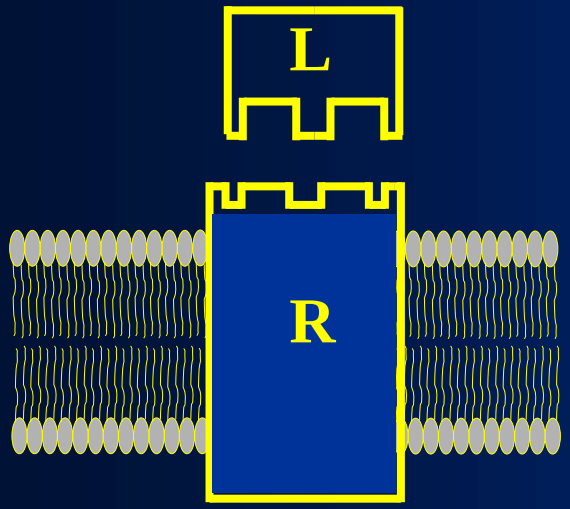
(Amplification)

Cellular actions

SOME SURFACE MEMBRANE RECEPTORS:

- have enzymatic activity
- are linked to enzymes
- are linked to G proteins
- are ligand-gated ion channels

ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR



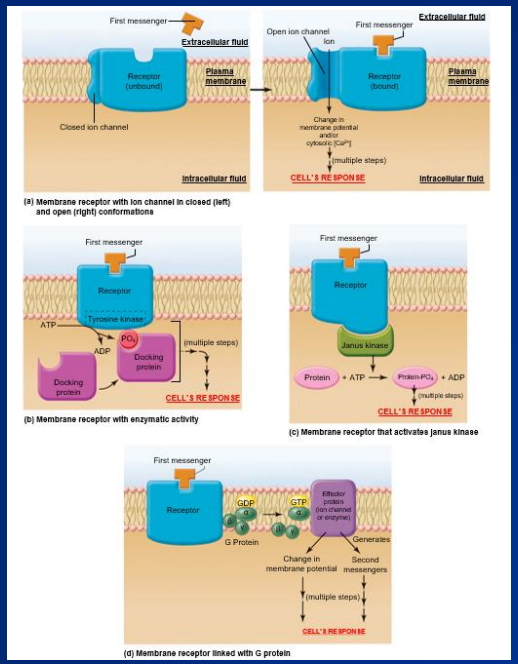
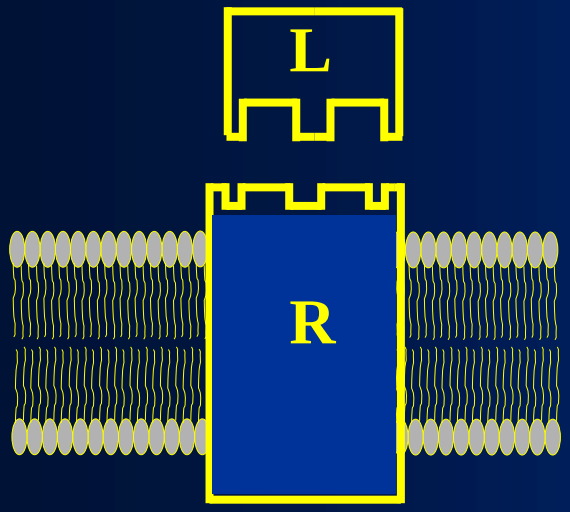


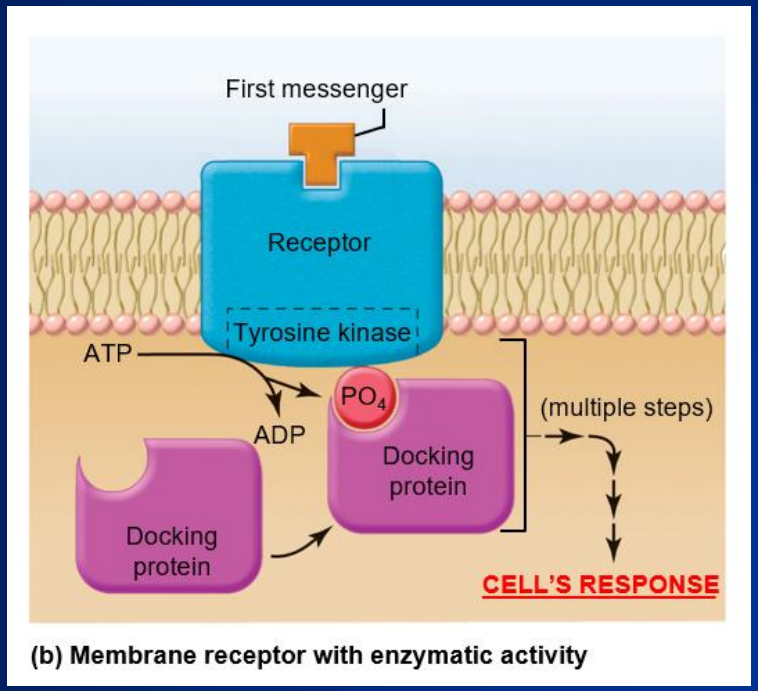
Fig. 5-5

SOME SURFACE MEMBRANE RECEPTORS:

- have enzymatic activity (often tyrosine kinase activity)
- are linked to enzymes
- are linked to G proteins
- are ligand-gated ion channels

ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR

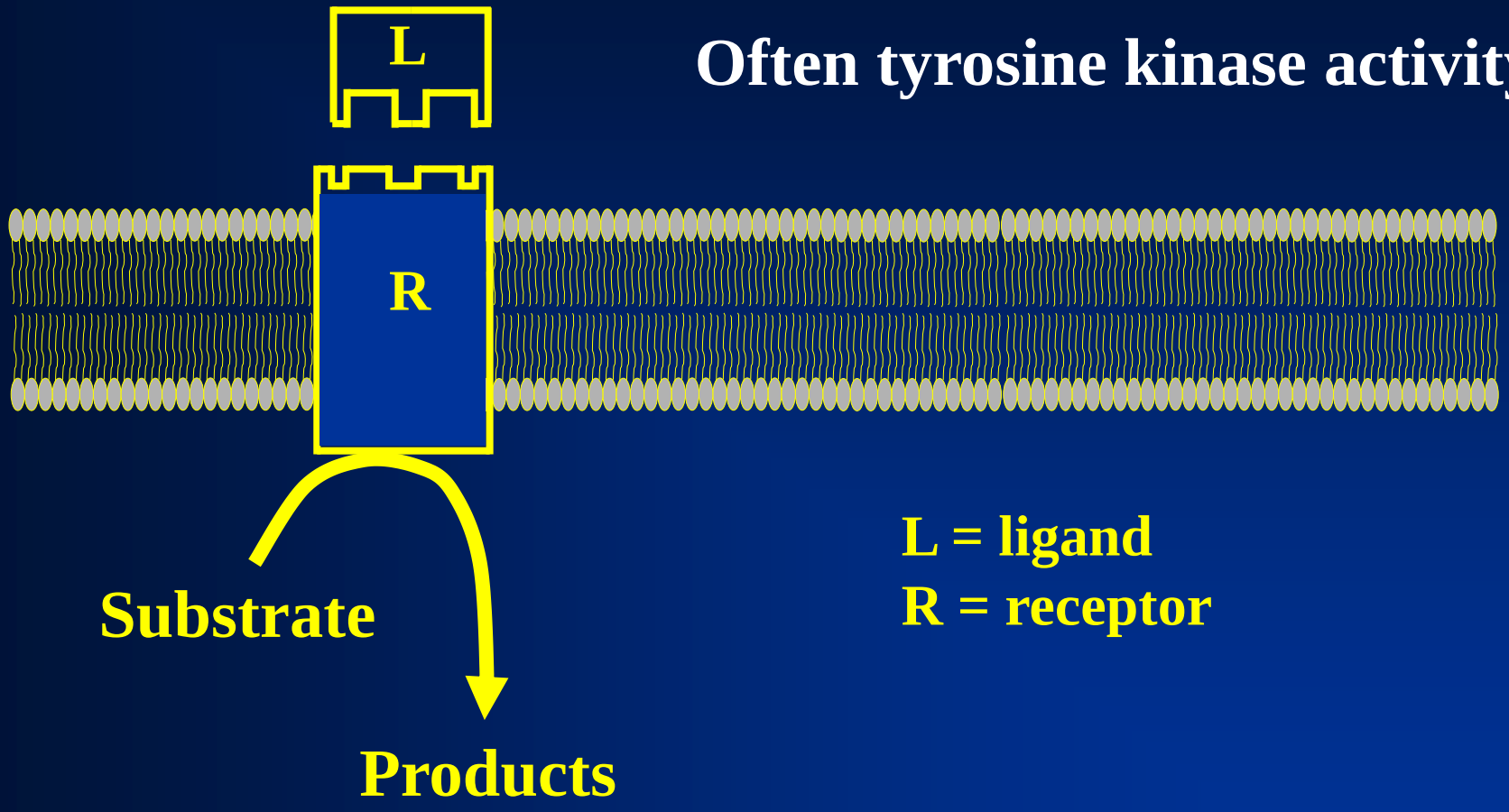




(b) Membrane receptor with enzymatic activity

Fig. 5-5b

SOME RECEPTORS HAVE ENZYMATIC ACTIVITY

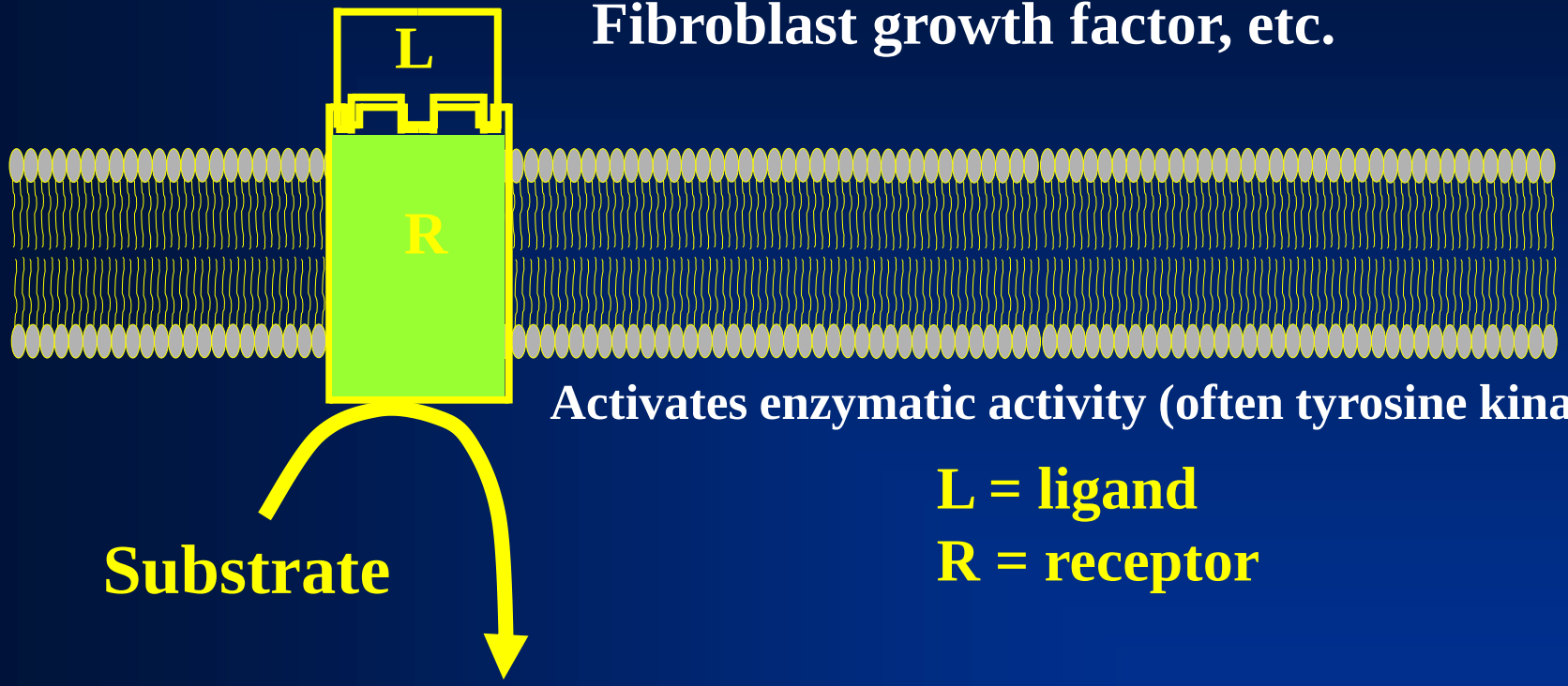


Often tyrosine kinase activity

L = ligand
R = receptor

SOME RECEPTORS HAVE ENZYMATIC ACTIVITY

Insulin, Epidermal growth factor, Fibroblast growth factor, etc.



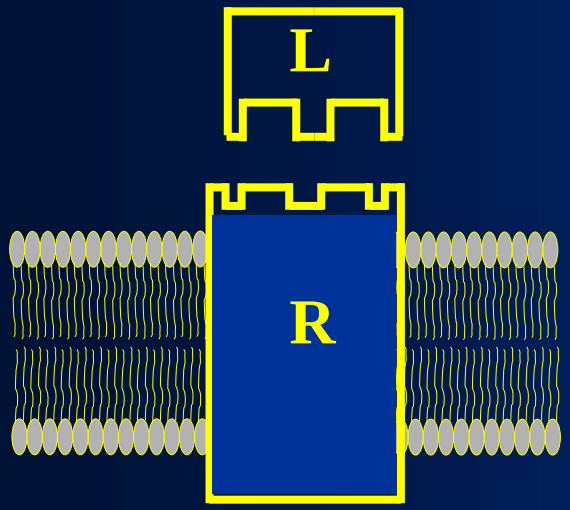
Activates enzymatic activity (often tyrosine kinase)

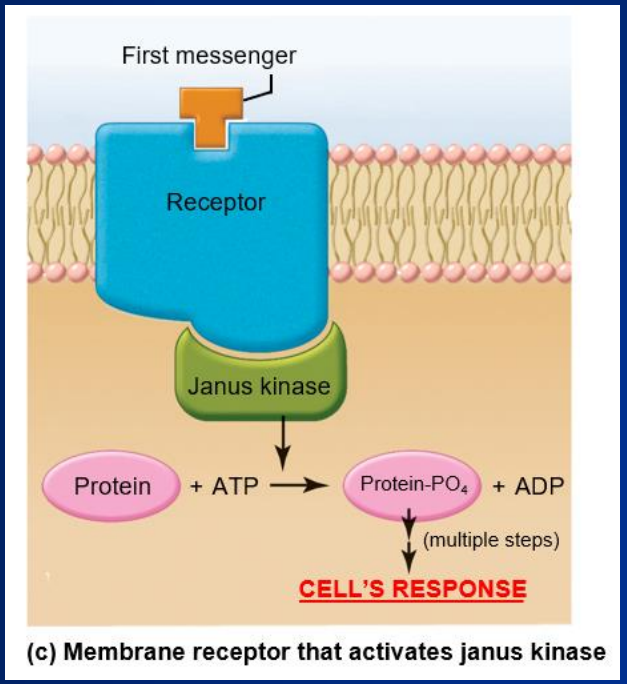
L = ligand
R = receptor

SOME SURFACE MEMBRANE RECEPTORS:

- have enzymatic activity
- are linked to enzymes (e.g., JAK/STAT pathways)
- are linked to G proteins
- are ligand-gated ion channels

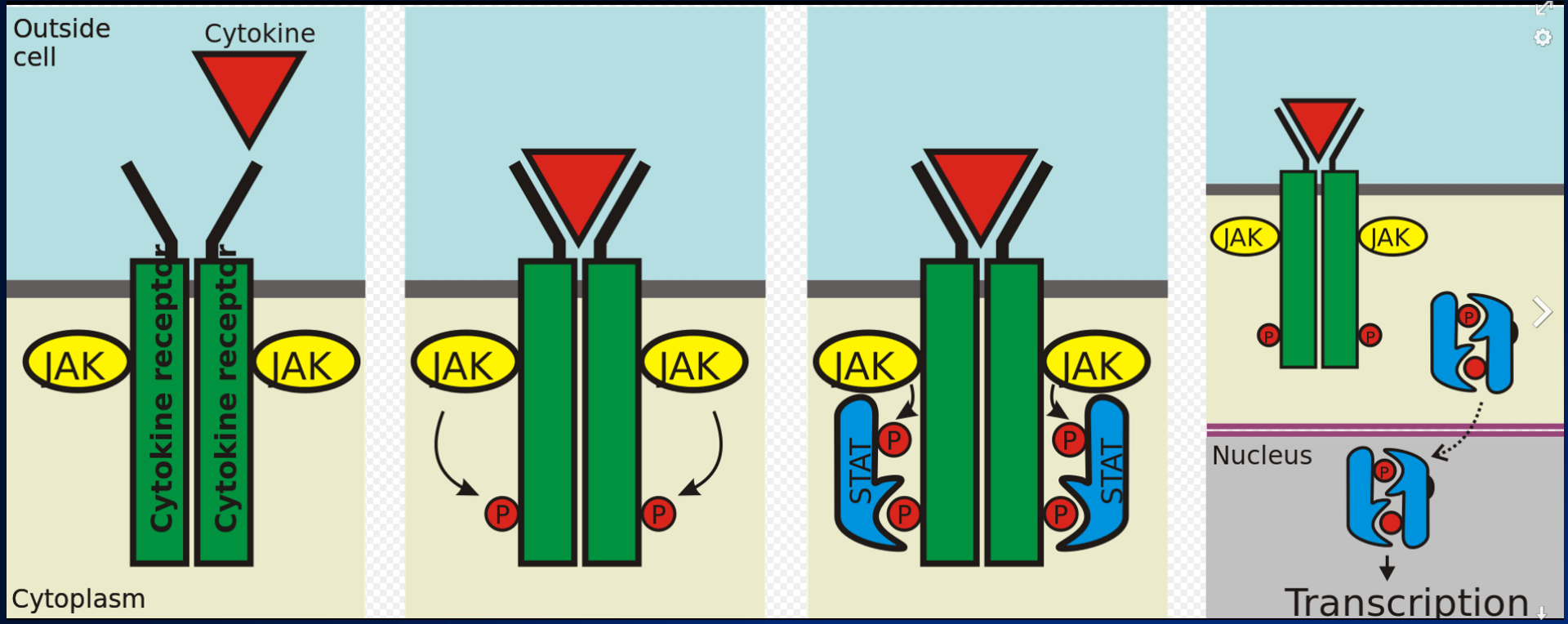
ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR





(c) Membrane receptor that activates janus kinase

Fig. 5-5c



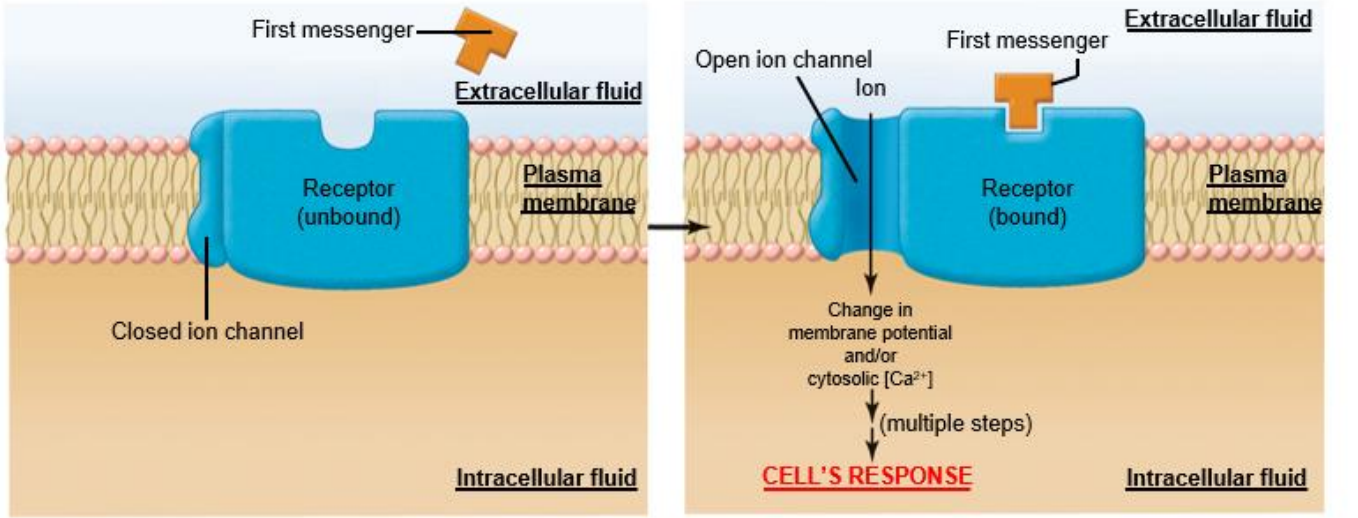
JAK-STAT pathways

- **Ligand binding activates JAK (JAnus Kinase), which phosphorylates receptor**
- **Phosphoryl subunits are transferred to and activate STATs (Signal Transducers and Activators of Transcription proteins)**
- **STATS dimerize and enter nucleus to bind to specific genes and regulate gene expression**
- **JAK-STAT pathways are utilized for many processes, including immunity, cell division, cellular apoptosis and tumorigenesis.**

SOME SURFACE MEMBRANE RECEPTORS:

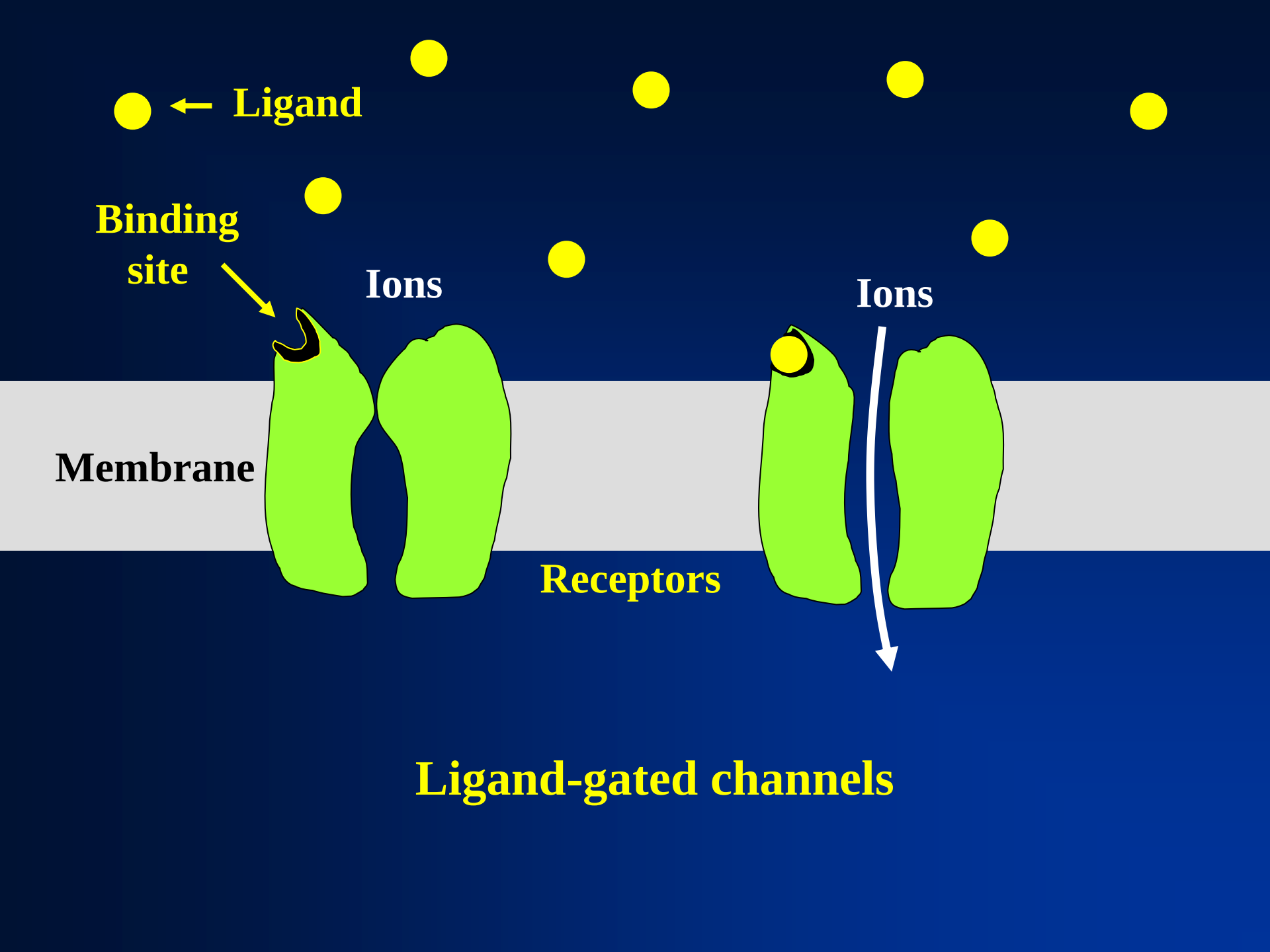
- **have enzymatic activity**
- **are linked to enzymes**
- **are linked to G proteins**
- **are ligand-gated ion channels**

ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR



(a) Membrane receptor with ion channel in closed (left) and open (right) conformations

Fig. 5-5a

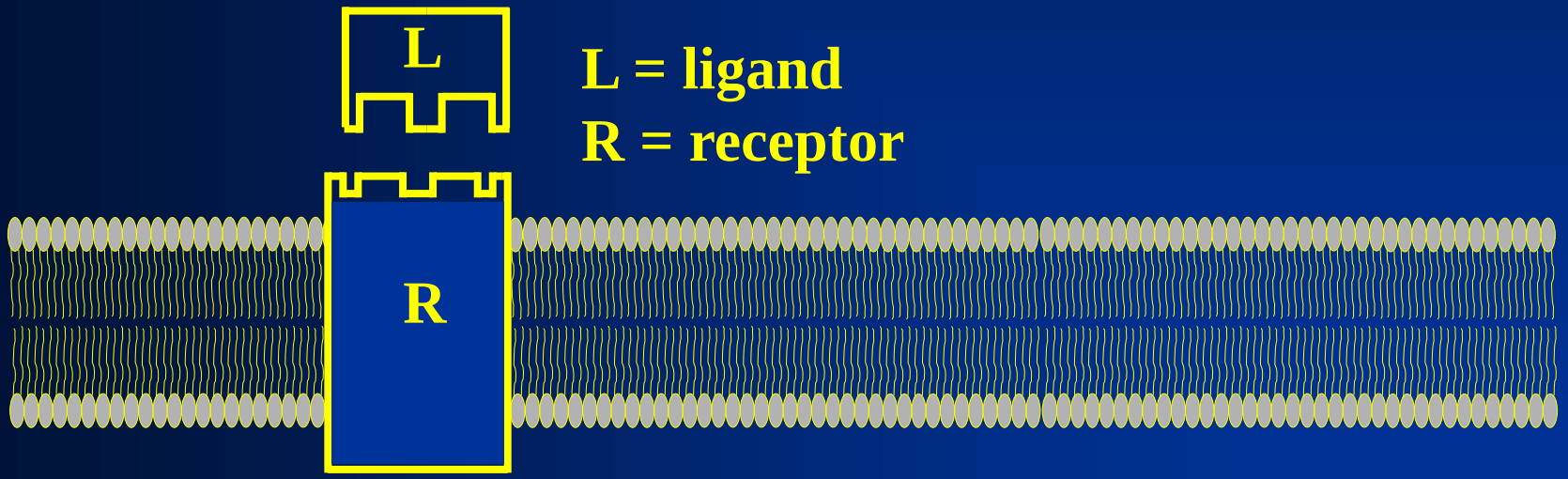


Ligand-gated channels

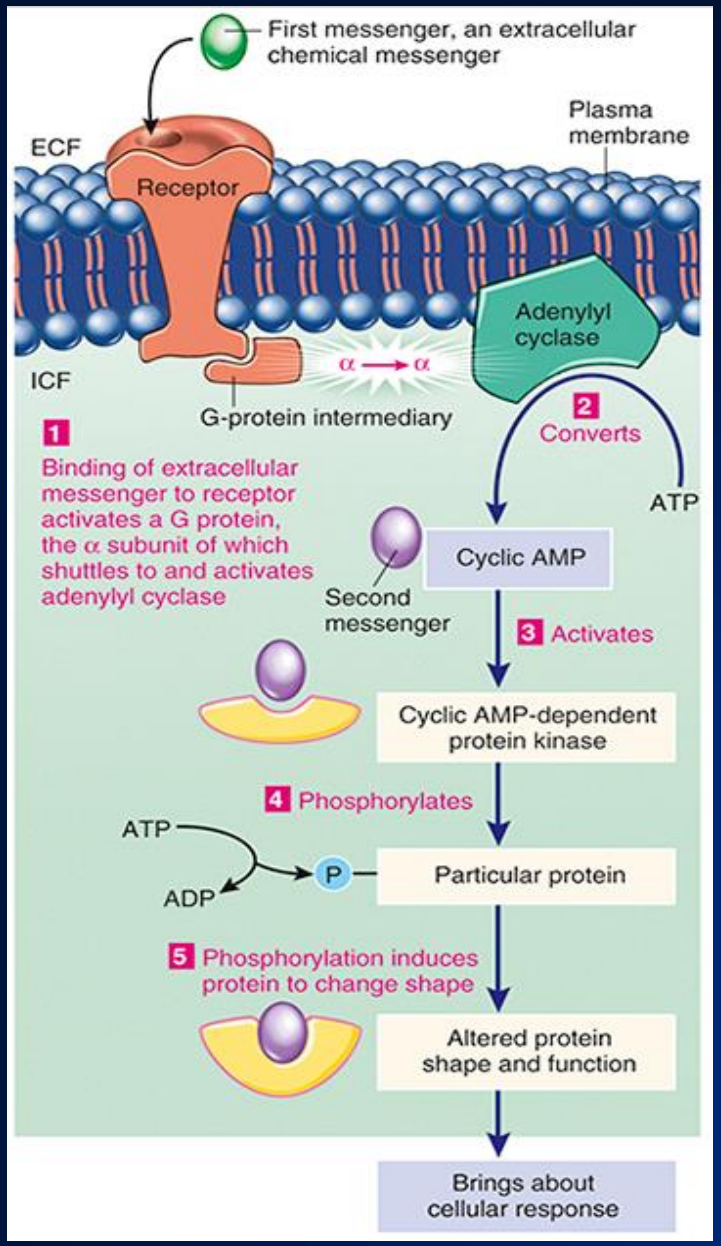
SOME SURFACE MEMBRANE RECEPTORS:

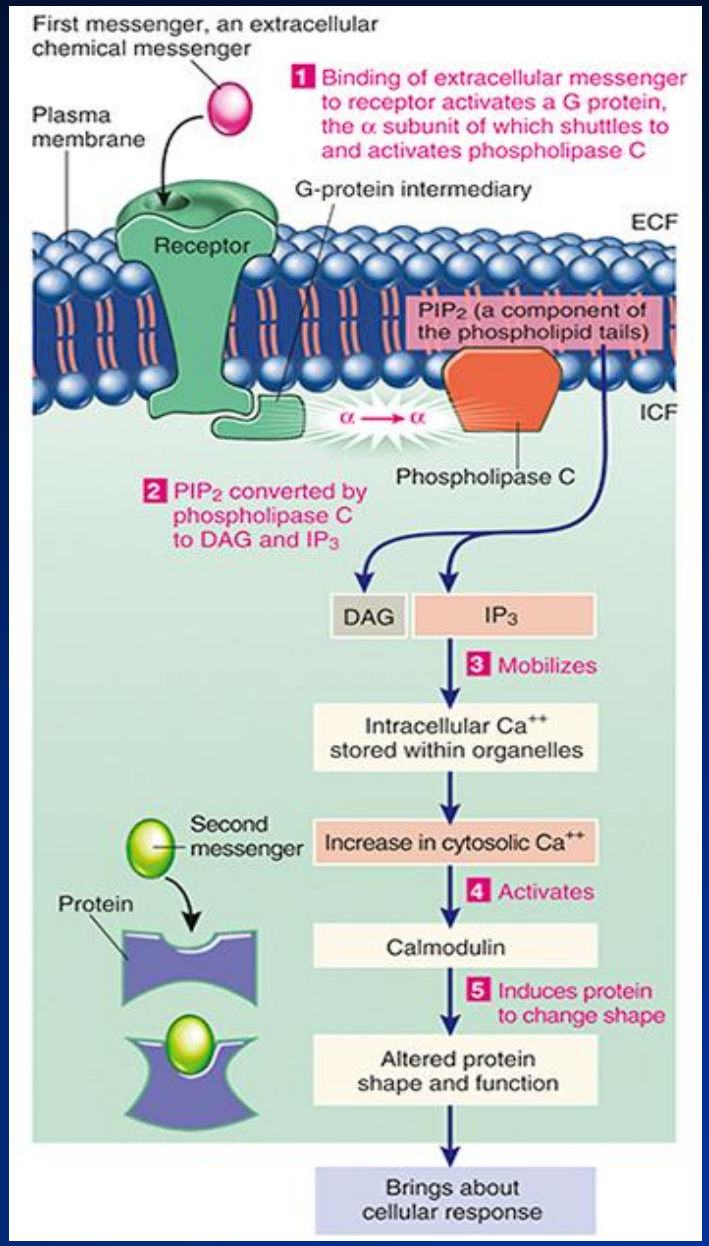
- **have enzymatic activity**
- **are linked to enzymes**
- **are linked to G proteins**
- **are ligand-gated ion channels**

ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR



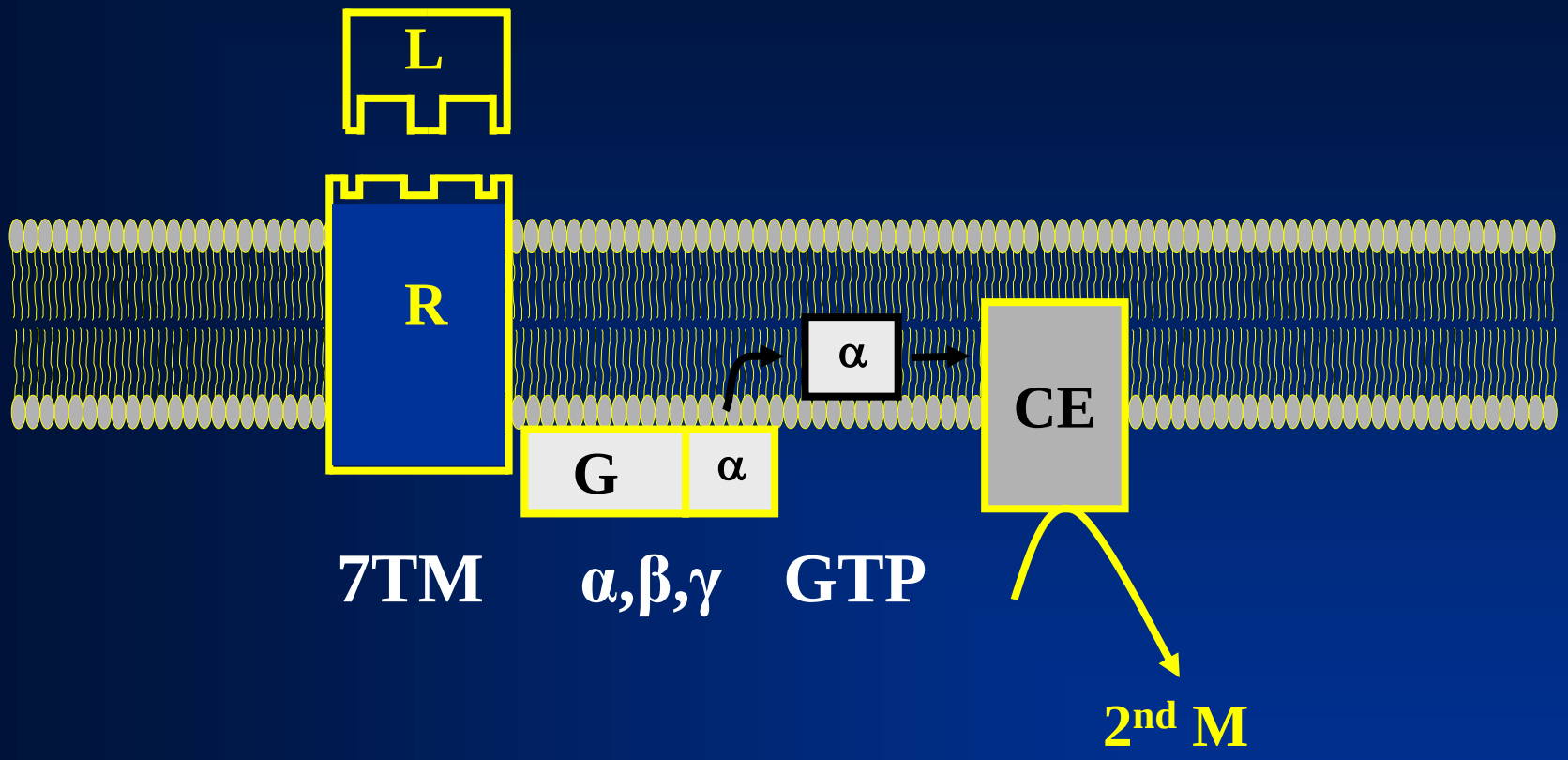
L = ligand
R = receptor





cAMP/PKA pathway **IP3/Ca++ pathway**

Clin Path textbook to come

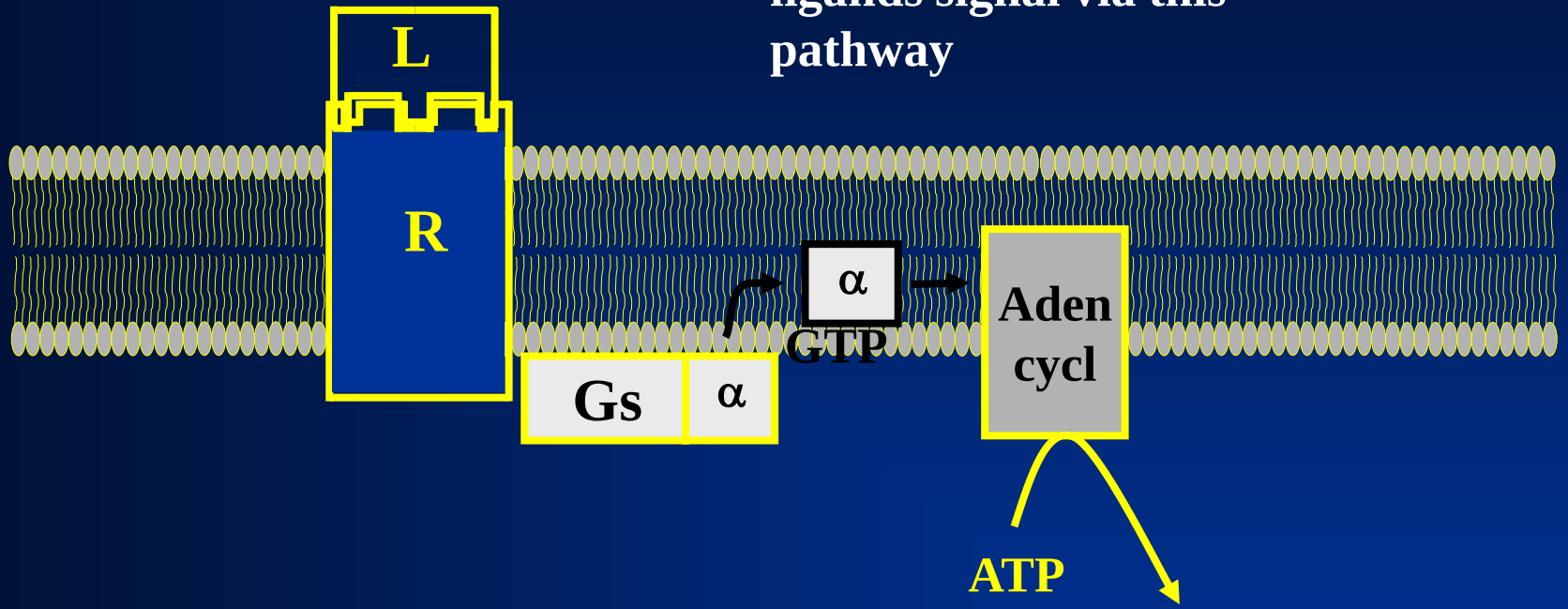


L = ligand G = G protein

R = receptor CE = catalytic enzyme

2nd M = second messenger

Norepi and epi (via βAR), ACTH, LH, FSH, TSH, glucagon, gastrin, PTH, etc., etc.; more than 200 ligands signal via this pathway

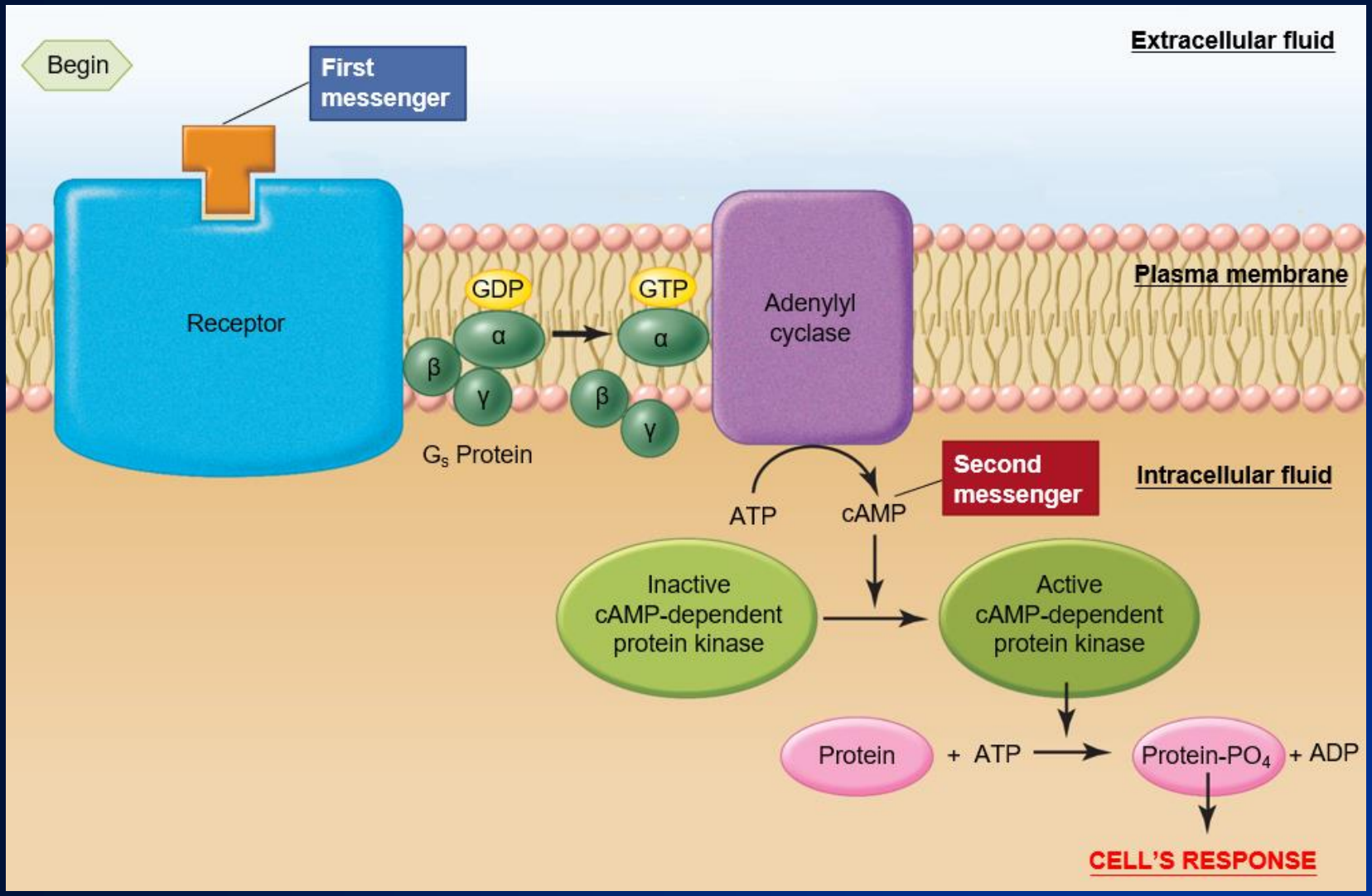


THE ADENYLATE CYCLASE/ cAMP SIGNAL TRANSDUCTION PATHWAY

cAMP- dependent Protein kinase A (inactive) → cAMP-dependent Protein kinase A (active)

cAMP/PKA tutorial will be placed on CANVAS

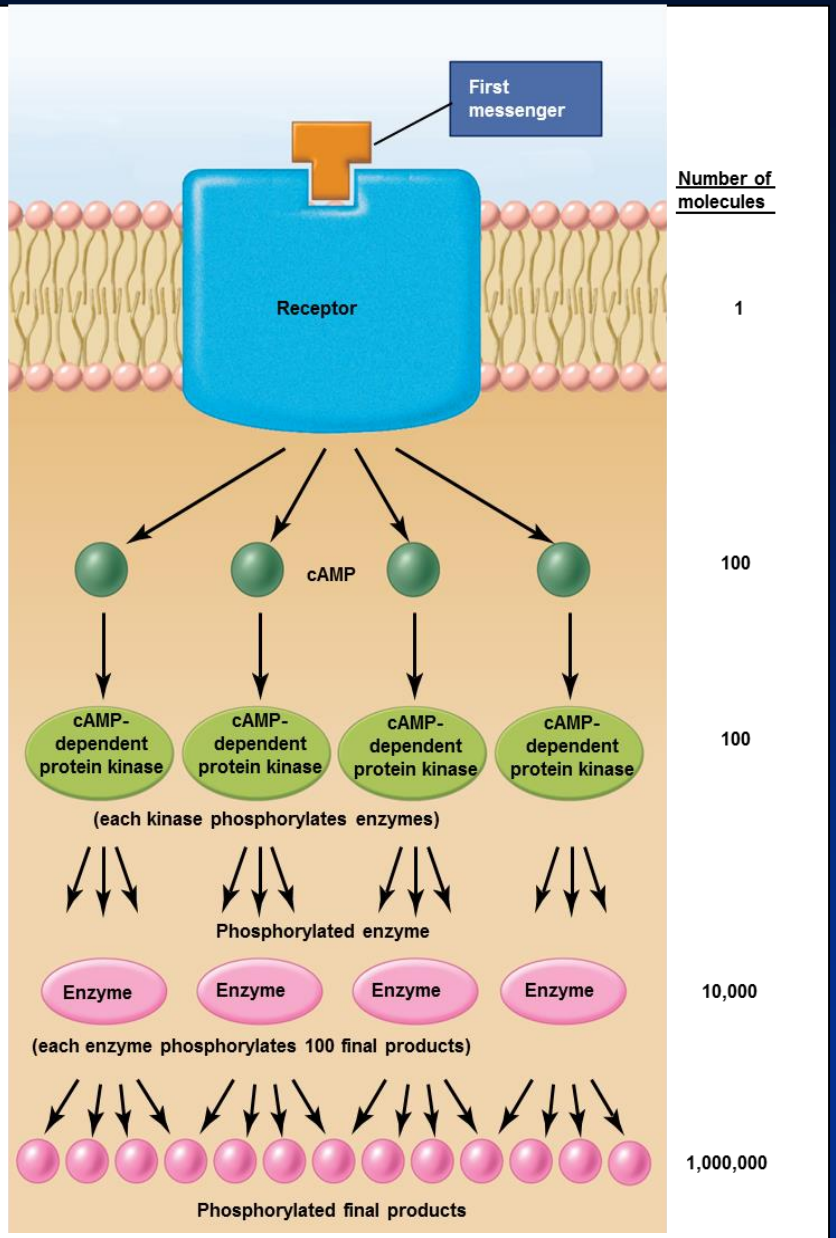
Fig 5-6



The cyclic AMP second messenger system.

From **Vander's Human Physiology** 15th Edition

Signal Amplification Cascade

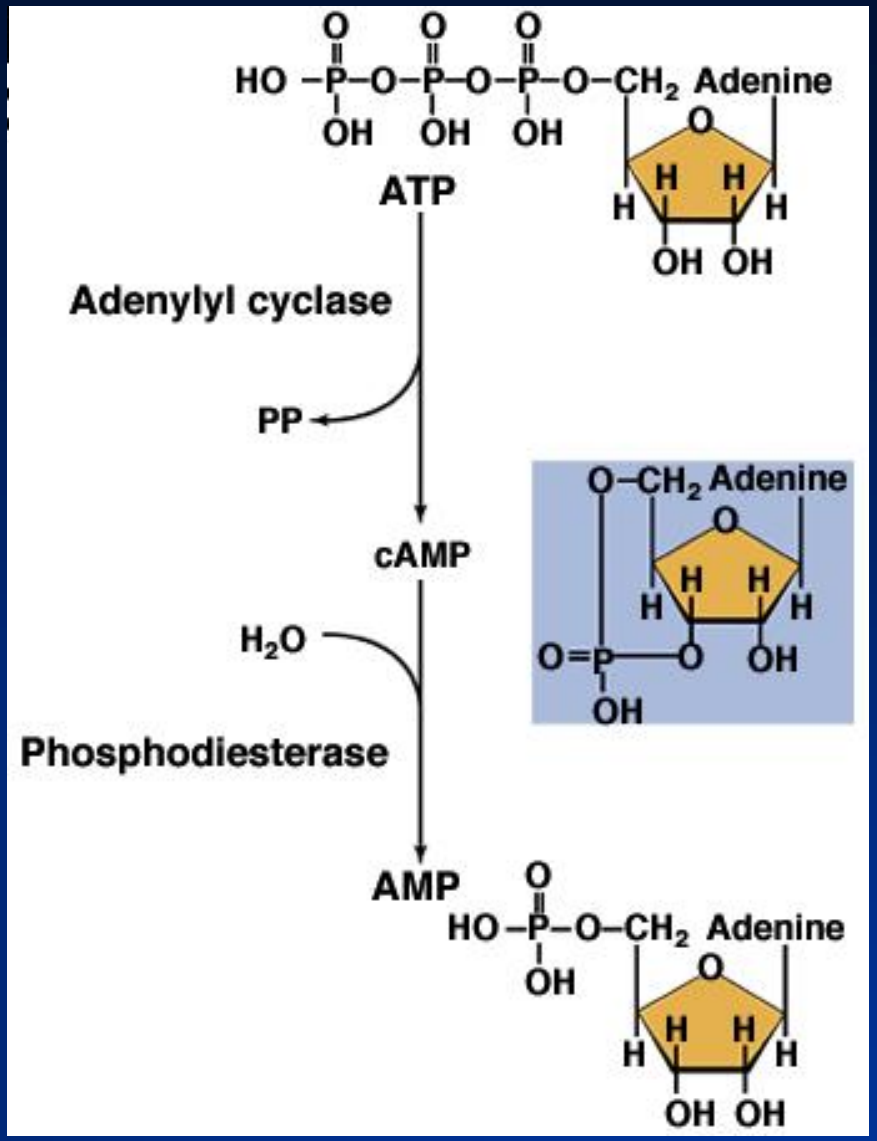


The cAMP system rapidly amplifies the response capacity of cells: here, one "first messenger" led to the formation of one million product molecules.

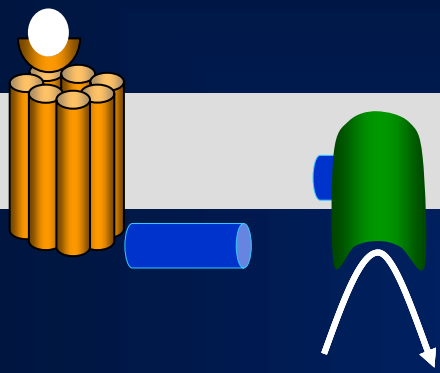
From
Vander's Human Physiology
16th Edition
Fig. 5.7

Adenylyl cyclase forms cAMP, a "second messenger" that activates enzymes used in cellular responses.

The phosphodiesterase enzymes "terminate" the second messenger cAMP.



cAMP/PKA tutorial will be placed on CANVAS



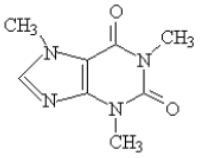
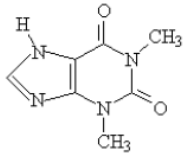
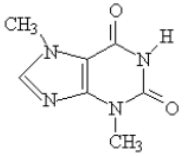
Plasma membrane

cAMP (active) → AMP (inactive)

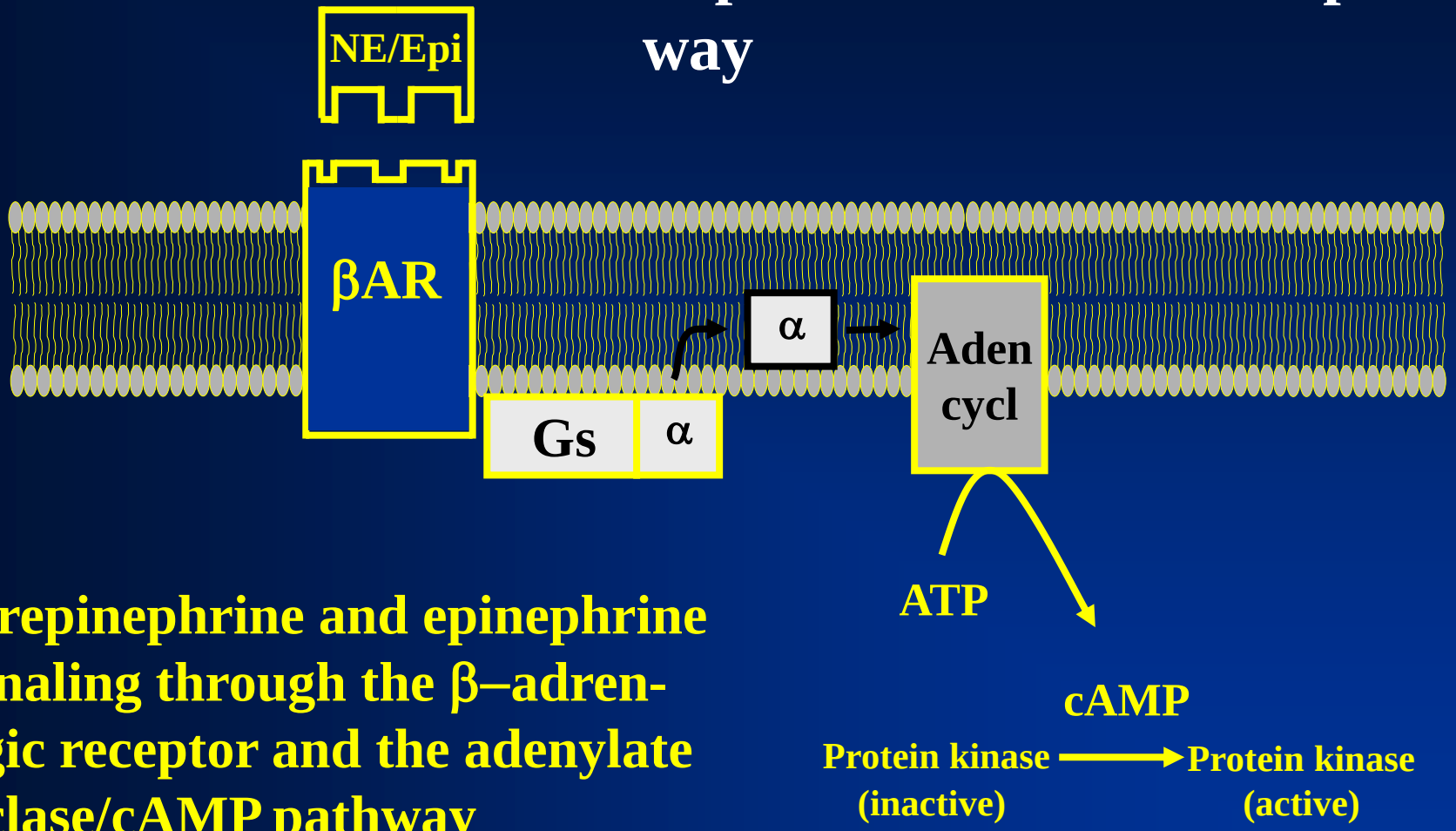
Phosphodiesterase (PDE)

(-) methyl xanthines

Phosphodiesterase (PDE) is inhibited by the actions of the methyl xanthines. These include caffeine, theophylline and theobromine. What are the implications of this?

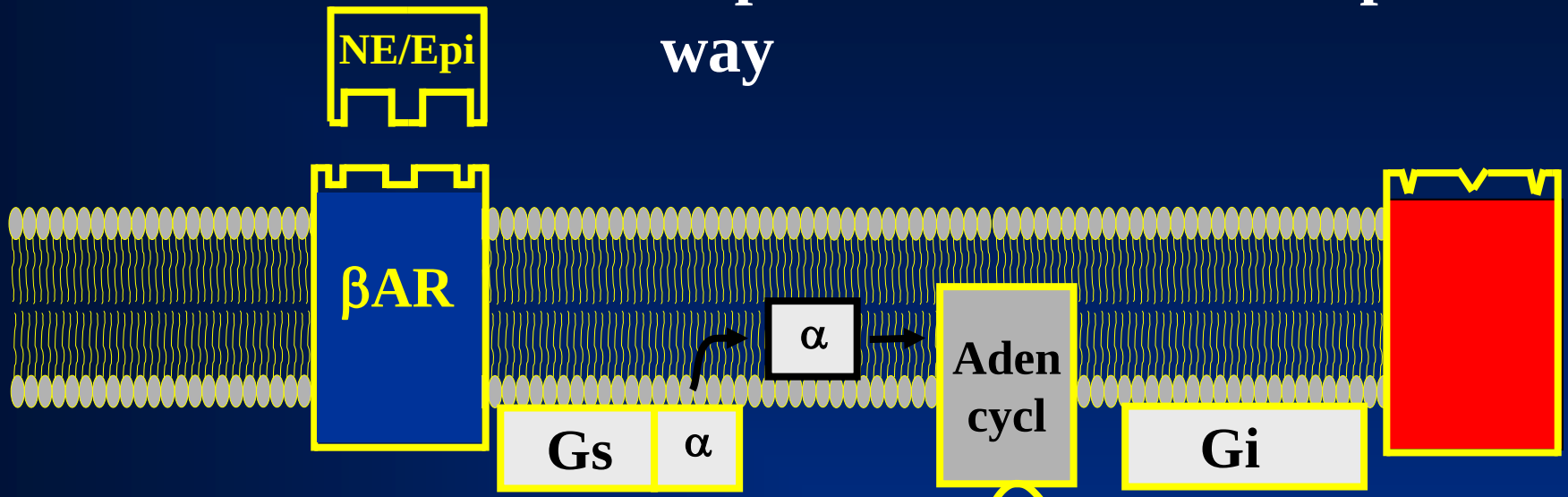
Caffeine		coffee, colas, etc.
Theophylline		tea
Theobromine		chocolate

There are specific drugs that impact each level of this path-way



Norepinephrine and epinephrine signaling through the β–adren-ergic receptor and the adenylate cyclase/cAMP pathway

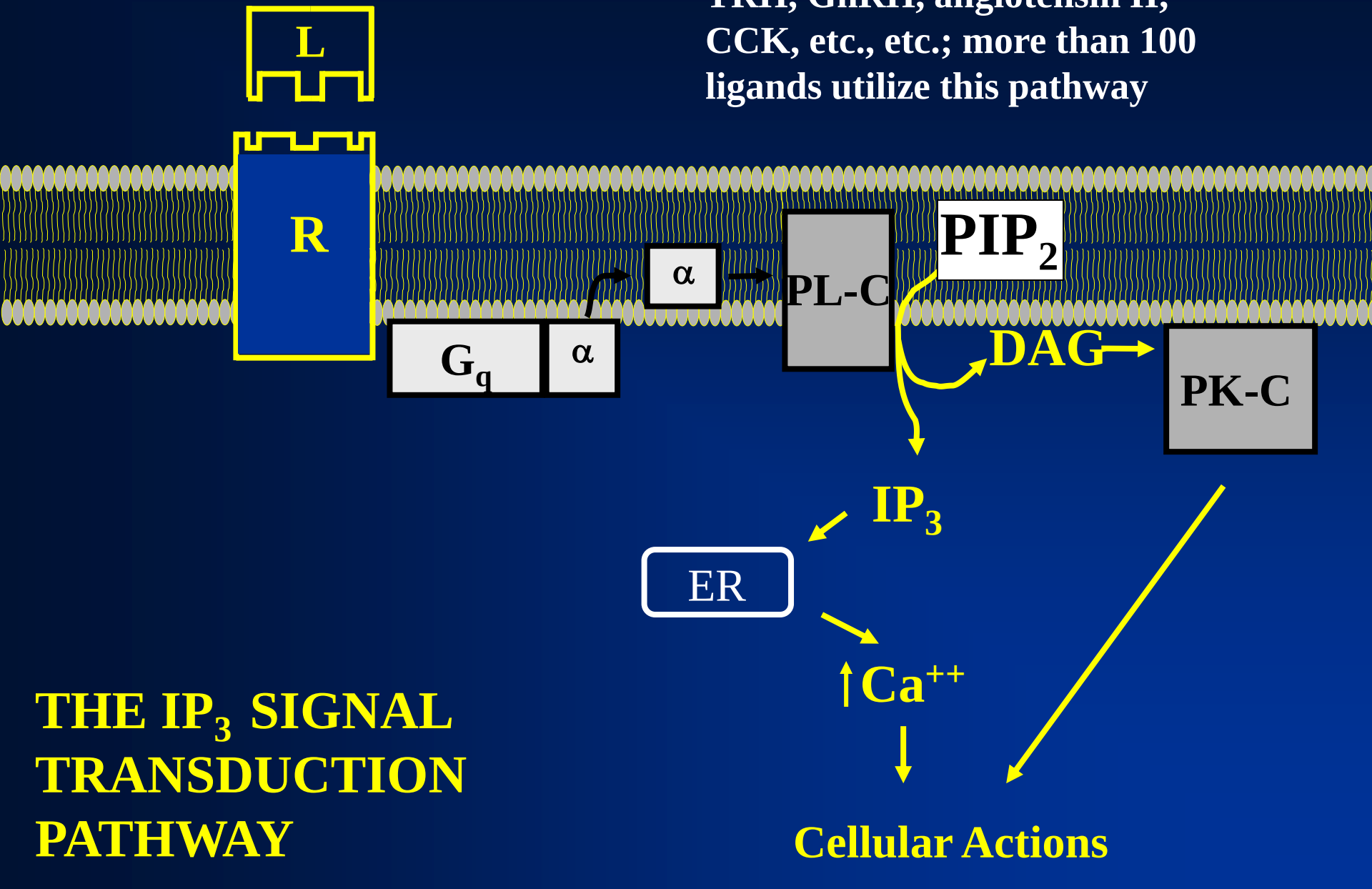
There are specific drugs that impact each level of this path-way



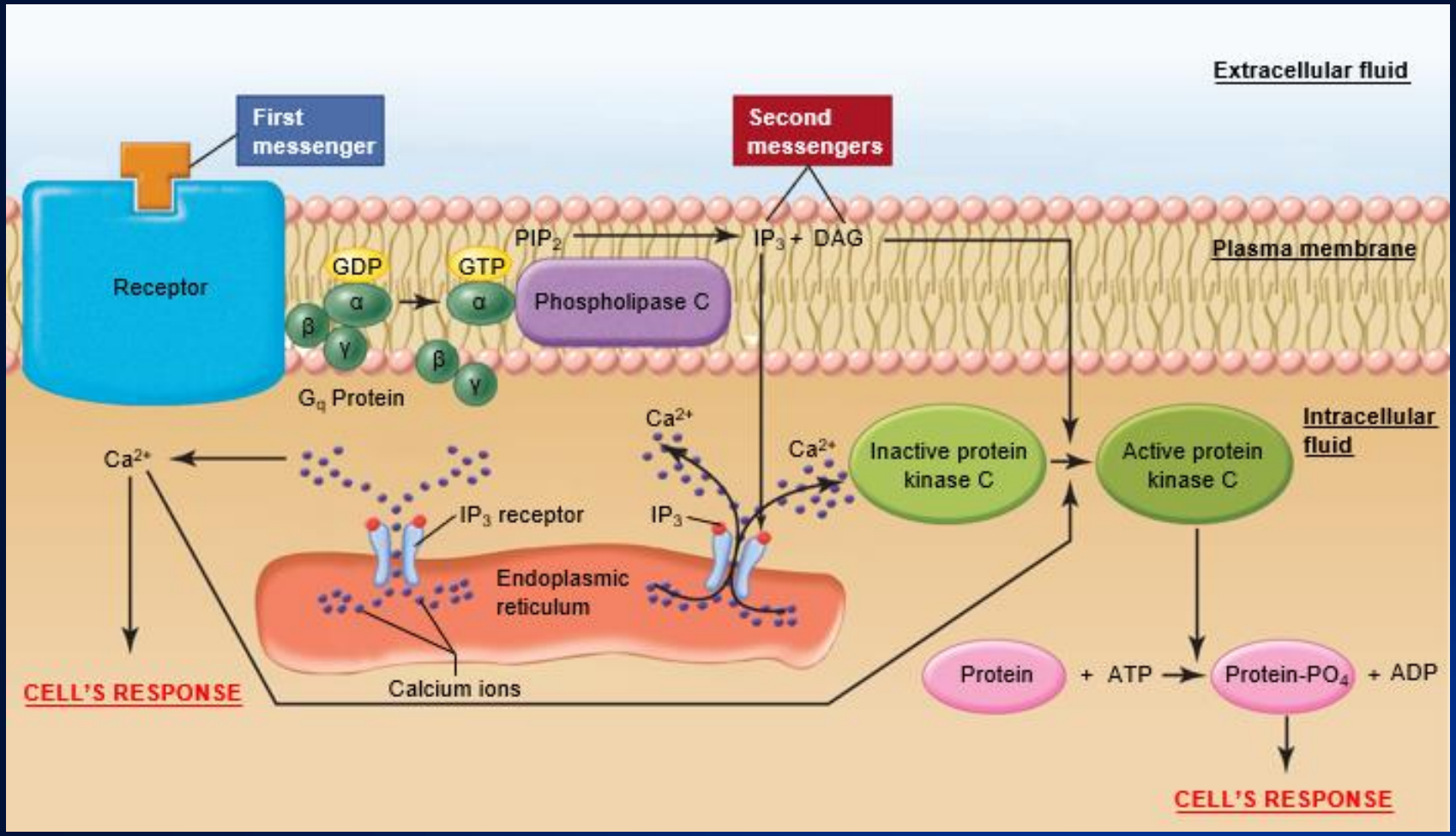
Gi(nhibitory) stim-ulation inactitives adenylate cyclase

Norepinephrine and epinephrine signaling through the β–adren-ergic receptor and the adenylate cyclase/cAMP pathway

Norepi and epi (via αAR), TRH, GnRH, angiotensin II, CCK, etc., etc.; more than 100 ligands utilize this pathway



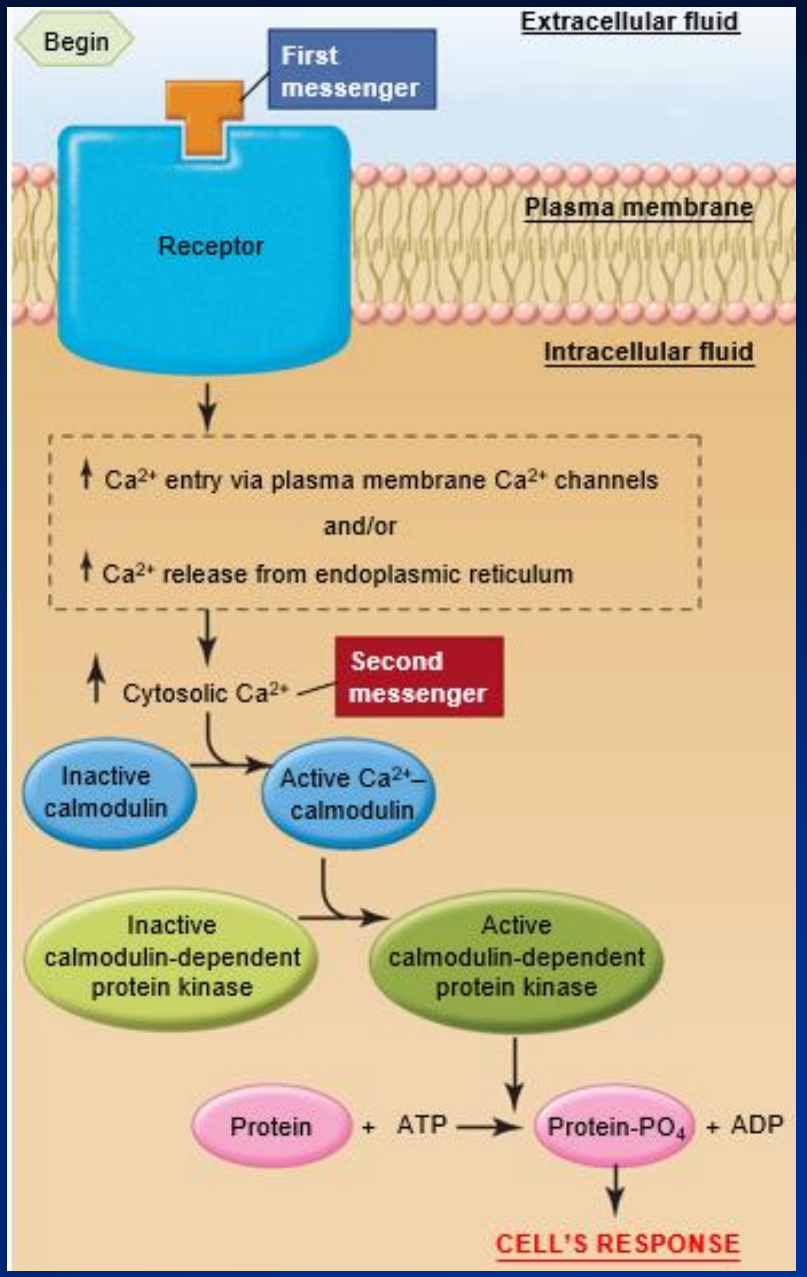
THE IP_3 SIGNAL TRANSDUCTION PATHWAY

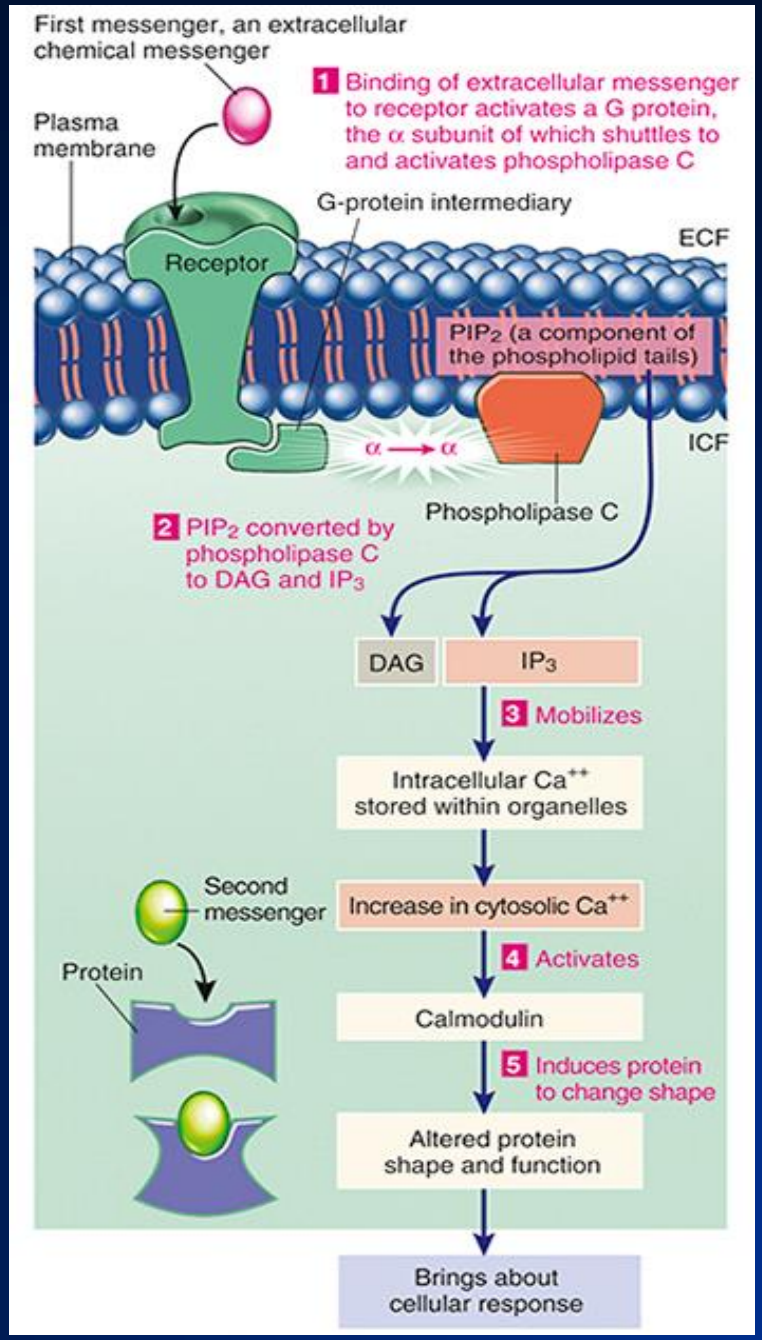


THE IP_3 SIGNAL TRANSDUCTION PATHWAY

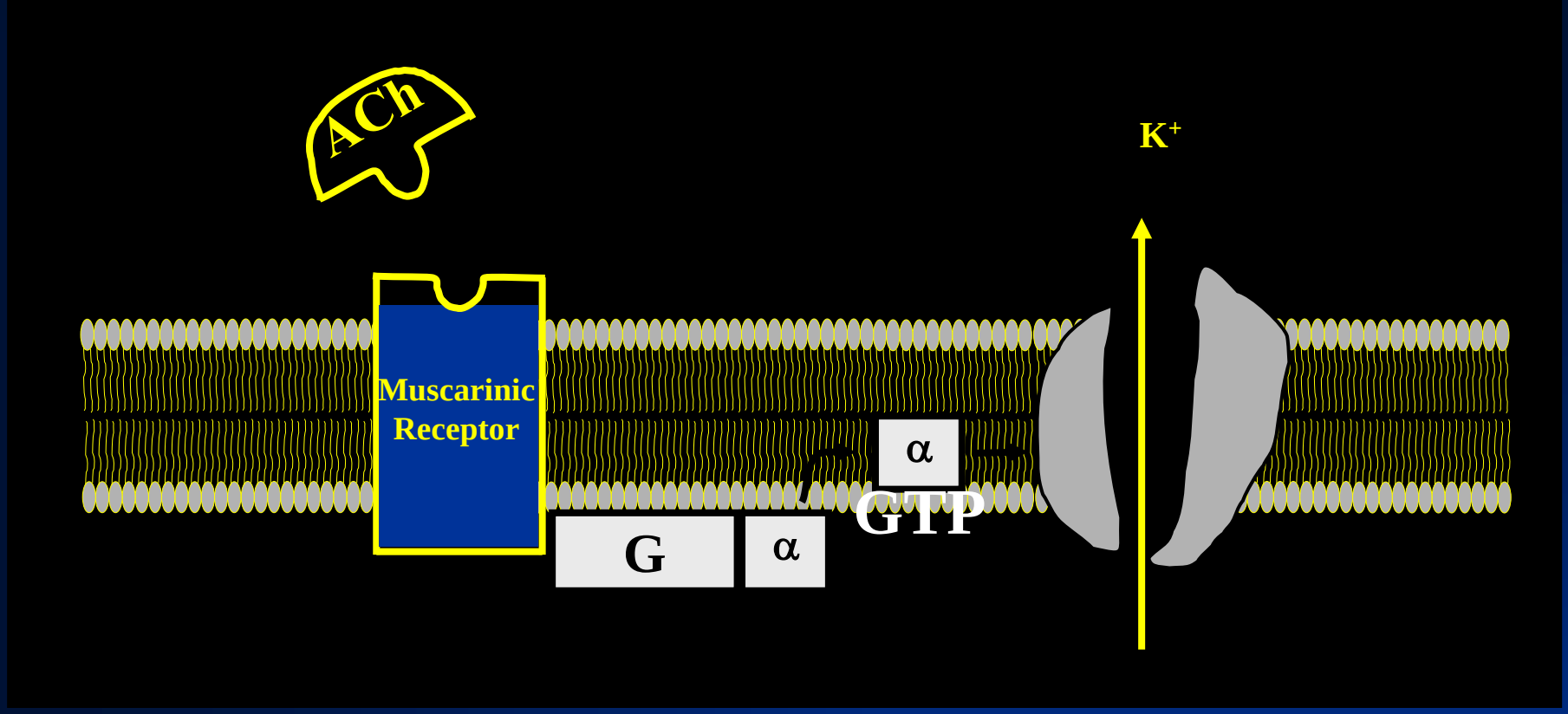
Elevating cytosolic calcium regulates many cellular functions, often through its association with the calium binding protein calmodulin.

The IP_3 SIGNAL TRANSDUCTION PATHWAY is just one way in which cellular Ca++ can be increased quickly to regulate cellular activity.





IP3/Ca++ pathway



SIGNAL TRANSDUCTION VIA THE MUSCARINIC RECEPTOR IN THE HEART

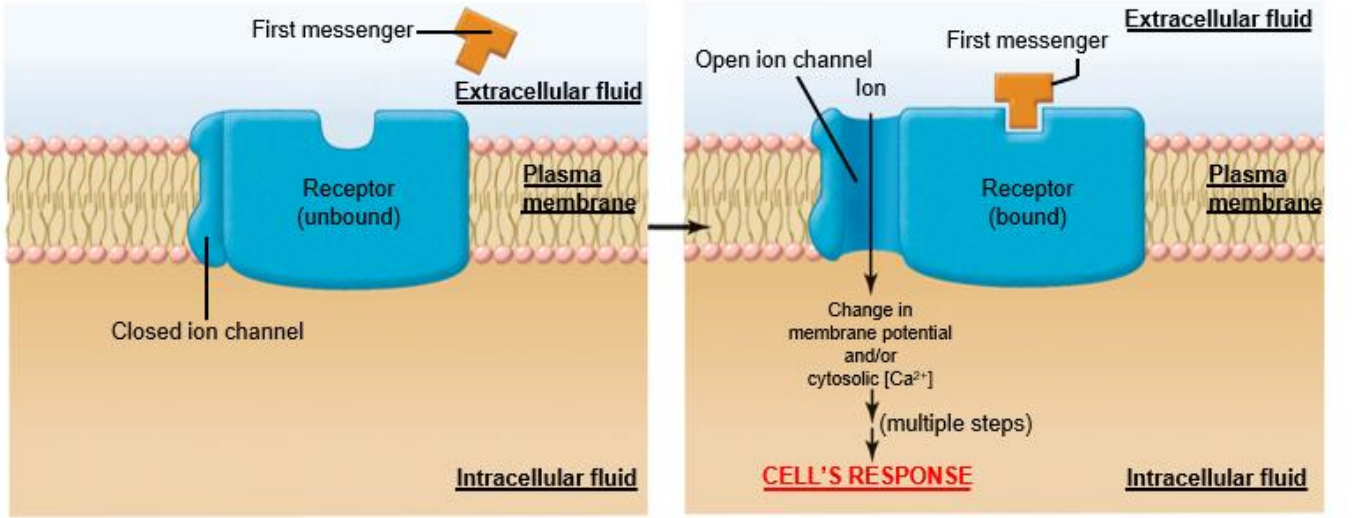
is a G protein mediated event

Vagal parasympathetic activity to the heart slows heart rate

SOME SURFACE MEMBRANE RECEPTORS:

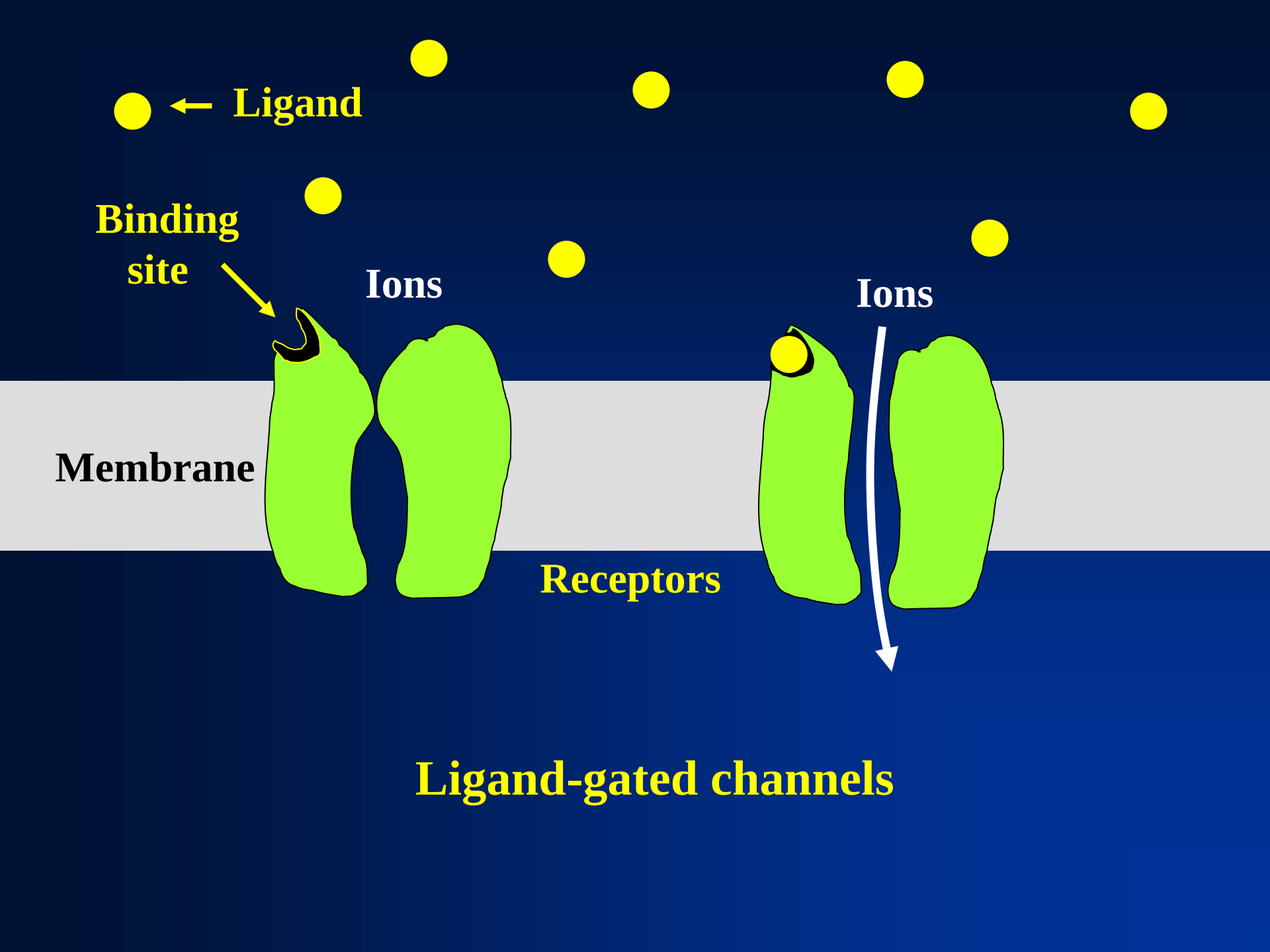
- **have enzymatic activity**
- **are linked to enzymes**
- **are linked to G proteins**
- **are ligand-gated ion channels**

ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR



(a) Membrane receptor with ion channel in closed (left) and open (right) conformations

Fig. 5-5a

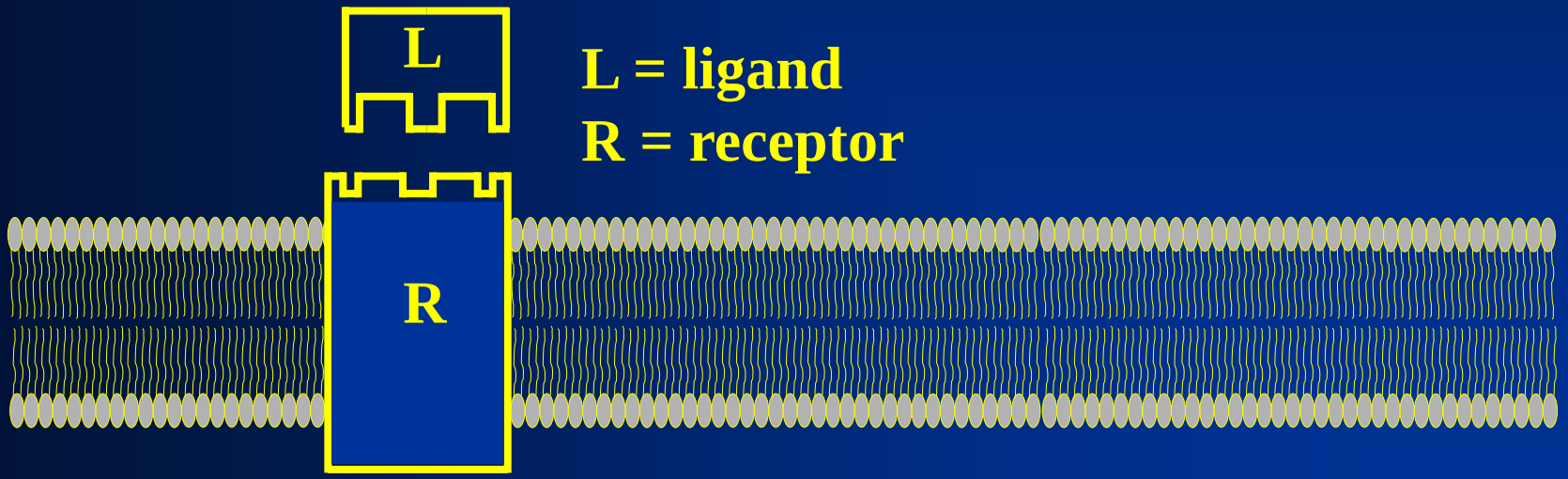


Ligand-gated channels

SOME SURFACE MEMBRANE RECEPTORS:

- **have enzymatic activity**
- **are linked to enzymes**
- **are linked to G proteins**
- **are ligand-gated ion channels**

ALL FUNCTION TO TRANSMIT AN EXTRA-CELLULAR MESSAGE TO THE CELL INTERIOR

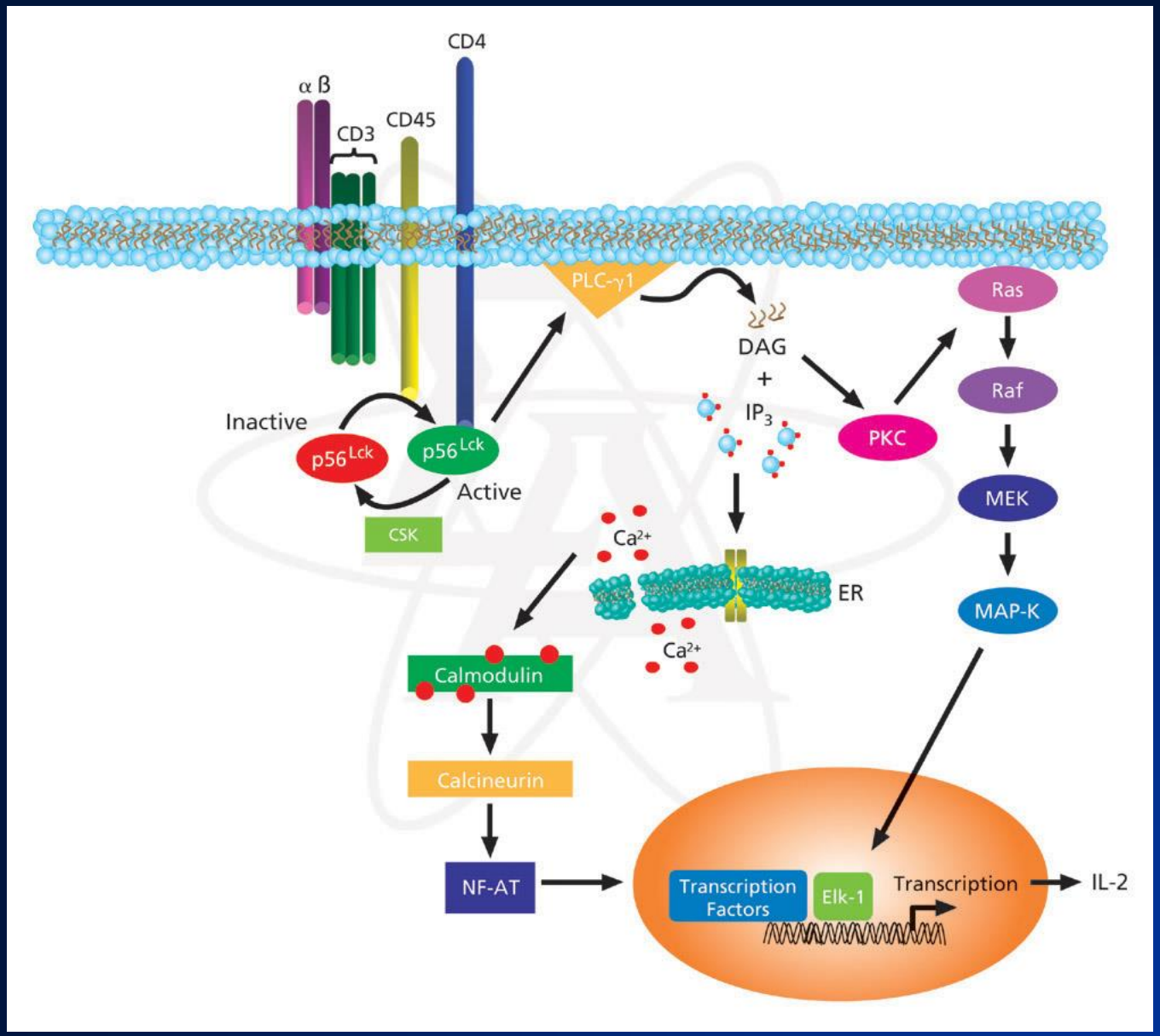


L = ligand
R = receptor

And there are many more …

T Cell Receptor (TCR) Signaling

SIGMA-ALDRICH



Signal Transduction Pathways

1. **Pathways initiated by lipid-soluble messengers** – can cross plasma membrane (e.g., steroid and thyroid hormones, vitamin D, etc.)
2. **Pathways initiated by water-soluble messengers** – cannot cross plasma membrane (Membrane signal transduction; e.g., peptide hormones, cytokines, neurotransmitters, growth factors, etc.)

2. **Pathways initiated by water-soluble messengers (ligands)**

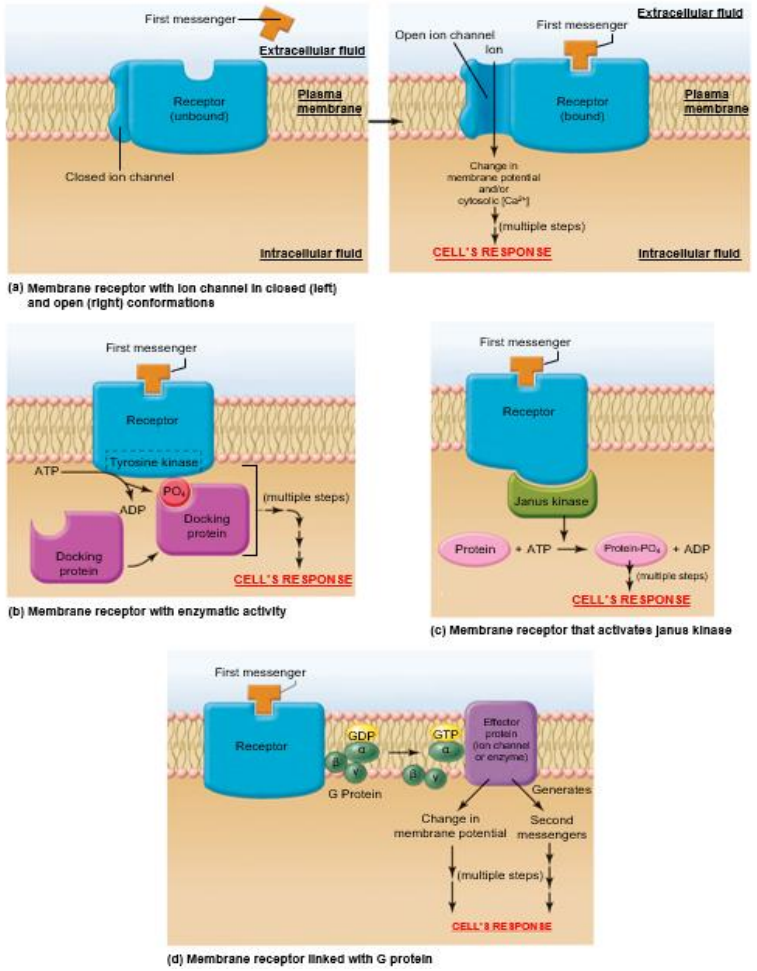
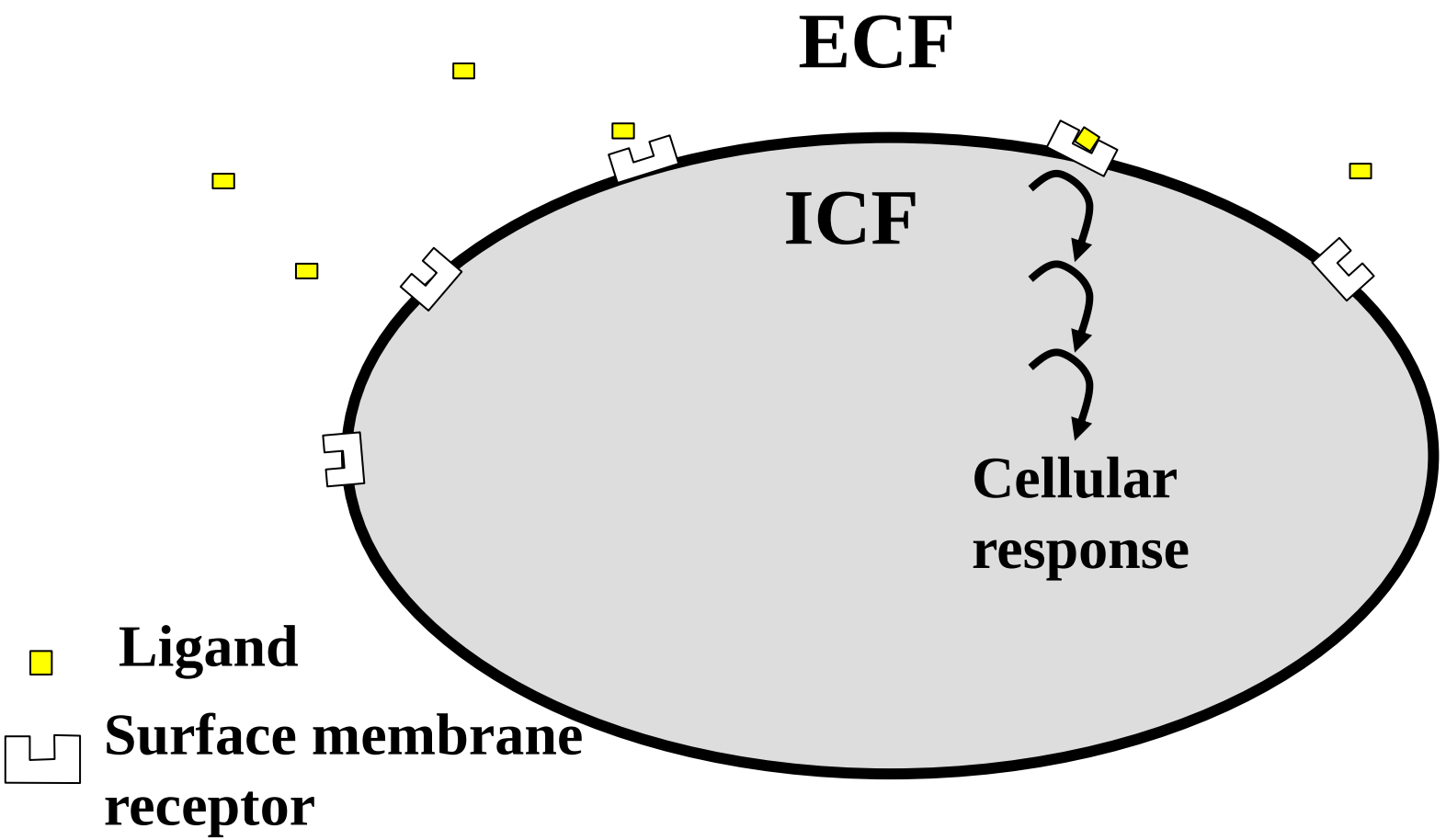


Fig. 5-5



Many ligands such as peptide hormones and neurotransmitters communicate with their target cells via surface membrane receptors; ligand-receptor binding activates a transmembrane signal transduction process that elicits a cellular response

Signal Transduction Pathways

1. **Pathways initiated by lipid-soluble messengers** – can cross plasma membrane (e.g., steroid and thyroid hormones, vitamin D, etc.)
2. **Pathways initiated by water-soluble messengers** – cannot cross plasma membrane (Membrane signal transduction; e.g., peptide hormones, cytokines, neurotransmitters, growth factors, etc.)

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1. **Pathways initiated by lipid-soluble messengers**

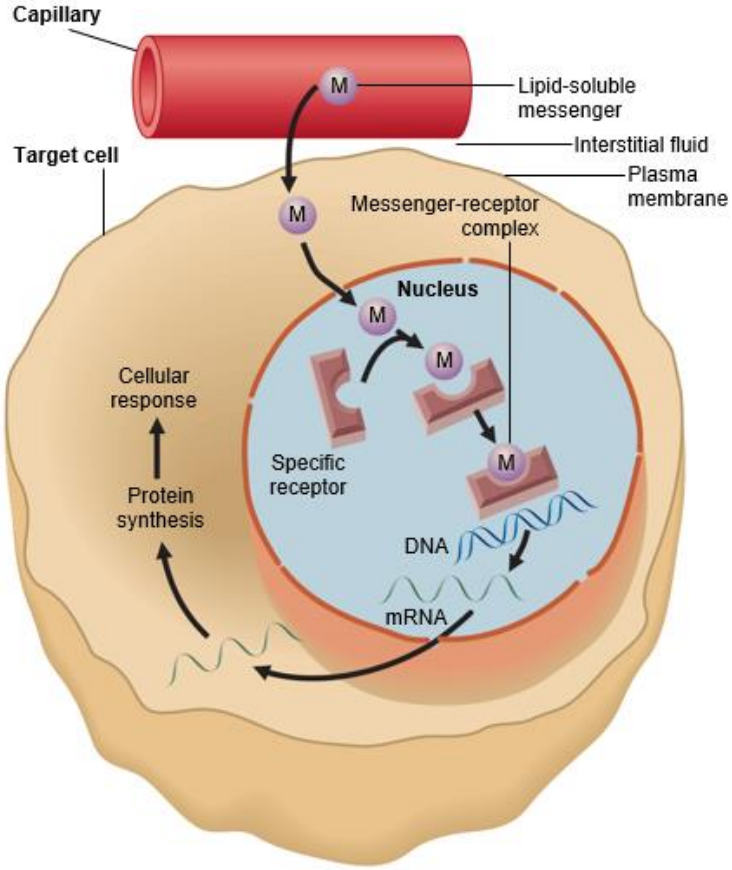
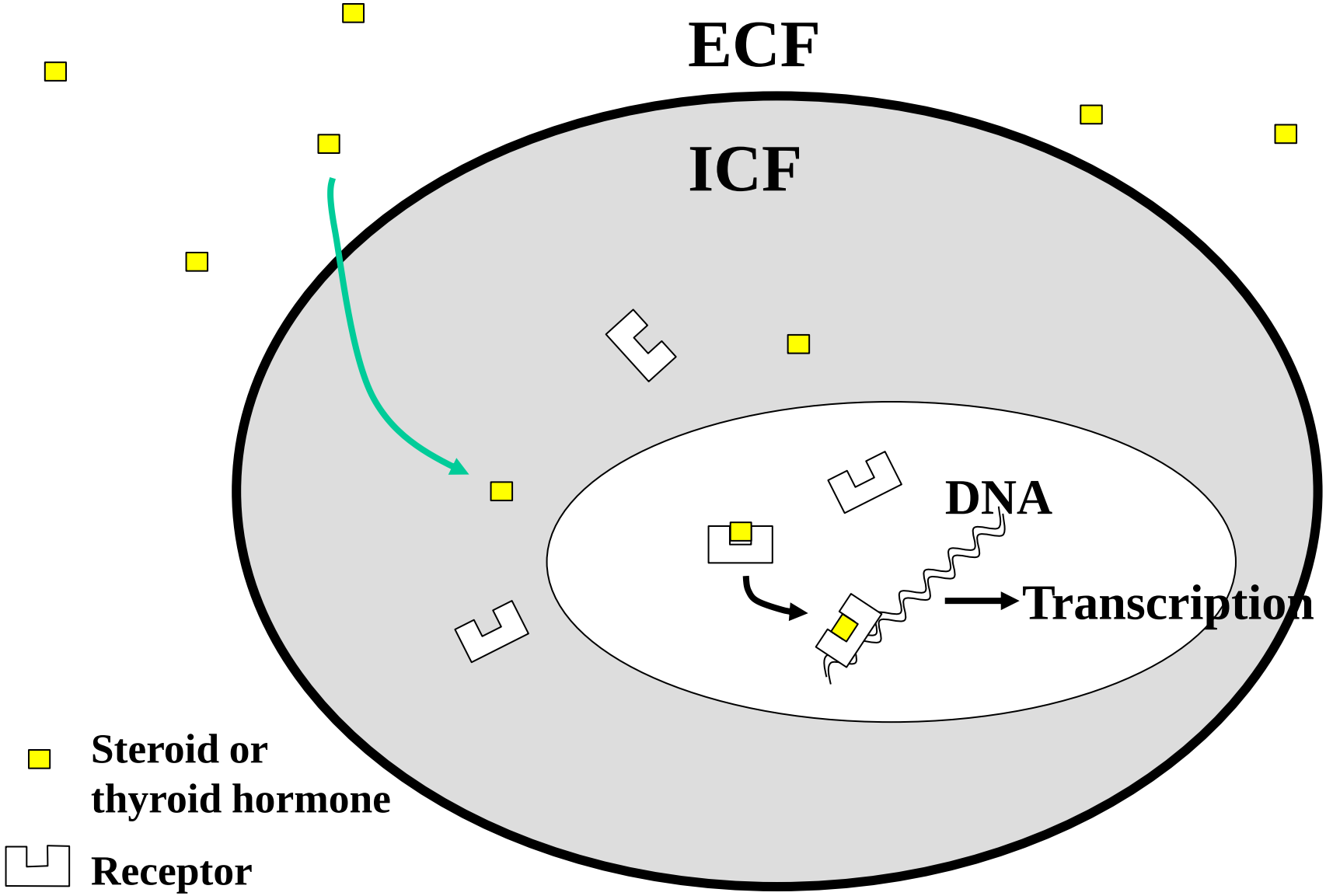


Fig. 5-4



Steroid and thyroid hormones bind cytosolic and/or nuclear receptors and modulate gene transcription

GENERAL RELATIONSHIPS/PROPERTIES/ACTIONS

	Peptide hormones & neurotransmitters	**Steroid hormones & thyroid hormone**
Precursors	Peptide synthesis (PH) Varies (NT)	Cholesterol (SH) Tyrosine (TH)
Physical chemistry	Charged, polar, -OH groups	Uncharged, neutral
Characteristics	Hydophilic, lipophobic	Hydrophobic, lipophilic
Solubility in ECF	Freely soluble	Insoluble
Transport in blood	Free; unbound	Bound to plasma proteins
Cross membrane?	No	Yes
Receptor location	Membrane surface	Cytosolic and nuclear
2nd messengers or membrane signal transduction?	Yes	No
Mode of action (general)	Turn on/off existing machinery **(Sometimes gene transcription)**	Synthesis of new machinery (gene transcription)
Response time	Rapid **(Sometimes delayed)**	Delayed
Duration of action	Shorter acting **(Sometimes longer)**	Longer acting

FUNDAMENTAL PHYSIOLOGICAL PRINCIPLES (Part I)

I. Introduction to physiology and homeostasis
   - A. Physiology – Definitions
   - B. Homeostasis
   - C. Fluid compartments
   - D. Homeostatically regulated factors
   - E. How/why are factors regulated
   - F. Homeostatic hmmmmms

II. Membrane dynamics (structure, function, signaling)
   - A. Gradients drive physiological processes
   - B. Membrane organization
   - C. Membrane transport and translocation
   - D. Signal transduction

III. Membrane potentials
   - A. Membrane potentials and gradients
   - B. Types of membrane potentials
   - C. Resting membrane potentials
   - D. Communication using membrane potentials
   - E. Determinates of membrane and equilibrium potentials
   - F. Graded and action potentials

IV. Nerve impulse propagation, synapses and neural modulation
   - A. Think communication!
   - B. Nerve impulse propagation and conduction
   - C. Synapses
   - D. Neural modulation



IV. Membrane Potentials

A. Membrane potentials are electrical differences across a membrane

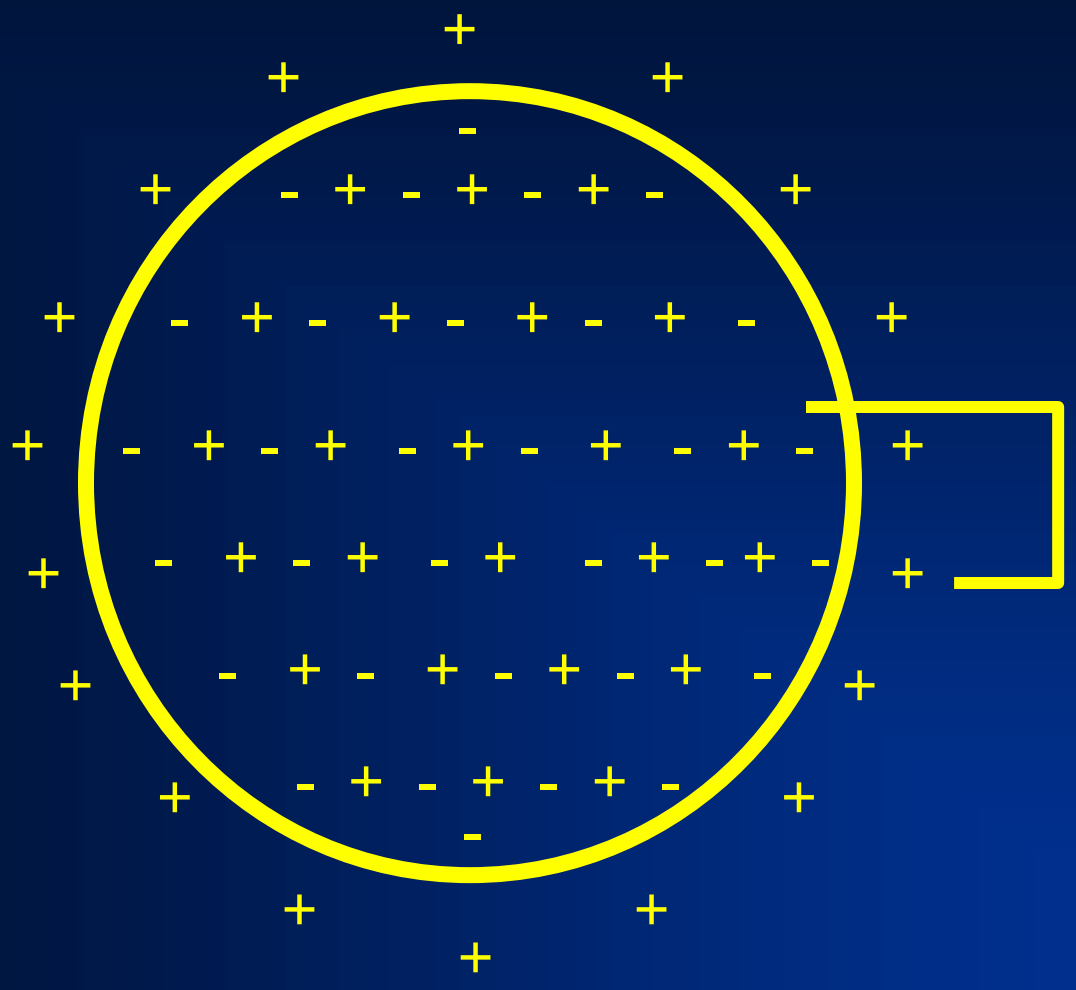
*(***think of them as electrical gradients***)* ***GRADIENTS DRIVE PHYSIOLOGICAL PROCESSES***

B. There are several types of membrane potentials, including (but not limited to):

1. **Resting potential**
2. **Equilibrium potential**
3. **Graded potential**
4. **Action potential**
5. **Excitatory post-synaptic potentials (EPSPs)**
6. **Inhibitory post-synaptic potentials (IPSPs)**

C. Resting cells have membrane potentials (electrical differences across the membrane; polar) with interior of a resting cell being electronegative relative to surrounding extracellular fluid (extracellular fluid, by convention, is considered to be electrically neutral, i.e., 0 mV)

1. **The degree of electronegativity is dictated by cell type and cell size**
2. **Large motor nerves and skeletal muscle fibers, cardiac muscle: -90 mV**
3. **Small nerves, CNS nerves and smooth muscle fibers: -40 mV to –60 mV**
4. **Red blood cells: -10mV to –20 mV**


-70 mV
ECF = 0 mV

D. Some cells can communicate with one another by rapidly changing their membrane potentials

1. Nerves, muscles and some glandular cells have excitable membranes that can change membrane potential rapidly and propagate the change rapidly along the surface of the membrane (e.g., nerve impulse propagation, skeletal muscle depolarization)

E. What determines resting membrane potential?

1. Na+-K+ ATPase exchange pump (- 4 mV)

2. "Trapped" intracellular anions

a) Proteins

b) Phosphates

3. Ionic concentration gradients and selective permeability

a) Examples of K+ and Na+

(1) What if an electrically neutral cell is permeable only to K+?

(2) What if an electrically neutral cell is permeable only to Na+?

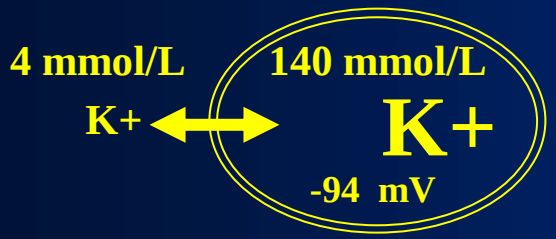
EQUILIBRIUM POTENTIAL FOR K+



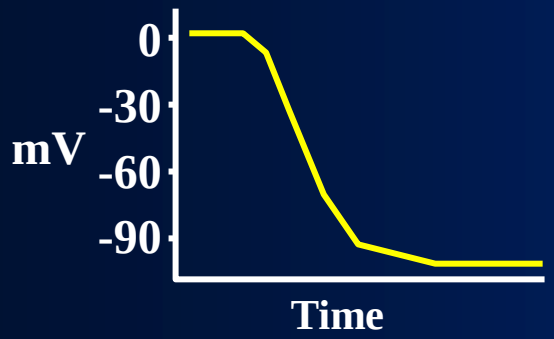
What would happen if an electrically neutral cell suddenly became permeable to K+ only?



K+ would efflux out of the cell, going down its chemical concentration gradient; as more and more positive charges exit the cell, the cell interior becomes increasingly electronegative



The developed negative electrical charge becomes so great it "holds" onto the positively charged K+, inhibiting any further net efflux of K+; this electro-chemical equilibrium point is called the *equilibrium potential* (Nerst potential); the equilibrium potential for K+ (at physiological [ICF] and [ECF]) is -94 mV

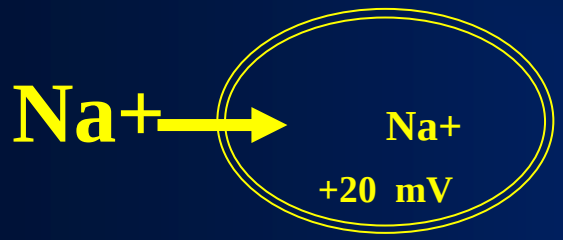


Therefore, during periods of maximum membrane permeability for K+, the membrane potential moves toward K+'s equilibrium potential (i.e., -94 mV)

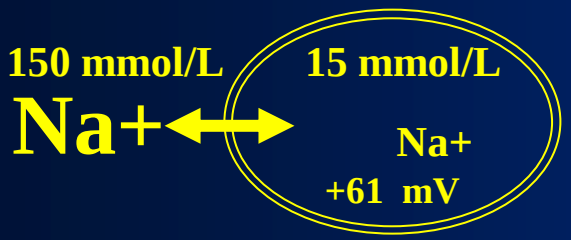
EQUILIBRIUM POTENTIAL FOR Na+



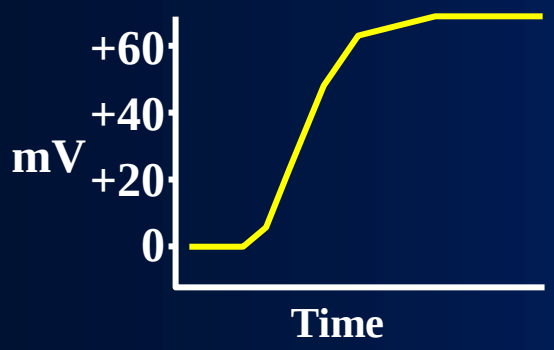
What would happen if an electrically neutral cell suddenly became permeable to Na+ only?



Na+ would influx into the cell, going down its chemical concentration gradient; as more and more positive charges enter the cell, the cell interior becomes increasingly electropositive



The developed positive electrical charge becomes so great it "repels" additional positively charged Na+, inhibiting any further net influx of Na+; this electro-chemical equilibrium point is called the *equilibrium potential* (Nerst potential); the equilibrium potential for Na+ (at physiological [ICF] and [ECF]) is +61 mV



Therefore, during periods of maximum membrane permeability for Na+, the membrane potential moves toward Na+'s equilibrium potential (i.e., +61 mV)

(3) Nernst equation – estimates the equilibrium potential (mV) for an ion at a given concentration gradient; the electromotive force required to balance the force of the chemical concentration diffusion gradient; i.e., the developed electrical potential is equal to and opposite of the chemical concentration diffusion potential

$$\text{Equilibrium potential}_{\text{Ion } X} = \frac{\pm 61}{z} \log \frac{[X]_{\text{outside}}}{[X]_{\text{inside}}}$$

+/- sign : use (+61) for positively charged cations and (-61) for negatively charged anions
z : valence number; 1 for monovalent ions (Na+, Cl-), 2 for divalent ions (Ca++)
[X]outside : concentration of ion X in the ECF
[X]inside : concentration of ion X in the ICF

b) Nernst equilibrium potentials for common ions:

		Concentration (mmol/L)			Nernst
Ion	***+/- sign***	***[ECF]***	***[ICF]***	***z valence#***	**Potential *(mV)***
Na+	***+ 61***	***150***	***15***	***1***	***+ 61***
K+	***+ 61***	***4***	***140***	***1***	***- 94***
Cl-	***- 61***	***120 (110)***	***4 (10)***	***1***	***- 90 (-63)***
Ca^{++}	***+ 61***	***2***	***0.0001***	***2***	***+131***

Remember:

log [10]/[1] = log 10 = +1

log [1]/[1] = log 1 = 0

log [1]/[10] = log 0.1 = -1

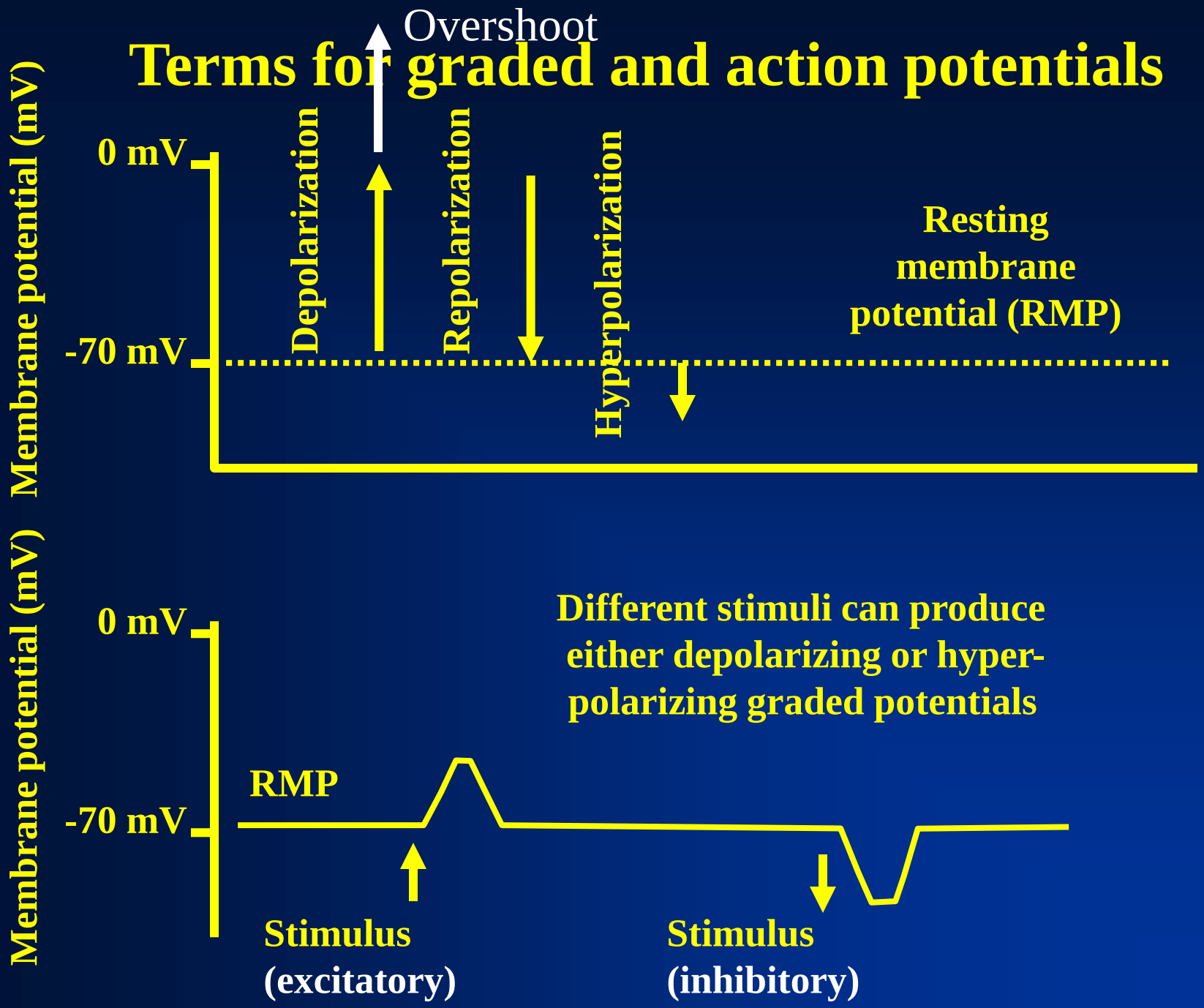
What happens if Na+ channels suddenly open?

What happens if K+ channels suddenly open?

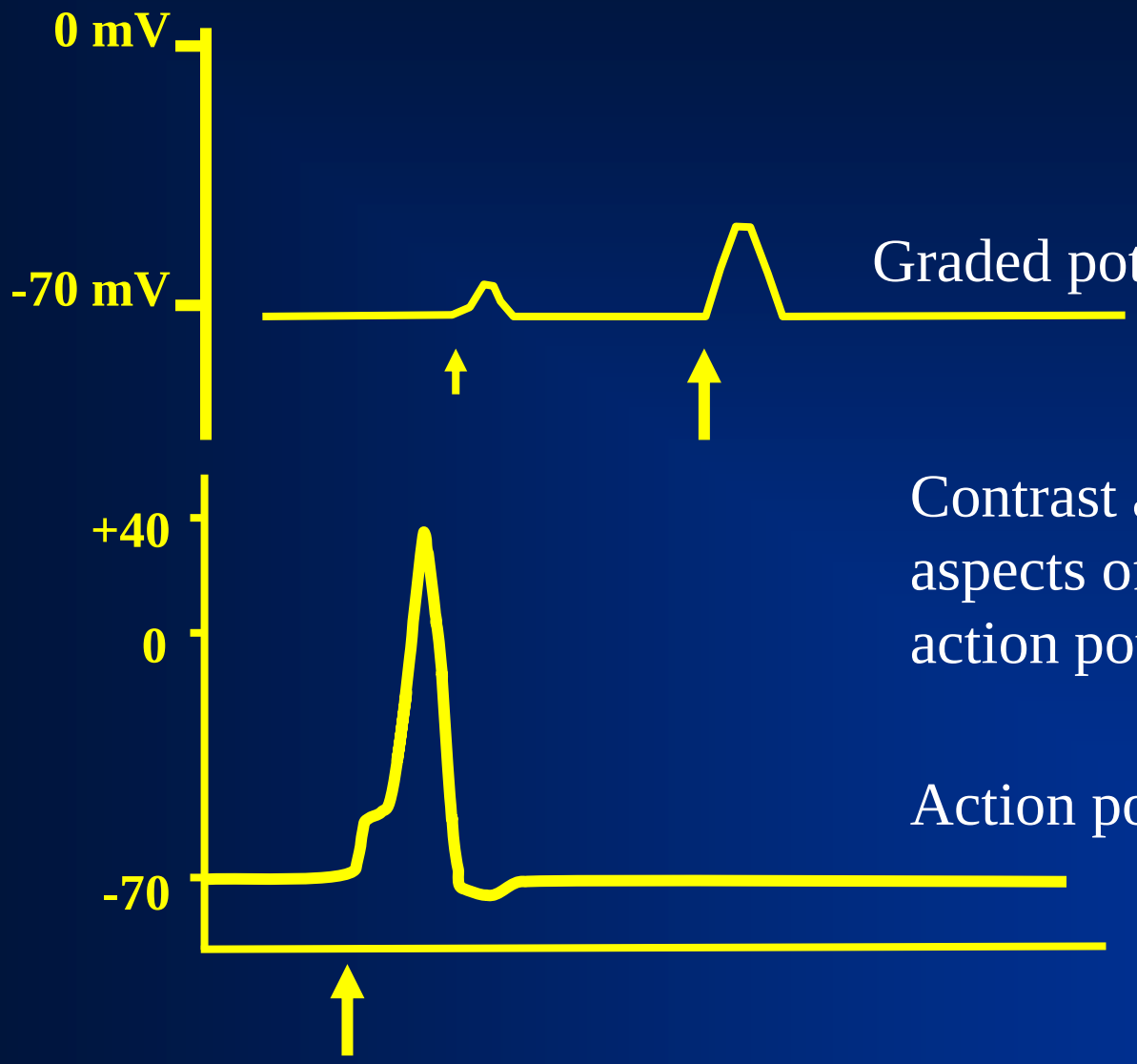
What happens if K+ channels suddenly open but stay open only briefly?



Terms for graded and action potentials



Graded and Action Potentials: Where, when, how

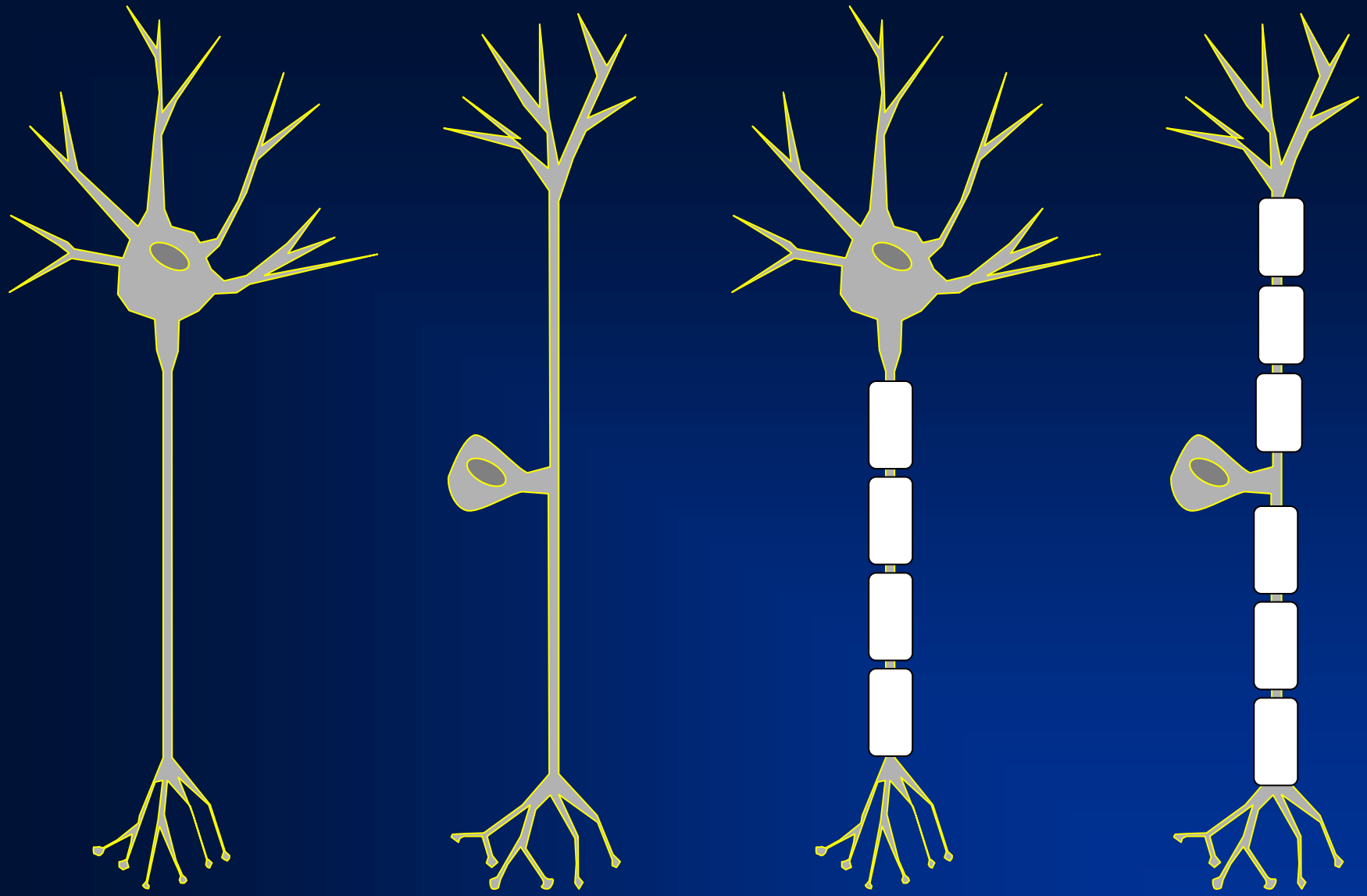


Contrast and compare all aspects of graded and action potentials

Action potential

	Graded potentials	Action potentials
Location on the neuron?	_____	_____
Yes or No: Varies with stimulus strength?	_____	_____
Due to ligand or voltage gated channels?	_____	_____
Yes or No: Do they have a threshold?	_____	_____
Yes or No: Are they all-or-none?	_____	_____
Yes or No: Do they have refractory periods?	_____	_____
Yes or No: Can they under-go summation?	_____	_____
Yes or No: Are they blocked by local anesthetics?	_____	_____

What other contrast and compare similarities and differences can you identify?

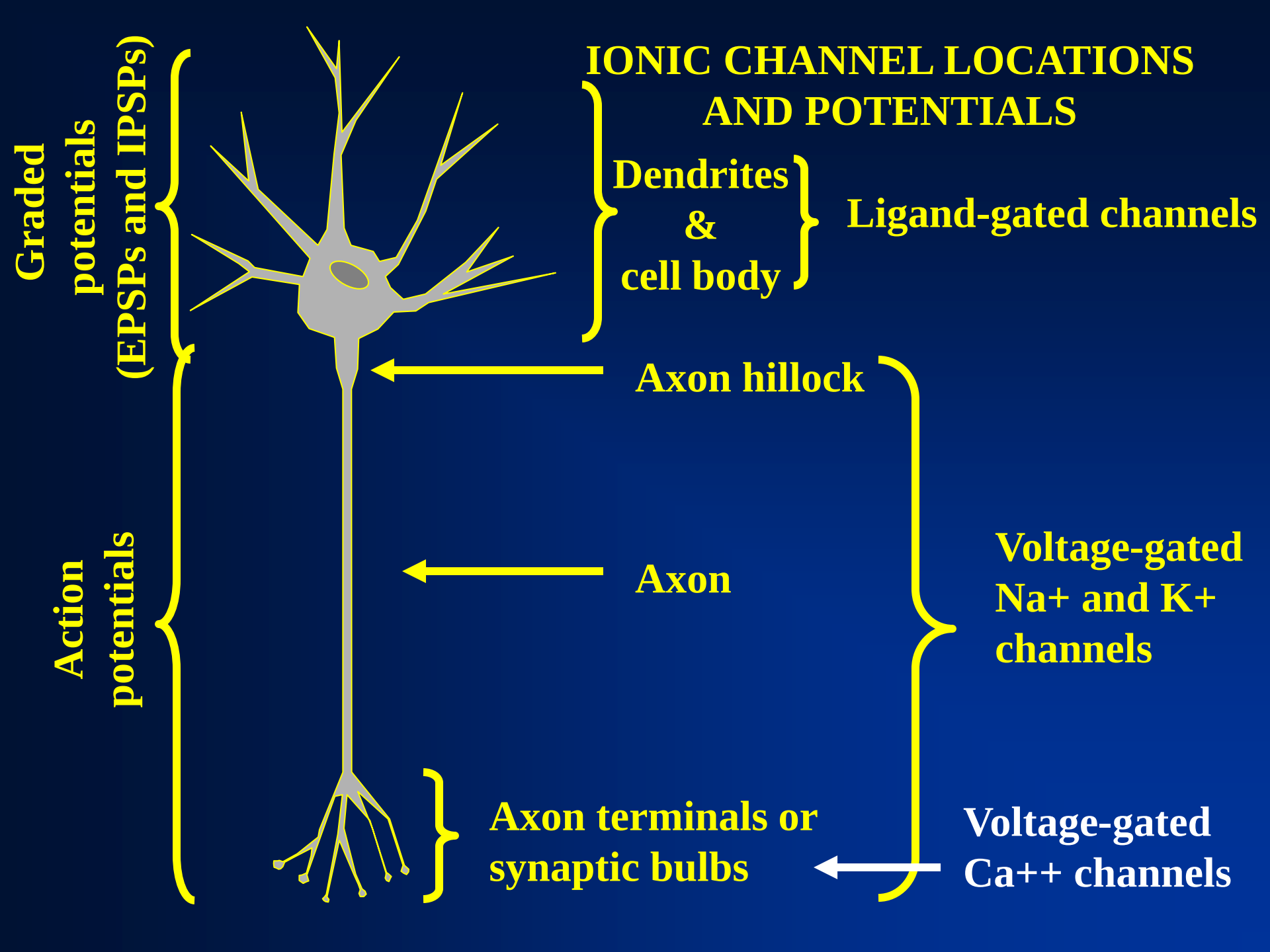


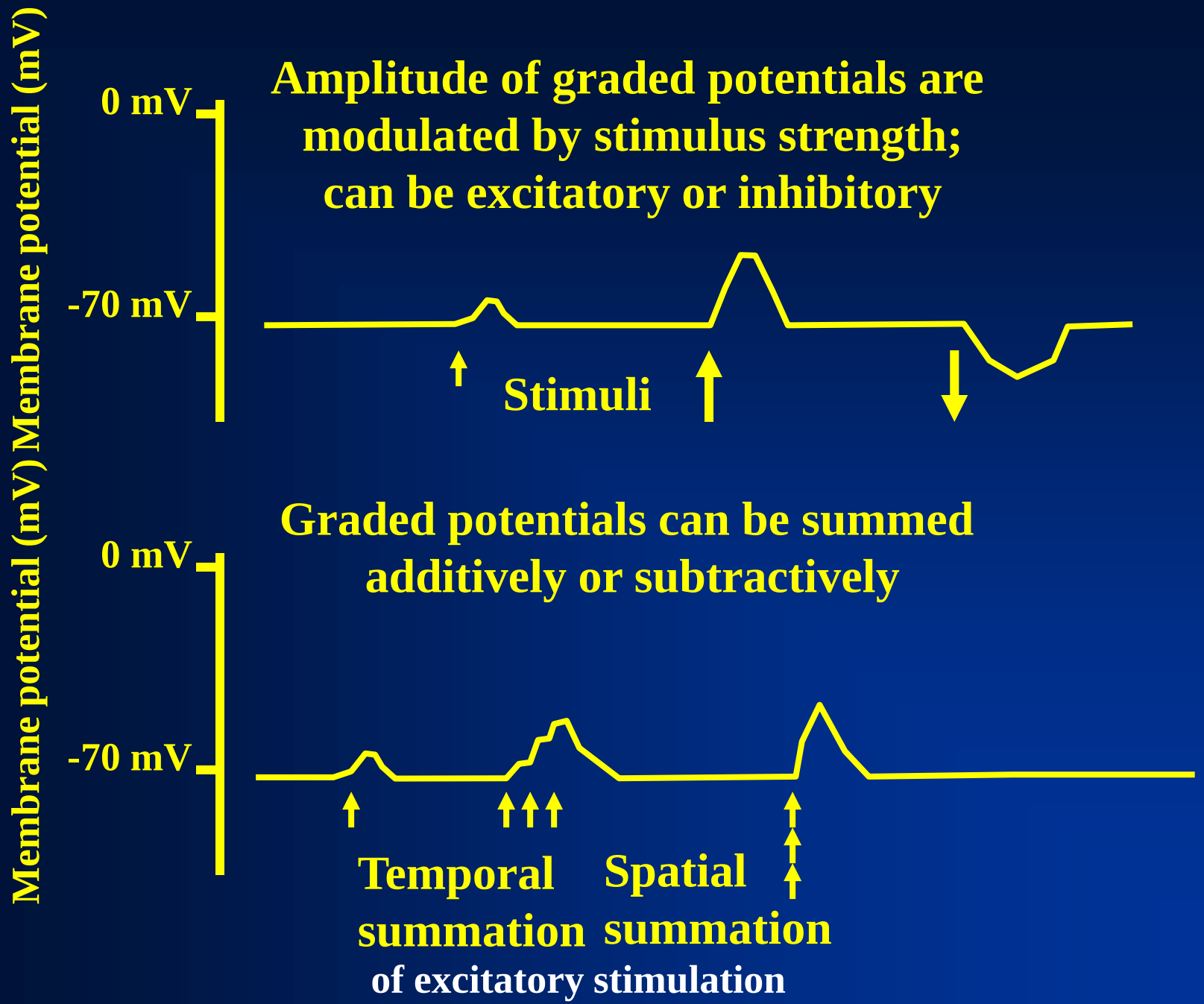
Unmyelinated motor neuron

Unmyelinated sensory neuron

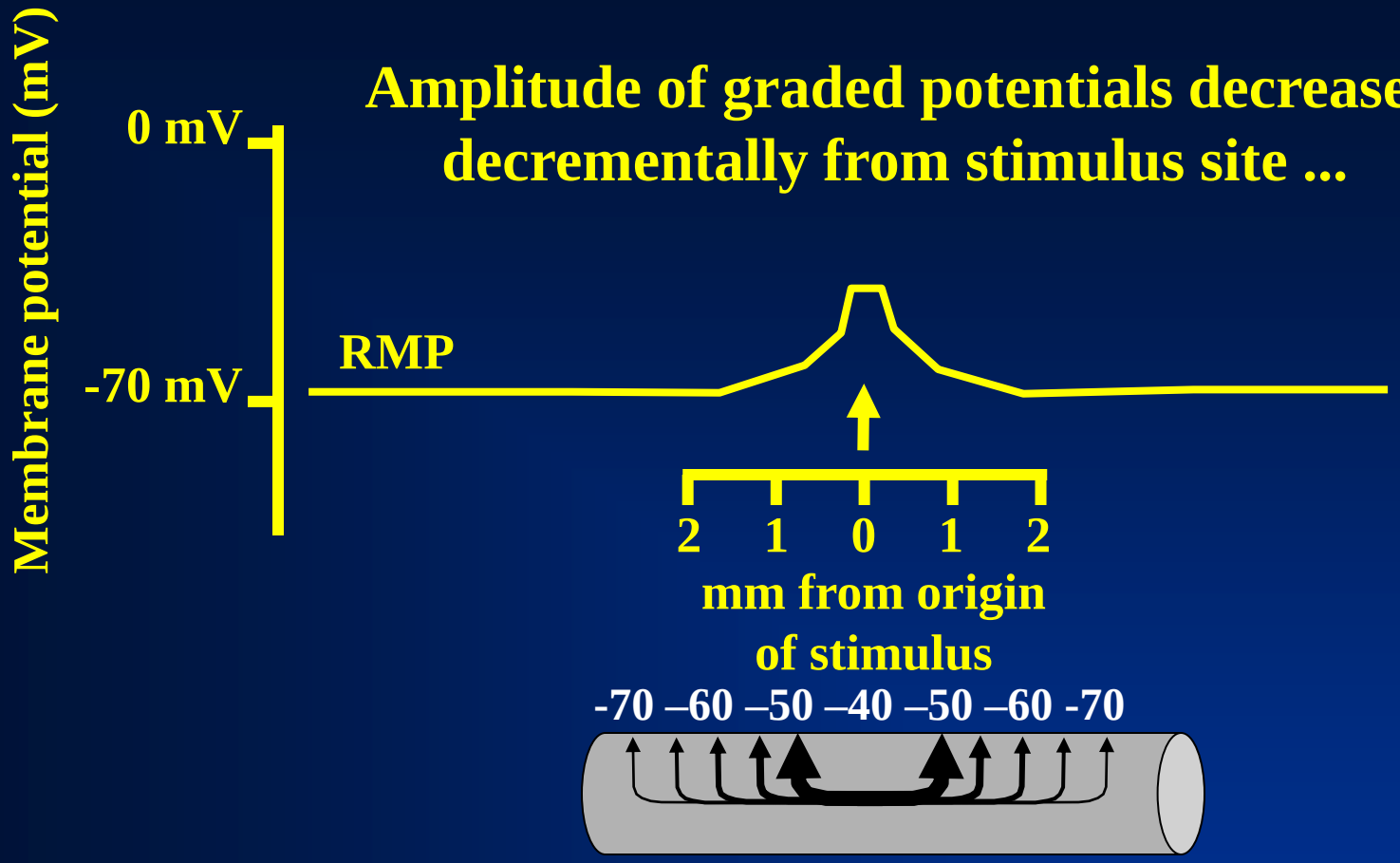
Myelinated motor neuron

Myelinated sensory neuron

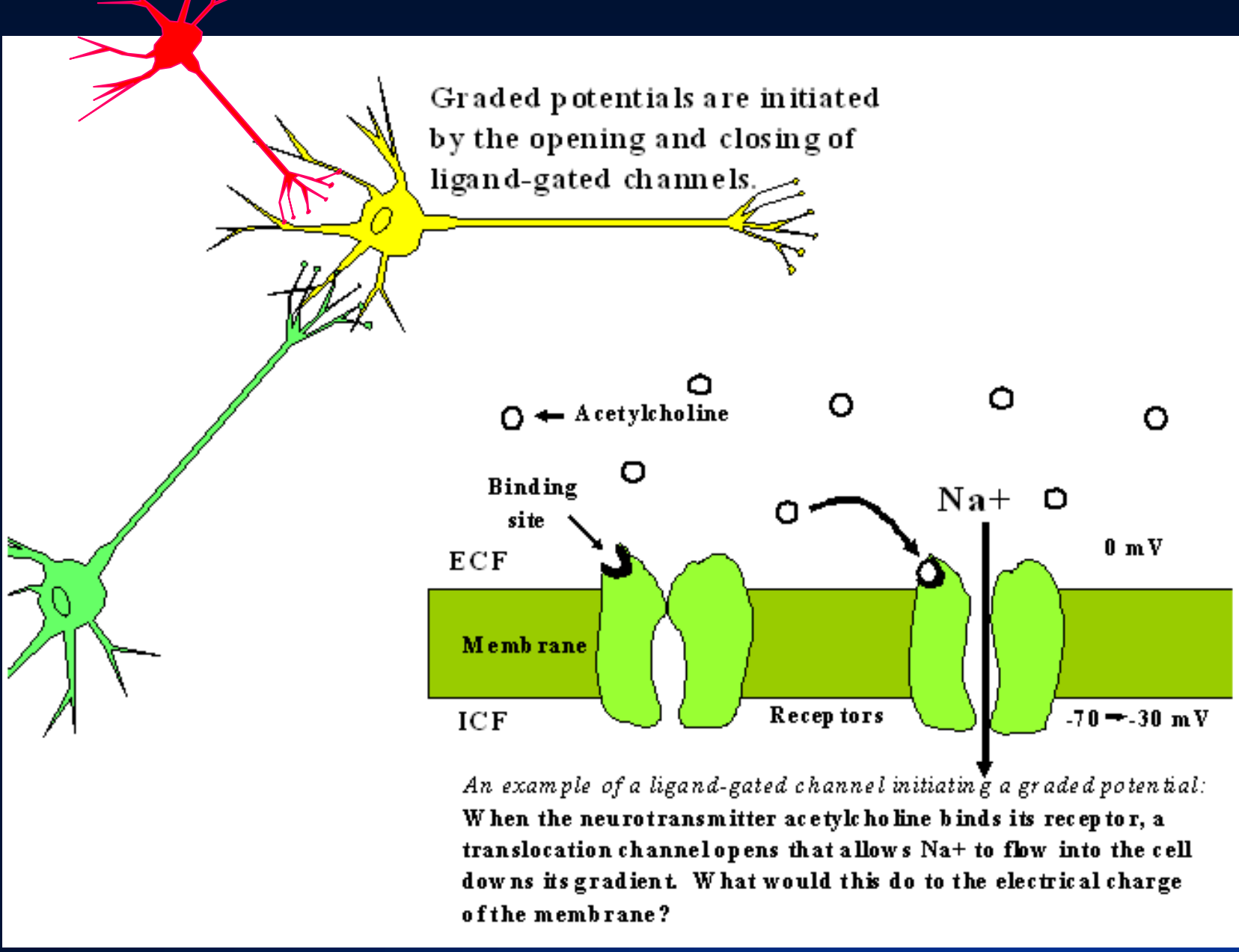

IONIC CHANNEL LOCATIONS AND POTENTIALS
Graded potentials (EPSPs and IPSPs)
Action potentials
Dendrites & cell body
Ligand-gated channels
Axon hillock
Axon
Voltage-gated Na+ and K+ channels
Axon terminals or synaptic bulbs
Voltage-gated Ca++ channels


Membrane potential (mV)
0 mV
-70 mV
Amplitude of graded potentials are modulated by stimulus strength; can be excitatory or inhibitory
Stimuli
Membrane potential (mV)
0 mV
-70 mV
Graded potentials can be summed additively or subtractively
Temporal summation
Spatial summation
of excitatory stimulation

Amplitude of graded potentials decrease decrementally from stimulus site ...



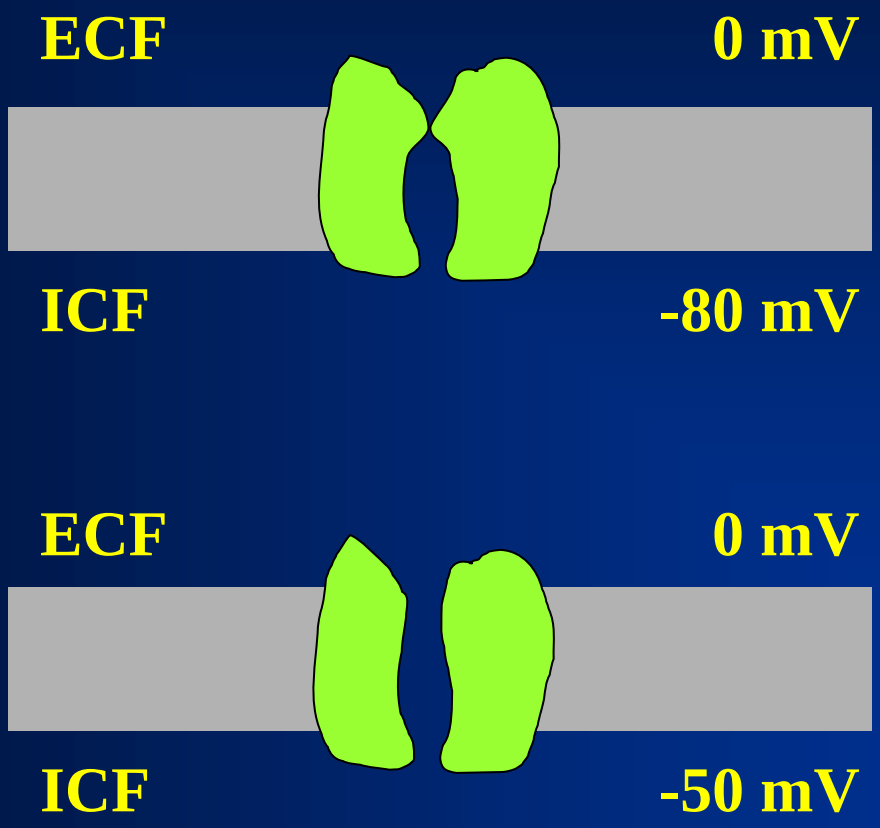
… because charge leaks across the membrane, reducing the spreading current



An example of a ligand-gated channel initiating a graded potential: **When the neurotransmitter acetylcholine binds its receptor, a translocation channel opens that allows Na+ to flow into the cell downs its gradient. What would this do to the electrical charge of the membrane?**

What would happen if a specific ligand would bind a receptor that opens a ligand-gated K+ channel?

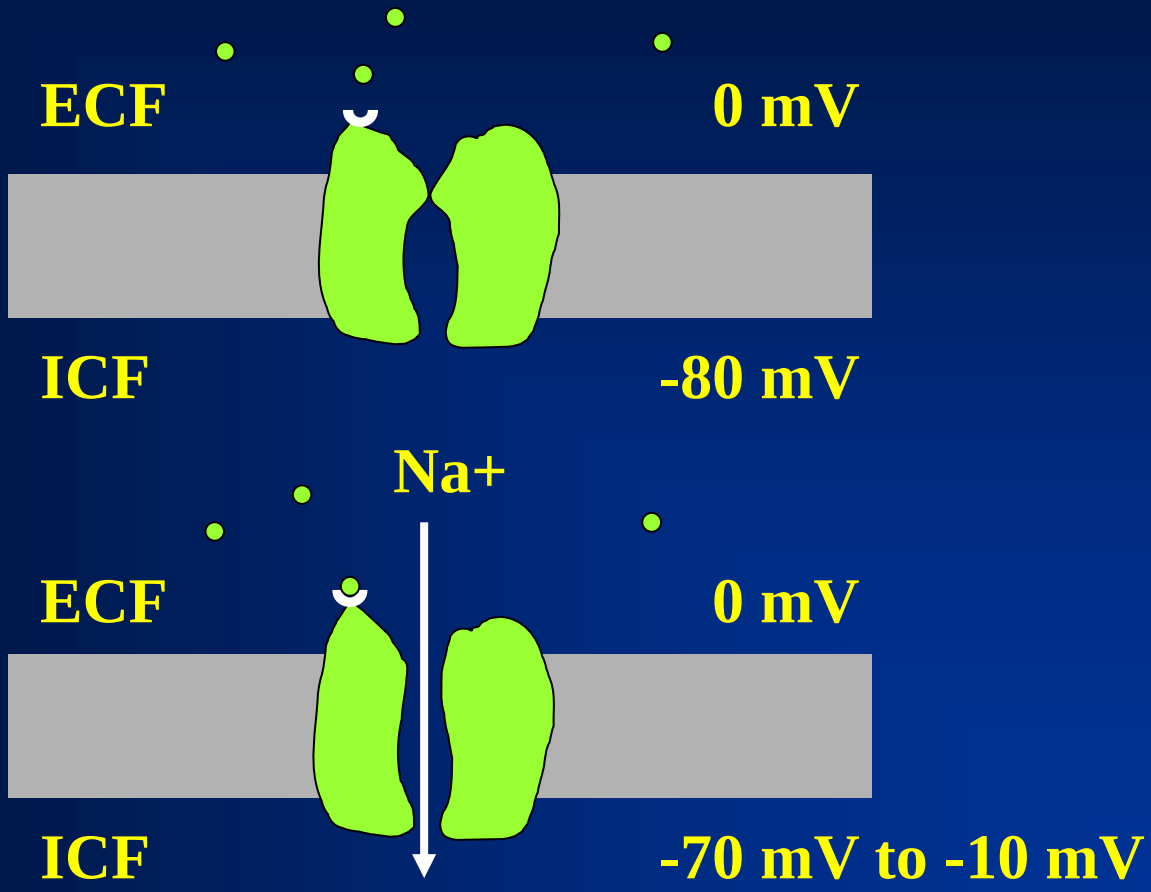
Action potentials can only be generated in excitable membranes having regions of voltage-gated (or sensitive) channels



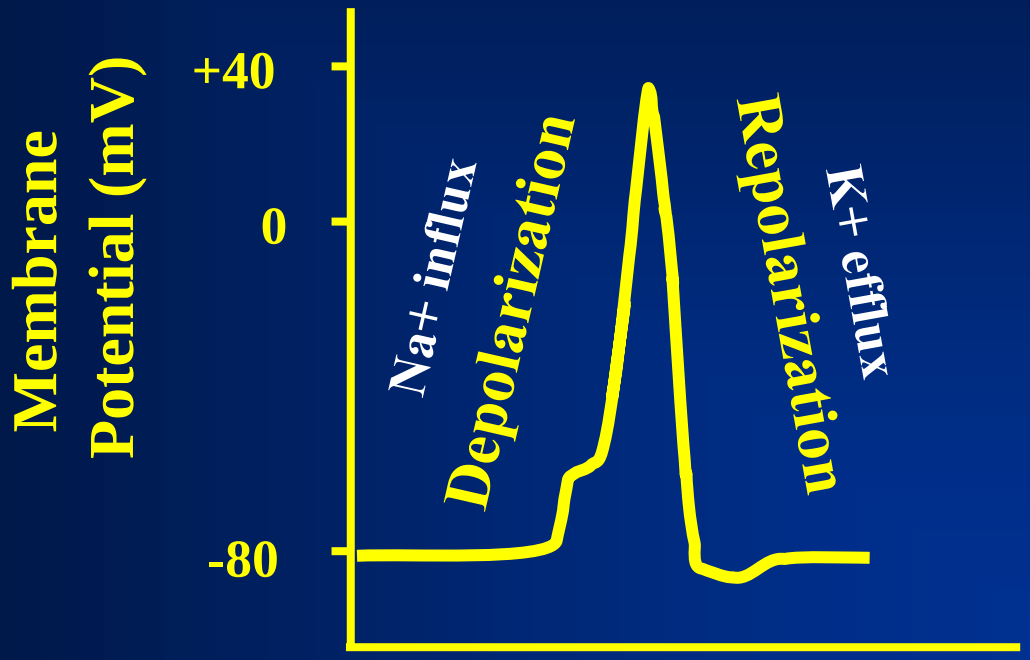
This channel protein "senses" the voltage change and opens

Opens when membrane voltage hits threshold

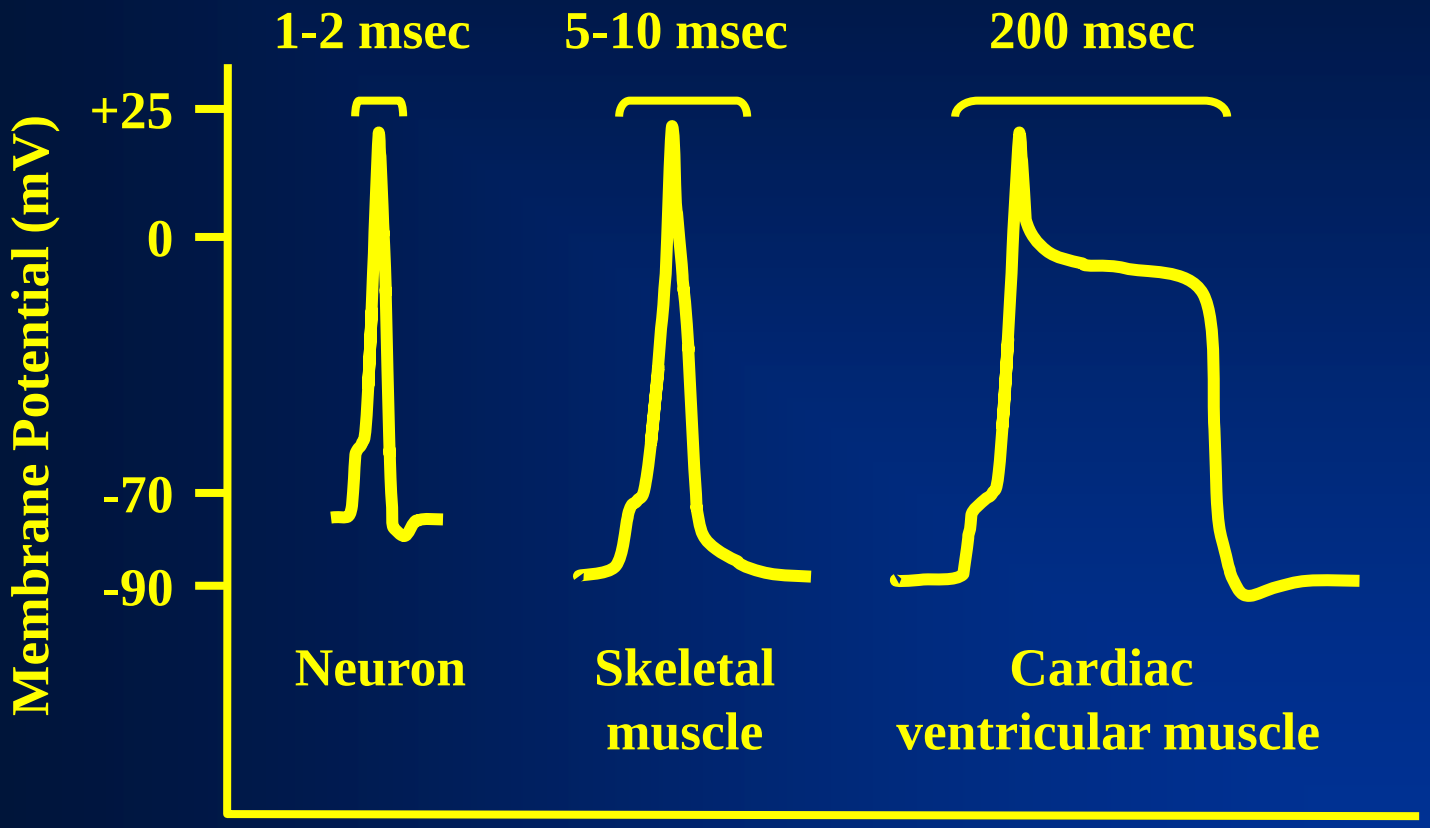
Contrast voltage-gated channels to ligand-gated channels; ligand binds receptor and opens channel pores



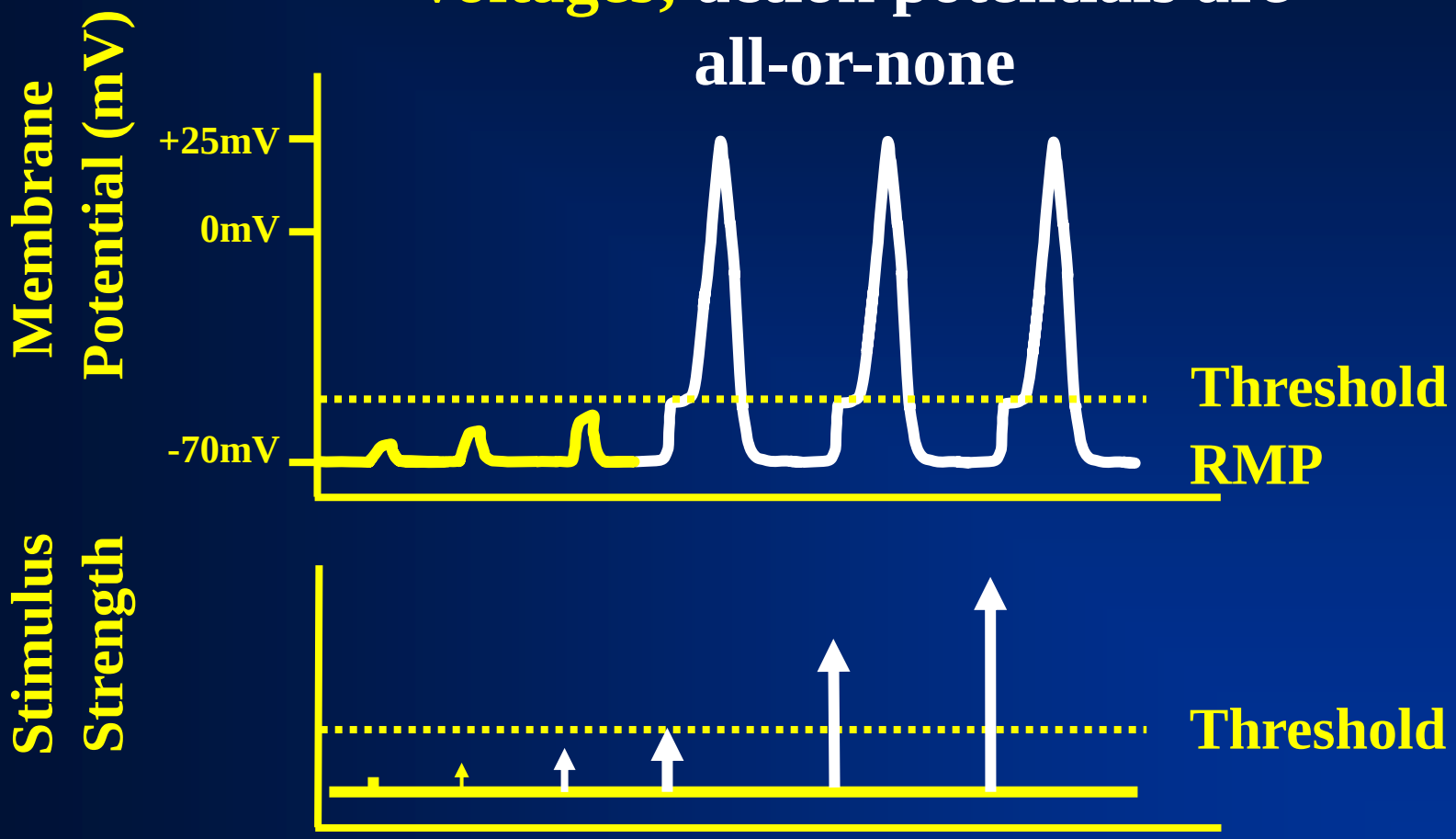
Actions potentials are characterized by a rapid and complete depolarization of the membrane



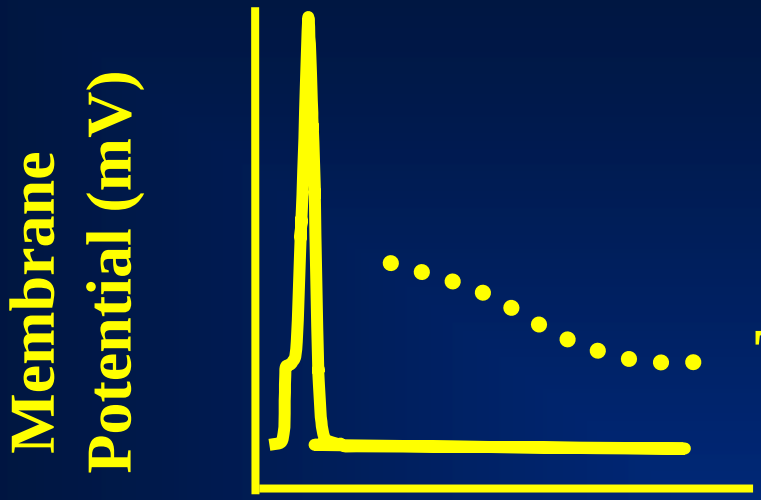
Membrane repolarization rate and action potential duration are dependent on cell type



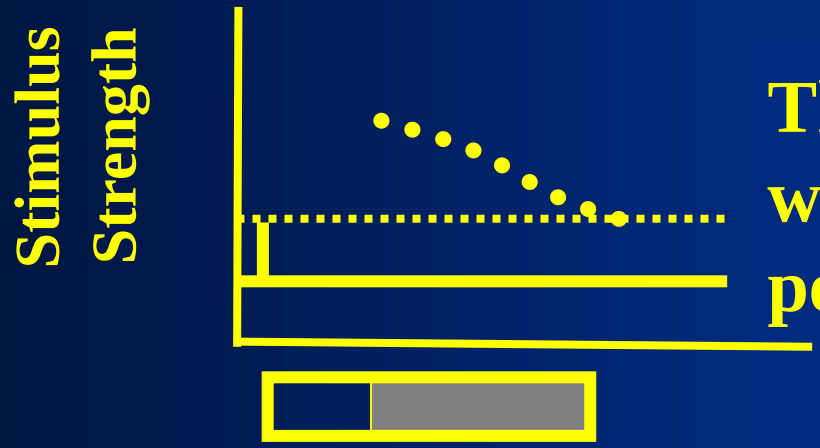
Action potentials have threshold voltages; action potentials are all-or-none



Action potentials have refractory periods and therefore cannot be summed



Threshold



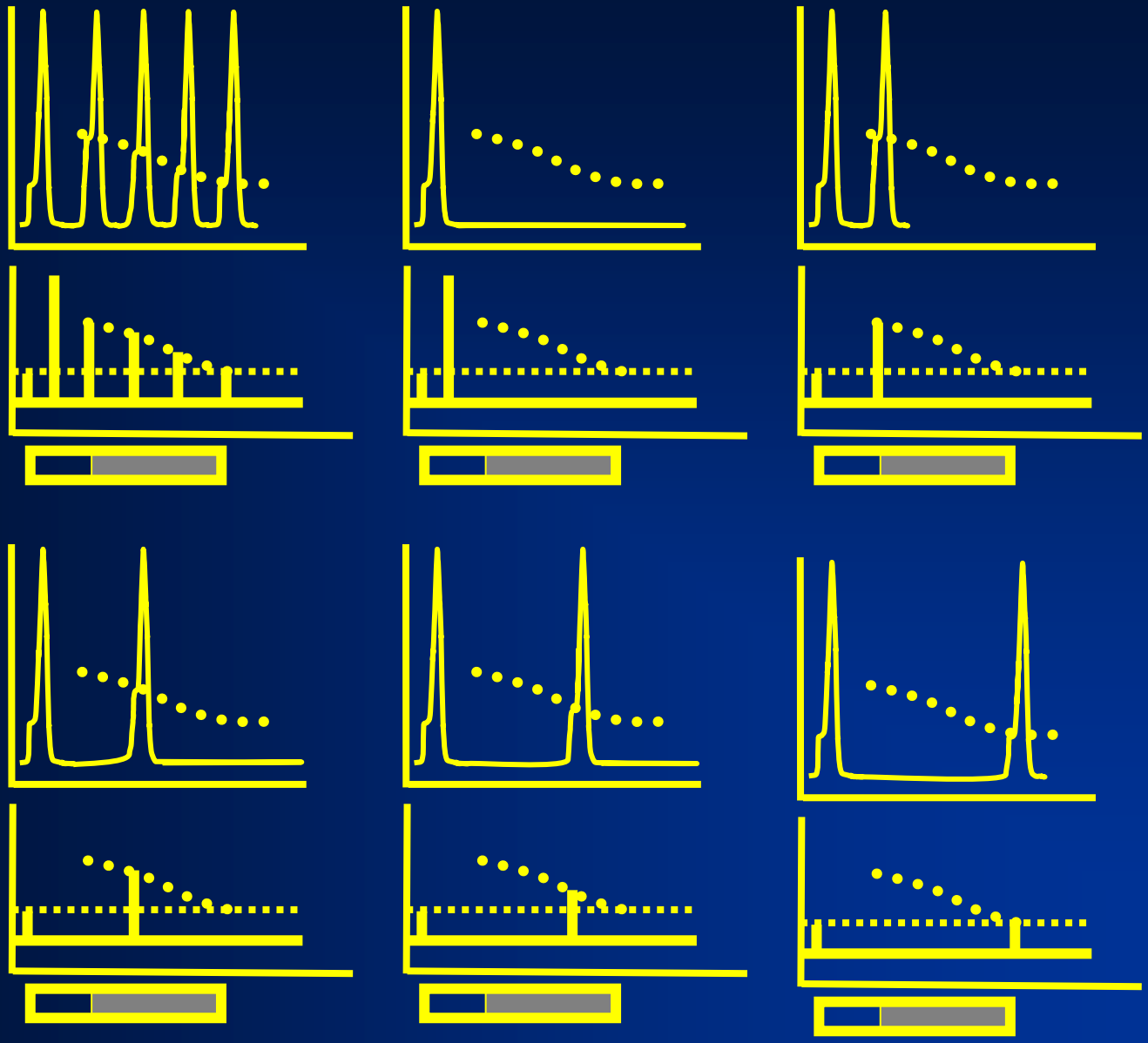
Threshold varies with refractory period

Relative Refractory Period

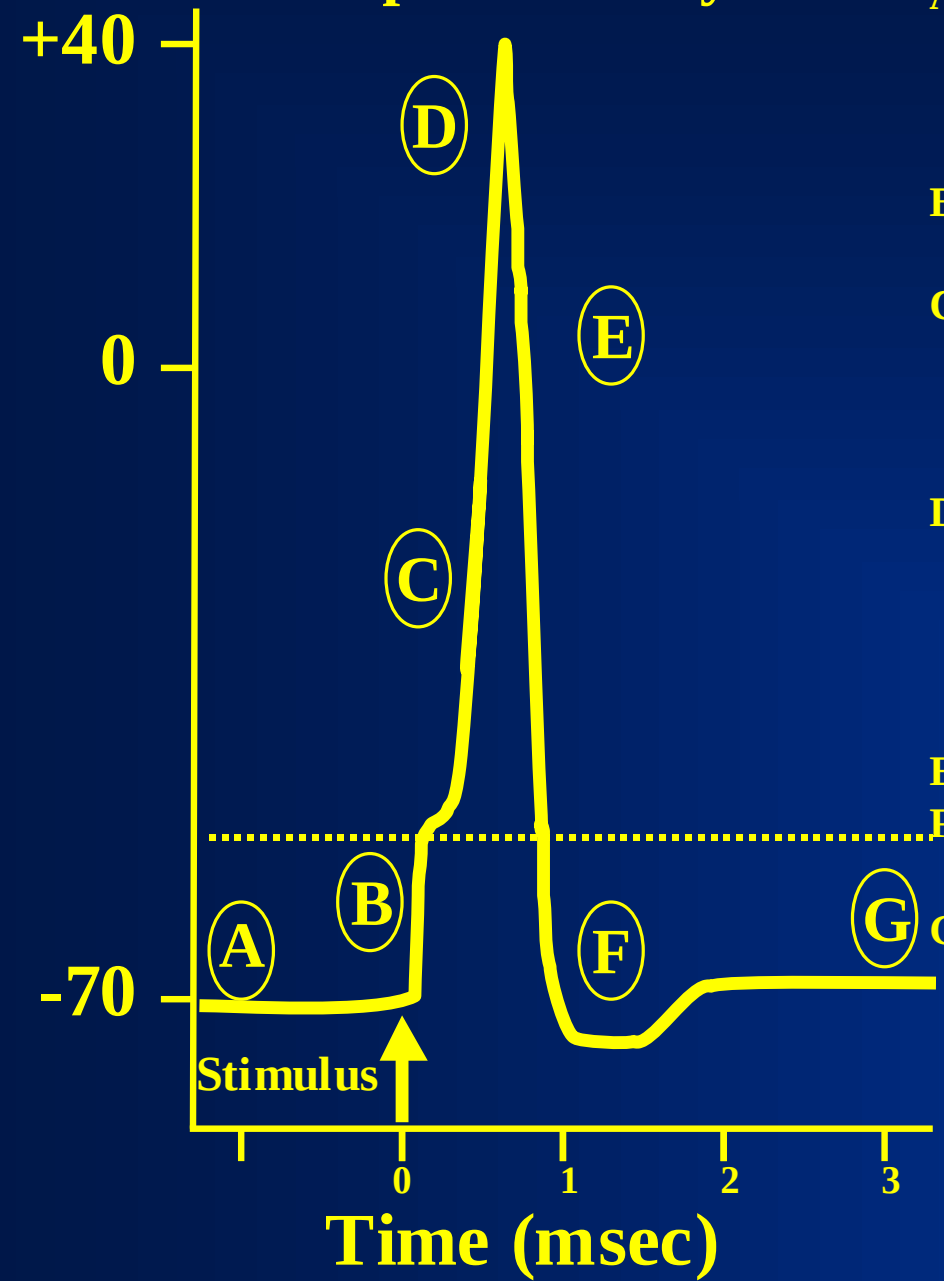
Absolute Refractory Period

(ARP ends and RRP begins with opening of *h gate*)

Threshold varies over the course of the refractory period



Action potentials are generated by rapid alterations in membrane permeability



A - the *resting membrane potential* of this cell is -70 mV (the membrane is *polarized*); voltage-gated channels for both Na+ and K+ are closed

B - a stimulus is applied to the membrane driving the membrane potential to threshold

C - at threshold voltage, *fast voltage-gated Na+ channels open;* Na+ rushes into cell, resulting in *membrane depolarization* and additional Na+ channel opening

D - Na+ channels close, but not before the membrane potential actually becomes positive (*overshoot*); *voltage-gated K+ channels open* and K+ rushes out of the cell

E - the membrane *repolarizes* as K+ exits cell

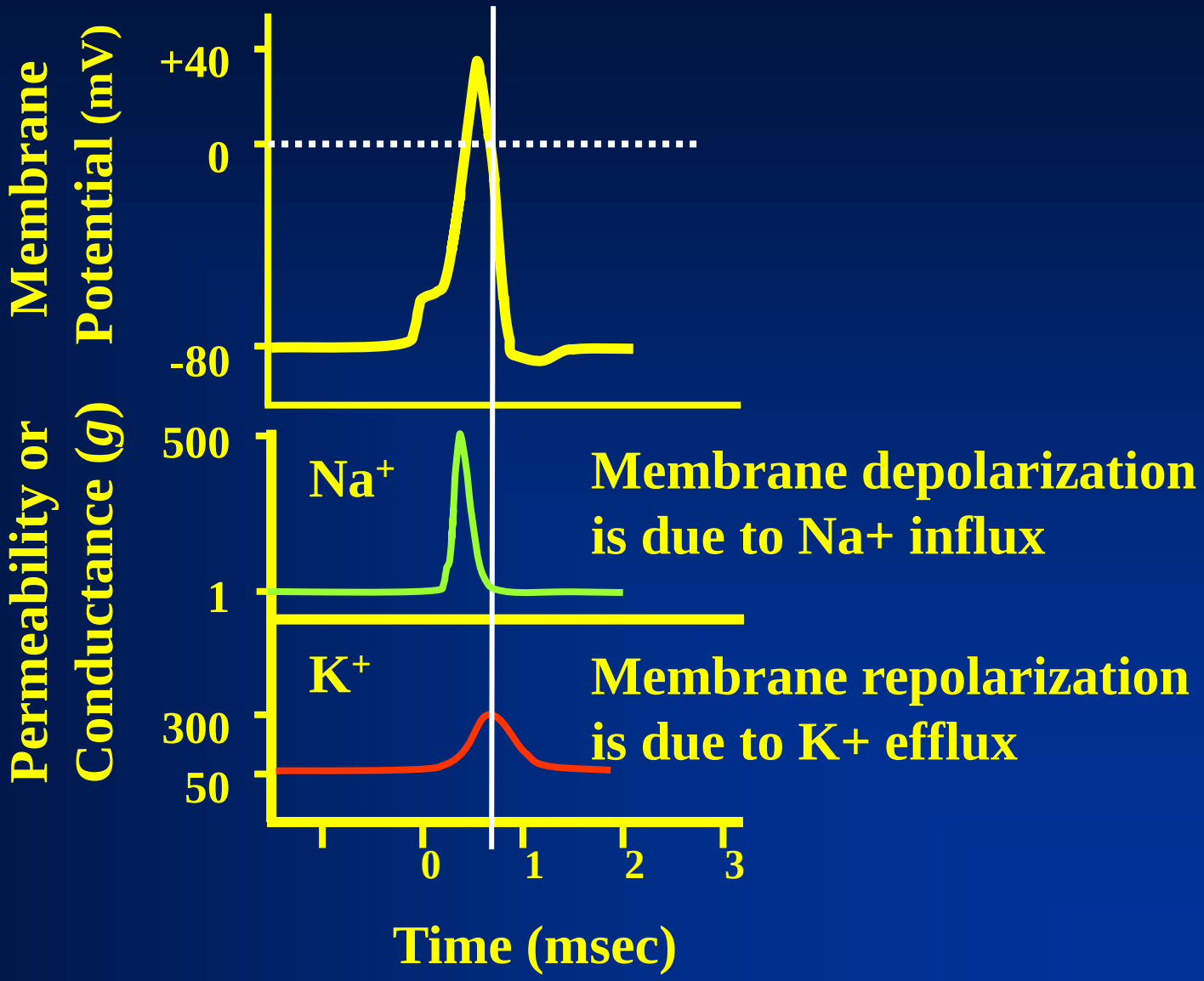
F - a transient *hyperpolarization* often occurs before all K+ channels close completely

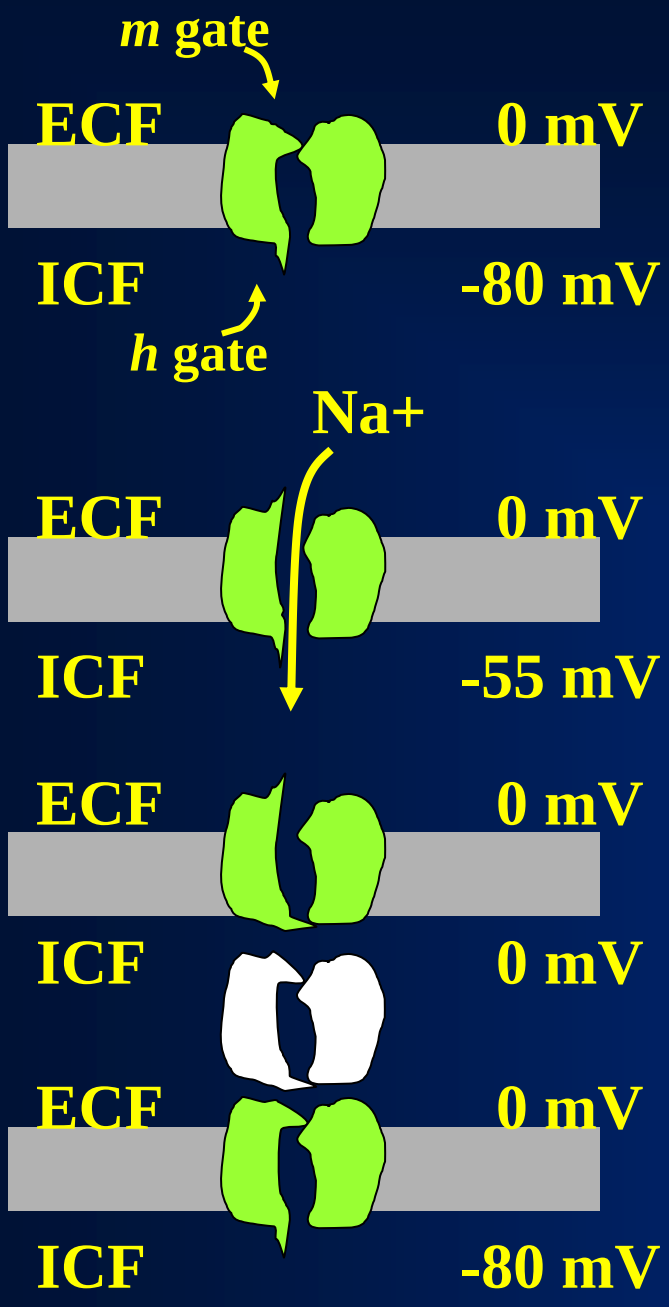
G - with Na+ and K+ channels both closed, the membrane returns to its resting potential with the aid of the Na+-K+ ATPase exchange pump

c) Ionic permeabilities

(1) The resting membrane is 50-100 times more permeable to K+ than Na+

(2) Therefore, the resting membrane potential is closer to the K+ equilibrium potential than the Na+ equilibrium potential





Na+ channel gating

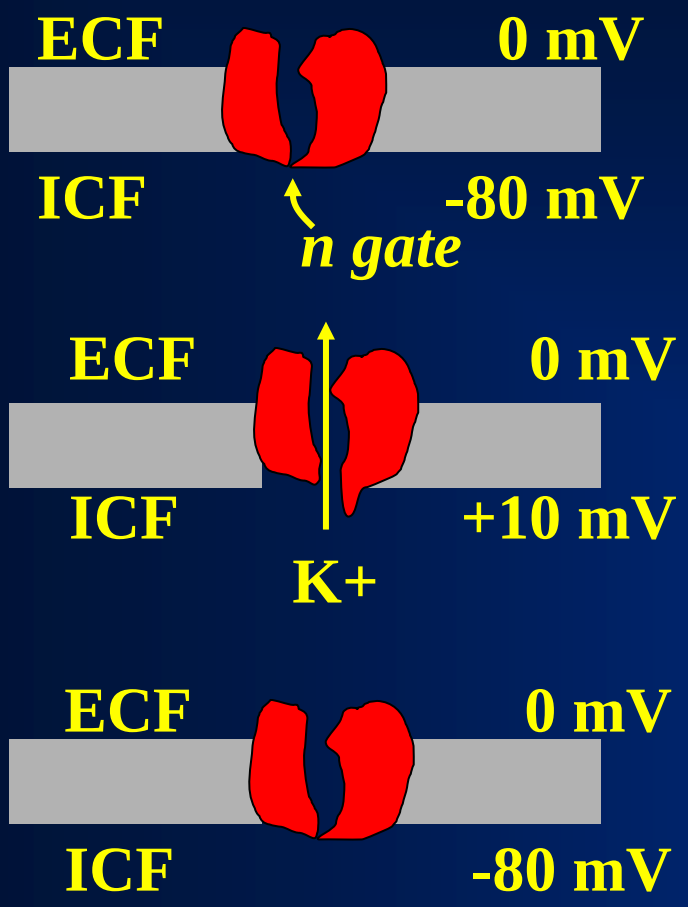
At the resting membrane potential, the *m* or activation gate is closed and the *h* or inactivation gate is open

At threshold, the *m* gate springs open, allowing Na+ to flow into the cell (beginning of ARP)

As the membrane potential approaches complete depolarization, the *h* gate closes, bringing Na+ influx to a halt

With membrane repolarization, the *m* gate closes and then the *h* gate reopens (end of ARP and beginning of RRP)

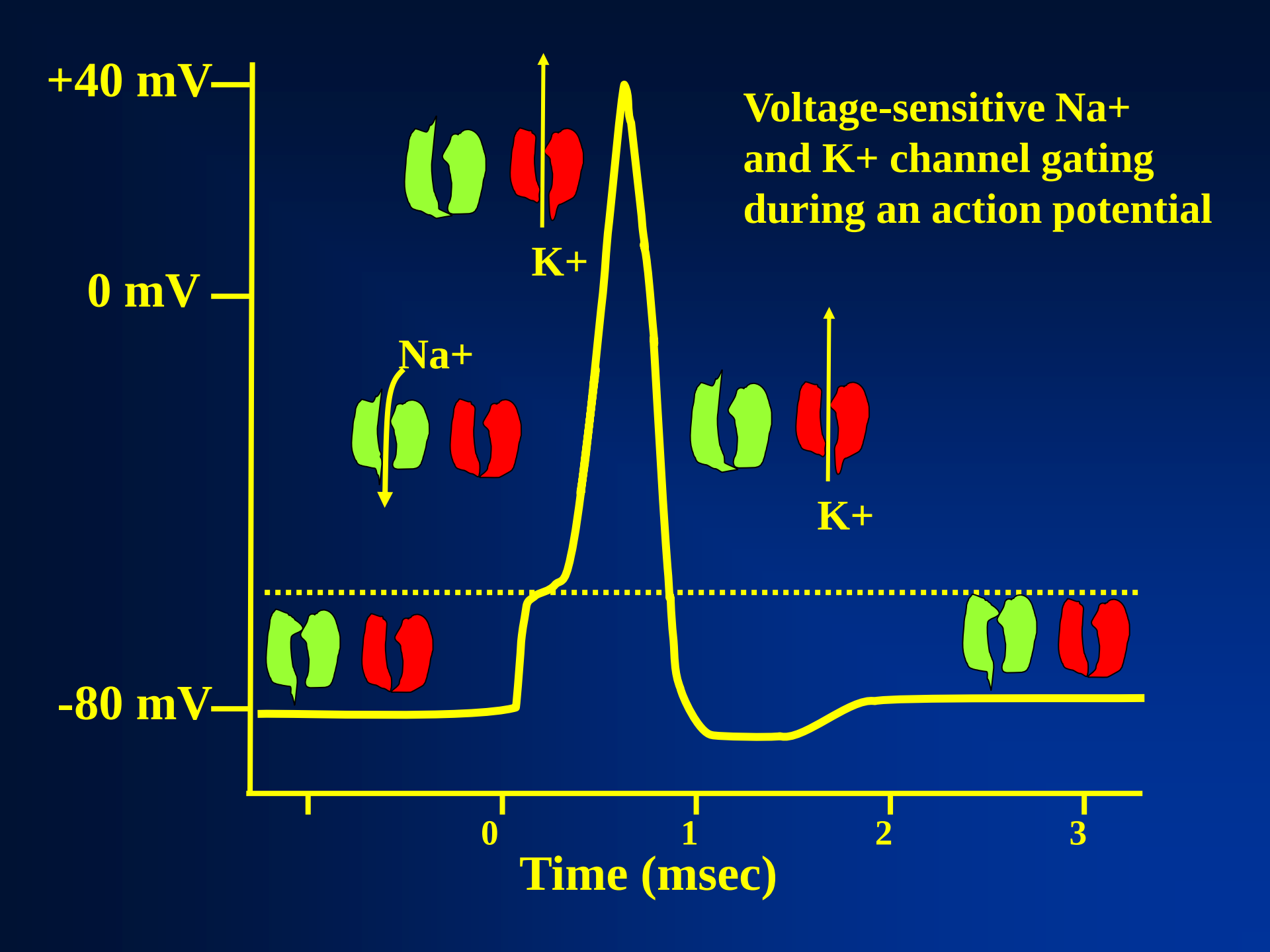
K+ channel gating



At the resting membrane potential the *n* gate of the K+ channel is closed

As the membrane depolarizes to overshoot, the *n* gates open, initiating K+ efflux and membrane repolarization

After the membrane is restore to its resting potential, the *n* gate closes


Voltage-sensitive Na+ and K+ channel gating during an action potential
+40 mV
0 mV
-80 mV
K+
Na+
K+
0
1
2
3
Time (msec)

CLINICAL CORRELATION:
Hyponatremia

Graded potentials and action potentials (and many other physiologic processes) are made possible by chemical concentration gradients. What happens when a simple but vital chemical concentration gradient becomes disordered? For example, what happens with hyponatremia?

Hypo/natr/em/ia – the condition of low blood sodium

Case of Dr. N.C.

- 82 year old male; mild to moderate heart and kidney impairment
- Over the course of two days, became increasingly disoriented, mentally sluggish, speech impaired, less and less responsive
- Admitted to hospital, soon lost consciousness
- Tests proved negative for stroke, subdural hemotoma, etc.
- Simple blood test determined his plasma sodium was 115 mM
- Hyponatremia (2nd to renal impairment of sodium reabsorption) leading to neural dysfunction and moderate brain edema

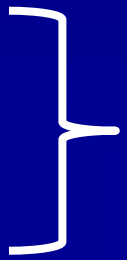
FUNDAMENTAL PHYSIOLOGICAL PRINCIPLES (Part I)

I. Introduction to physiology and homeostasis
   - A. Physiology – Definitions
   - B. Homeostasis
   - C. Fluid compartments
   - D. Homeostatically regulated factors
   - E. How/why are factors regulated
   - F. Homeostatic hmmmmms

II. Membrane dynamics (structure, function, signaling)
   - A. Gradients drive physiological processes
   - B. Membrane organization
   - C. Membrane transport and translocation
   - D. Signal transduction

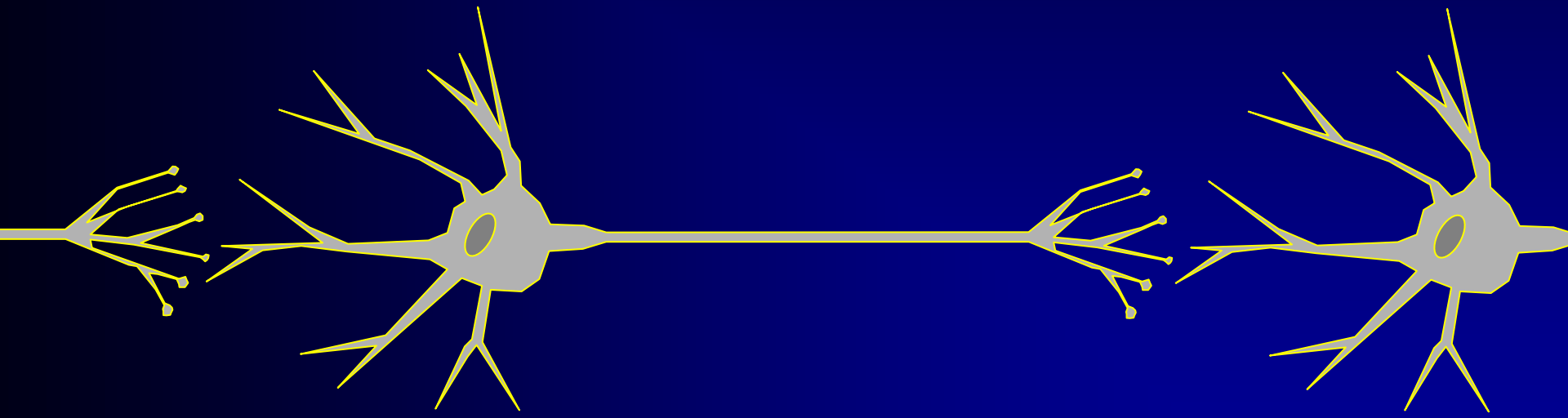
III. Membrane potentials
   - A. Membrane potentials and gradients
   - B. Types of membrane potentials
   - C. Resting membrane potentials
   - D. Communication using membrane potentials
   - E. Determinates of membrane and equilibrium potentials
   - F. Graded and action potentials

IV. Nerve impulse propagation, synapses and neural modulation
   - **A. Think communication!**
   - **B. Nerve impulse propagation and conduction**
   - **C. Synapses**
   - **D. Neural modulation**



What do neurons do and how do they do it?

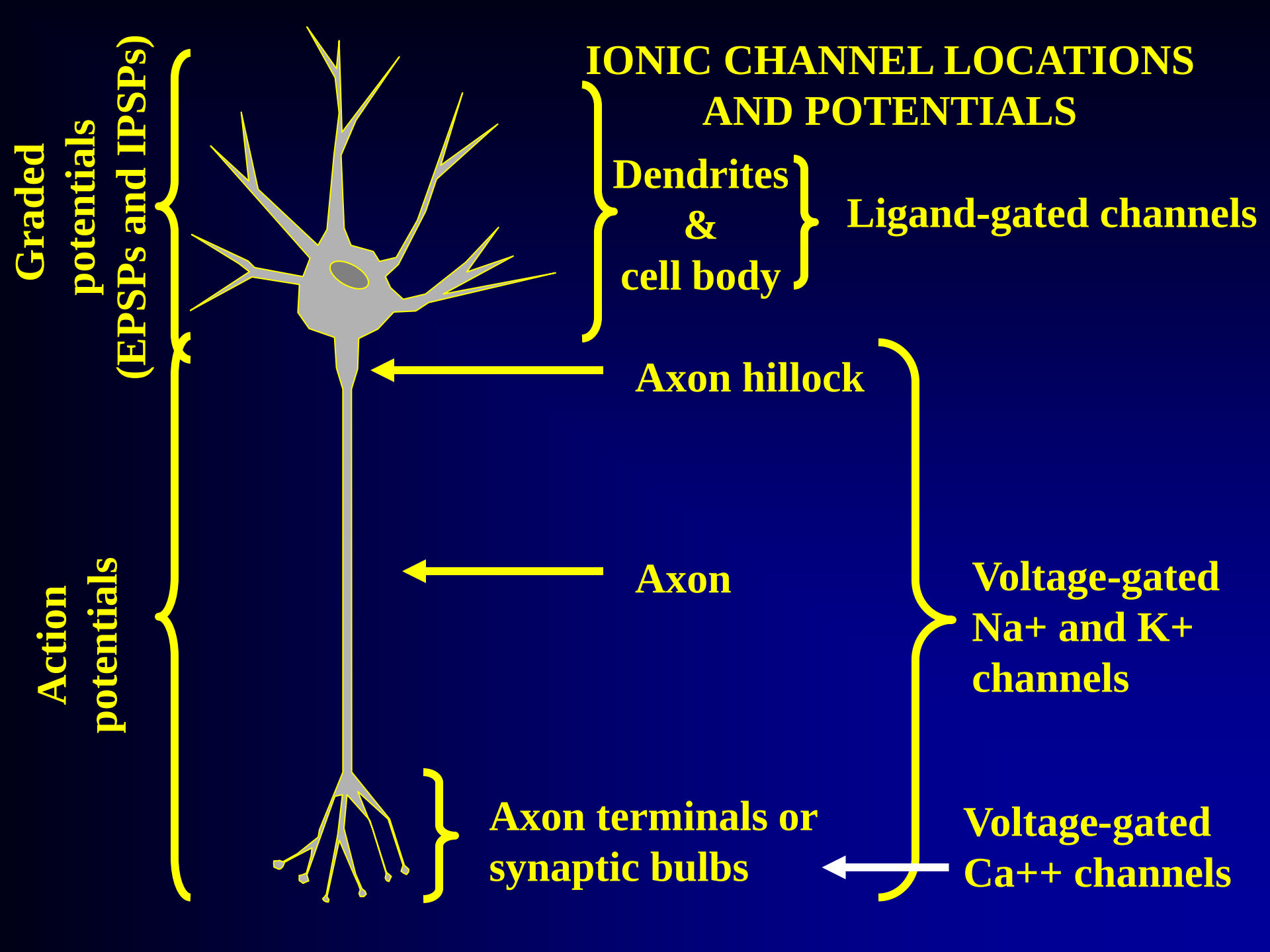
Think communication*: Impulse propagation and synapses*
Membrane potentials*: Graded and action*

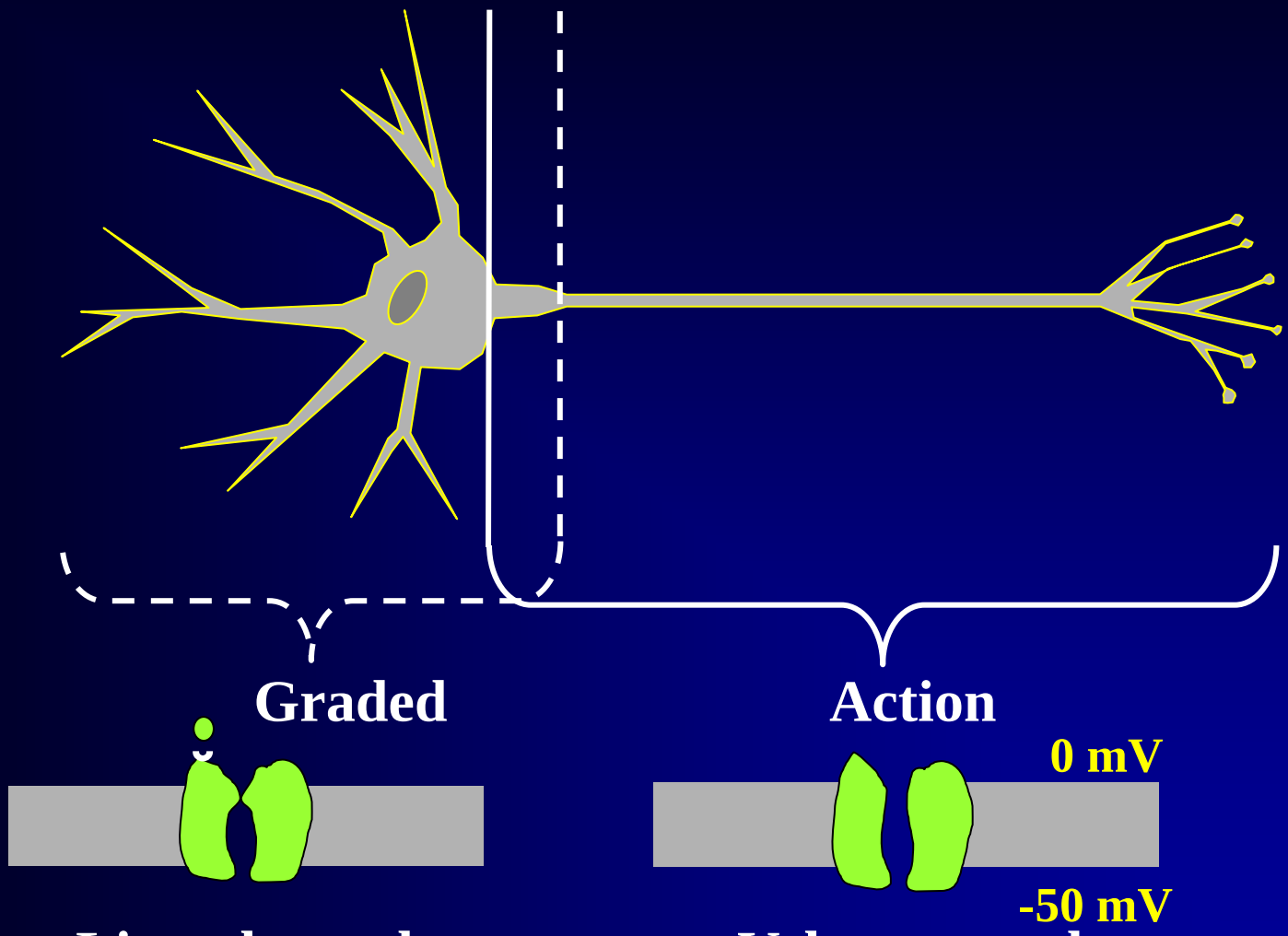


Synapse **Propagation** **Synapse**

PERIPHERAL NERVE FIBER CLASSIFICATION

Fiber Designation		Diameter (μm)	Conduction (m/sec)	Afferent (receptor)	Efferent (effector organ)
Aα	I	12-20	70-120	Muscle spindle Golgi tendon organ	α-motoneuron to skeletal muscle
Aβ	II	6-12	35-70	Mechanoreceptors	
Aγ		2-8	12-50		γ-motoneuron to muscle spindle
Aδ	III	1-6	3-35	Pain receptors (acute-sharp) Thermoreceptors Mechanoreceptors	
B		1-3	3-15		Pregangl. ANS
C	IV	1	0.5-2	Pain receptors (slow) Thermoreceptors Mechanoreceptors	Postgangl. ANS

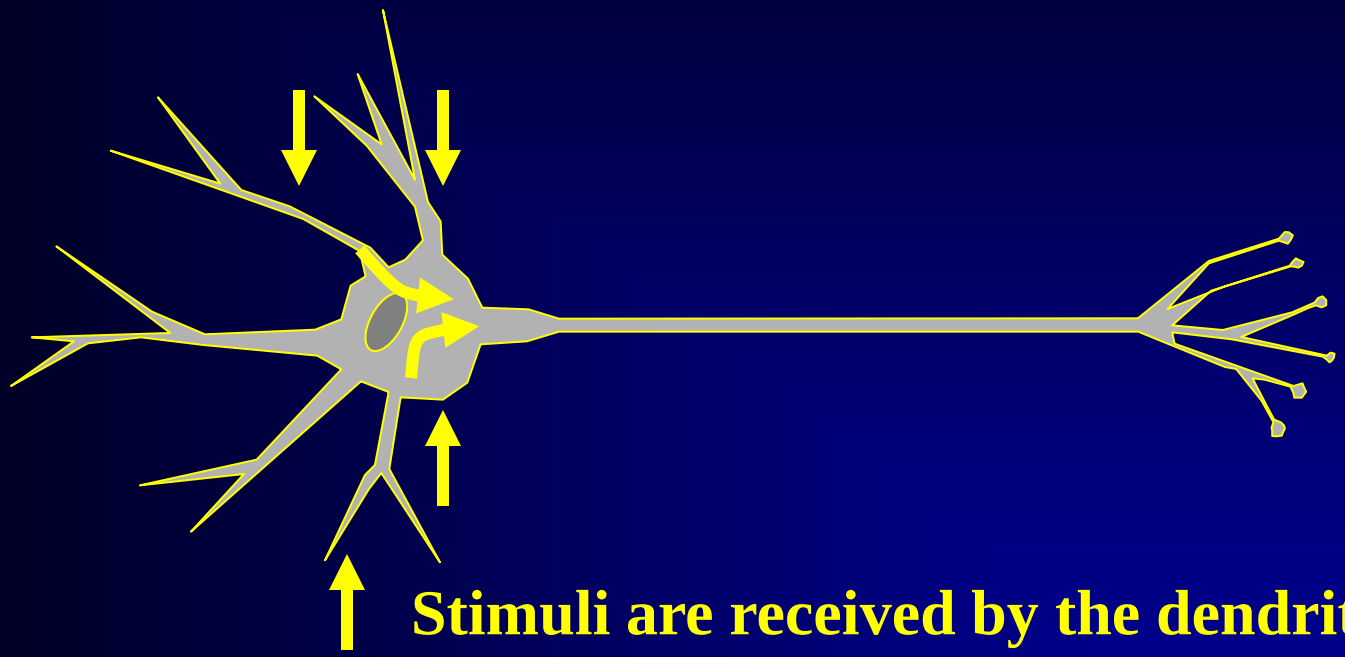

IONIC CHANNEL LOCATIONS AND POTENTIALS
Graded potentials (EPSPs and IPSPs)
Action potentials
Dendrites & cell body
Ligand-gated channels
Axon hillock
Axon
Voltage-gated Na+ and K+ channels
Axon terminals or synaptic bulbs
Voltage-gated Ca++ channels



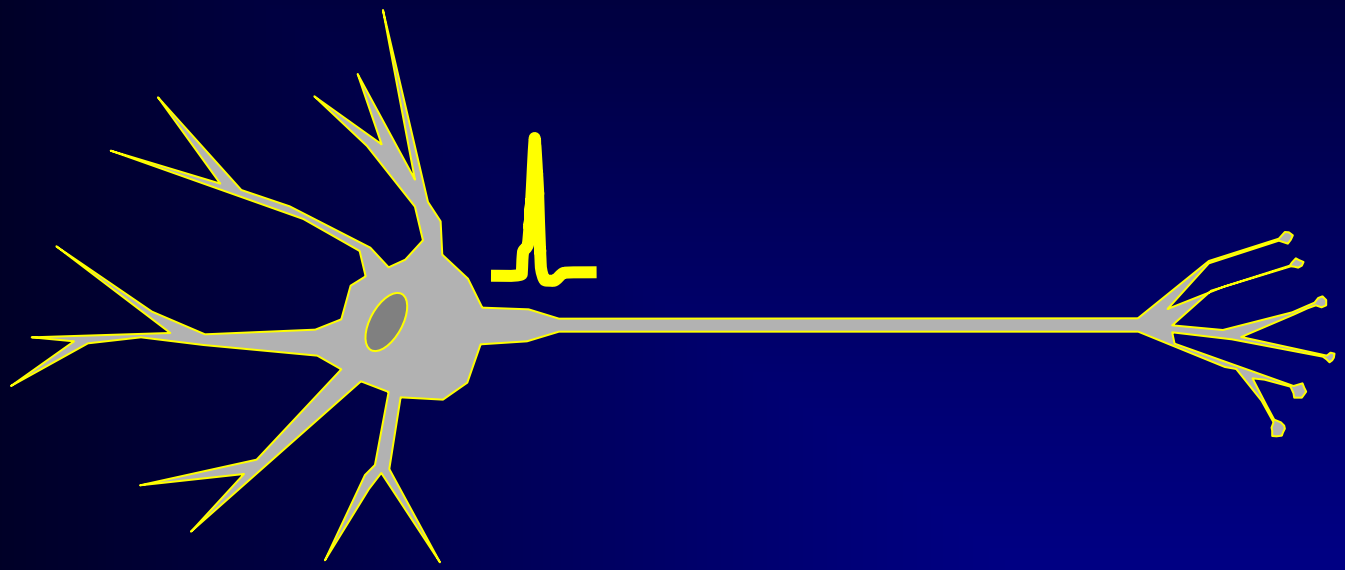
Ligand-gated channels initiate graded potentials

Voltage-gated channels initiate action potentials

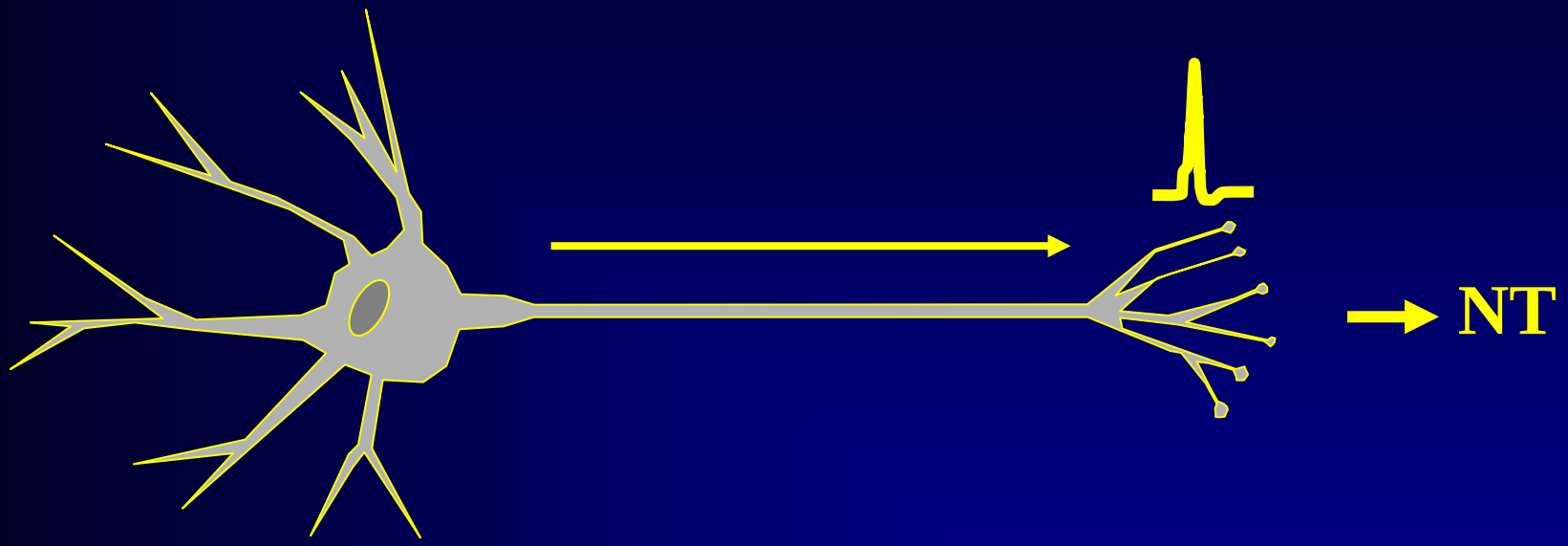
Action potential initiation and propagation



Stimuli are received by the dendrites and cell body, generating graded potentials which may spread to the axon hillock

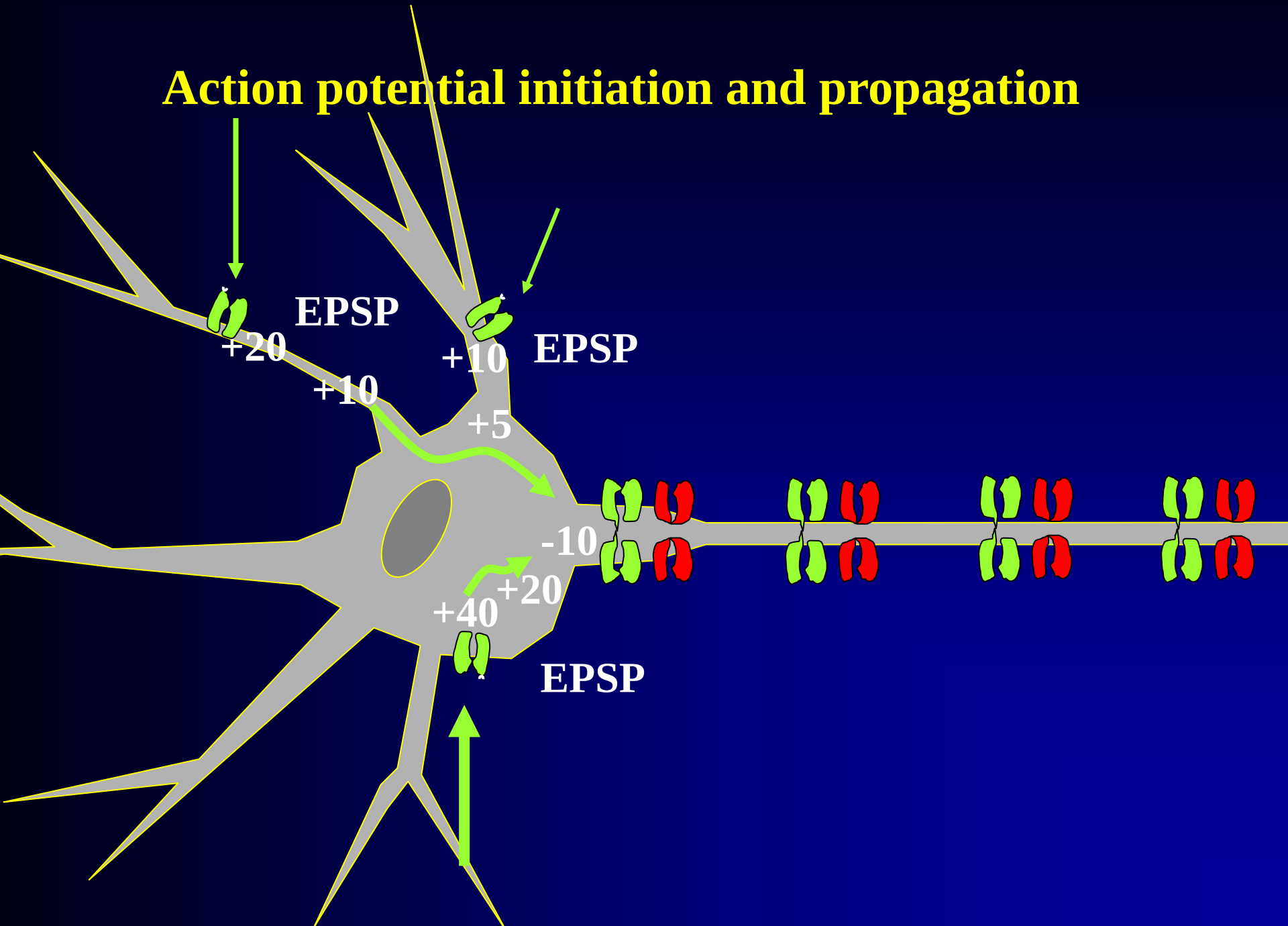


If incoming graded potentials depolarize the hillock membrane to threshold, an action potential will result at the hillock (due of the presence of voltage-gated Na+ and K+ channels)

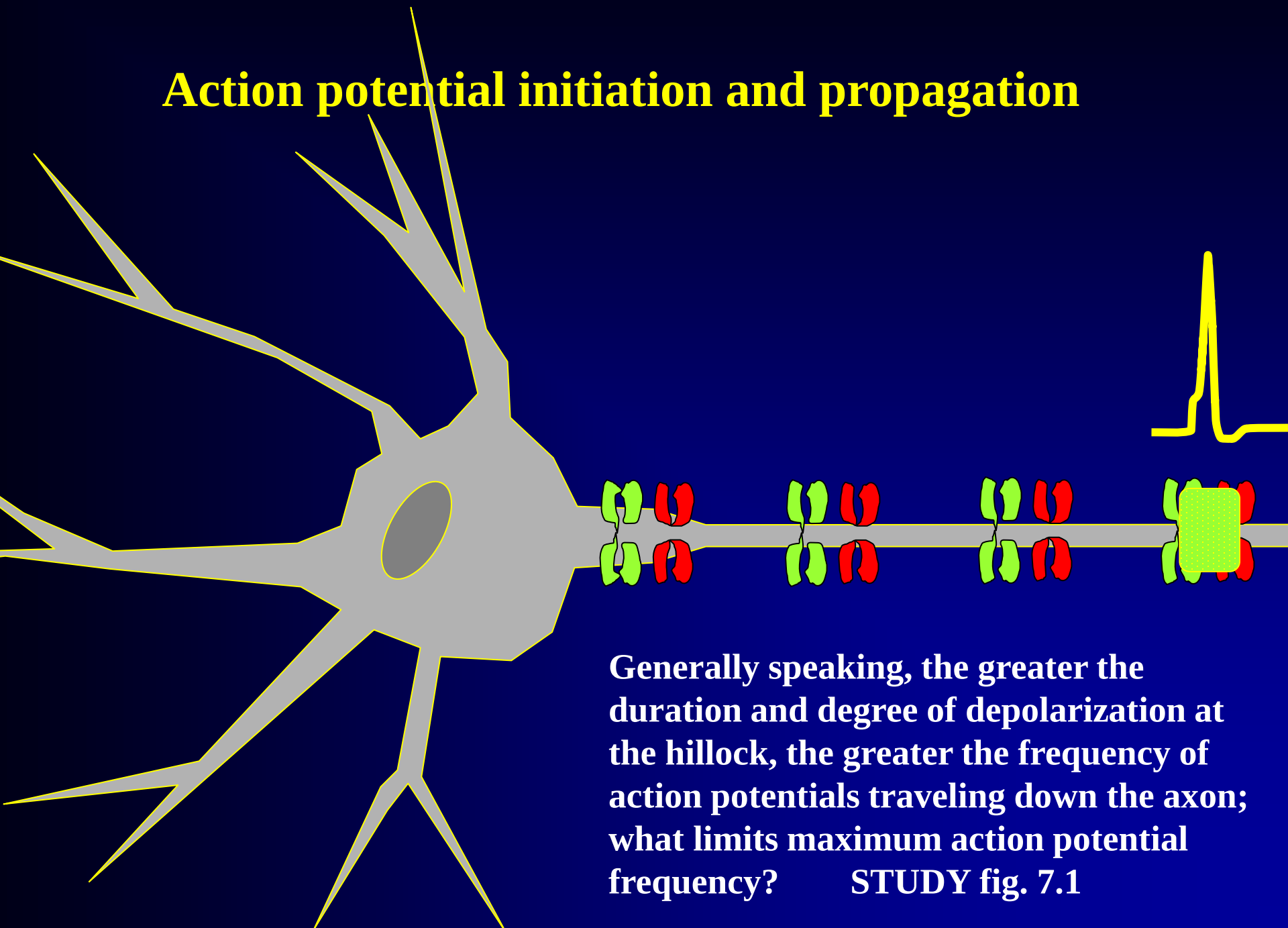


Depolarization of the nerve terminals results in the release of neurotransmitter (NT) from the neuron, leading to a synapse with another neuron or an end-target organ (muscle or gland)

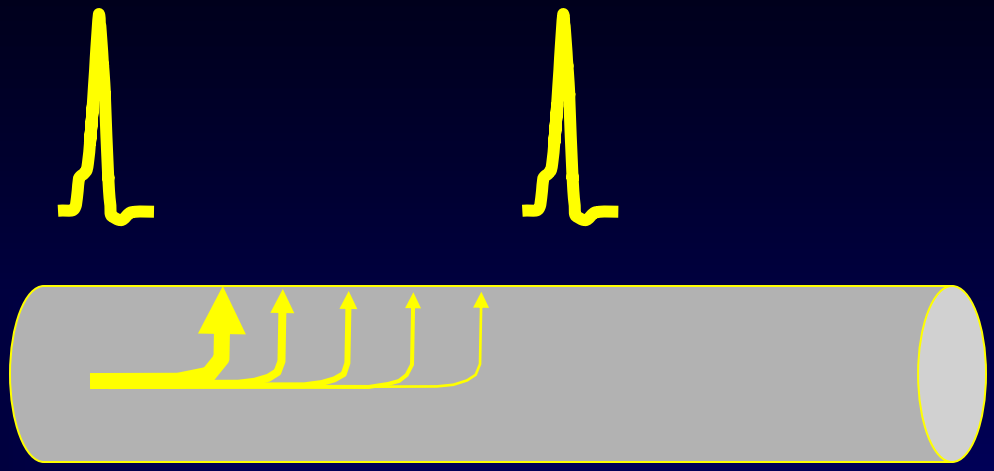
Action potential initiation and propagation



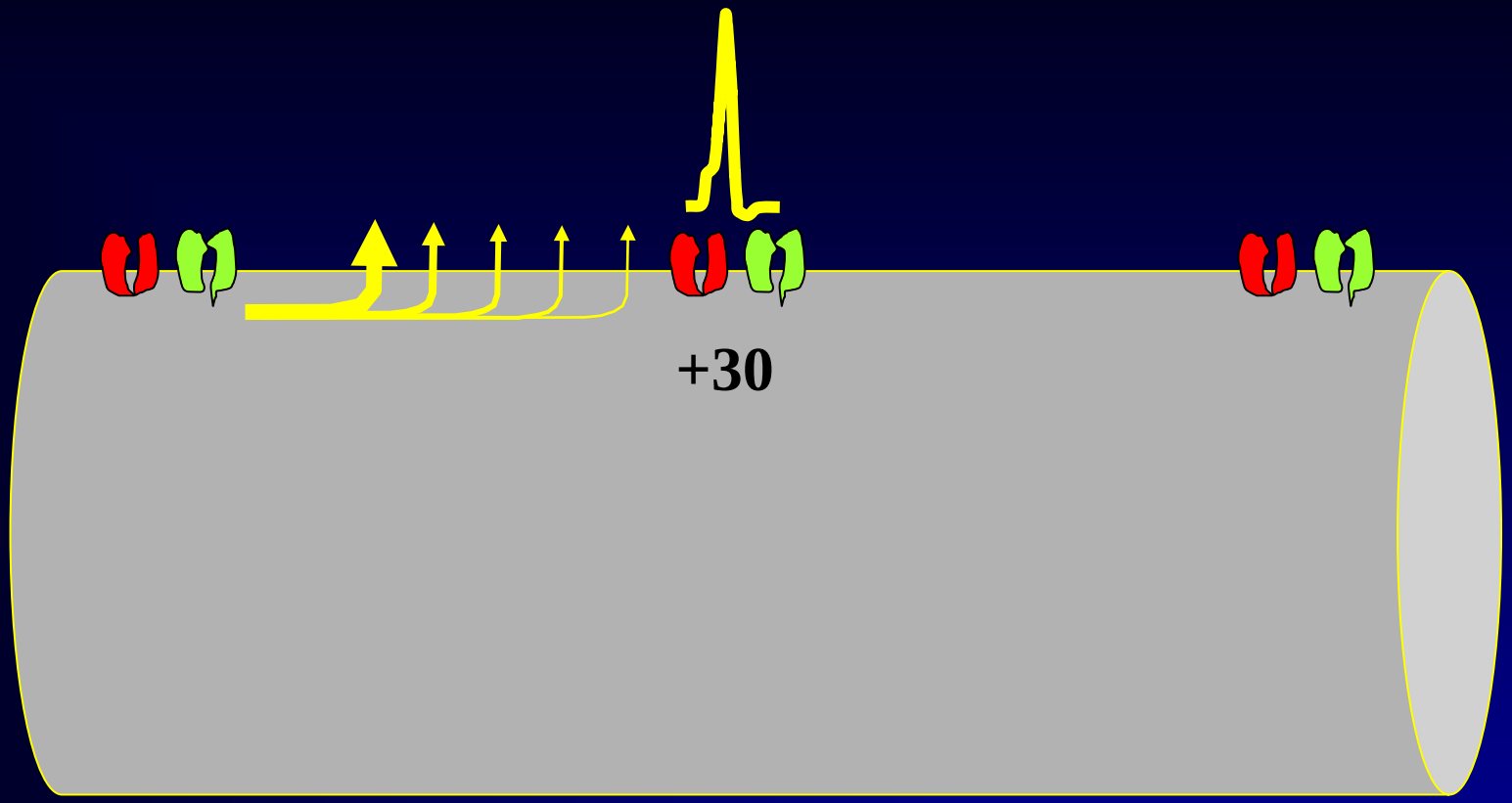
Action potential initiation and propagation



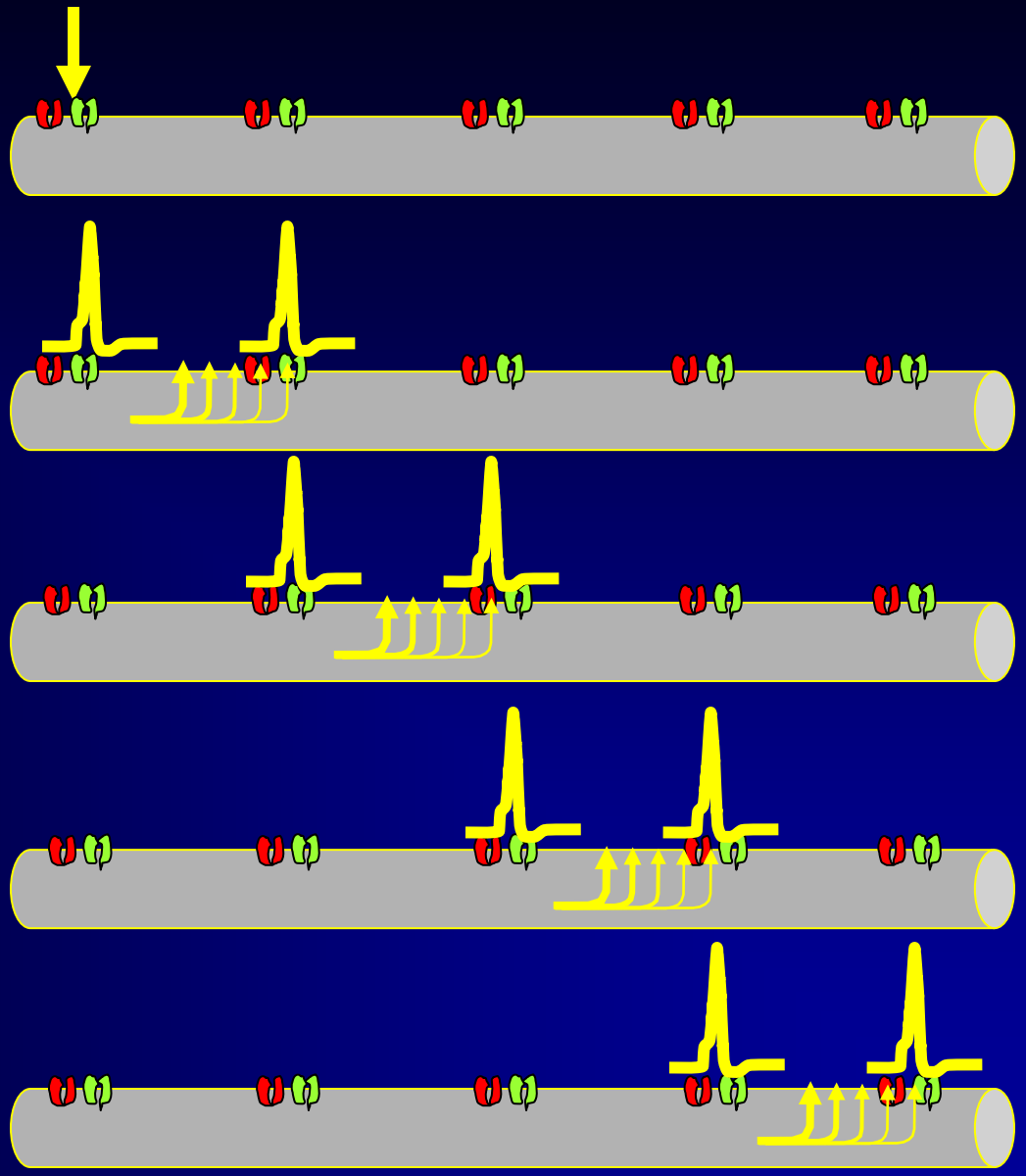
Generally speaking, the greater the duration and degree of depolarization at the hillock, the greater the frequency of action potentials traveling down the axon; what limits maximum action potential frequency? STUDY fig. 7.1



The hillock action potential spreads a depolarizing current down the adjacent axon, opening local voltage-gated channels and generating an action potential; this process is repeated down the length of the axon; depolarizing current can leak from the membrane as the current runs down the axon


+30

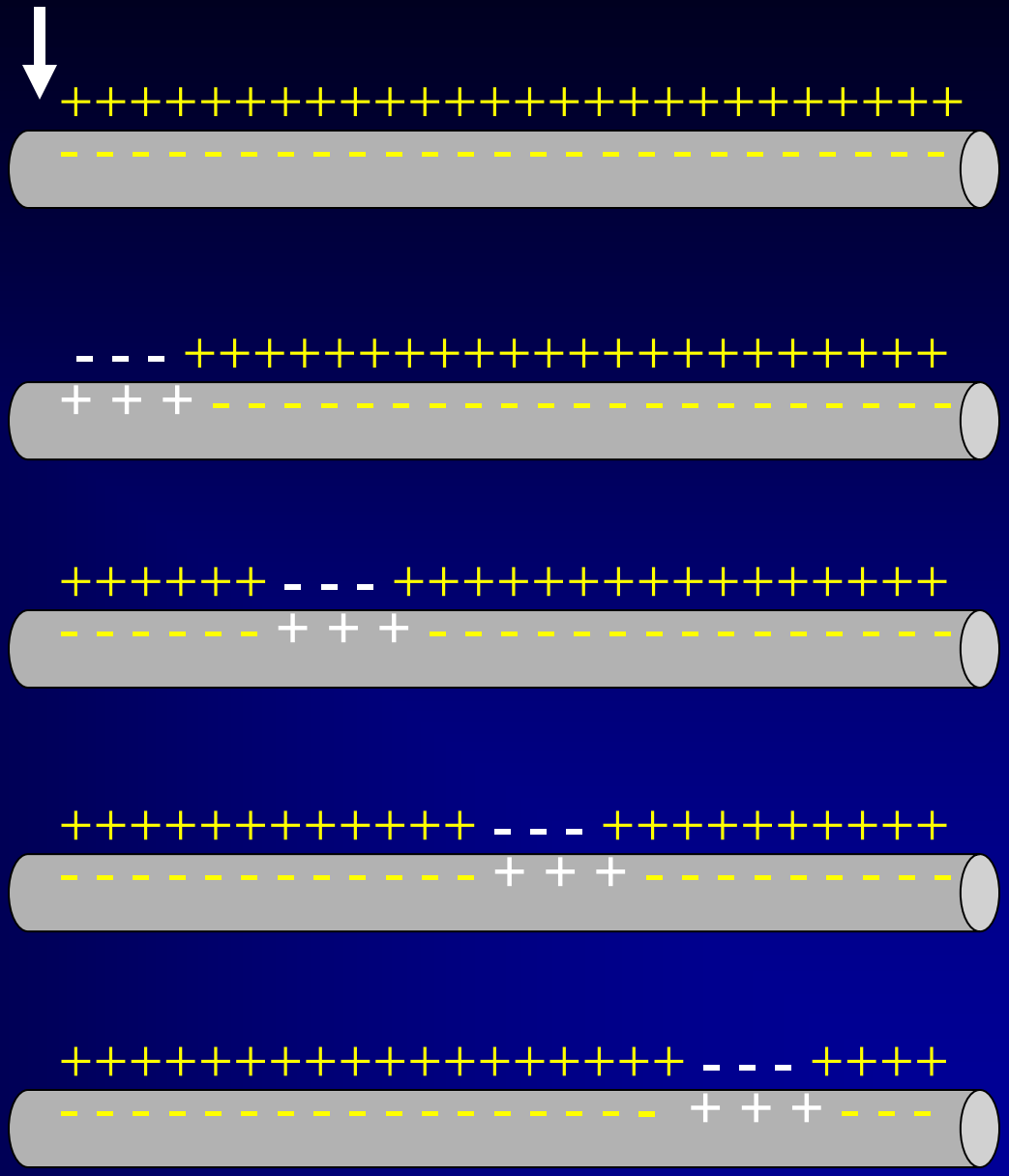
Stimulus



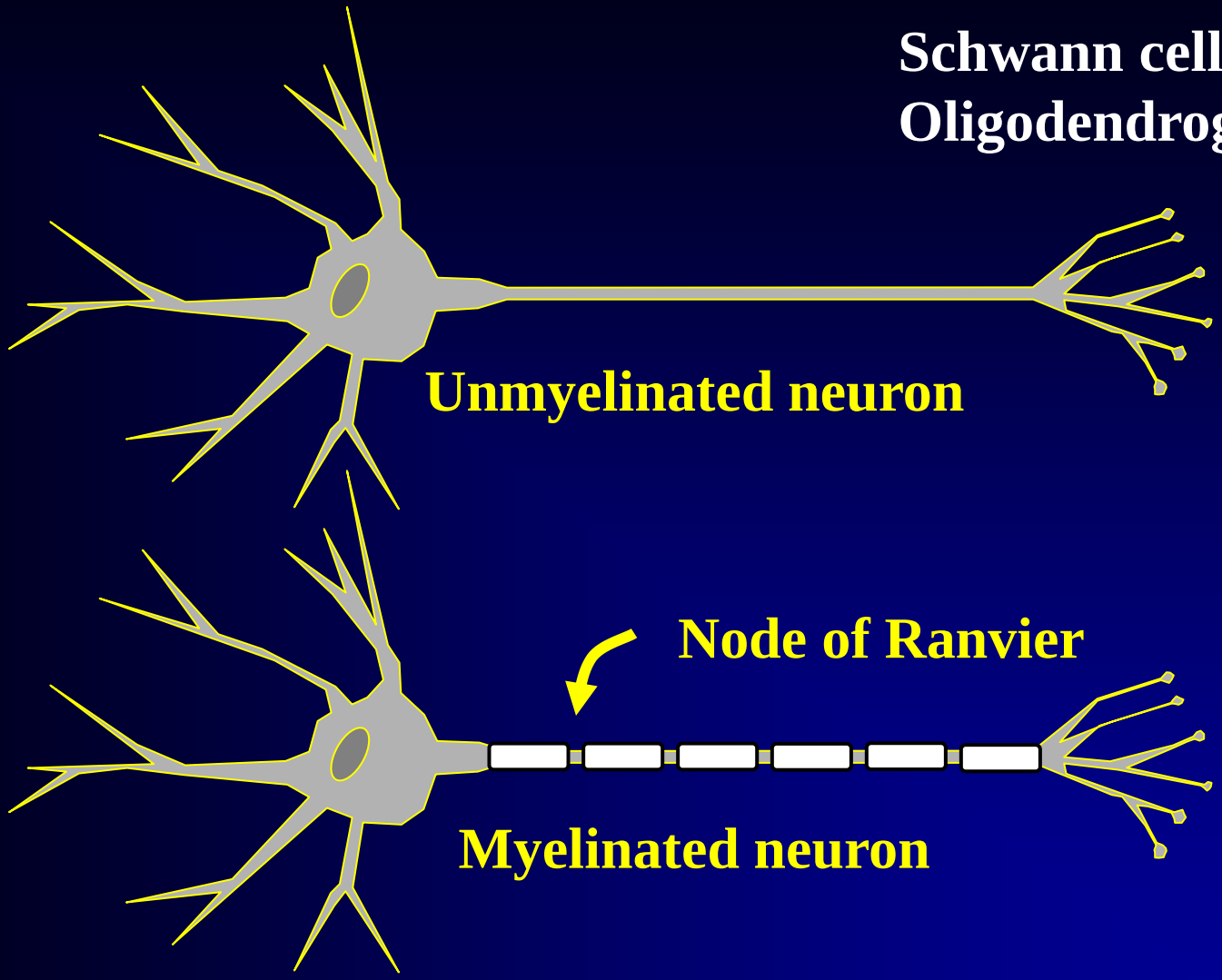
Nerve impulse propagation can be thought of as a wave of depolarization (action potentials) moving down the length of the axon.

Stimulus

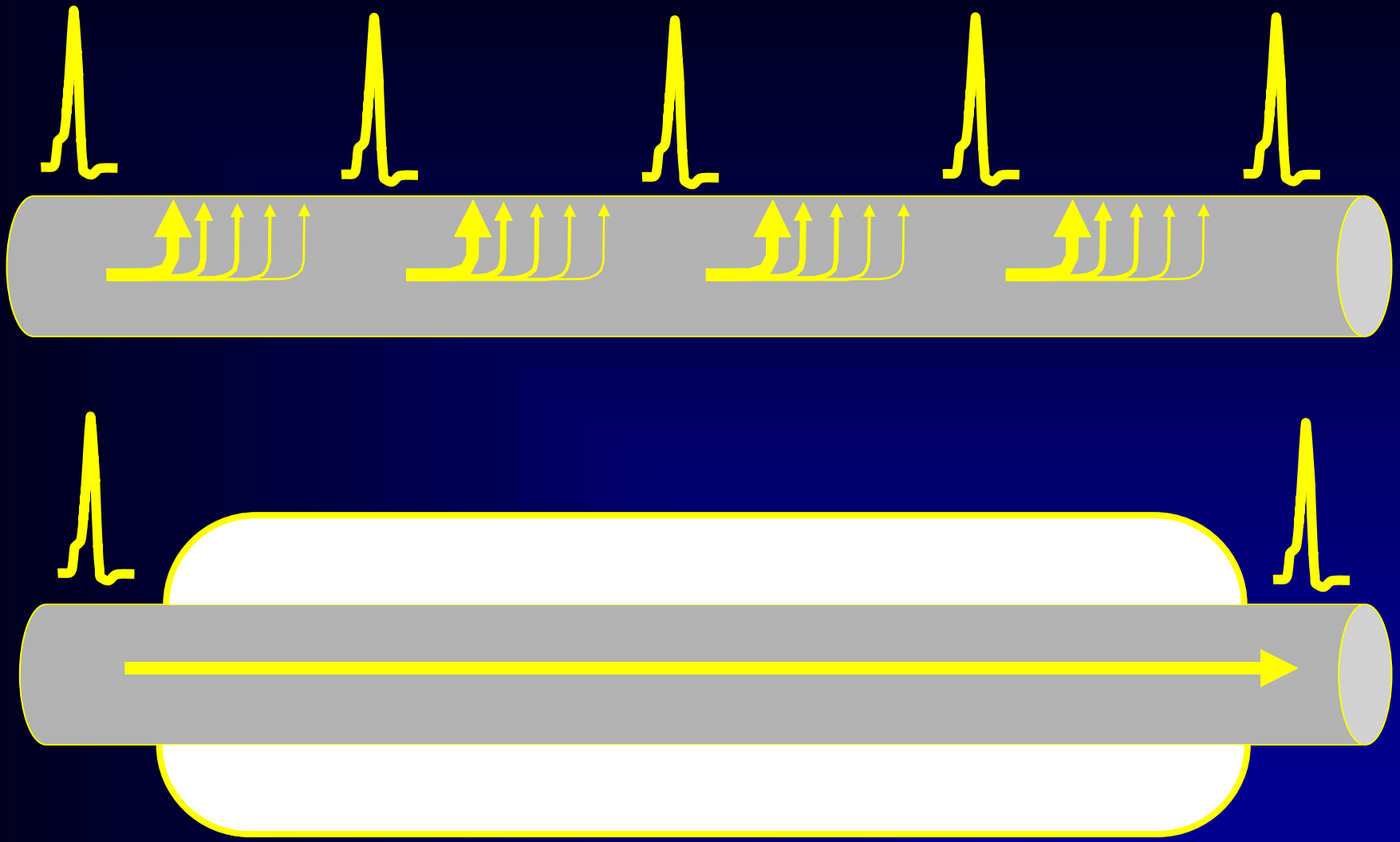
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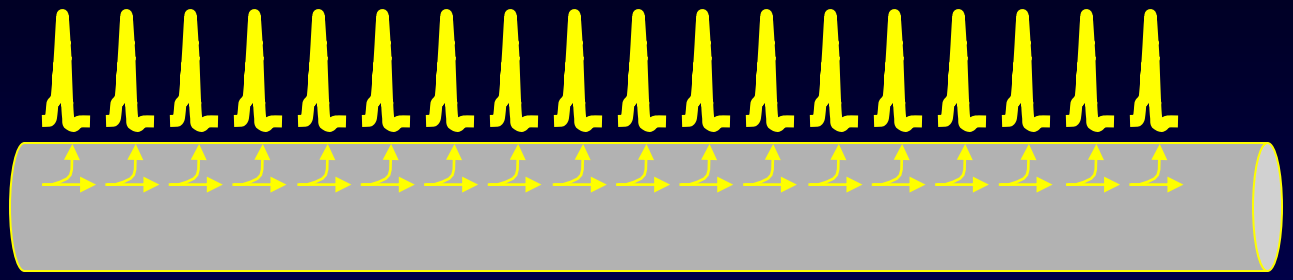
Schwann cells – PNS
Oligodendroglia - CNS



Effects of myelination



Myelination of neurons decreases current leakage 5000-fold



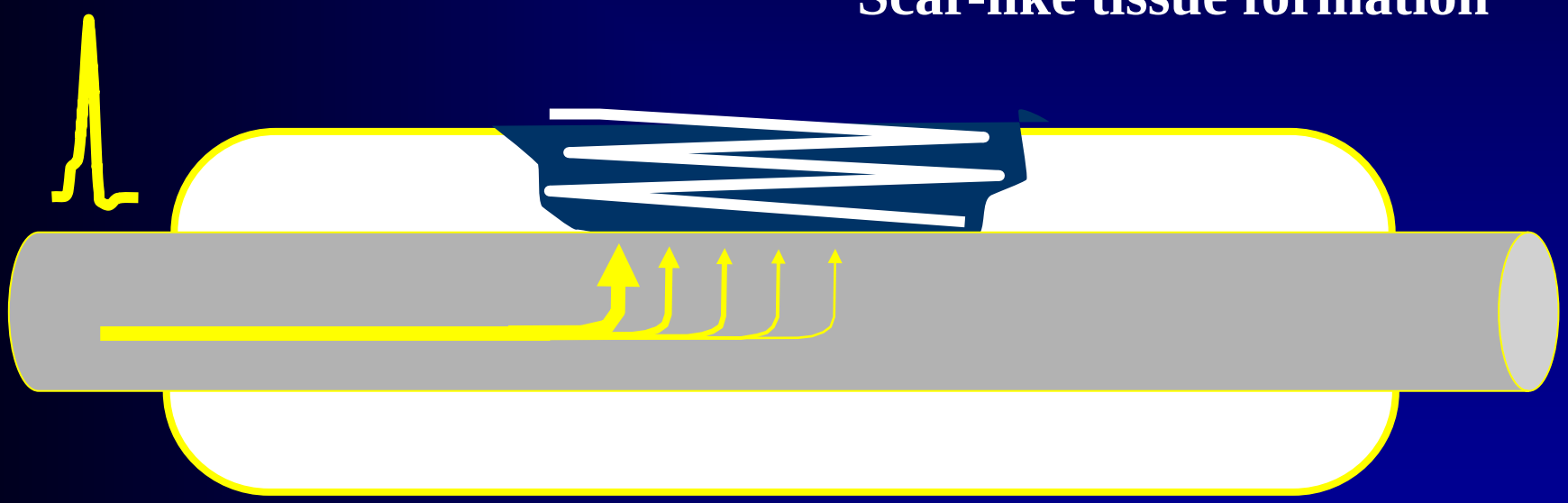
Unmyelinated axon



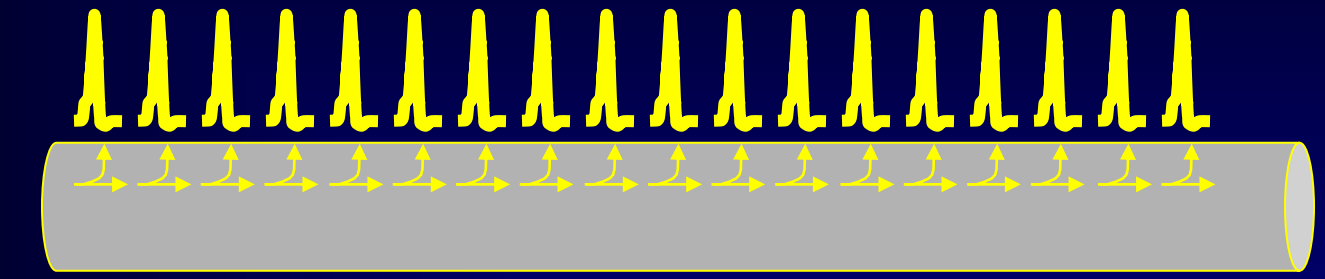
Myelinated axon

Action potentials occur along the entire length of an unmyelinated axon, but only at the nodes of a myelinated axon (saltatory conduction); this increases nerve conduction velocity 5-50-fold and decreases the energy requirement for conduction; nerve regeneration

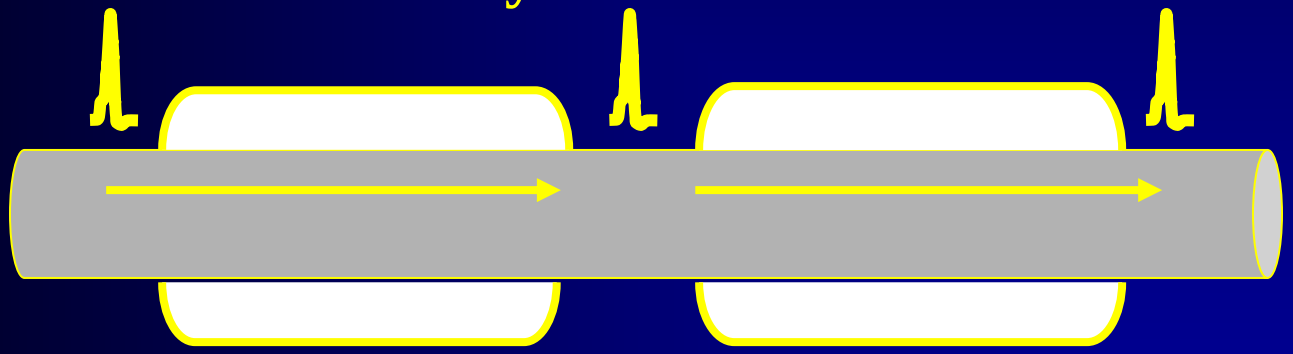
Demyelinating diseases: Multiple Sclerosis (Autoimmune attack on myelin sheath)



Local anesthetics (e.g., Lidocaine), voltage-gated Na+ channels and impulse propagation



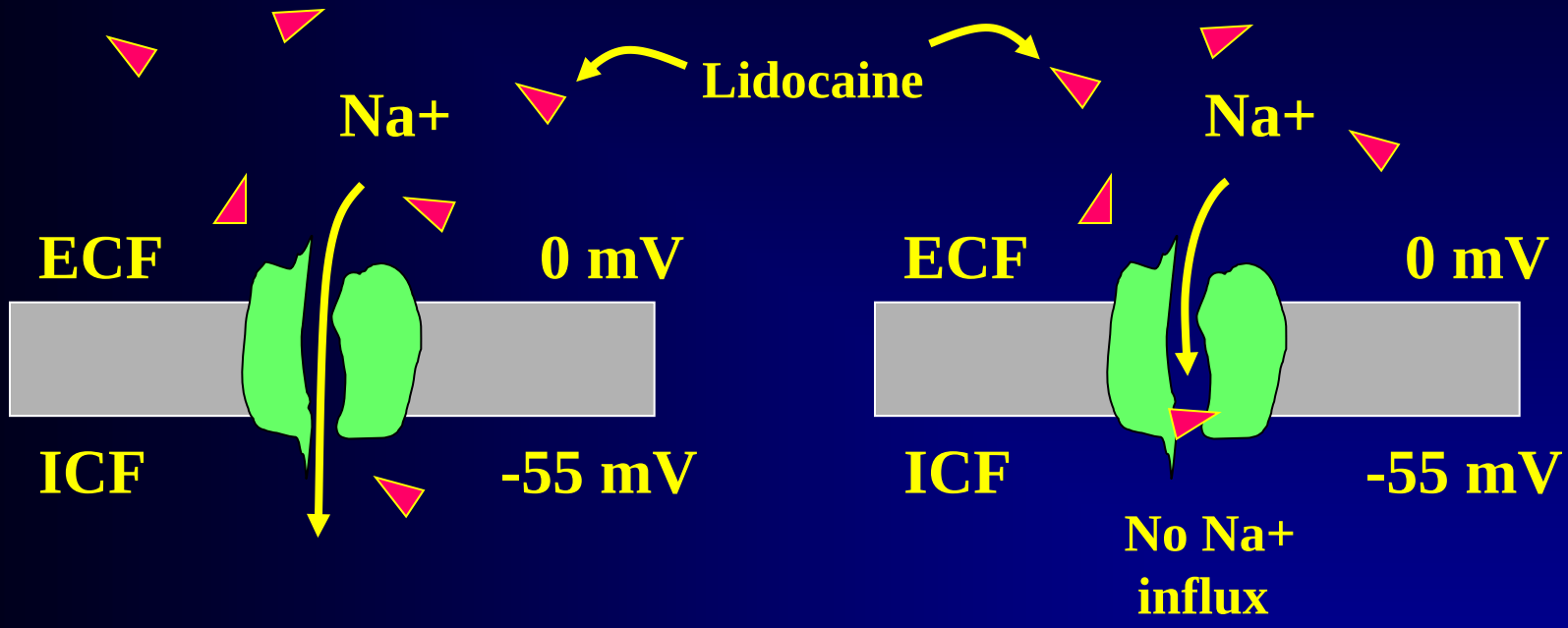
Unmyelinated axon



Myelinated axon

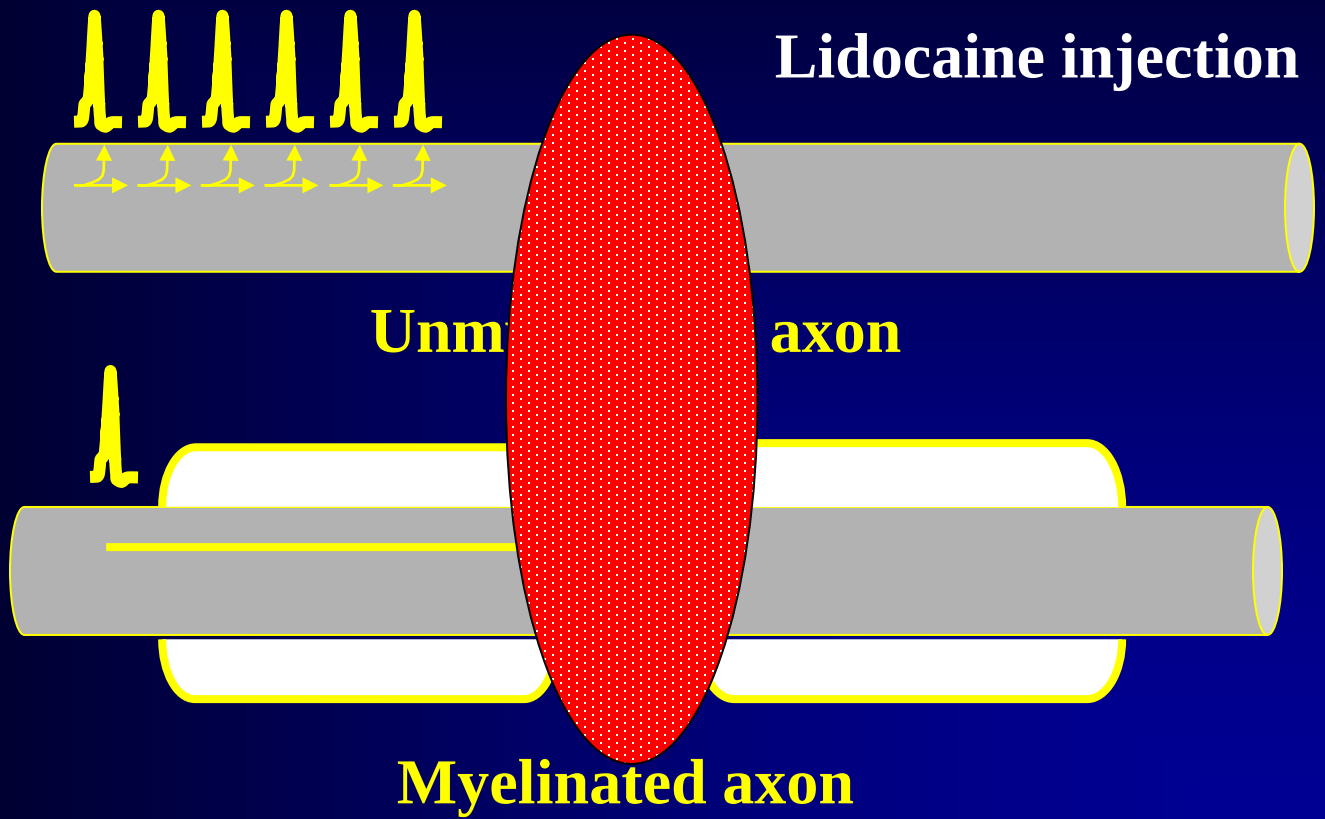
Lidocaine inhibits voltage-gated sodium channels. Consequences on action potentials? Propagation?

Local anesthetics (e.g., Lidocaine), voltage-gated Na+ channels and impulse propagation

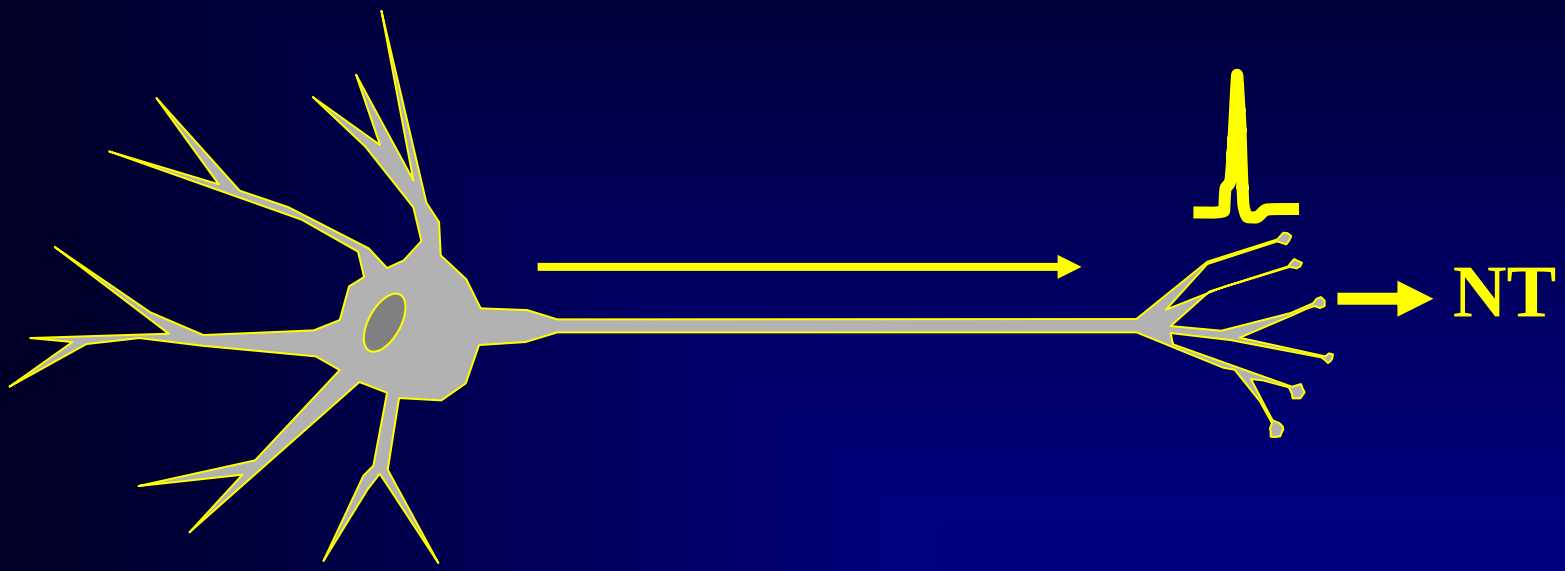


Lidocaine inhibits voltage-gated sodium channels. Consequences on action potentials? Propagation?

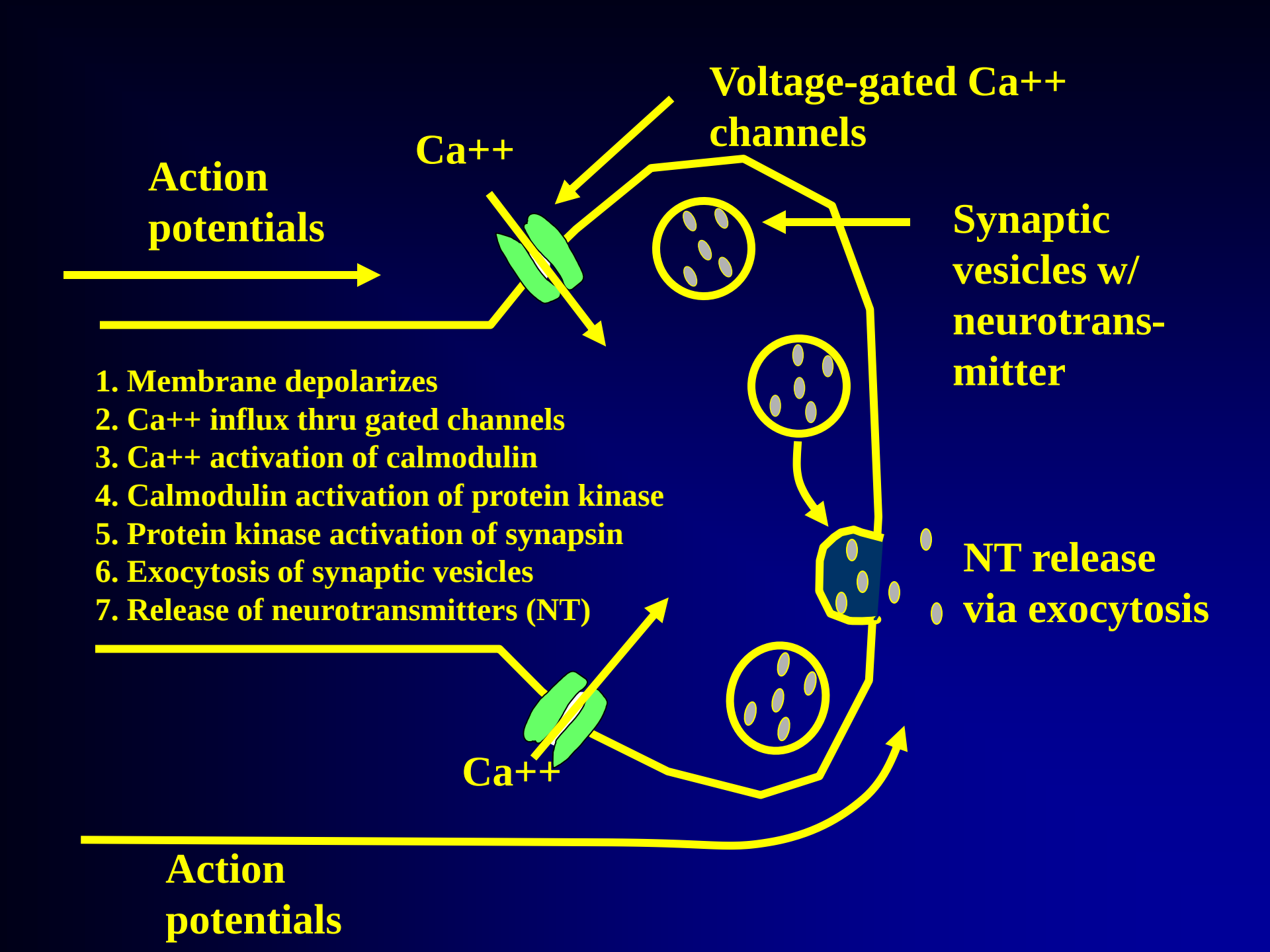
Local anesthetics (e.g., Lidocaine), voltage-gated Na+ channels and impulse propagation

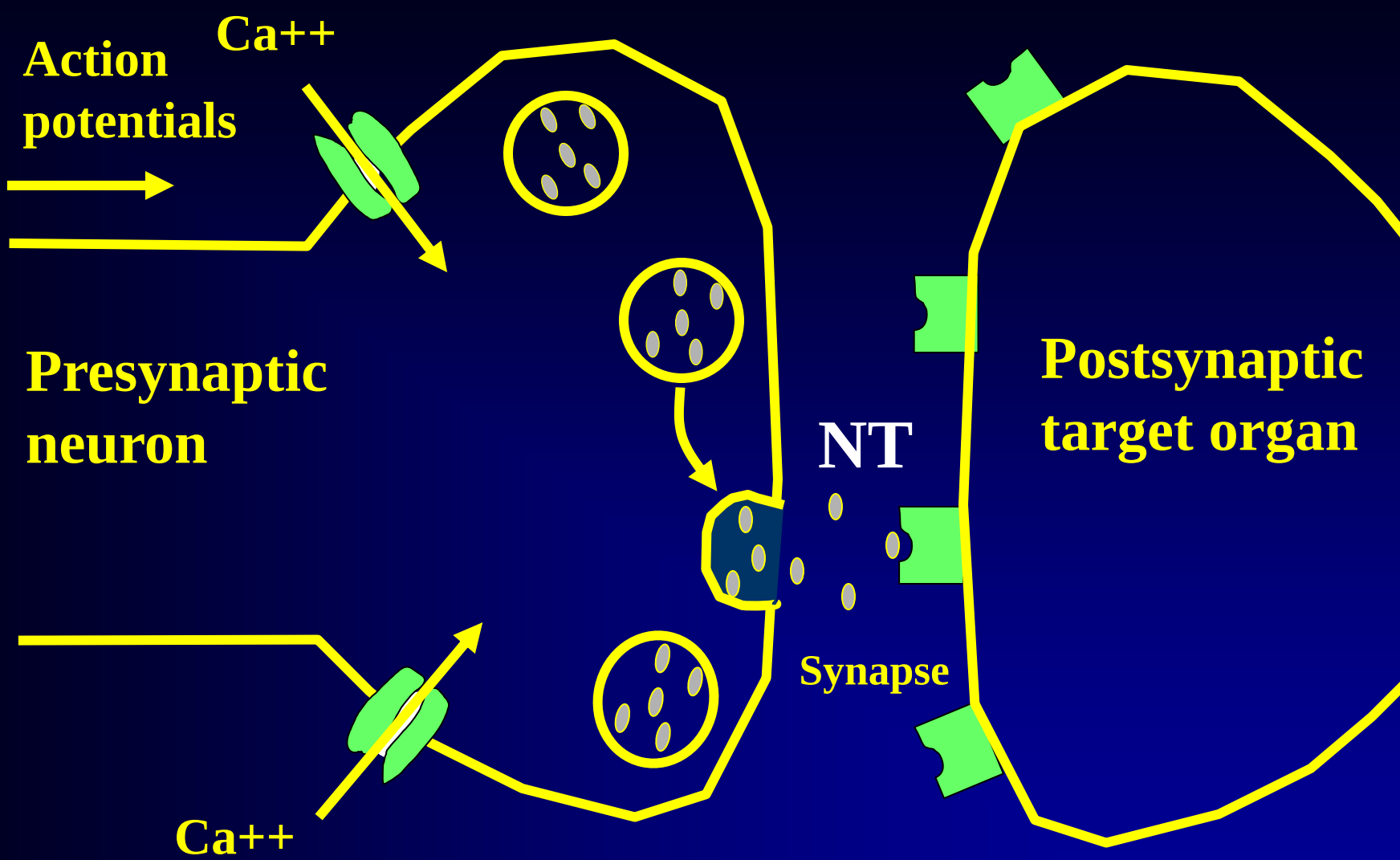


Lidocaine inhibits voltage-gated sodium channels. Consequences on action potentials? Propagation?



Depolarization of the nerve terminals results in the release of neurotransmitter (NT) from the neuron, leading to a synapse with another neuron or an end-target organ (muscle or gland)


Voltage-gated Ca++ channels
Ca++
Action potentials
Synaptic vesicles w/ neurotrans-mitter
1. Membrane depolarizes
2. Ca++ influx thru gated channels
3. Ca++ activation of calmodulin
4. Calmodulin activation of protein kinase
5. Protein kinase activation of synapsin
6. Exocytosis of synaptic vesicles
7. Release of neurotransmitters (NT)
NT release via exocytosis
Ca++
Action potentials



Neurotransmitters and neuromodulators

b. Neurotransmitters

(1) In addition to well known neurotransmitters such as acetylcholine and norepinephrine, more than 50 other chemical substances have been identified as synaptic neurotransmitters

(a.) Rapid acting small molecule neurotransmitters

Ach, NE, E, dopamine, serotonin, histamine, GABA, glycine, glutamate, aspartate, nitric oxide, etc.

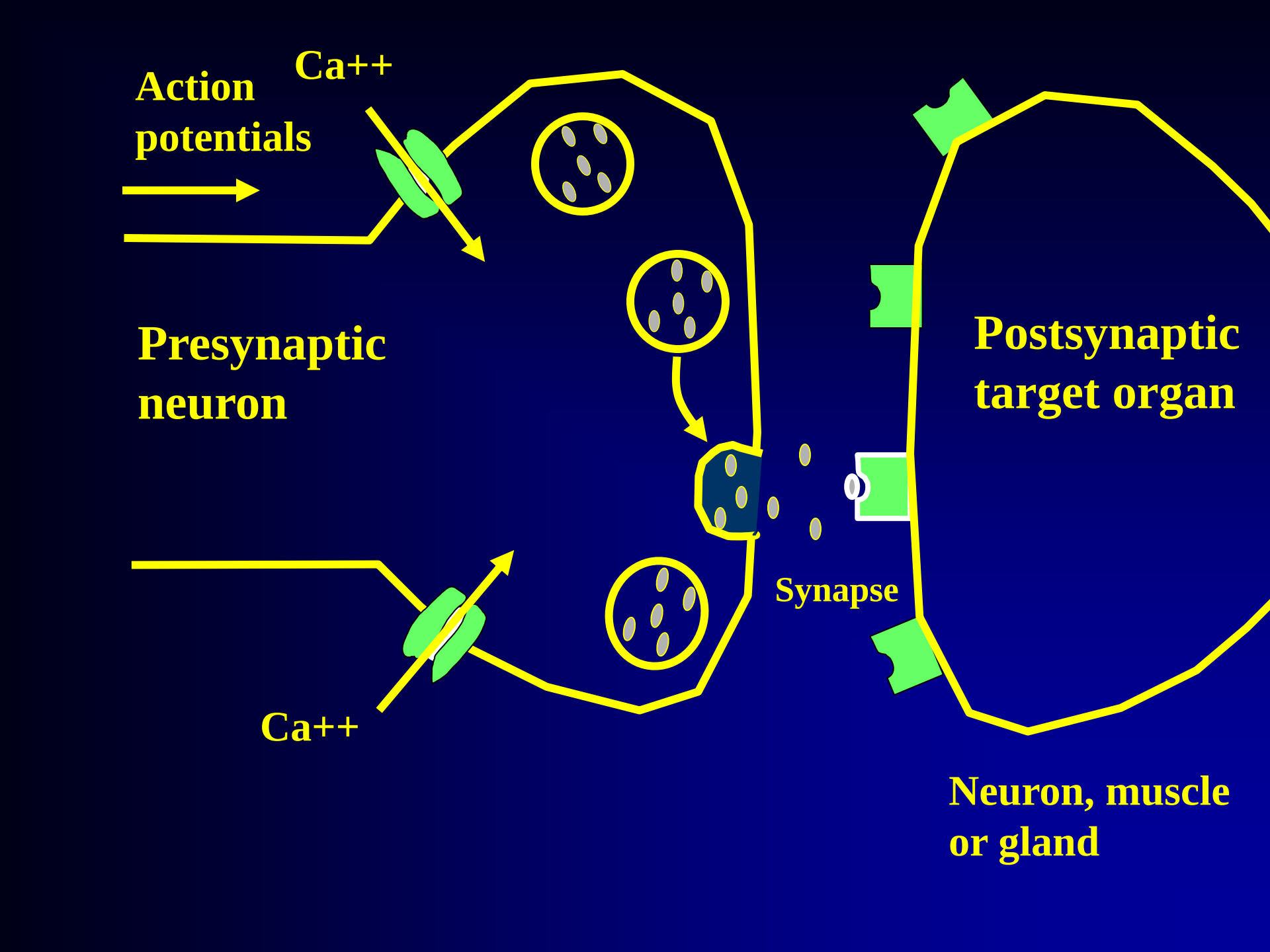
(b.) Slow acting neuropeptide neurotransmitters

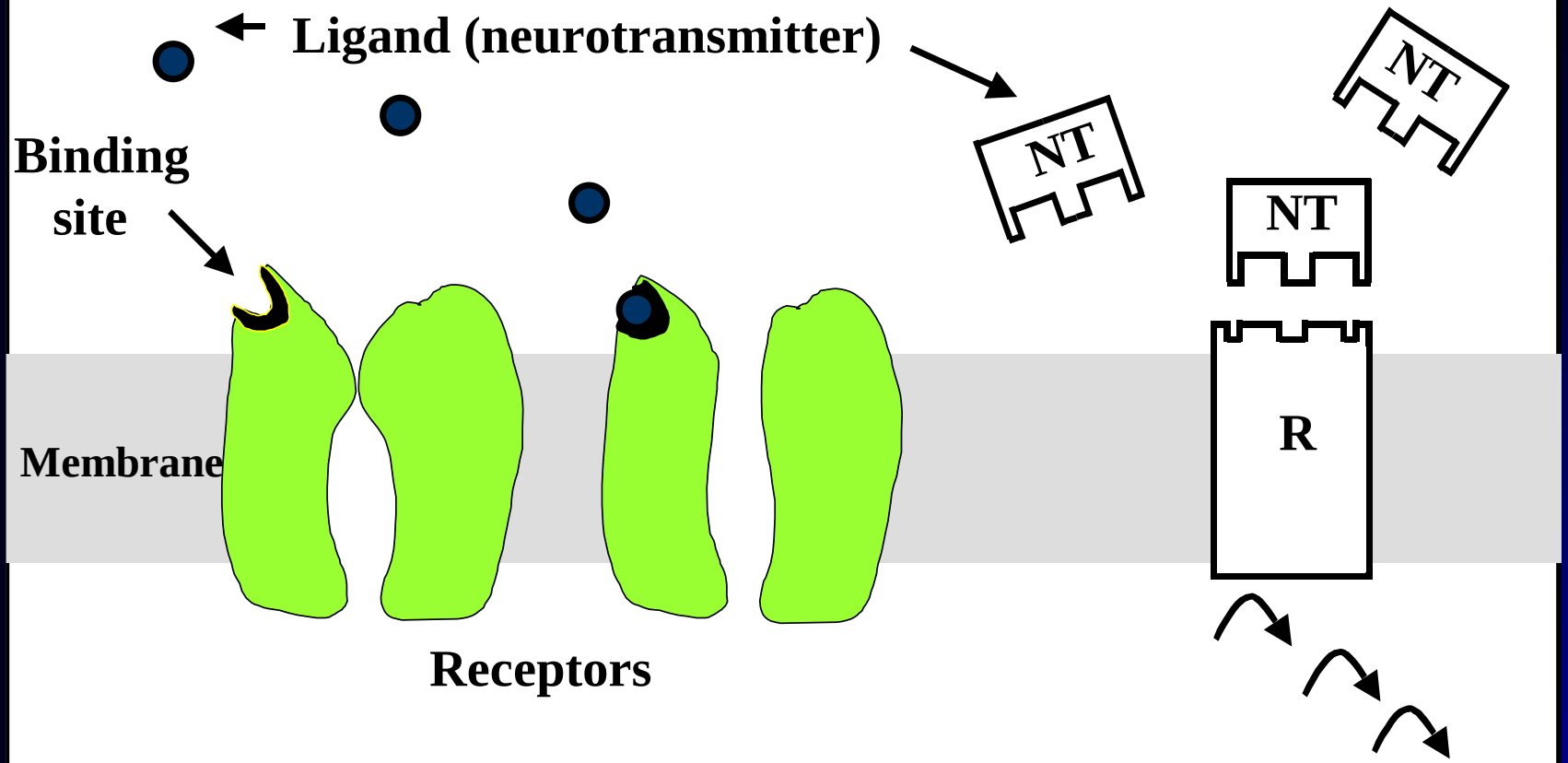
Hypothalamic releasing hormones, pituitary peptides, enkephalin, substance P, bradykinins, GI hormones, etc.

(2) Synaptic terminals from one neuron release only one small molecule fast acting neurotransmitter, but may also release one or more of the slow acting neuropeptides

(3) Some rapid acting small molecule neurotransmitters and slow acting neuropeptide neurotransmitters are excitatory at the post-synaptic membrane and some are inhibitory

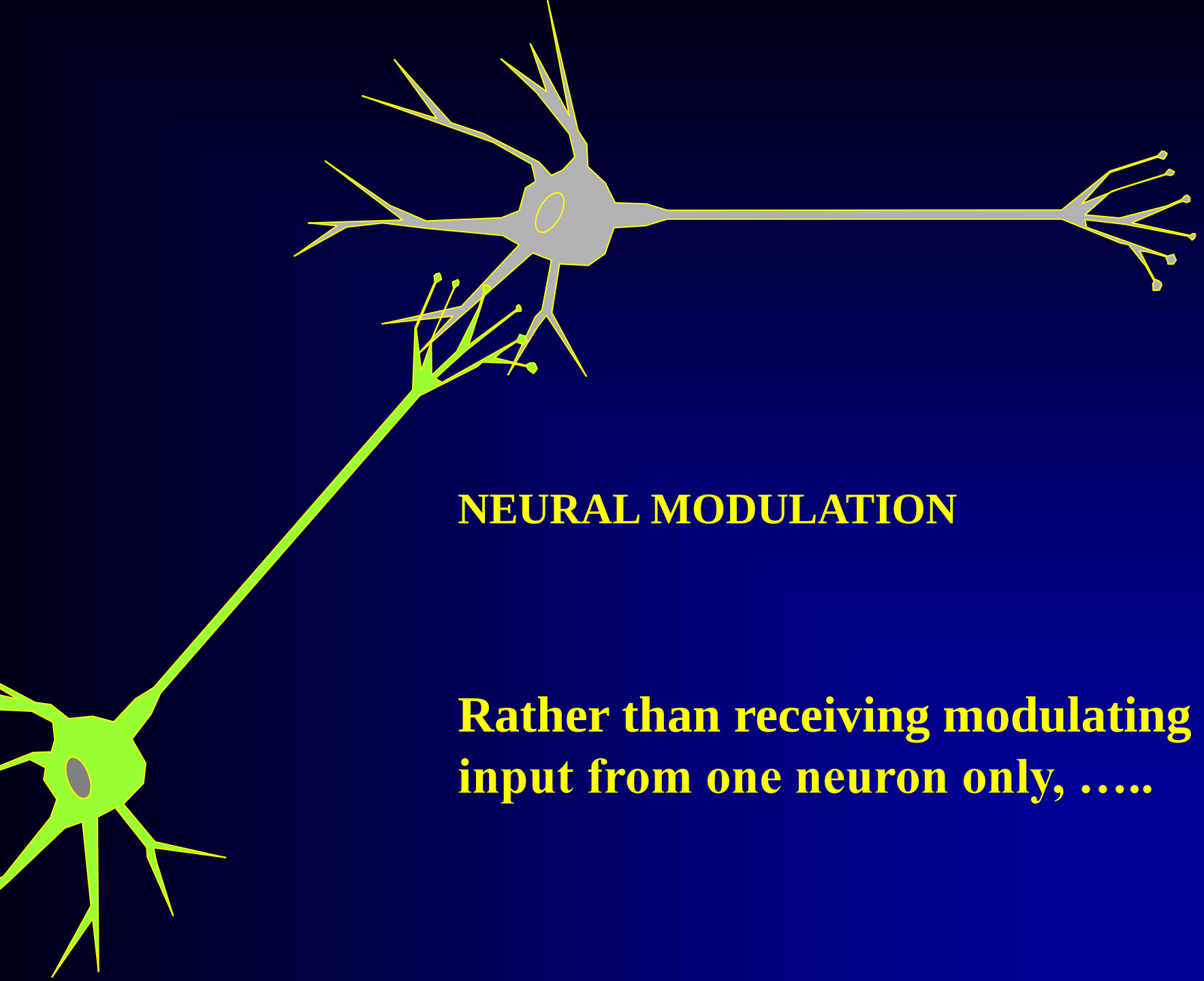
(4) Slow acting neuropeptides are generally extremely potent (1000-fold more than the small, fast acting molecules) and exert long-lasting post-synaptic effects (hours, days or years)


Action potentials
Ca++
Presynaptic neuron
Postsynaptic target organ
Synapse
Ca++
Neuron, muscle or gland



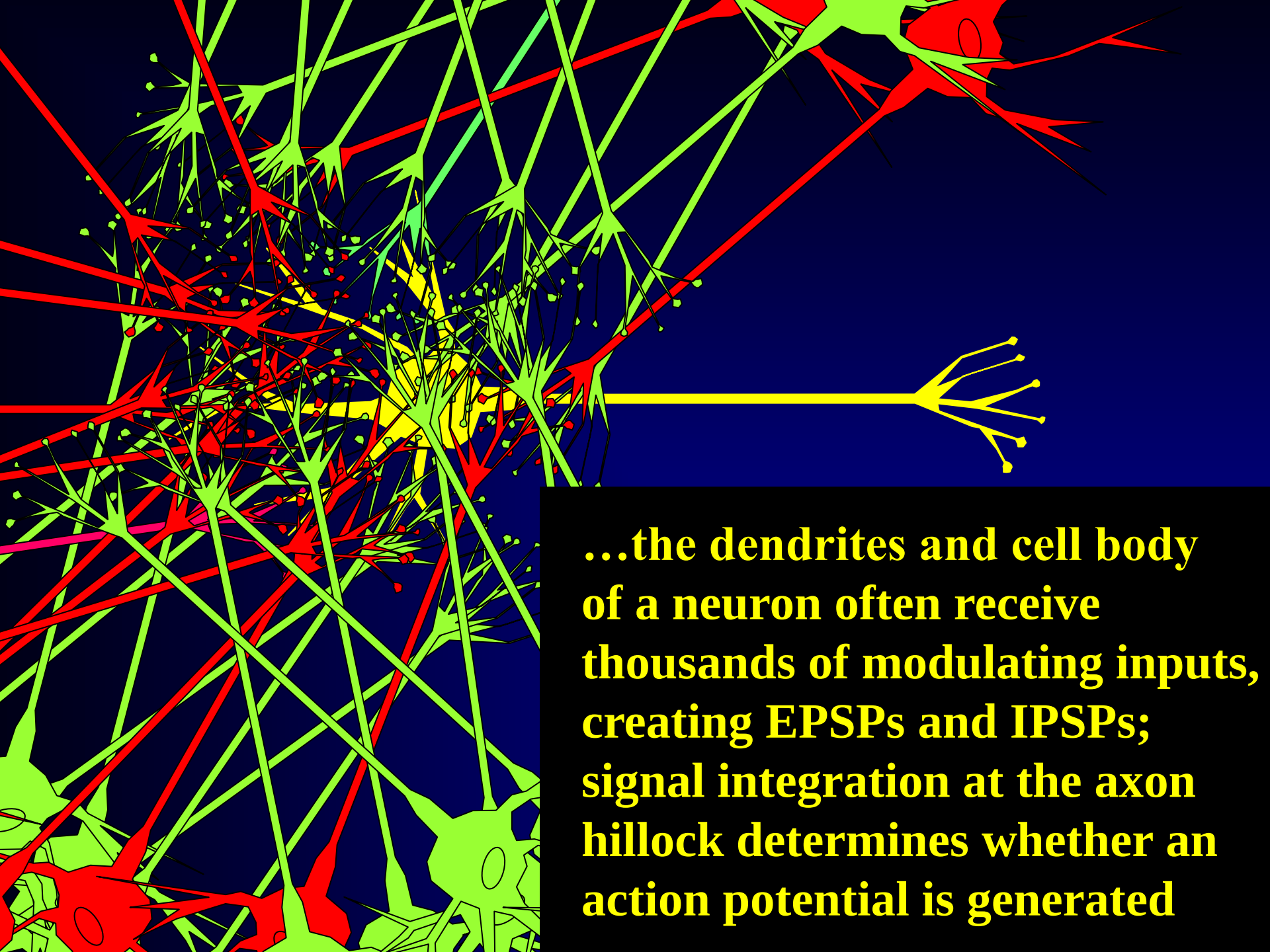
Neurotransmitters bind receptors that:

Open (close) membrane channels OR (in)activate G-protein-linked signal transduction mechanisms OR (in)activate receptor enzymatic activity

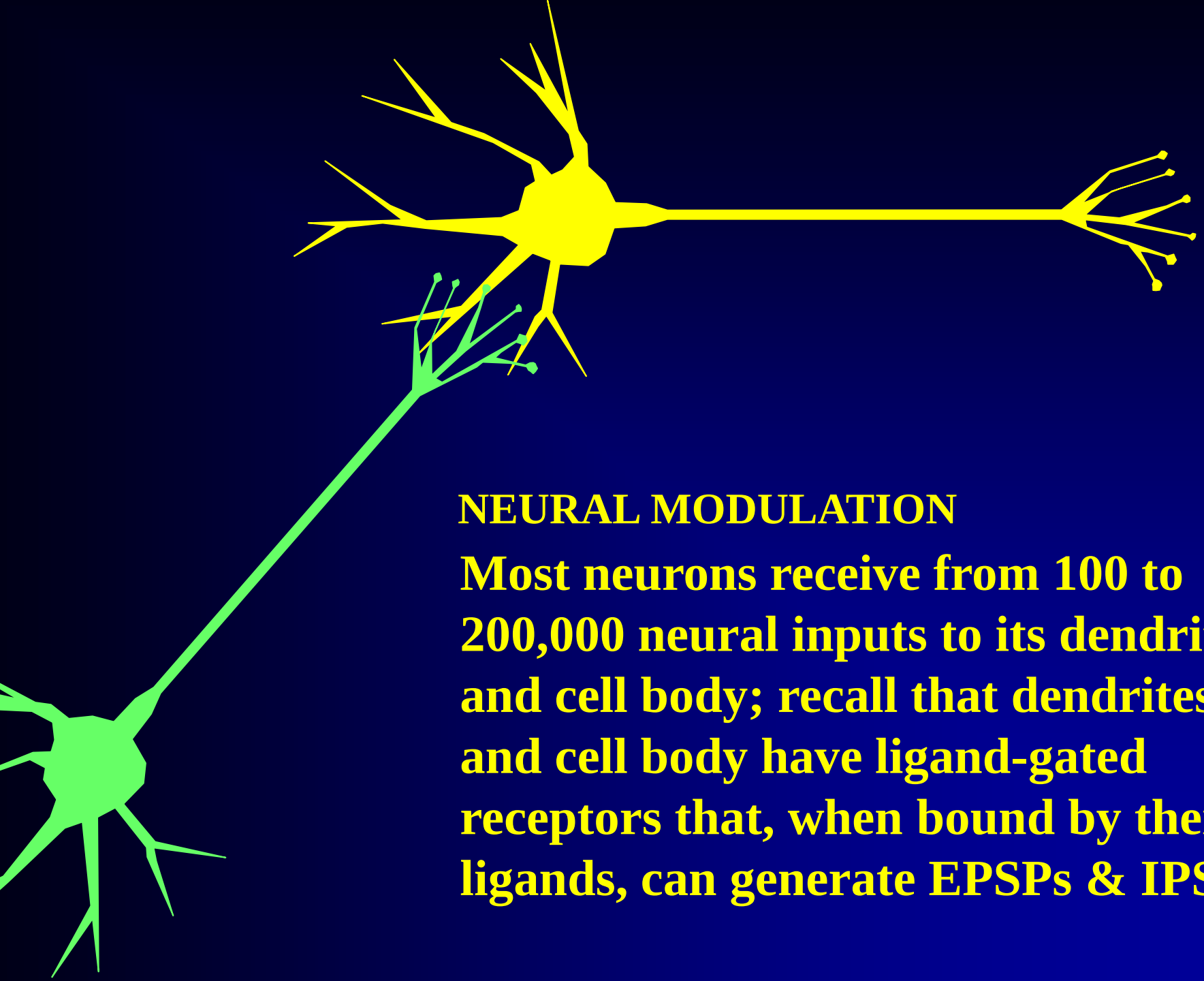


NEURAL MODULATION

Rather than receiving modulating input from one neuron only, …..

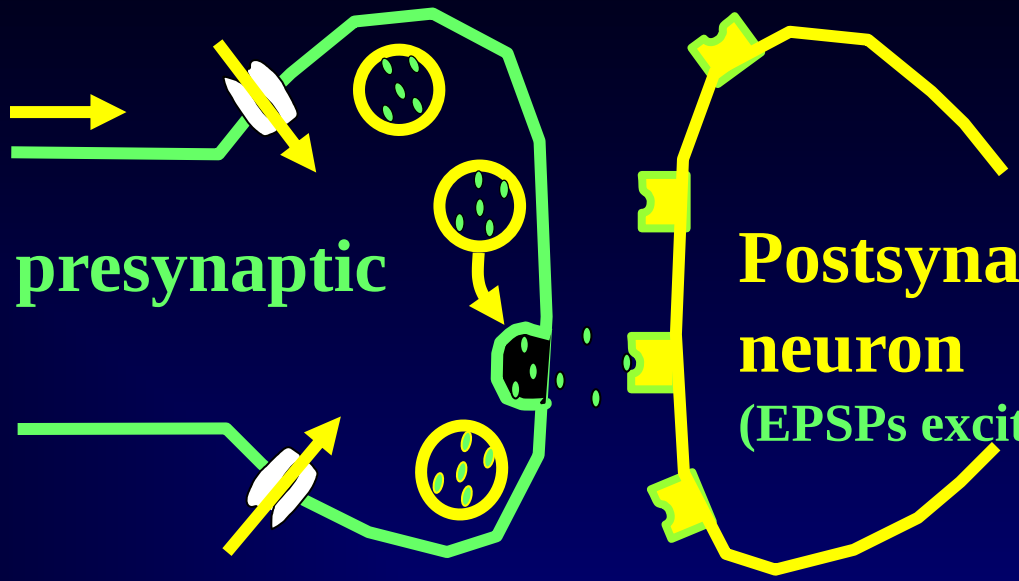


…the dendrites and cell body of a neuron often receive thousands of modulating inputs, creating EPSPs and IPSPs; signal integration at the axon hillock determines whether an action potential is generated



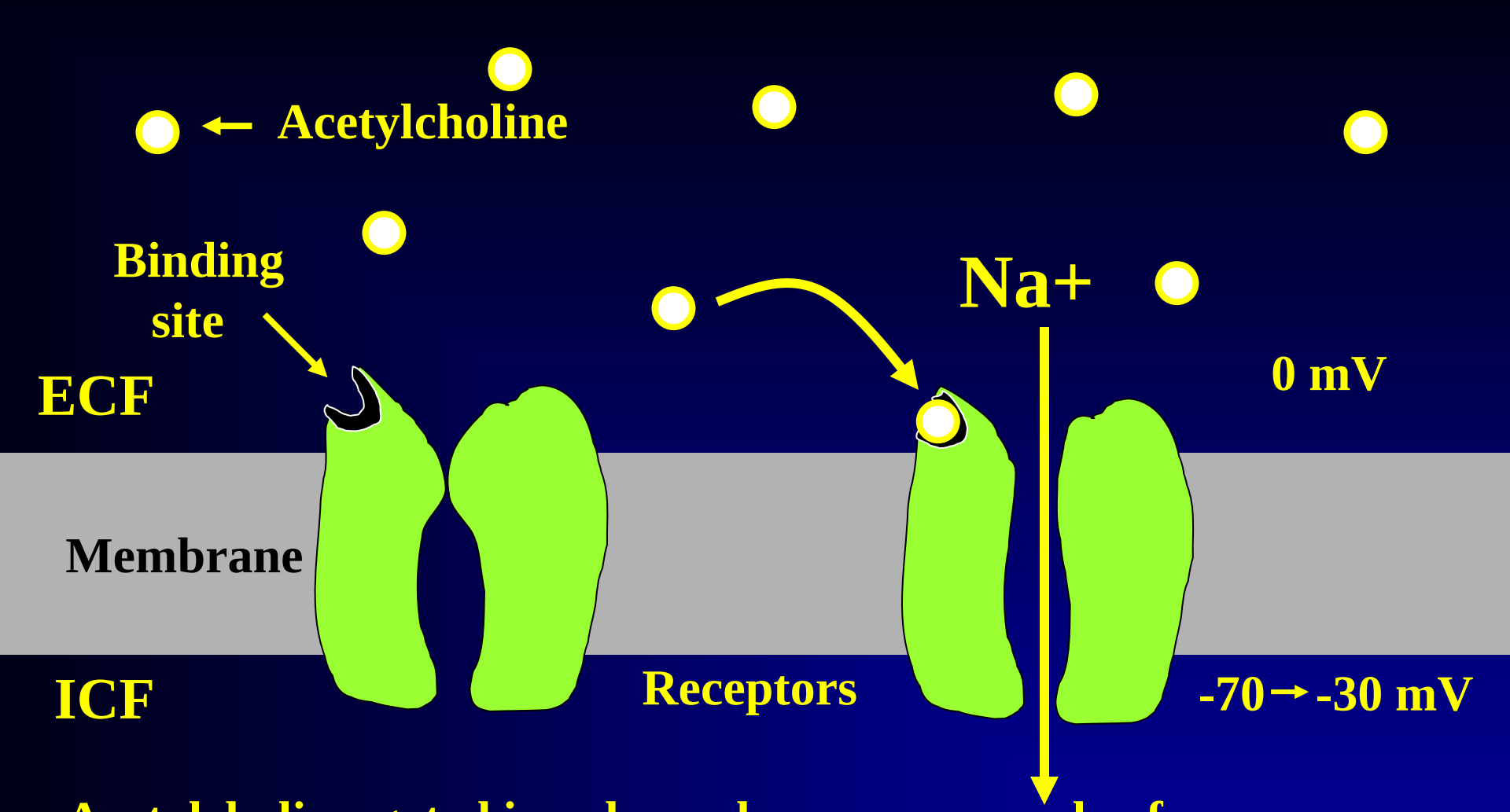
NEURAL MODULATION

Most neurons receive from 100 to 200,000 neural inputs to its dendrites and cell body; recall that dendrites and cell body have ligand-gated receptors that, when bound by their ligands, can generate EPSPs & IPSPs

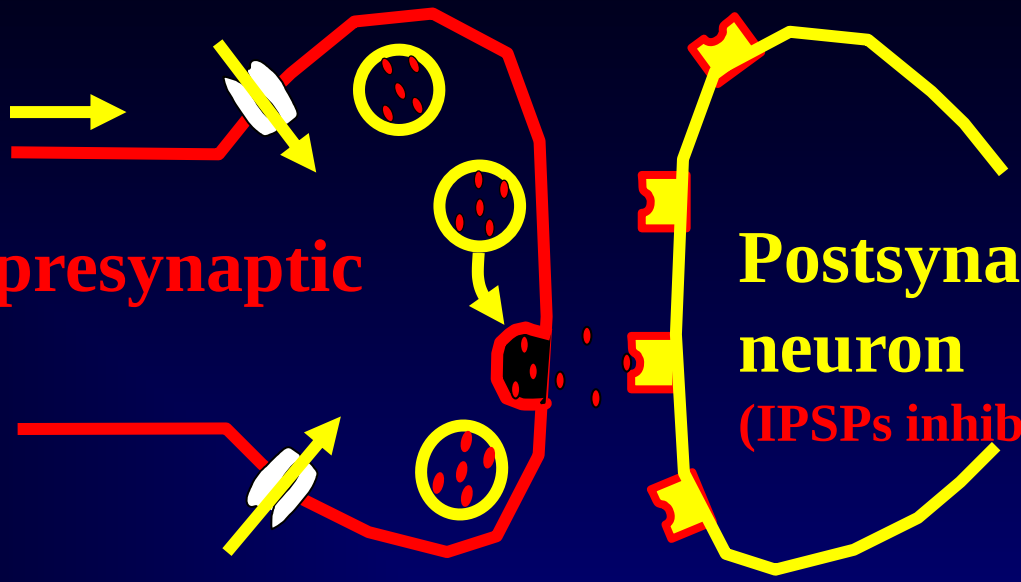


EPSPs - excitatory post-synaptic potentials are graded potentials that drive the axon hillock closer to threshold (depolarize), making the neuron more excitable (i.e., easier to elicit an action potential)

IPSPs - inhibitory post-synaptic potentials are graded potentials that drive the axon hillock away from threshold (hyperpolerize), making the neuron less excitable (i.e., harder to elicit an action potential**)**

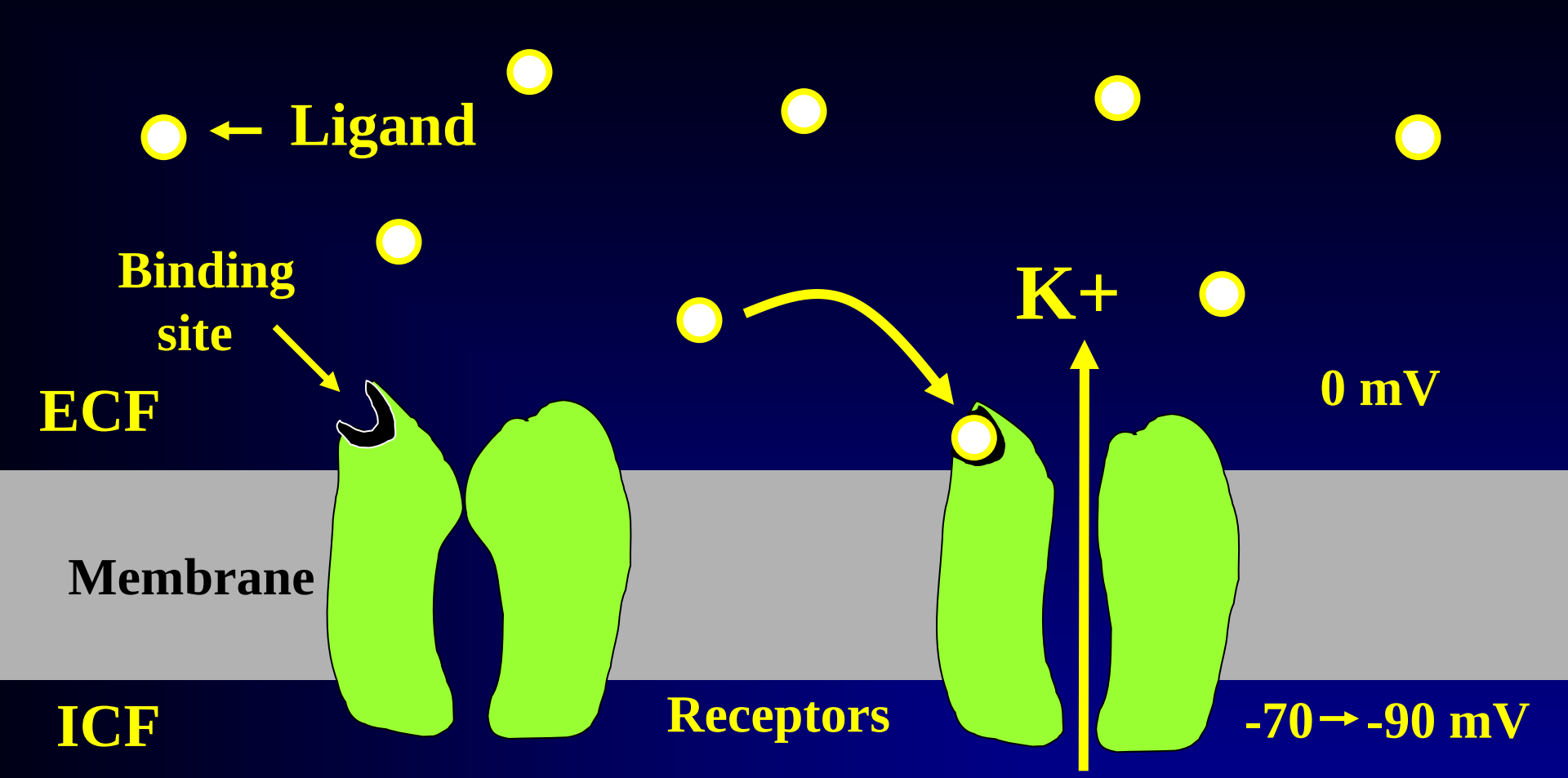


Acetylcholine-gated ion channels are an example of neuro-modulating ligand-gated channels; Ach binding its receptor opens a pore that allows Na+ influx; as more Na+ enters the cell, the membrane depolarizes closer to threshold (an EPSP); EPSPs can also be developed by ligands that close K+ or Cl- channels



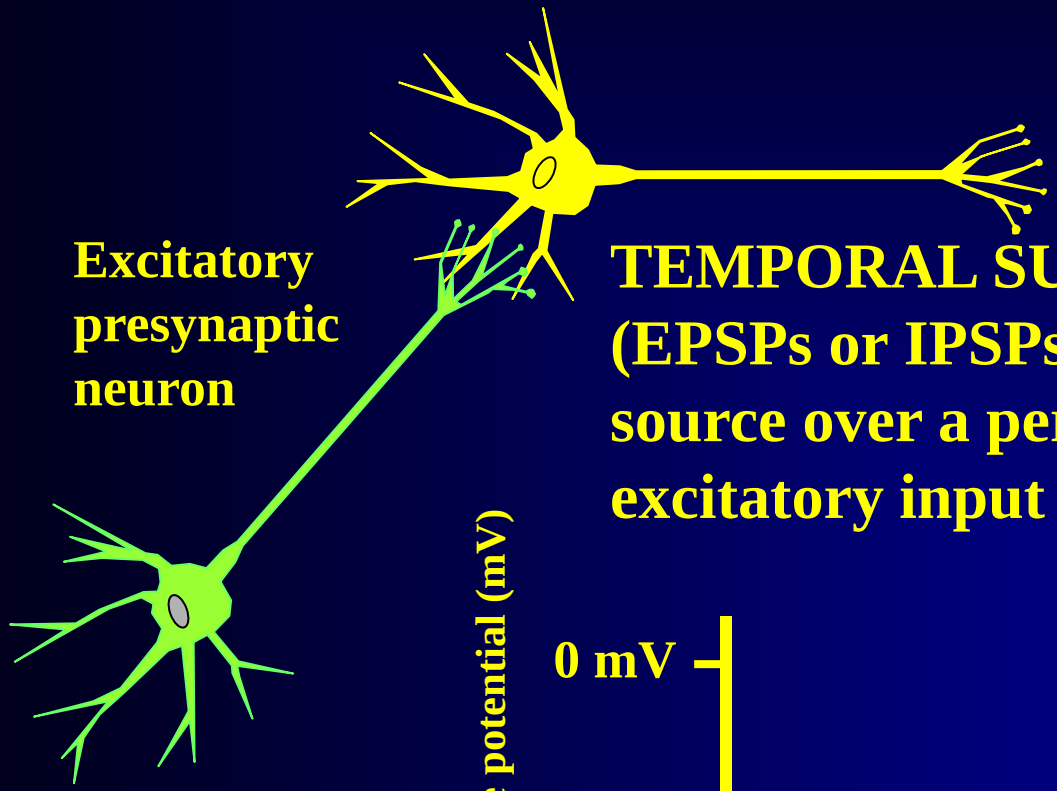
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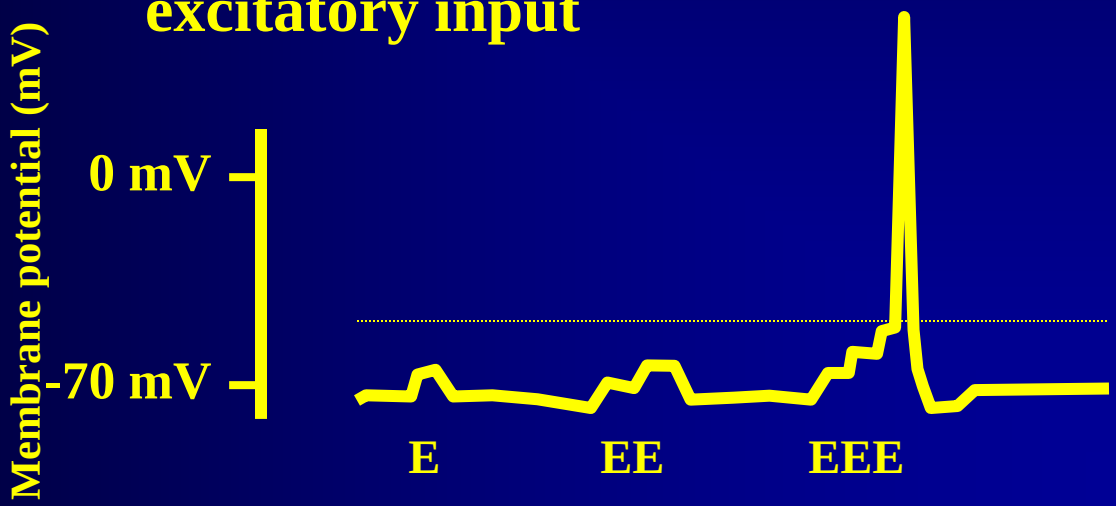


IPSPs can be generated by specific ligands binding receptors on post-synaptic membranes, causing specific K+ or Cl- channels to open; K+ efflux or Cl- influx would hyperpolerize the membrane, driving it away from threshold

SUMMATION



TEMPORAL SUMMATION - neural input (EPSPs or IPSPs) received from a single source over a period of time; e.g., a volley of excitatory input

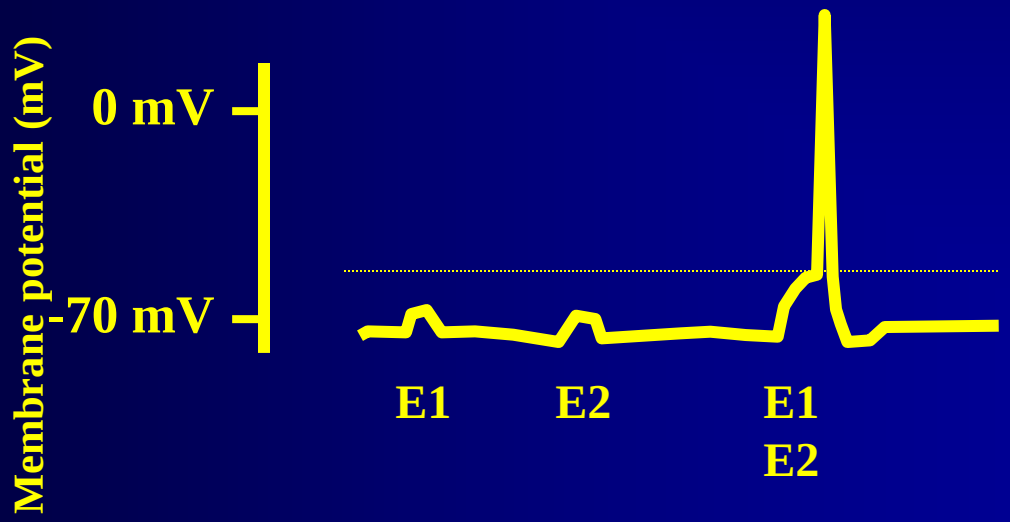


SUMMATION

Excitatory presynaptic neuron (E2)

Excitatory presynaptic neuron (E1)

SPATIAL SUMMATION - neural input (EPSPs or IPSPs) received from multiple sources that arrive at the same time

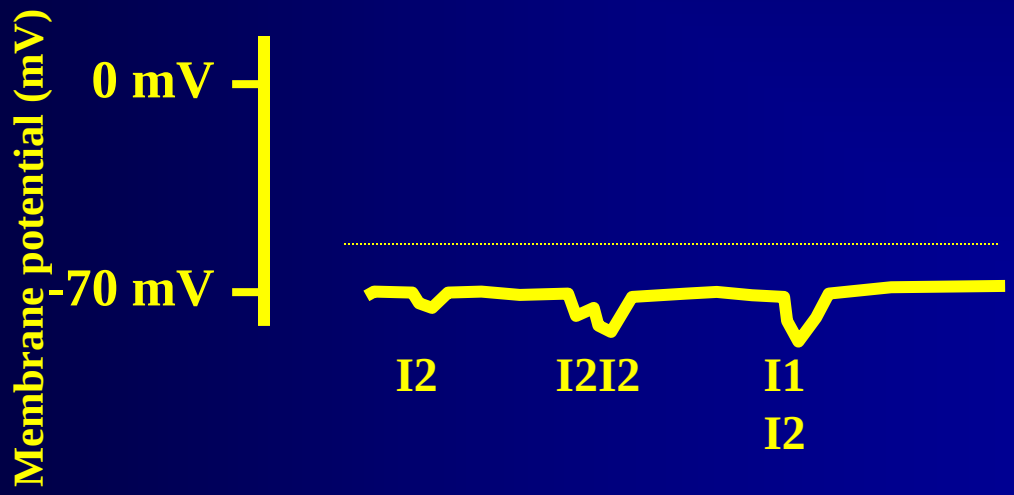


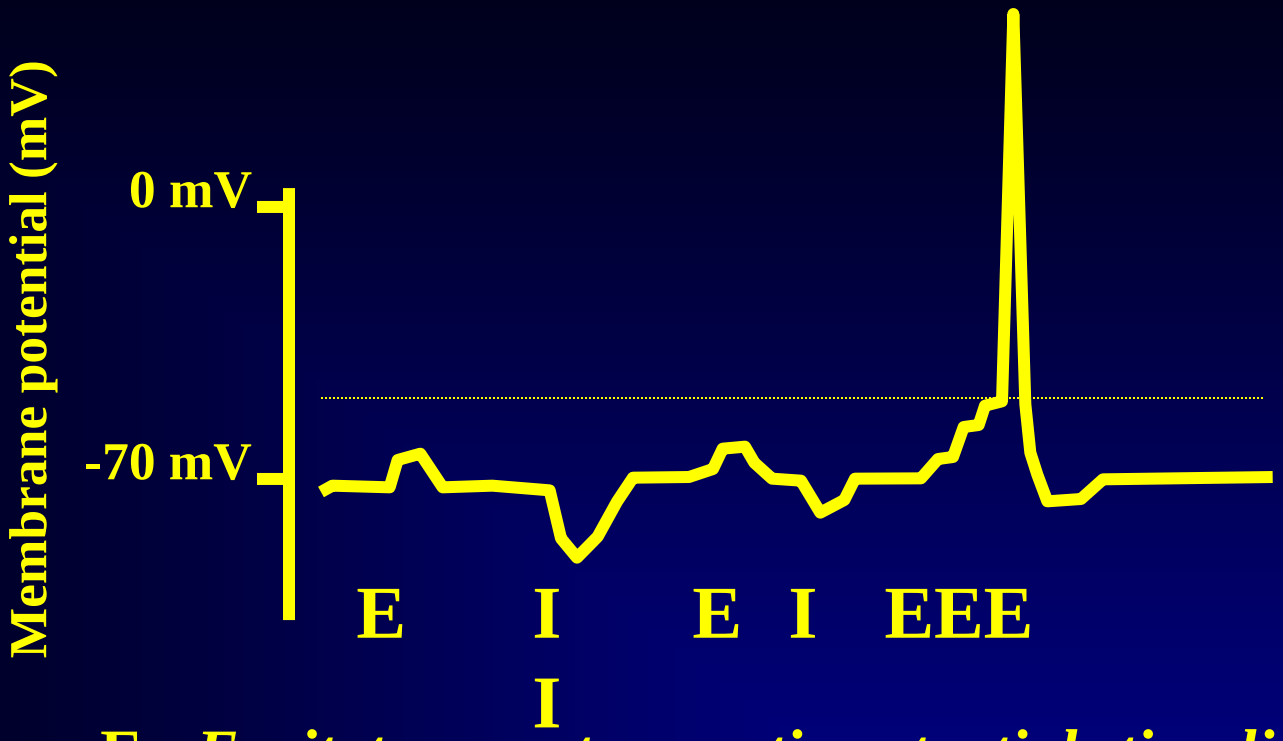
E.g., IPSP SUMMATION

Inhibitory presynaptic neuron (I2)

Inhibitory presynaptic neuron (I1)

TEMPORAL and SPATIAL SUMMATION - IPSPs can also be summed; IPSPs hyper-polarize the membrane away from threshold

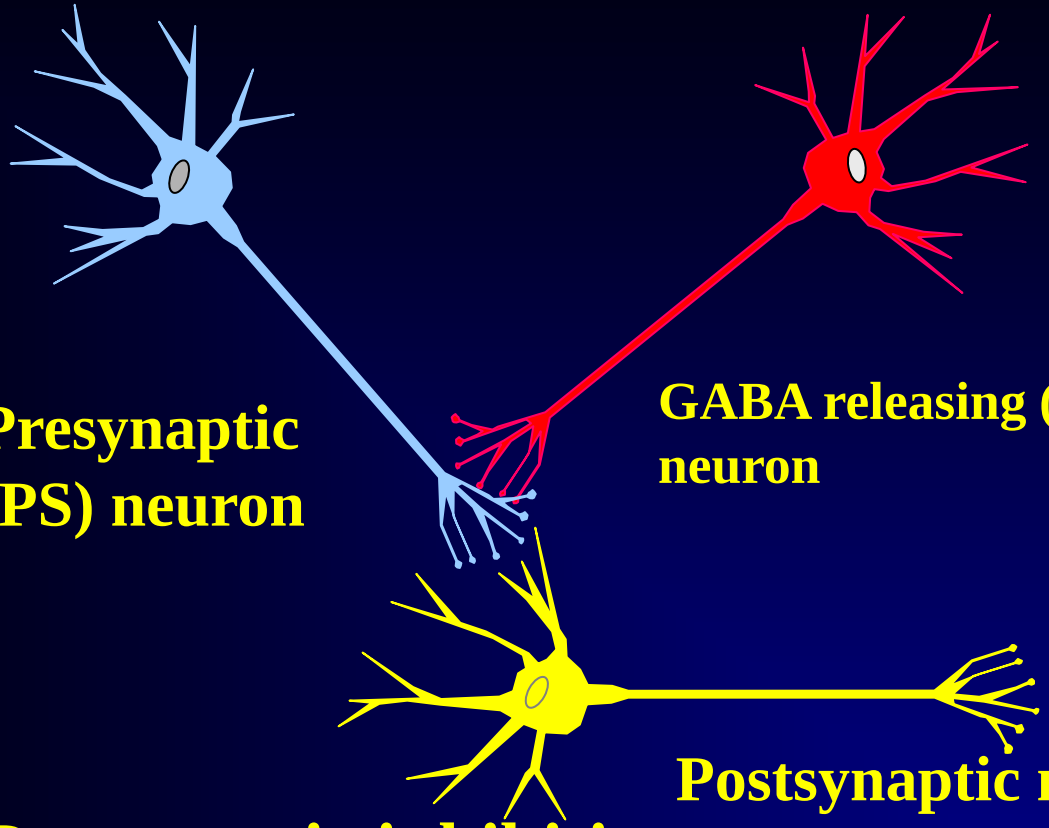




E - *Excitatory post-synaptic potential stimuli*

I - *Inhibitory post-synaptic potential stimuli*

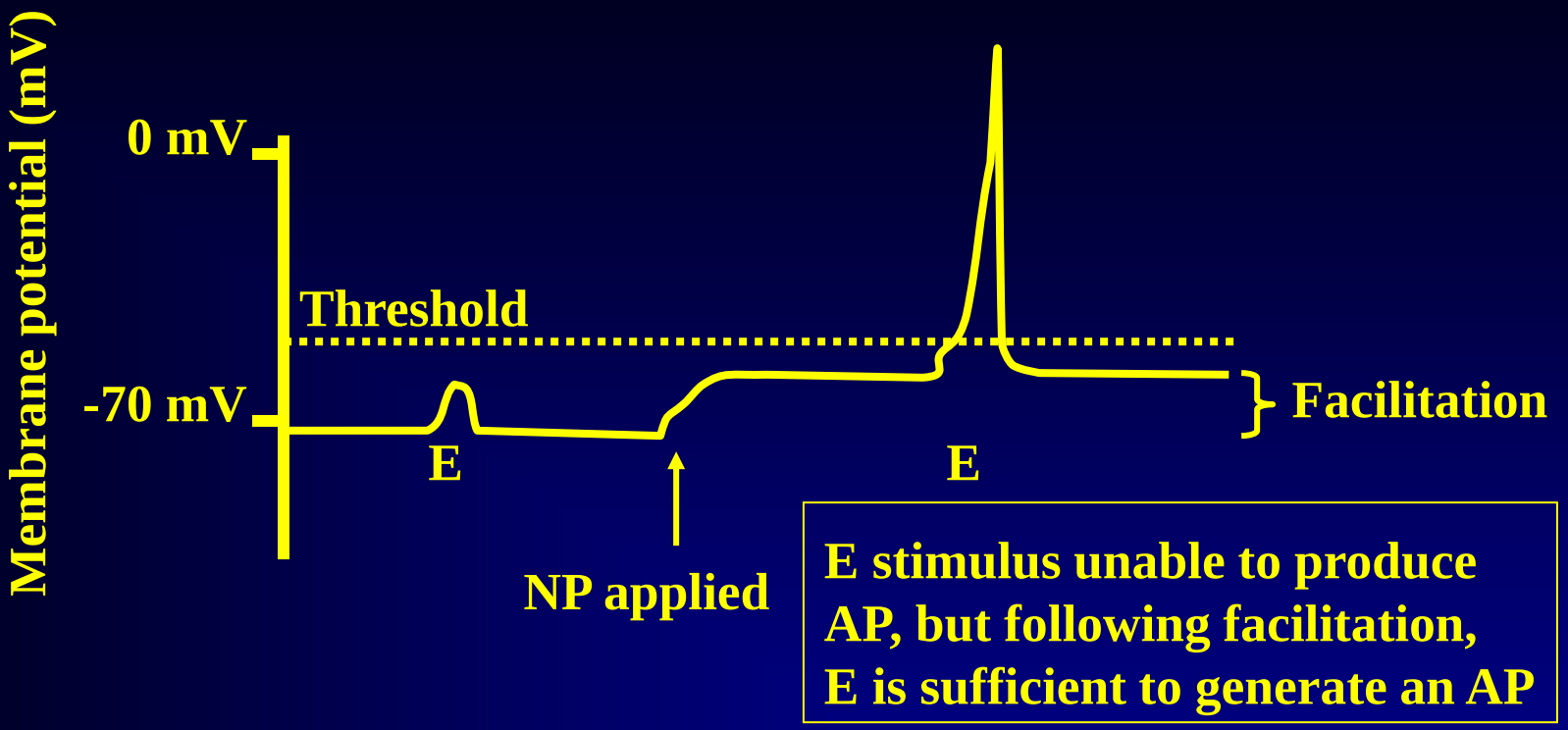
EPSP and IPSP inputs produce graded potentials that are integrated at the axon hillock, determining whether an action potential will result; *remember*, most neurons receive 100 to 200,000 neural inputs, some temporal, some spatial, some excitatory and some inhibitory



Presynaptic inhibition - Presynaptic inhibition neurons release a neuromodulator (usually GABA) at the *nerve terminals* of a presynaptic (PS) neuron; GABA opens Cl- channels on the PS neuron's terminals, preventing neuro-transmitter release from the PS neuron

Inhibition of an excitatory PS neuron = Postsynaptic inhibition

Inhibition of an inhibitory PS neuron = Postsynaptic excitation



Facilitation of neurons - Certain neuropeptides (NP) can regulate the longterm membrane permeability for select ions; this "re-sets" the resting membrane potential (RMP); if the RMP is re-set to a potential that is nearer to threshold, the neuron is said to be "facilitated" and can by excited to generate action potentials (AP) more easily; e.g., learning new material, pain transmission pathways

Defacilitation of neurons